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(54) **NOVEL MODULATORS OF THE
5-HYDROXYTRYPTAMINE RECEPTOR 7
AND THEIR METHOD OF USE**

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(57)

ABSTRACT

Pharmaceutical compositions of the invention comprise
functionalized lactone derivatives having a disease-modify-
ing action in the treatment of diseases associated with
dysregulation of 5-hydroxytryptamine receptor 7 activity.

NOVEL MODULATORS OF THE 5-HYDROXYTRYPTAMINE RECEPTOR 7 AND THEIR METHOD OF USE

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 17/709,847, filed Mar. 31, 2022, which is a continuation application of U.S. patent application Ser. No. 17/015,696, filed on Sep. 9, 2020, which issued as U.S. Pat. No. 11,319,327 on May 3, 2022, which is a continuation application of U.S. patent application Ser. No. 16/349,811, filed on May 14, 2019, which issued as U.S. Pat. No. 10,858,368 on Dec. 8, 2020, which is a 35 U.S.C. § 371 National Stage Application of International Application No. PCT/US2017/061677, filed Nov. 15, 2017, which claims the benefit of U.S. provisional application No. 62/422,344, filed Nov. 15, 2016, which are each herein incorporated by reference in their entireties.

STATEMENT OF FEDERALLY FUNDED RESEARCH

[0002] This invention was made with government support under HHSN-271-2008-00025-C awarded by the National Institutes of Health. The government has certain rights in the invention.

FIELD OF INVENTION

[0003] Embodiments of the invention are directed to novel compounds useful as modulators of 5-hydroxytryptamine receptor 7 (5-HT₇) activity and their method of use. Embodiments are further directed to a novel chemotype useful for the treatment diseases that are associated with dysregulation of 5-hydroxytryptamine receptor 7 activity.

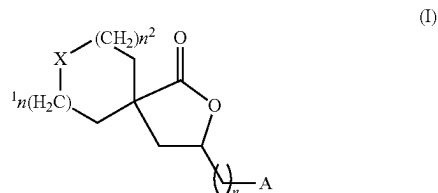
BACKGROUND OF THE INVENTION

[0004] Serotonin was discovered in the late 1940s and is present in both the peripheral and central nervous systems [Physiol. Res, 60 (2011) 15-25; Psychopharmacology 213 (2011) 167-169]. Serotonin or 5-hydroxytryptamine (5-HT) is a monoamine neurotransmitter of the indolalkylamine group that acts at synapses of nerve cells. Seven distinct families of serotonin receptors have been identified and at least 20 subpopulations have been cloned on the basis of sequence similarity, signal transduction coupling and pharmacological characteristics. The seven families of 5-HT receptor are named 5-HT₁, 5-HT₂, 5-HT₃, 5-HT₄, 5-HT₅, 5-HT₆, and 5-HT₇ and each of these receptors in turn has subfamilies or subpopulations. The signal transduction mechanism for all seven families have been studied and it is known that activation of 5-HT₁ and 5-HT₅ receptors causes a decrease in intracellular cAMP whereas activation of 5-HT₂, 5-HT₃, 5-HT₄, 5-HT₆, and 5-HT₇ results in an increase in intracellular IP₃ and DAG. The 5-HT pathways in the brain are important targets for drug development in the area of CNS disorders. The neurotransmitter binds to its a G-protein coupled receptor and is involved in a wide variety of actions including cognition, mood, anxiety, attention, appetite, cardiovascular function, vasoconstriction, sleep (ACS Medicinal Chemistry Letters, 2011, 2, 929-932; Physiological Research, 2011, 60, 15-25), inflammatory bowel disease (IBD), and intestinal inflammation (WO 2012058769, Khan, W. I., et. al. Journal of Immunology,

2013, 190, 4795-4804), epilepsy, seizure disorders (Epilepsy Research (2007) 75, 39), drug addiction, and alcohol addiction (Hauser, S. R. et. al. Frontiers in Neuroscience, 2015, 8, 1-9) among others.

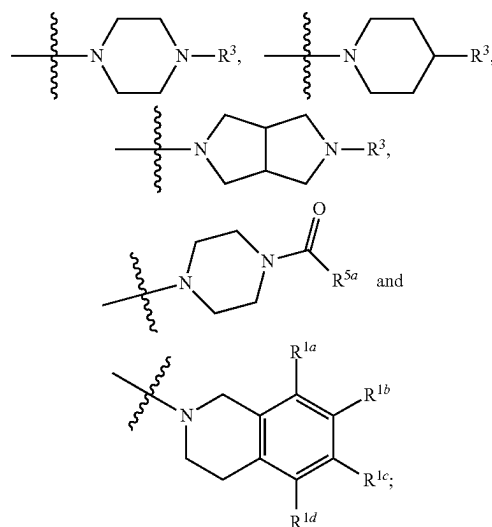
BRIEF SUMMARY OF THE INVENTION

[0005] The present invention is directed toward novel 5-hydroxytryptamine receptor 7 (5-HT₇) activity modulators, compounds of formula (I),



[0006] Including hydrates, solvates, pharmaceutically acceptable salts, prodrugs and complexes thereof, wherein:

[0007] A is selected from a group consisting of

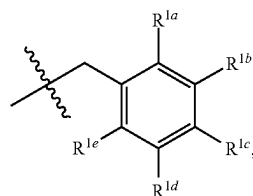


[0008] X is selected from the group consisting of O, S, SO, SO₂, NR;

[0009] n¹ is 0, 1, 2;

[0010] n² is 0, 1, 2;

[0011] R is selected from the group consisting of H, C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, C₃₋₇ cycloalkyl,



COR^2 , CO_2R^{2a} , $\text{CONR}^{2b}\text{R}^{2c}$, $\text{SO}_2\text{NR}^{2b}\text{R}^{2c}$, and SO_2R^{2a} ; R^{1a} , R^{1b} , R^{1c} , R^{1d} , and R^{1e} are at each occurrence independently selected from the group consisting of H, OH, NO_2 , halogen, CN, C_{1-6} linear alkyl, C_{3-7} branched alkyl, C_{3-7} cycloalkyl, C_{1-6} linear alkoxy, C_{3-7} branched alkoxy, C_{3-7} cycloalkoxy, C_{1-6} linear haloalkyl, C_{3-7} branched haloalkyl, C_{1-6} linear haloalkoxy, $-\text{S}(\text{C}_{1-6} \text{ linear alkyl})$, $\text{S}(\text{C}_{3-7} \text{ branched alkyl})$, $-\text{S}(\text{C}_{3-7} \text{ cycloalkyl})$, COR^6 , CO_2R^7 , $\text{CONR}^{8a}\text{R}^{8b}$, $\text{SO}_2\text{NR}^{8a}\text{R}^{8b}$, $\text{NR}^{9a}\text{R}^{9b}$, $\text{NR}^{9a}\text{COR}^{10}$, $\text{NR}^{9a}\text{SO}_2\text{R}^{11}$, and $\text{NR}^{9a}\text{SO}_2\text{NR}^{12a}\text{R}^{12b}$;

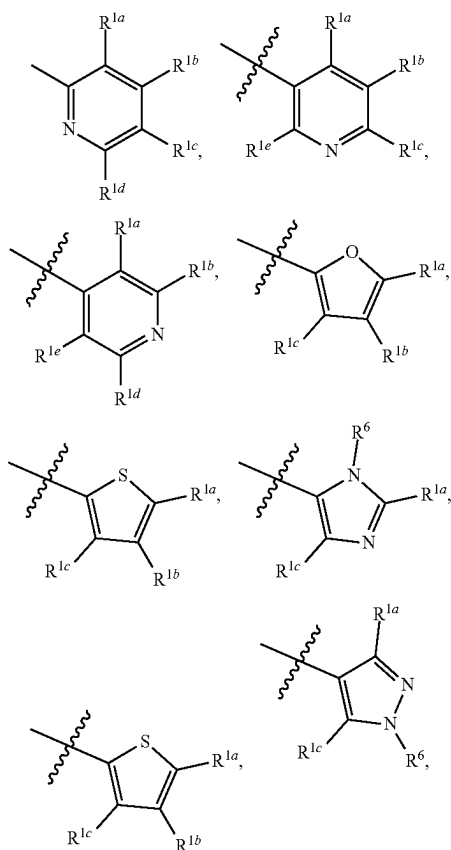
[0012] R^2 is selected from the group consisting of H, C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;

[0013] R^{2a} is selected from the group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;

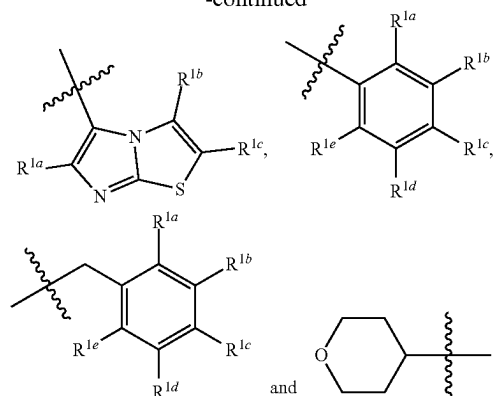
[0014] R^{2b} is selected from the group consisting of H, C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;

[0015] R^{2c} is selected from the group consisting of H, C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;

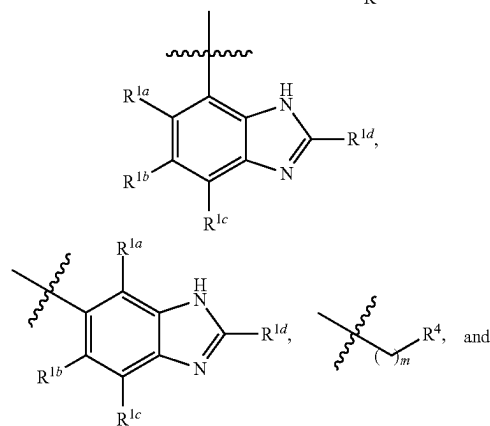
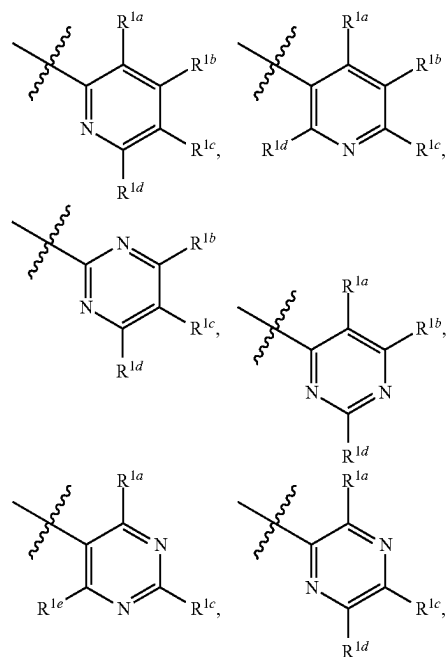
[0016] R^{21} is selected from the group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, C_{3-7} cycloalkyl, C_{1-6} linear haloalkyl, C_{3-7} branched haloalkyl, $-(\text{CH}_2)_q\text{CN}$, $-(\text{CH}_2)_q\text{SO}_2\text{R}^{13}$, $-(\text{CH}_2)_q\text{OR}^{14}$,

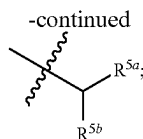


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[0017] R^3 is selected from a group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, C_{3-7} cycloalkyl, optionally substituted aryl,





- [0018]** R⁴ is an optionally substituted aryl;
[0019] R^{5a} and R^{5b} are each independently optionally substituted aryl;
[0020] R⁶ is at each occurrence independently selected from the group consisting of H, C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, and C₃₋₇ cycloalkyl;
[0021] R⁷ is at each occurrence independently selected from the group consisting of C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, and C₃₋₇ cycloalkyl;
[0022] R⁸ is at each occurrence independently selected from the group consisting of H, C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, and C₃₋₇ cycloalkyl;
[0023] R^{8b} is at each occurrence independently selected from the group consisting of H, C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, and C₃₋₇ cycloalkyl;
[0024] R^{9a} is at each occurrence independently selected from the group consisting of H, C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, and C₃₋₇ cycloalkyl;
[0025] R^{9b} is at each occurrence independently selected from the group consisting of H, C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, and C₃₋₇ cycloalkyl;
[0026] R¹⁰ is at each occurrence independently selected from the group consisting of H, C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, and C₃₋₇ cycloalkyl;
[0027] R¹¹ is at each occurrence independently selected from the group consisting of C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, and C₃₋₇ cycloalkyl;
[0028] R^{12a} is at each occurrence independently selected from the group consisting of C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, and C₃₋₇ cycloalkyl;
[0029] R^{12b} is at each occurrence independently selected from the group consisting of C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, and C₃₋₇ cycloalkyl;
[0030] R¹³ is selected from the group consisting of C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, and C₃₋₇ cycloalkyl;
[0031] R¹⁴ is selected from the group consisting of C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, and C₃₋₇ cycloalkyl;
[0032] n is 1, 2, or 3;
[0033] m is 1 or 2;
[0034] and q is 1, 2, or 3;
[0035] The present invention further relates to compositions comprising: an effective amount of one or more compounds according to the present invention and an excipient.
[0036] The present invention also relates to a method for treating or preventing diseases that involve dysregulation of 5-hydroxytryptamine receptor 7 activity, including, for example, circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine, neuropathic pain, peripheral pain, allodynia, thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder, attention deficit/hyperactivity disorder, anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar disorder, inflammatory bowel disease (IBD), intestinal inflammation,

epilepsy, seizure disorders, drug addiction, and alcohol addiction said method comprising administering to a subject an effective amount of a compound or composition according to the present invention.

[0037] The present invention yet further relates to a method for treating or preventing diseases that involve dysregulation of 5-hydroxytryptamine receptor 7 activity, including, for example, circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine, neuropathic pain, peripheral pain, allodynia, thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder, attention deficit/hyperactivity disorder, anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar, disorder inflammatory bowel disease (IBD), intestinal inflammation, epilepsy, seizure disorders, drug addiction, and alcohol addiction wherein said method comprises administering to a subject a composition comprising an effective amount of one or more compounds according to the present invention and an excipient.

[0038] The present invention also relates to a method for treating or preventing diseases or conditions associated with circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine, neuropathic pain, peripheral pain, allodynia, thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder, attention deficit/hyperactivity disorder, anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar disorder, inflammatory bowel disease (IBD), intestinal inflammation, epilepsy, seizure disorders, drug addiction, alcohol addiction and diseases that involve dysregulation of 5-hydroxytryptamine receptor 7 activity. Said methods comprise administering to a subject an effective amount of a compound or composition according to the present invention.

[0039] The present invention yet further relates to a method for treating or preventing diseases or conditions associated with circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine, neuropathic pain, peripheral pain, allodynia, thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder, attention deficit/hyperactivity disorder, anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, and bipolar disorder, inflammatory bowel disease (IBD), intestinal inflammation, epilepsy, seizure disorders, drug addiction, alcohol addiction and diseases that involve dysregulation of 5-hydroxytryptamine receptor 7 activity, wherein said method comprises administering to a subject a composition comprising an effective amount of one or more compounds according to the present invention and an excipient.

[0040] The present invention also relates to a method for treating or preventing diseases or conditions associated with dysregulation of 5-hydroxytryptamine receptor 7 activity. Said methods comprise administering to a subject an effective amount of a compound or composition according to the present invention.

[0041] The present invention yet further relates to a method for treating or preventing diseases or conditions associated with dysregulation of 5-hydroxytryptamine receptor 7 activity, wherein said method comprises administering to a subject a composition comprising an effective amount of one or more compounds according to the present invention and an excipient.

[0042] The present invention further relates to a process for preparing the 5-hydroxytryptamine receptor 7 activity modulators of the present invention.

[0043] These and other objects, features, and advantages will become apparent to those of ordinary skill in the art from a reading of the following detailed description and the appended claims. All percentages, ratios and proportions herein are by weight, unless otherwise specified. All temperatures are in degrees Celsius ($^{\circ}$ C.) unless otherwise specified. All documents cited are in relevant part, incorporated herein by reference; the citation of any document is not to be construed as an admission that it is prior art with respect to the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0044] There is evidence that suggests a role for the 5-HT₇ receptor in a number of medical disorders. 5-HT₇ receptor activity modulators are likely to have a beneficial effect on patients suffering from these disorders. The disorders in which 5-HT₇ dysregulation plays a role and modulation of 5-HT₇ receptor activity by a therapeutic agent may be a viable approach to therapeutic relief include, but are not limited to, circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine (Vanhoenacker, P. et al. Trends in Pharmacological Sciences, 2000, 21, 2, 70-77), neuropathic pain, peripheral pain, allodynia (EP1875899), thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder (WO20100197700) attention deficit/hyperactivity disorder (ADHD) (WO20100069390), anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar disorder (WO20040229874), inflammatory bowel disease (IBD), intestinal inflammation (WO 2012058769, Khan, W. I., et. al. Journal of Immunology, 2013, 190, 4795-4804), epilepsy, seizure disorders (Epilepsy Research (2007) 75, 39), drug addiction, and alcohol addiction (Hauser, S. R. et. al. Frontiers in Neuroscience, 2015, 8, 1-9).

[0045] There is a long felt need for new 5-HT₇ modulators that will provide therapeutic relief from patients suffering from diseases associated with dysregulation of 5-hydroxytryptamine receptor 7 activity. The invention addresses the need to identify novel 5-HT₇ modulators capable of treating disease associated with dysregulation of 5-hydroxytryptamine receptor 7 activity. The present invention addresses the need to develop new therapeutic agents for the treatment and prevention of circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine, neuropathic pain, peripheral pain, allodynia, thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder, attention deficit/hyperactivity disorder, anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, pre-

menstrual dysphonic disorder, seasonal affective disorder, bipolar disorder, inflammatory bowel disease (IBD), intestinal inflammation epilepsy, seizure disorders, drug addiction, and alcohol addiction.

[0046] The 5-hydroxytryptamine receptor 7 activity modulators of the present invention are capable of treating and preventing diseases associated with dysregulation of 5-hydroxytryptamine receptor 7 activity, for example circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine, neuropathic pain, peripheral pain, allodynia, thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder, attention deficit/hyperactivity disorder, anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar disorder, inflammatory bowel disease (IBD), intestinal inflammation, epilepsy, seizure disorders, drug addiction, and alcohol addiction. It has been discovered that the 5-hydroxytryptamine receptor 7 play a role in a number of medical disorders, and therefore, 5-HT₇ receptor activity modulators are likely to have a beneficial effect on patients suffering from these disorders. The disorders in which 5-HT₇ dysregulation plays a role and modulation of 5-HT₇ receptor activity by a therapeutic agent may be a viable approach to therapeutic relief include, but are not limited to, circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine (Vanhoenacker, P. et. al. Trends in Pharmacological Sciences, 2000, 21, 2, 70-77), neuropathic pain, peripheral pain, allodynia (EP1875899), thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder (WO20100197700) attention deficit/hyperactivity disorder (ADHD) (WO20100069390), anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar disorder (WO20040229874), inflammatory bowel disease (IBD), intestinal inflammation (WO 2012058769) epilepsy, seizure disorders (Epilepsy Research (2007) 75, 39), drug addiction, and alcohol addiction (Hauser, S. R. et. al. Frontiers in Neuroscience, 2015, 8, 1-9).

[0047] Without wishing to be limited by theory, it is believed that 5-hydroxytryptamine receptor 7 receptor activity modulators of the present invention can ameliorate, abate, otherwise cause to be controlled, diseases associated with dysregulation of 5-hydroxytryptamine receptor 7 activity. The diseases include, but are not limited to circadian rhythm disorder, depression, schizophrenia, neurogenic inflammation, hypertension, peripheral, vascular diseases, migraine, neuropathic pain, peripheral pain, allodynia, thermoregulation disorder, learning disorder, memory disorder, hippocampal signaling disorder, sleep disorder, attention deficit/hyperactivity disorder, anxiety, avoidant personality disorder, premature ejaculation, eating disorder, premenstrual syndrome, premenstrual dysphonic disorder, seasonal affective disorder, bipolar disorder, inflammatory bowel disease (IBD), intestinal inflammation, epilepsy, seizure disorders, drug addiction, and alcohol addiction.

[0048] Throughout the description, where compositions are described as having, including, or comprising specific components, or where processes are described as having, including, or comprising specific process steps, it is con-

templated that compositions of the present teachings also consist essentially of, or consist of, the recited components, and that the processes of the present teachings also consist essentially of, or consist of, the recited processing steps.

[0049] In the application, where an element or component is said to be included in and/or selected from a list of recited elements or components, it should be understood that the element or component can be any one of the recited elements or components and can be selected from a group consisting of two or more of the recited elements or components.

[0050] The use of the singular herein includes the plural (and vice versa) unless specifically stated otherwise. In addition, where the use of the term “about” is before a quantitative value, the present teachings also include the specific quantitative value itself, unless specifically stated otherwise.

[0051] It should be understood that the order of steps or order for performing certain actions is immaterial so long as the present teachings remain operable. Moreover, two or more steps or actions can be conducted simultaneously.

[0052] As used herein, the term “halogen” shall mean chlorine, bromine, fluorine and iodine.

[0053] As used herein, unless otherwise noted, “alkyl” and/or “aliphatic” whether used alone or as part of a substituent group refers to straight and branched carbon chains having 1 to 20 carbon atoms or any number within this range, for example 1 to 6 carbon atoms or 1 to 4 carbon atoms. Designated numbers of carbon atoms (e.g. C₁₋₆) shall refer independently to the number of carbon atoms in an alkyl moiety or to the alkyl portion of a larger alkyl-containing substituent. Non-limiting examples of alkyl groups include methyl, ethyl, n-propyl, iso-propyl, n-butyl, sec-butyl, iso-butyl, tert-butyl, and the like. Alkyl groups can be optionally substituted. Non-limiting examples of substituted alkyl groups include hydroxymethyl, chloromethyl, trifluoromethyl, aminomethyl, 1-chloroethyl, 2-hydroxyethyl, 1,2-difluoroethyl, 3-carboxypropyl, and the like. In substituent groups with multiple alkyl groups such as (C₁₋₆alkyl)₂amino, the alkyl groups may be the same or different.

[0054] As used herein, the terms “alkenyl” and “alkynyl” groups, whether used alone or as part of a substituent group, refer to straight and branched carbon chains having 2 or more carbon atoms, preferably 2 to 20, wherein an alkenyl chain has at least one double bond in the chain and an alkynyl chain has at least one triple bond in the chain. Alkenyl and alkynyl groups can be optionally substituted. Nonlimiting examples of alkenyl groups include ethenyl, 3-propenyl, 1-propenyl (also 2-methylethenyl), isopropenyl (also 2-methylethen-2-yl), buten-4-yl, and the like. Nonlimiting examples of substituted alkenyl groups include 2-chloroethenyl (also 2-chlorovinyl), 4-hydroxybuten-1-yl, 7-hydroxy-7-methyloct-4-en-2-yl, 7-hydroxy-7-methyloct-3,5-dien-2-yl, and the like. Nonlimiting examples of alkynyl groups include ethynyl, prop-2-ynyl (also propargyl), propyn-1-yl, and 2-methyl-hex-4-yn-1-yl. Nonlimiting examples of substituted alkynyl groups include, 5-hydroxy-5-methylhex-3-ynyl, 6-hydroxy-6-methylhept-3-yn-2-yl, 5-hydroxy-5-ethylhept-3-ynyl, and the like.

[0055] As used herein, “cycloalkyl,” whether used alone or as part of another group, refers to a non-aromatic carbon-containing ring including cyclized alkyl, alkenyl, and alkynyl groups, e.g., having from 3 to 14 ring carbon atoms, preferably from 3 to 7 or 3 to 6 ring carbon atoms, or even

3 to 4 ring carbon atoms, and optionally containing one or more (e.g., 1, 2, or 3) double or triple bond. Cycloalkyl groups can be monocyclic (e.g., cyclohexyl) or polycyclic (e.g., containing fused, bridged, and/or spiro ring systems), wherein the carbon atoms are located inside or outside of the ring system. Any suitable ring position of the cycloalkyl group can be covalently linked to the defined chemical structure. Cycloalkyl rings can be optionally substituted. Nonlimiting examples of cycloalkyl groups include: cyclopropyl, 2-methyl-cyclopropyl, cyclopropenyl, cyclobutyl, 2,3-dihydroxycyclobutyl, cyclobutenyl, cyclopentyl, cyclopentenyl, cyclopentadienyl, cyclohexyl, cyclohexenyl, cycloheptyl, cyclooctanyl, decalanyl, 2,5-dimethylcyclopentyl, 3,5-dichlorocyclohexyl, 4-hydroxycyclohexyl, 3,3,5-trimethylcyclohex-1-yl, octahydropentalenyl, octahydro-1H-indenyl, 3a,4,5,6,7,7a-hexahydro-3H-inden-4-yl, decahydroazulenyl; bicyclo[6.2.0]decanyl, decahydronaphthalenyl, and dodecahydro-1H-fluorenyl. The term “cycloalkyl” also includes carbocyclic rings which are bicyclic hydrocarbon rings, non-limiting examples of which include, bicyclo-[2.1.1]hexanyl, bicyclo[2.2.1]heptanyl, bicyclo[3.1.1]heptanyl, 1,3-dimethyl[2.2.1]heptan-2-yl, bicyclo[2.2.2]octanyl, and bicyclo[3.3.3]undecanyl.

[0056] “Haloalkyl” is intended to include both branched and straight-chain saturated aliphatic hydrocarbon groups having the specified number of carbon atoms, substituted with 1 or more halogen. Haloalkyl groups include perhaloalkyl groups, wherein all hydrogens of an alkyl group have been replaced with halogens (e.g., —CF₃, —CF₂CF₃). Haloalkyl groups can optionally be substituted with one or more substituents in addition to halogen. Examples of haloalkyl groups include, but are not limited to, fluoromethyl, dichloroethyl, trifluoromethyl, trichloromethyl, pentafluoroethyl, and pentachloroethyl groups.

[0057] The term “alkoxy” refers to the group —O-alkyl, wherein the alkyl group is as defined above. Alkoxy groups optionally may be substituted. The term C₃-C₆ cyclic alkoxy refers to a ring containing 3 to 6 carbon atoms and at least one oxygen atom (e.g., tetrahydrofuran, tetrahydro-2H-pyran). C₃-C₆ cyclic alkoxy groups optionally may be substituted.

[0058] The term “haloalkoxy” refers to the group —O-haloalkyl, wherein the haloalkyl group is as defined above. Examples of haloalkoxy groups include, but are not limited to, fluoromethoxy, difluoromethoxy, trifluoromethoxy, and pentafluoroethoxyl.

[0059] The term “aryl,” wherein used alone or as part of another group, is defined herein as an unsaturated, aromatic monocyclic ring of 6 carbon members or to an unsaturated, aromatic polycyclic ring of from 10 to 14 carbon members. Aryl rings can be, for example, phenyl or naphthyl ring each optionally substituted with one or more moieties capable of replacing one or more hydrogen atoms. Non-limiting examples of aryl groups include: phenyl, naphthyl-1-yl, naphthyl-2-yl, 4-fluorophenyl, 2-hydroxyphenyl, 3-methylphenyl, 2-amino-4-fluorophenyl, 2-(N,N-diethylamino)phenyl, 2-cyanophenyl, 2,6-di-tert-butylphenyl, 3-methoxyphenyl, 8-hydroxynaphthyl-2-yl, 4,5-dimethoxynaphthyl-1-yl, and 6-cyano-naphthyl-1-yl. Aryl groups also include, for example, phenyl or naphthyl rings fused with one or more saturated or partially saturated carbon rings (e.g., bicyclo[4.2.0]octa-1,3,5-trienyl, indanyl), which can be substituted at one or more carbon atoms of the aromatic and/or saturated or partially saturated rings.

[0060] The term “arylalkyl” or “aralkyl” refers to the group-alkyl-aryl, where the alkyl and aryl groups are as defined herein. Aralkyl groups of the present invention are optionally substituted. Examples of arylalkyl groups include, for example, benzyl, 1-phenylethyl, 2-phenylethyl, 3-phenylpropyl, 2-phenylpropyl, fluorenylmethyl and the like.

[0061] The terms “heterocyclic” and/or “heterocycle” and/or “heterocyclyl,” whether used alone or as part of another group, are defined herein as one or more ring having from 3 to 20 atoms wherein at least one atom in at least one ring is a heteroatom selected from nitrogen (N), oxygen (O), or sulfur (S), and wherein further the ring that includes the heteroatom is non-aromatic. In heterocycle groups that include 2 or more fused rings, the non-heteroatom bearing ring may be aryl (e.g., indolyl, tetrahydroquinolyl, chromanyl). Exemplary heterocycle groups have from 3 to 14 ring atoms of which from 1 to 5 are heteroatoms independently selected from nitrogen (N), oxygen (O), or sulfur (S). One or more N or S atoms in a heterocycle group can be oxidized. Heterocycle groups can be optionally substituted.

[0062] Non-limiting examples of heterocyclic units having a single ring include: diazirinyl, aziridinyl, urazolyl, azetidyl, pyrazolidinyl, imidazolidinyl, oxazolidinyl, isoxazolinyl, isoxazolyl, thiazolidinyl, isothiazolyl, isothiazolinyl, oxathiazolidinonyl, oxazolidinonyl, hydantoinyl, tetrahydrofuranyl, pyrrolidinyl, morpholinyl, piperazinyl, piperidinyl, dihydropyranyl, tetrahydropyranyl, piperidin-2-onyl (valerolactam), 2,3,4,5-tetrahydro-1H-azepinyl, 2,3-dihydro-1H-indole, and 1,2,3,4-tetrahydro-quinoline. Non-limiting examples of heterocyclic units having 2 or more rings include: hexahydro-1H-pyrroliziny, 3a,4,5,6,7,7a-hexahydro-1H-benzo[d]imidazolyl, 3a,4,5,6,7,7a-hexahydro-1H-indolyl, 1,2,3,4-tetrahydroquinolyl, chromanyl, isochromanyl, indolyl, isoindolyl, and decahydro-1H-cycloocta[b]pyrrolyl.

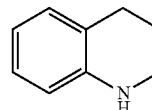
[0063] The term “heteroaryl,” whether used alone or as part of another group, is defined herein as one or more rings having from 5 to 20 atoms wherein at least one atom in at least one ring is a heteroatom chosen from nitrogen (N), oxygen (O), or sulfur (S), and wherein further at least one of the rings that includes a heteroatom is aromatic. In heteroaryl groups that include 2 or more fused rings, the non-heteroatom bearing ring may be a carbocycle (e.g., 6,7-Dihydro-5H-cyclopentapyrimidine) or aryl (e.g., benzofuranyl, benzothiophenyl, indolyl). Exemplary heteroaryl groups have from 5 to 14 ring atoms and contain from 1 to 5 ring heteroatoms independently selected from nitrogen (N), oxygen (O), or sulfur (S). One or more N or S atoms in a heteroaryl group can be oxidized. Heteroaryl groups can be substituted. Non-limiting examples of heteroaryl rings containing a single ring include: 1,2,3,4-tetrazolyl, [1,2,3]triazolyl, [1,2,4]triazolyl, triazinyl, thiazolyl, 1H-imidazolyl, oxazolyl, furanyl, thiophenyl, pyrimidinyl, 2-phenylpyrimidinyl, pyridinyl, 3-methylpyridinyl, and 4-dimethylamino-pyridinyl. Non-limiting examples of heteroaryl rings containing 2 or more fused rings include: benzofuranyl, benzothiophenyl, benzoxazolyl, benzthiazolyl, benztriazolyl, cinnolyl, naphthyridinyl, phenanthridinyl, 7H-puriny, 9H-puriny, 6-amino-9H-puriny, 5H-pyrrolo[3,2-d]pyrimidinyl, 7H-pyrrolo[2,3-d]pyrimidinyl, pyrido[2,3-d]pyrimidinyl, 2-phenylbenzo[d]thiazolyl, 1H-indolyl, 4,5,6,7-tetrahydro-1H-indolyl, quinoxalyl, 5-methylquinoxalyl, quinazolyl, quinolyl, 8-hydroxy-

quinolyl, 1H-benzo[d]imidazol-2(3H)-onyl, 1H-benzo[d]imidazolyl, and isoquinolyl.

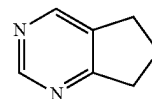
[0064] One non-limiting example of a heteroaryl group as described above is C₁-C₅ heteroaryl, which has 1 to 5 carbon ring atoms and at least one additional ring atom that is a heteroatom (preferably 1 to 4 additional ring atoms that are heteroatoms) independently selected from nitrogen (N), oxygen (O), or sulfur (S). Examples of C₁-C₅ heteroaryl include, but are not limited to, triazinyl, thiazol-2-yl, thiazol-4-yl, imidazol-1-yl, 1H-imidazol-2-yl, 1H-imidazol-4-yl, isoxazolin-5-yl, furan-2-yl, furan-3-yl, thiophen-2-yl, thiophen-4-yl, pyrimidin-2-yl, pyrimidin-4-yl, pyrimidin-5-yl, pyridin-2-yl, pyridin-3-yl, and pyridin-4-yl.

[0065] Unless otherwise noted, when two substituents are taken together to form a ring having a specified number of ring atoms (e.g., R² and R³ taken together with the nitrogen (N) to which they are attached to form a ring having from 3 to 7 ring members), the ring can have carbon atoms and optionally one or more (e.g., 1 to 3) additional heteroatoms independently selected from nitrogen (N), oxygen (O), or sulfur (S). The ring can be saturated or partially saturated and can be optionally substituted.

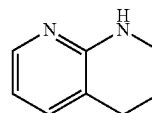
[0066] For the purposes of the present invention fused ring units, as well as spirocyclic rings, bicyclic rings and the like, which comprise a single heteroatom will be considered to belong to the cyclic family corresponding to the heteroatom containing ring. For example, 1,2,3,4-tetrahydroquinoline having the formula:



is, for the purposes of the present invention, considered a heterocyclic unit. 6,7-Dihydro-5H-cyclopentapyrimidine having the formula:



is, for the purposes of the present invention, considered a heteroaryl unit. When a fused ring unit contains heteroatoms in both a saturated and an aryl ring, the aryl ring will predominate and determine the type of category to which the ring is assigned. For example, 1,2,3,4-tetrahydro-[1,8]naphthyridine having the formula:



is, for the purposes of the present invention, considered a heteroaryl unit.

[0067] Whenever a term or either of their prefix roots appear in a name of a substituent the name is to be interpreted as including those limitations provided herein.

For example, whenever the term “alkyl” or “aryl” or either of their prefix roots appear in a name of a substituent (e.g., arylalkyl, alkylamino) the name is to be interpreted as including those limitations given above for “alkyl” and “aryl.”

[0068] The term “substituted” is used throughout the specification. The term “substituted” is defined herein as a moiety, whether acyclic or cyclic, which has one or more hydrogen atoms replaced by a substituent or several (e.g., 1 to 10) substituents as defined herein below. The substituents are capable of replacing one or two hydrogen atoms of a single moiety at a time. In addition, these substituents can replace two hydrogen atoms on two adjacent carbons to form said substituent, new moiety or unit. For example, a substituted unit that requires a single hydrogen atom replacement includes halogen, hydroxyl, and the like. A two hydrogen atom replacement includes carbonyl, oximino, and the like. A two hydrogen atom replacement from adjacent carbon atoms includes epoxy, and the like. The term “substituted” is used throughout the present specification to indicate that a moiety can have one or more of the hydrogen atoms replaced by a substituent. When a moiety is described as “substituted” any number of the hydrogen atoms may be replaced. For example, difluoromethyl is a substituted C_1 alkyl; trifluoromethyl is a substituted C_1 alkyl; 4-hydroxyphenyl is a substituted aromatic ring; (N,N-dimethyl-5-amino)octanyl is a substituted C_8 alkyl; 3-guanidinopropyl is a substituted C_3 alkyl; and 2-carboxypyridinyl is a substituted heteroaryl.

[0069] The variable groups defined herein, e.g., alkyl, alkenyl, alkynyl, cycloalkyl, alkoxy, aryloxy, aryl, heterocycle and heteroaryl groups defined herein, whether used alone or as part of another group, can be optionally substituted. Optionally substituted groups will be so indicated.

[0070] The following are non-limiting examples of substituents which can substitute for hydrogen atoms on a moiety: halogen (chlorine (C_1), bromine (Br), fluorine (F) and iodine(I)), $-\text{CN}$, $-\text{NO}_2$, oxo ($=\text{O}$), $-\text{OR}^{15}$, $-\text{SR}^{15}$, $-\text{N}(\text{R}^{15})_2$, $-\text{NR}^{15}\text{C}(\text{O})\text{R}^{15}$, $-\text{SO}_2\text{R}^{15}$, $-\text{SO}_2\text{OR}^{15}$, $-\text{SO}_2\text{N}(\text{R}^{15})_2$, $-\text{C}(\text{O})\text{R}^{15}$, $-\text{C}(\text{O})\text{OR}^{15}$, $-\text{C}(\text{O})\text{N}(\text{R}^{15})_2$, C_{1-6} alkyl, C_{1-6} haloalkyl, C_{1-6} alkoxy, C_{2-8} alkenyl, C_{2-8} alkynyl, C_{3-14} cycloalkyl, aryl, heterocycle, or heteroaryl, wherein each of the alkyl, haloalkyl, alkenyl, alkynyl, alkoxy, cycloalkyl, aryl, heterocycle, and heteroaryl groups is optionally substituted with 1-10 (e.g., 1-6 or 1-4) groups selected independently from halogen, $-\text{CN}$, $-\text{NO}_2$, oxo, and R^{15} , wherein R^{15} , at each occurrence, independently is hydrogen, $-\text{OR}^{16}$, $-\text{SR}^{16}$, $-\text{C}(\text{O})\text{R}^{16}$, $-\text{C}(\text{O})\text{OR}^{16}$, $-\text{C}(\text{O})\text{N}(\text{R}^{16})_2$, $-\text{SO}_2\text{R}^{16}$, $-\text{S}(\text{O})_{20}\text{R}^{16}$, $-\text{N}(\text{R}^{16})_2$, $-\text{NR}^{16}\text{C}(\text{O})\text{R}^{16}$, C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-8} alkenyl, C_{2-8} alkynyl, cycloalkyl (e.g., C_{3-6} cycloalkyl), aryl, heterocycle, or heteroaryl, or two R^{15} units taken together with the atom(s) to which they are bound form an optionally substituted carbocycle or heterocycle wherein said carbocycle or heterocycle has 3 to 7 ring atoms; wherein R^{16} , at each occurrence, independently is hydrogen, C_{1-6} alkyl, C_{1-6} haloalkyl, C_{2-8} alkenyl, C_{2-8} alkynyl, cycloalkyl (e.g., C_{3-6} cycloalkyl), aryl, heterocycle, or heteroaryl, or two R^{16} units taken together with the atom(s) to which they are bound form an optionally substituted carbocycle or heterocycle wherein said carbocycle or heterocycle preferably has 3 to 7 ring atoms.

[0071] In some embodiments, the substituents are selected from

- [0072]** i) $-\text{OR}^{17}$; for example, $-\text{OH}$, $-\text{OCH}_3$, $-\text{OCH}_2\text{CH}_3$, $-\text{OCH}_2\text{CH}_2\text{CH}_3$;
- [0073]** ii) $-\text{C}(\text{O})\text{R}^{17}$; for example, $-\text{COCH}_3$, $-\text{COCH}_2\text{CH}_3$, $-\text{COCH}_2\text{CH}_2\text{CH}_3$;
- [0074]** iii) $-\text{C}(\text{O})\text{OR}^{17}$; for example, $-\text{CO}_2\text{CH}_3$, $-\text{CO}_2\text{CH}_2\text{CH}_3$, $-\text{CO}_2\text{CH}_2\text{CH}_2\text{CH}_3$;
- [0075]** iv) $-\text{C}(\text{O})\text{N}(\text{R}^{17})_2$; for example, $-\text{CONH}_2$, $-\text{CONHCH}_3$, $-\text{CON}(\text{CH}_3)_2$;
- [0076]** v) $-\text{N}(\text{R}^{17})_2$; for example, $-\text{NH}_2$, $-\text{NHCH}_3$, $-\text{N}(\text{CH}_3)_2$, $-\text{NH}(\text{CH}_2\text{CH}_3)$;
- [0077]** vi) halogen: $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, and $-\text{I}$;
- [0078]** vii) $-\text{CH}_e\text{X}_g$; wherein X is halogen, m is from 0 to 2, $e+g=3$; for example, $-\text{CH}_2\text{F}$, $-\text{CHF}_2$, $-\text{CF}_3$, $-\text{CCl}_3$, or $-\text{CBr}_3$;
- [0079]** viii) $-\text{SO}_2\text{R}^{17}$; for example, $-\text{SO}_2\text{H}$, $-\text{SO}_2\text{CH}_3$; $-\text{SO}_2\text{C}_6\text{H}_5$;
- [0080]** ix) C_1 - C_6 linear, branched, or cyclic alkyl;
- [0081]** x) Cyano
- [0082]** xi) Nitro;
- [0083]** xii) $\text{N}(\text{R}^{17})\text{C}(\text{O})\text{R}^{17}$;
- [0084]** xiii) Oxo ($=\text{O}$);
- [0085]** xiv) Heterocycle; and
- [0086]** xv) Heteroaryl.

wherein each R^{17} is independently hydrogen, optionally substituted C_1 - C_6 linear or branched alkyl (e.g., optionally substituted C_1 - C_4 linear or branched alkyl), or optionally substituted C_3 - C_6 cycloalkyl (e.g. optionally substituted C_3 - C_4 cycloalkyl); or two R^{17} units can be taken together to form a ring comprising 3-7 ring atoms. In certain aspects, each R^{17} is independently hydrogen, C_1 - C_6 linear or branched alkyl optionally substituted with halogen or C_3 - C_6 cycloalkyl or C_3 - C_6 cycloalkyl.

[0087] At various places in the present specification, substituents of compounds are disclosed in groups or in ranges. It is specifically intended that the description include each and every individual subcombination of the members of such groups and ranges. For example, the term “ C_{1-6} alkyl” is specifically intended to individually disclose C_1 , C_2 , C_3 , C_4 , C_5 , C_6 , C_1 - C_6 , C_1 - C_5 , C_1 - C_4 , C_1 - C_3 , C_1 - C_2 , C_2 - C_6 , C_2 - C_5 , C_2 - C_4 , C_2 - C_3 , C_3 - C_6 , C_3 - C_5 , C_3 - C_4 , C_4 - C_6 , C_4 - C_5 , and C_5 - C_6 , alkyl.

[0088] For the purposes of the present invention the terms “compound,” “analog,” and “composition of matter” stand equally well for the 5-hydroxytryptamine receptor 7 activity modulators described herein, including all enantiomeric forms, diastereomeric forms, salts, and the like, and the terms “compound,” “analog,” and “composition of matter” are used interchangeably throughout the present specification.

[0089] Compounds described herein can contain an asymmetric atom (also referred as a chiral center), and some of the compounds can contain one or more asymmetric atoms or centers, which can thus give rise to optical isomers (enantiomers) and diastereomers. The present teachings and compounds disclosed herein include such enantiomers and diastereomers, as well as the racemic and resolved, enantiomerically pure R and S stereoisomers, as well as other mixtures of the R and S stereoisomers and pharmaceutically acceptable salts thereof. Optical isomers can be obtained in pure form by standard procedures known to those skilled in the art, which include, but are not limited to, diastereomeric salt formation, kinetic resolution, and asymmetric synthesis. The present teachings also encompass cis and trans isomers of compounds containing alkenyl moieties (e.g., alkenes and

imines). It is also understood that the present teachings encompass all possible regioisomers, and mixtures thereof, which can be obtained in pure form by standard separation procedures known to those skilled in the art, and include, but are not limited to, column chromatography, thin-layer chromatography, and high-performance liquid chromatography.

[0090] Pharmaceutically acceptable salts of compounds of the present teachings, which can have an acidic moiety, can be formed using organic and inorganic bases. Both mono and polyanionic salts are contemplated, depending on the number of acidic hydrogens available for deprotonation. Suitable salts formed with bases include metal salts, such as alkali metal or alkaline earth metal salts, for example sodium, potassium, or magnesium salts; ammonia salts and organic amine salts, such as those formed with morpholine, thiomorpholine, piperidine, pyrrolidine, a mono-, di- or tri-lower alkylamine (e.g., ethyl-, tert-butyl-, diethyl-, diisopropyl-, triethyl-, tributyl- or dimethylpropylamine), or a mono-, di-, or trihydroxy lower alkylamine (e.g., mono-, di- or triethanolamine). Specific non-limiting examples of inorganic bases include NaHCO_3 , Na_2CO_3 , KHCO_3 , K_2CO_3 , Cs_2CO_3 , LiOH , NaOH , KOH , NaH_2PO_4 , Na_2HPO_4 , and Na_3PO_4 . Internal salts also can be formed. Similarly, when a compound disclosed herein contains a basic moiety, salts can be formed using organic and inorganic acids. For example, salts can be formed from the following acids: acetic, propionic, lactic, benzenesulfonic, benzoic, camphorsulfonic, citric, tartaric, succinic, dichloroacetic, ethenesulfonic, formic, fumaric, gluconic, glutamic, hippuric, hydrobromic, hydrochloric, isethionic, lactic, maleic, malic, malonic, mandelic, methanesulfonic, mucic, naphthalenesulfonic, nitric, oxalic, pantoic, pantothenic, phosphoric, phthalic, propionic, succinic, sulfuric, tartaric, toluenesulfonic, and camphorsulfonic as well as other known pharmaceutically acceptable acids.

[0091] When any variable occurs more than one time in any constituent or in any formula, its definition in each occurrence is independent of its definition at every other occurrence (e.g., in $\text{N}(\text{R}^9)_2$, each R^9 may be the same or different than the other). Combinations of substituents and/or variables are permissible only if such combinations result in stable compounds.

[0092] The terms “treat” and “treating” and “treatment” as used herein, refer to partially or completely alleviating, inhibiting, ameliorating and/or relieving a condition from which a patient is suspected to suffer.

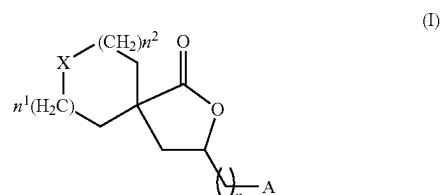
[0093] As used herein, “therapeutically effective” and “effective dose” refer to a substance or an amount that elicits a desirable biological activity or effect.

[0094] Except when noted, the terms “subject” or “patient” are used interchangeably and refer to mammals such as human patients and non-human primates, as well as experimental animals such as rabbits, rats, and mice, and other animals. Accordingly, the term “subject” or “patient” as used herein means any mammalian patient or subject to which the compounds of the invention can be administered. In an exemplary embodiment of the present invention, to identify subject patients for treatment according to the methods of the invention, accepted screening methods are employed to determine risk factors associated with a targeted or suspected disease or condition or to determine the status of an existing disease or condition in a subject. These screening methods include, for example, conventional work-ups to determine risk factors that may be associated with the

targeted or suspected disease or condition. These and other routine methods allow the clinician to select patients in need of therapy using the methods and compounds of the present invention.

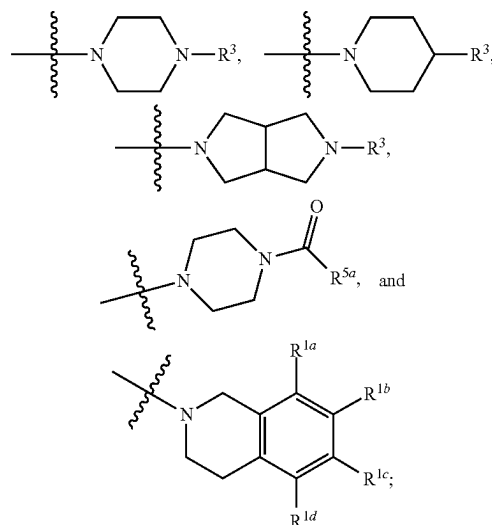
The 5-Hydroxytryptamine Receptor 7 Activity Modulators

[0095] The 5-hydroxytryptamine receptor 7 activity modulators of the present invention include all enantiomeric and diastereomeric forms thereof having the formula



Including hydrates, solvates, pharmaceutically acceptable salts, prodrugs and complexes thereof, wherein:

[0096] A is selected from a group consisting of

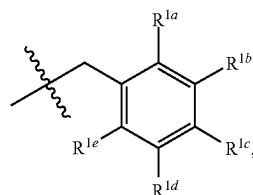


[0097] X is selected from the group consisting of O, S, SO, SO_2 , NR;

[0098] n^1 is 0, 1, 2;

[0099] n^2 is 0, 1, 2;

[0100] R is selected from the group consisting of H, C_{1-6} linear alkyl, C_{3-7} branched alkyl, C_{3-7} cycloalkyl,



COR^2 , CO_2R^{2a} , $\text{CONR}^{2b}\text{R}^{2c}$, $\text{SO}_2\text{NR}^{2b}\text{R}^{2c}$, and SO_2R^{2d} ;

[0101] R^{1a} , R^{1b} , R^{1c} , R^{1d} , and R^{1e} are at each occurrence independently selected from the group consisting of H, OH, NO_2 , halogen, CN, C_{1-6} linear alkyl, C_{3-7} branched alkyl, C_{3-7} cycloalkyl, C_{1-6} linear alkoxy, C_{3-7} branched alkoxy, C_{3-7} cycloalkoxy, C_{1-6} linear haloalkyl, C_{3-7} branched haloalkyl, C_{1-6} linear haloalkoxy, $-\text{S}(\text{C}_{1-6} \text{ linear alkyl})$, $\text{S}(\text{C}_{3-7} \text{ branched alkyl})$, $-\text{S}(\text{C}_{3-7} \text{ cycloalkyl})$, COR^6 , CO_2R^7 , $\text{CONR}^{8a}\text{R}^{8b}$, $\text{SO}_2\text{NR}^{8a}\text{R}^{8b}$, $\text{NR}^{9a}\text{R}^{9b}$, $\text{NR}^{9a}\text{COR}^{10}$, $\text{NR}^{9a}\text{SO}_2\text{R}^{11}$, and $\text{NR}^{9a}\text{SO}_2\text{NR}^{12a}\text{R}^{12b}$;

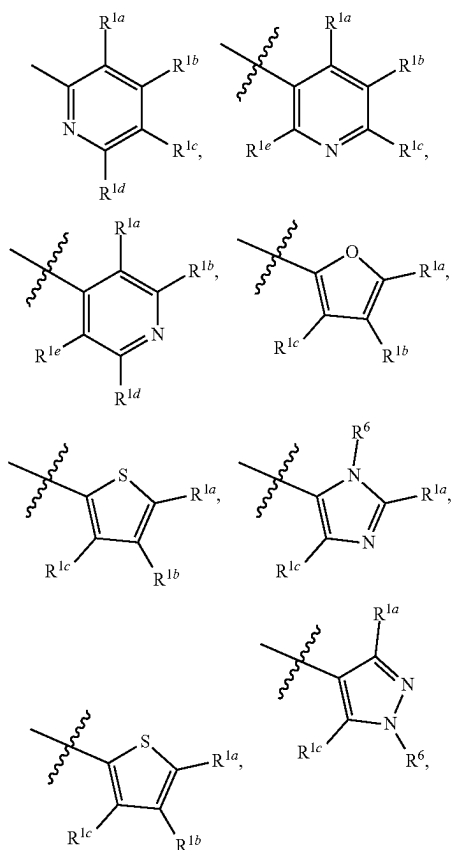
[0102] R^2 is selected from the group consisting of H, C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;

[0103] R^{2a} is selected from the group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;

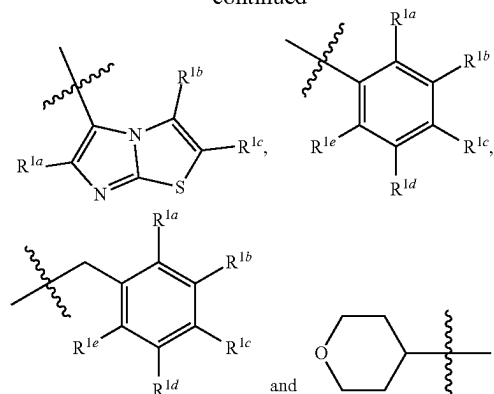
[0104] R^{2b} is selected from the group consisting of H, C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;

[0105] R^{2c} is selected from the group consisting of H, C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;

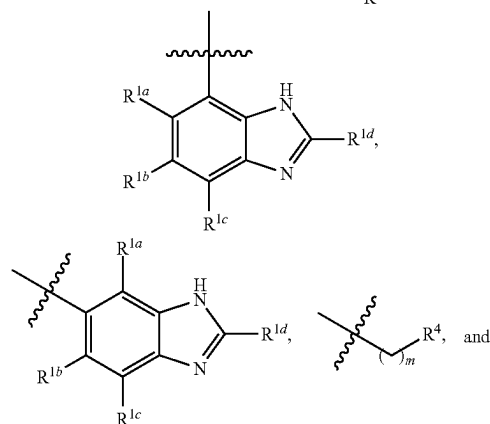
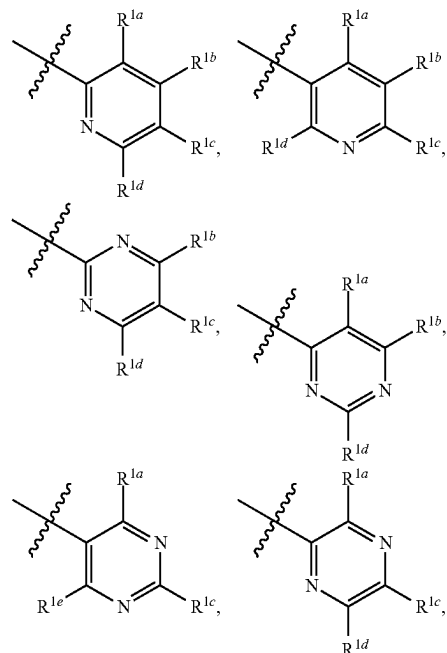
[0106] R^{2d} is selected from the group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, C_{3-7} cycloalkyl, C_{1-6} linear haloalkyl, C_{3-7} branched haloalkyl, $-(\text{CH}_2)_q\text{CN}$, $-(\text{CH}_2)_q\text{SO}_2\text{R}^{13}$, $-(\text{CH}_2)_q\text{OR}^{14}$,

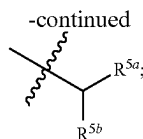


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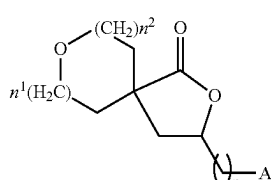


[0107] R^3 is selected from a group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, C_{3-7} cycloalkyl, optionally substituted aryl,





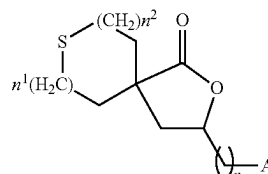
- [0108] R^4 is an optionally substituted aryl;
- [0109] R^{5a} and R^{5b} are each independently optionally substituted aryl;
- [0110] R^6 is at each occurrence independently selected from the group consisting of H, C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;
- [0111] R^7 is at each occurrence independently selected from the group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;
- [0112] R^{8a} is at each occurrence independently selected from the group consisting of H, C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;
- [0113] R^{8b} is at each occurrence independently selected from the group consisting of H, C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;
- [0114] R^{9a} is at each occurrence independently selected from the group consisting of H, C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;
- [0115] R^{9b} is at each occurrence independently selected from the group consisting of H, C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;
- [0116] R^{10} is at each occurrence independently selected from the group consisting of H, C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;
- [0117] R^{11} is at each occurrence independently selected from the group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;
- [0118] R^{12a} is at each occurrence independently selected from the group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;
- [0119] R^{12b} is at each occurrence independently selected from the group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;
- [0120] R^{13} is selected from the group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;
- [0121] R^{14} is selected from the group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;
- [0122] n is 1, 2, or 3;
- [0123] m is 1 or 2;
- [0124] and q is 1, 2, or 3;
- [0125] The embodiments of the present invention include compounds having formula (II):



(II)

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

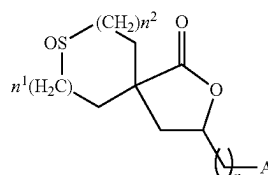
- [0126] The embodiments of the present invention include compounds having formula (III):



(III)

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

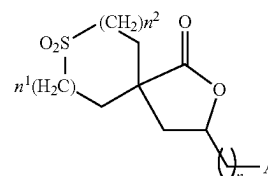
- [0127] The embodiments of the present invention include compounds having formula (IV):



(IV)

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

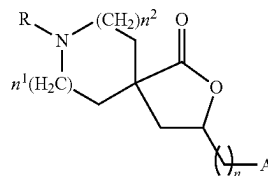
- [0128] The embodiments of the present invention include compounds having formula (V):



(V)

Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

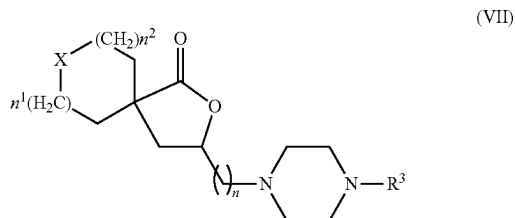
- [0129] The embodiments of the present invention include compounds having formula (VI):



(VI)

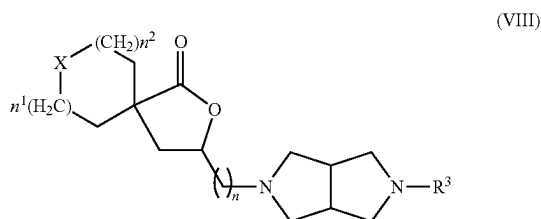
Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0130] The embodiments of the present invention include compounds having formula (VII):



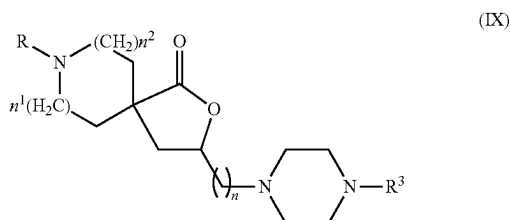
Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0131] The embodiments of the present invention include compounds having formula (VIII):



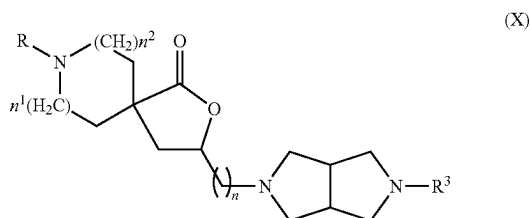
Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0132] The embodiments of the present invention include compounds having formula (IX):



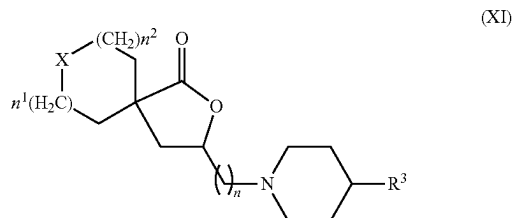
Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0133] The embodiments of the present invention include compounds having formula (X):



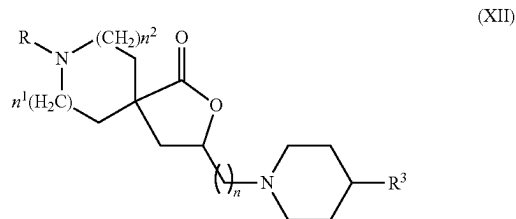
Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0134] The embodiments of the present invention include compounds having formula (XI):



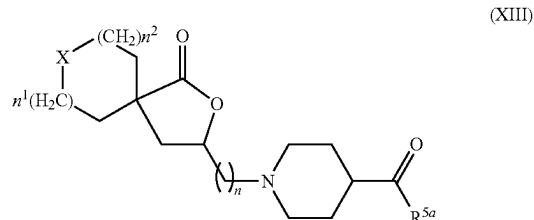
Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0135] The embodiments of the present invention include compounds having formula (XII):



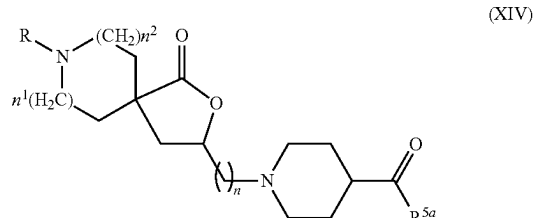
Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0136] The embodiments of the present invention include compounds having formula (XIII):



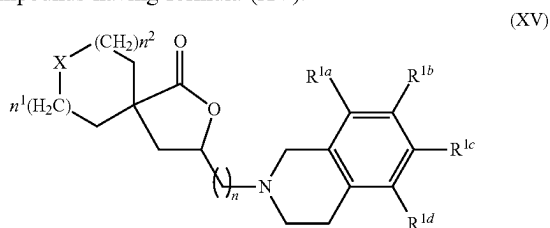
Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0137] The embodiments of the present invention include compounds having formula (XIV):



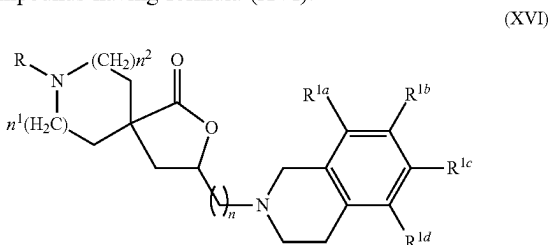
Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0138] The embodiments of the present invention include compounds having formula (XV):



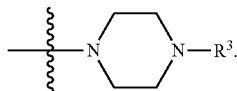
Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

[0139] The embodiments of the present invention include compounds having formula (XVI):

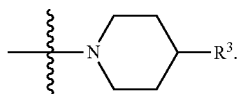


Including hydrates, solvates, enantiomers, diastereomers, pharmaceutically acceptable salts, and complexes thereof.

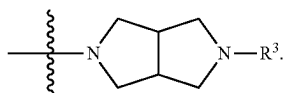
[0140] In some embodiments A is



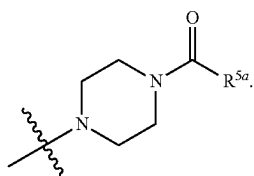
[0141] In some embodiments A is



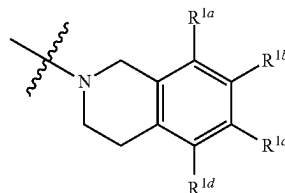
[0142] In some embodiments A is



[0143] In some embodiments A is



[0144] In some embodiments A is



[0145] In some embodiments X is O.

[0146] In some embodiments X is S.

[0147] In some embodiments X is SO.

[0148] In some embodiments X is SO₂.

[0149] In some embodiments X is NR.

[0150] In some embodiments n¹ is 0.

[0151] In some embodiments n¹ is 1.

[0152] In some embodiments n¹ is 2.

[0153] In some embodiments n¹ is 0.

[0154] In some embodiments n² is 1.

[0155] In some embodiments n² is 2.

[0156] In some embodiments n² is H.

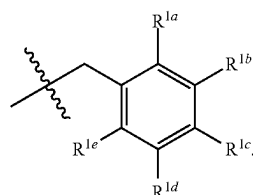
[0157] In some embodiments R is H.

[0158] In some embodiments R is C₁₋₆ linear alkyl.

[0159] In some embodiments R is C₃₋₇ branched alkyl.

[0160] In some embodiments R is C₃₋₇ cycloalkyl.

[0161] In some embodiments R is,



[0162] In some embodiments R is COR².

[0163] In some embodiments R is CO₂R^{2a}.

[0164] In some embodiments R is CONR^{2b}R^{2c}.

[0165] In some embodiments R is SO₂NR^{2b}R^{2c}.

[0166] In some embodiments R is SO₂R^{2d}.

[0167] In some embodiments R^{1a} is H.

[0168] In some embodiments R^{1a} is OH.

[0169] In some embodiments R^{1a} is NO₂.

[0170] In some embodiments R^{1a} is halogen.

[0171] In some embodiments R^{1a} is CN.

[0172] In some embodiments R^{1a} is C₁₋₆ linear alkyl.

[0173] In some embodiments R^{1a} is C₃₋₇ branched alkyl.

[0174] In some embodiments R^{1a} is C₃₋₇ cycloalkyl.

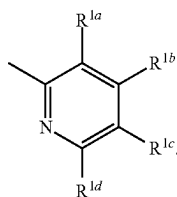
[0175] In some embodiments R^{1a} is C₁₋₆ linear alkoxy.

[0176] In some embodiments R^{1a} is C₃₋₇ branched alkoxy.

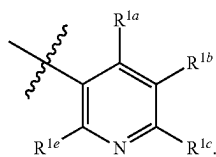
[0177] In some embodiments R^{1a} is C₃₋₇ cycloalkoxy.

- [0178] In some embodiments R^{1a} is C_{1-6} linear haloalkyl.
 [0179] In some embodiments R^{1a} is C_{3-7} branched haloalkyl.
 [0180] In some embodiments R^{1a} is C_{1-6} linear haloalkoxy.
 [0181] In some embodiments R^{1a} is $-(C_{1-6} \text{ linear alkyl})$.
 [0182] In some embodiments R^{1a} is $S(C_{3-7} \text{ branched alkyl})$.
 [0183] In some embodiments R^{1a} is $-(S(C_{3-7} \text{ cycloalkyl}))$.
 [0184] In some embodiments R^{1a} is COR^6 .
 [0185] In some embodiments R^{1a} is CO_2R^7 .
 [0186] In some embodiments R^{1a} is $CONR^{8a}R^{8b}$.
 [0187] In some embodiments R^{1a} is $SO_2NR^{8a}R^{8b}$.
 [0188] In some embodiments R^{1a} is $NR^{9a}R^{9b}$.
 [0189] In some embodiments R^{1a} is $NR^{9a}COR^{10}$.
 [0190] In some embodiments R^{1a} is $NR^{9a}SO_2R^{11}$.
 [0191] In some embodiments R^{1a} is $NR^{9a}SO_2NR^{12a}R^{12b}$.
 [0192] In some embodiments R^{1b} is H.
 [0193] In some embodiments R^{1b} is OH.
 [0194] In some embodiments R^{1b} is NO_2 .
 [0195] In some embodiments R^{1b} is halogen.
 [0196] In some embodiments R^{1b} is CN.
 [0197] In some embodiments R^{1b} is C_{1-6} linear alkyl.
 [0198] In some embodiments R^{1b} is C_{3-7} branched alkyl.
 [0199] In some embodiments R^{1b} is C_{3-7} cycloalkyl.
 [0200] In some embodiments R^{1b} is C_{1-6} linear alkoxy.
 [0201] In some embodiments R^{1b} is C_{3-7} branched alkoxy.
 [0202] In some embodiments R^{1b} is C_{3-7} cycloalkoxy.
 [0203] In some embodiments R^{1b} is C_{1-6} linear haloalkyl.
 [0204] In some embodiments R^{1b} is C_{3-7} branched haloalkyl.
 [0205] In some embodiments R^{1b} is C_{1-6} linear haloalkoxy.
 [0206] In some embodiments R^{1b} is $-(C_{1-6} \text{ linear alkyl})$.
 [0207] In some embodiments R^{1b} is $S(C_{3-7} \text{ branched alkyl})$.
 [0208] In some embodiments R^{1b} is $-(S(C_{3-7} \text{ cycloalkyl}))$.
 [0209] In some embodiments R^{1b} is COR^6 .
 [0210] In some embodiments R^{1b} is CO_2R^7 .
 [0211] In some embodiments R^{1b} is $CONR^{8a}R^{8b}$.
 [0212] In some embodiments R^{1b} is $SO_2NR^{8a}R^{8b}$.
 [0213] In some embodiments R^{1b} is $NR^{9a}R^{9b}$.
 [0214] In some embodiments R^{1b} is $NR^{9a}COR^{11}$.
 [0215] In some embodiments R^{1b} is $NR^{9a}SO_2R^{11}$.
 [0216] In some embodiments R^{1b} is $NR^{9a}SO_2NR^{12a}R^{12b}$.
 [0217] In some embodiments R^{1c} is H.
 [0218] In some embodiments R^{1c} is OH.
 [0219] In some embodiments R^{1c} is NO_2 .
 [0220] In some embodiments R^{1c} is halogen.
 [0221] In some embodiments R^{1c} is CN.
 [0222] In some embodiments R^{1c} is C_{1-6} linear alkyl.
 [0223] In some embodiments R^{1c} is C_{3-7} branched alkyl.
 [0224] In some embodiments R^{1c} is C_{3-7} cycloalkyl.
 [0225] In some embodiments R^{1c} is C_{1-6} linear alkoxy.
 [0226] In some embodiments R^{1c} is C_{3-7} branched alkoxy.
 [0227] In some embodiments R^{1c} is C_{3-7} cycloalkoxy.
 [0228] In some embodiments R^{1c} is C_{1-6} linear haloalkyl.
 [0229] In some embodiments R^{1c} is C_{3-7} branched haloalkyl.
 [0230] In some embodiments R^{1c} is C_{1-6} linear haloalkoxy.
 [0231] In some embodiments R^{1c} is $-(C_{1-6} \text{ linear alkyl})$.
 [0232] In some embodiments R^{1c} is $S(C_{3-7} \text{ branched alkyl})$.
 [0233] In some embodiments R^{1c} is $-(S(C_{3-7} \text{ cycloalkyl}))$.
 [0234] In some embodiments R^{1c} is COR^6 .
 [0235] In some embodiments R^{1c} is CO_2R^7 .
 [0236] In some embodiments R^{1c} is $CONR^{8a}R^{8b}$.
 [0237] In some embodiments R^{1c} is $SO_2NR^{8a}R^{8b}$.
 [0238] In some embodiments R^{1c} is $NR^{9a}R^{9b}$.
 [0239] In some embodiments R^{1c} is $NR^{9a}COR^{10}$.
 [0240] In some embodiments R^{1c} is $NR^{9a}SO_2R^{11}$.
 [0241] In some embodiments R^{1c} is $NR^{9a}SO_2NR^{12a}R^{12b}$.
 [0242] In some embodiments R^{1d} is H.
 [0243] In some embodiments R^{1d} is OH.
 [0244] In some embodiments R^{1d} is NO_2 .
 [0245] In some embodiments R^{1d} is halogen.
 [0246] In some embodiments R^{1d} is CN.
 [0247] In some embodiments R^{1d} is C_{1-6} linear alkyl.
 [0248] In some embodiments R^{1d} is C_{3-7} branched alkyl.
 [0249] In some embodiments R^{1d} is C_{3-7} cycloalkyl.
 [0250] In some embodiments R^{1d} is C_{1-6} linear alkoxy.
 [0251] In some embodiments R^{1d} is C_{3-7} branched alkoxy.
 [0252] In some embodiments R^{1d} is C_{3-7} cycloalkoxy.
 [0253] In some embodiments R^{1d} is C_{1-6} linear haloalkyl.
 [0254] In some embodiments R^{1d} is C_{3-7} branched haloalkyl.
 [0255] In some embodiments R^{1d} is C_{1-6} linear haloalkoxy.
 [0256] In some embodiments R^{1d} is $-(S(C_{1-6} \text{ linear alkyl}))$.
 [0257] In some embodiments R^{1d} is $S(C_{3-7} \text{ branched alkyl})$.
 [0258] In some embodiments R^{1d} is $-(S(C_{3-7} \text{ cycloalkyl}))$.
 [0259] In some embodiments R^{1d} is COR^6 .
 [0260] In some embodiments R^{1d} is CO_2R^7 .
 [0261] In some embodiments R^{1d} is $CONR^{8a}R^{8b}$.
 [0262] In some embodiments R^{1d} is $SO_2NR^{8a}R^{8b}$.
 [0263] In some embodiments R^{1d} is $NR^{9a}R^{9b}$.
 [0264] In some embodiments R^{1d} is $NR^{9a}COR^{10}$.
 [0265] In some embodiments R^{1d} is $NR^{9a}SO_2R^{11}$.
 [0266] In some embodiments R^{1d} is $NR^{9a}SO_2NR^{12a}R^{12b}$.
 [0267] In some embodiments R^{1e} is H.
 [0268] In some embodiments R^{1e} is OH.
 [0269] In some embodiments R^{1e} is NO_2 .
 [0270] In some embodiments R^{1e} is halogen.
 [0271] In some embodiments R^{1e} is CN.
 [0272] In some embodiments R^{1e} is C_{1-6} linear alkyl.
 [0273] In some embodiments R^{1e} is C_{3-7} branched alkyl.
 [0274] In some embodiments R^{1e} is C_{3-7} cycloalkyl.
 [0275] In some embodiments R^{1e} is C_{1-6} linear alkoxy.
 [0276] In some embodiments R^{1e} is C_{3-7} branched alkoxy.
 [0277] In some embodiments R^{1e} is C_{3-7} cycloalkoxy.
 [0278] In some embodiments R^{1e} is C_{1-6} linear haloalkyl.
 [0279] In some embodiments R^{1e} is C_{3-7} branched haloalkyl.
 [0280] In some embodiments R^{1e} is C_{1-6} linear haloalkoxy.
 [0281] In some embodiments R^{1e} is $-(S(C_{1-6} \text{ linear alkyl}))$.
 [0282] In some embodiments R^{1e} is $S(C_{3-7} \text{ branched alkyl})$.
 [0283] In some embodiments R^{1e} is $-(S(C_{3-7} \text{ cycloalkyl}))$.
 [0284] In some embodiments R^{1e} is COR^6 .
 [0285] In some embodiments R^{1e} is CO_2R^7 .
 [0286] In some embodiments R^{1e} is $CONR^{8a}R^{8b}$.
 [0287] In some embodiments R^{1e} is $SO_2NR^{8a}R^{8b}$.
 [0288] In some embodiments R^{1e} is $NR^{9a}R^{9b}$.
 [0289] In some embodiments R^{1e} is $NR^{9a}COR^{10}$.
 [0290] In some embodiments R^{1e} is $NR^{9a}SO_2R^{11}$.
 [0291] In some embodiments R^{1e} is $NR^{9a}SO_2NR^{12a}R^{12b}$.
 [0292] In some embodiments R^2 is H.

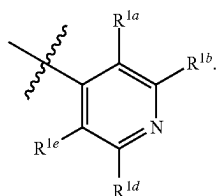
- [0293] In some embodiments R^2 is C_{1-6} linear alkyl.
 [0294] In some embodiments R^2 is, C_{3-7} branched alkyl.
 [0295] In some embodiments R^2 is C_{3-7} cycloalkyl.
 [0296] In some embodiments R^{2a} is C_{1-6} linear alkyl.
 [0297] In some embodiments R^{2a} is C_{3-7} branched alkyl.
 [0298] In some embodiments R^{2a} is C_{3-7} cycloalkyl.
 [0299] In some embodiments R^{2b} is H.
 [0300] In some embodiments R^{2b} is C_{1-6} linear alkyl.
 [0301] In some embodiments R^{2b} is, C_{3-7} branched alkyl.
 [0302] In some embodiments R^{2b} is C_{3-7} cycloalkyl.
 [0303] In some embodiments R^{2c} is H.
 [0304] In some embodiments R^{2c} is C_{1-6} linear alkyl.
 [0305] In some embodiments R^{2c} is, C_{3-7} branched alkyl.
 [0306] In some embodiments R^{2c} is C_{3-7} cycloalkyl.
 [0307] In some embodiments R^{2d} is C_{1-6} linear alkyl.
 [0308] In some embodiments R^{2d} is C_{3-7} branched alkyl.
 [0309] In some embodiments R^{2d} is C_{3-7} cycloalkyl.
 [0310] In some embodiments R^{2d} is C_{1-6} linear haloalkyl.
 [0311] In some embodiments R^{2d} is C_{3-7} branched haloalkyl.
 [0312] In some embodiments R^{2d} is $-(CH_2)_qCN$.
 [0313] In some embodiments R^{2d} is $-(CH_2)_qSO_2R^{13}$.
 [0314] In some embodiments R^{2d} is $-(CH_2)_qOR^{14}$.
 [0315] In some embodiments R^{2d} is



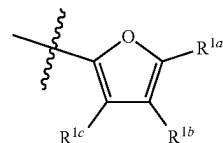
- [0316] In some embodiments R^{2d} is



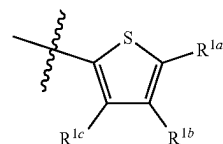
- [0317] In some embodiments R^{2d} is



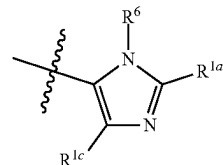
- [0318] In some embodiments R^{2d} is



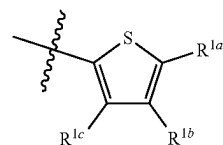
- [0319] In some embodiments R^{2d} is



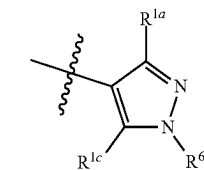
- [0320] In some embodiments R^{2d} is



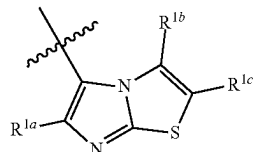
- [0321] In some embodiments R^{2d} is



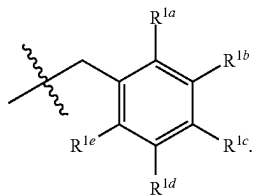
- [0322] In some embodiments R^{2d} is



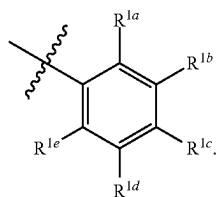
- [0323] In some embodiments R^{2d} is



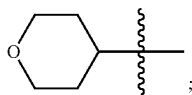
[0324] In some embodiments R^{2d} is



[0325] In some embodiments R^{2d} is



[0326] In some embodiments R^{2d} is



[0327] In some embodiments R^3 is C_{1-6} linear alkyl.

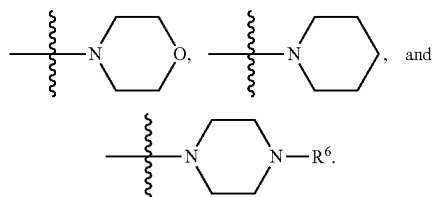
[0328] In some embodiments R^3 is C_{3-7} branched alkyl.

[0329] In some embodiments R^3 is C_{3-7} cycloalkyl.

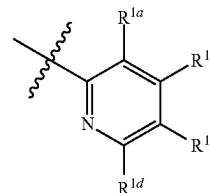
[0330] In some embodiments R^3 is optionally substituted aryl.

[0331] In some embodiments R^3 is phenyl.

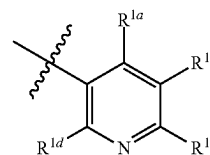
[0332] In some embodiments R^3 is an optionally aryl substituted with 1 to 4 units independently selected from the group consisting of OH, NO_2 , halogen, CN, C_{1-6} linear alkyl, C_{3-7} branched alkyl, C_{3-7} cycloalkyl, C_{1-6} linear alkoxy, C_{3-7} branched alkoxy, C_{3-7} cycloalkoxy, C_{1-6} linear haloalkyl, C_{3-7} branched haloalkyl, C_{1-6} linear haloalkoxy, $-S(C_{1-6}$ linear alkyl), $S(C_{3-7}$ branched alkyl), $-S(C_{3-7}$ cycloalkyl), COR^6 , CO_2R^7 , $CONR^{8a}R^{8b}$, $SO_2NR^{8a}R^{8b}$, $NR^{9a}R^{9b}$, $NR^{9a}COR^{10}$, $NR^{9a}SO_2R^{11}$, $NR^{9a}SO_2NR^{12a}R^{12b}$,



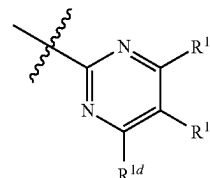
[0333] In some embodiments R^3 is



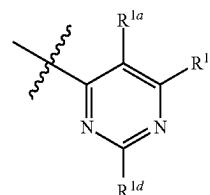
[0334] In some embodiments R^3 is



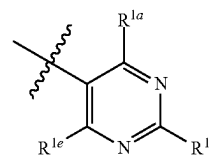
[0335] In some embodiments R^3 is



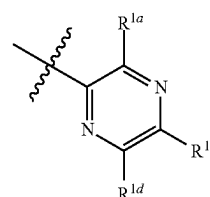
[0336] In some embodiments R^3 is



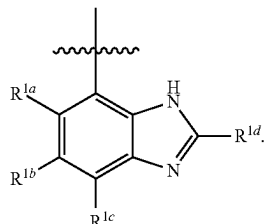
[0337] In some embodiments R^3 is



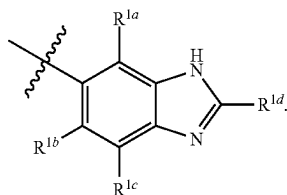
[0338] In some embodiments R^3 is



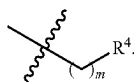
[0339] In some embodiments R^3 is



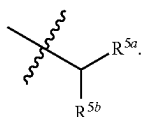
[0340] In some embodiments R^3 is



[0341] In some embodiments R^3 is



[0342] In some embodiments R^3 is



[0343] In some embodiments R^4 is optionally substituted aryl.

[0344] In some embodiments R^4 is an optionally substituted aryl substituted with 1 to 4 units independently selected from the group consisting of OH, NO₂, halogen, CN, C₁₋₆ linear alkyl, C₃₋₇ branched alkyl, C₃₋₇ cycloalkyl, C₁₋₆ linear alkoxy, C₃₋₇ branched alkoxy, C₃₋₇ cycloalkoxy, C₁₋₆ linear haloalkyl, C₃₋₇ branched haloalkyl, C₁₋₆ linear haloalkoxy, —S(C₁₋₆ linear alkyl), S(C₃₋₇ branched alkyl), —S(C₃₋₇ cycloalkyl), COR⁶, CO₂R⁷, CONR^{8a}R^{8b}, SO₂NR^{8a}R^{8b}, NR^{9a}R^{9b}, NR^{9a}COR¹⁰, NR^{9a}SO₂R¹¹, NR^{9a}SO₂NR^{12a}R^{12b},

[0345] In some embodiments R^5 is optionally substituted aryl.

[0346] In some embodiments R^5 is optionally substituted aryl.

[0347] In some embodiments R^6 is H.

[0348] In some embodiments R^6 is C₁₋₆ linear alkyl.

[0349] In some embodiments R^6 is C₃₋₇ branched alkyl.

[0350] In some embodiments R^6 is C₃₋₇ cycloalkyl.

[0351] In some embodiments R^7 is C₁₋₆ linear alkyl.

[0352] In some embodiments R^7 is C₃₋₇ branched alkyl.

[0353] In some embodiments R^7 is C₃₋₇ cycloalkyl.

[0354] In some embodiments R^{8a} is H.

[0355] In some embodiments R^8 is C₁₋₆ linear alkyl.

[0356] In some embodiments R^{8a} is C₃₋₇ branched alkyl.

[0357] In some embodiments R^{8a} is C₃₋₇ cycloalkyl.

[0358] In some embodiments R^{8b} is H.

[0359] In some embodiments R^{8b} is C₁₋₆ linear alkyl.

[0360] In some embodiments R^{8b} is C₃₋₇ branched alkyl.

[0361] In some embodiments R^{8b} is C₃₋₇ cycloalkyl.

[0362] In some embodiments R^{9a} is H.

[0363] In some embodiments R^{9a} is C₁₋₆ linear alkyl.

[0364] In some embodiments R^{9a} is C₃₋₇ branched alkyl.

[0365] In some embodiments R^{9a} is C₃₋₇ cycloalkyl.

[0366] In some embodiments R^{9b} is H.

[0367] In some embodiments R^{9b} is C₁₋₆ linear alkyl.

[0368] In some embodiments R^{9b} is C₃₋₇ branched alkyl.

[0369] In some embodiments R^{9b} is C₃₋₇ cycloalkyl.

[0370] In some embodiments R^{10} is H.

[0371] In some embodiments R^{10} is C₁₋₆ linear alkyl.

[0372] In some embodiments R^{10} is C₃₋₇ branched alkyl.

[0373] In some embodiments R^{10} is C₃₋₇ cycloalkyl.

[0374] In some embodiments R^{11} is C₁₋₆ linear alkyl.

[0375] In some embodiments R^{11} is C₃₋₇ branched alkyl.

[0376] In some embodiments R^{11} is C₃₋₇ cycloalkyl.

[0377] In some embodiments R^{10} is C₁₋₆ linear alkyl.

[0378] In some embodiments R^{10} is C₃₋₇ branched alkyl.

[0379] In some embodiments R^{10} is C₃₋₇ cycloalkyl.

[0380] In some embodiments R^{12a} is C₁₋₆ linear alkyl.

[0381] In some embodiments R^{12a} is C₃₋₇ branched alkyl.

[0382] In some embodiments R^{12a} is C₃₋₇ cycloalkyl.

[0383] In some embodiments R^{12b} is C₁₋₆ linear alkyl.

[0384] In some embodiments R^{12b} is C₃₋₇ branched alkyl.

[0385] In some embodiments R^{12b} is C₃₋₇ cycloalkyl.

[0386] In some embodiments R^{13} is C₁₋₆ linear alkyl.

[0387] In some embodiments R^{13} is C₃₋₇ branched alkyl.

[0388] In some embodiments R^{13} is C₃₋₇ cycloalkyl.

[0389] In some embodiments R^{14} is C₁₋₆ linear alkyl.

[0390] In some embodiments R^{14} is C₃₋₇ branched alkyl.

[0391] In some embodiments R^{14} is C₃₋₇ cycloalkyl.

[0392] In some embodiments n is 1.

[0393] In some embodiments n is 2.

[0394] In some embodiments n is 3.

[0395] In some embodiments m is 1.

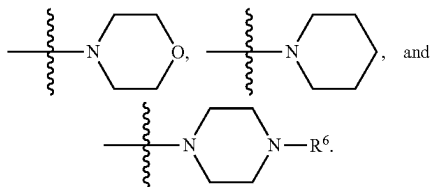
[0396] In some embodiments m is 2.

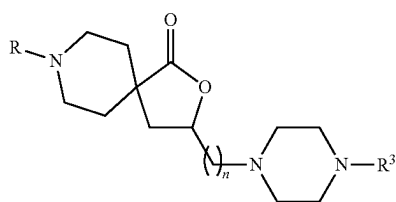
[0397] In some embodiments q is 1.

[0398] In some embodiments q is 2.

[0399] In some embodiments q is 3.

[0400] Exemplary embodiments include compounds having the formula (XVII)





or a pharmaceutically acceptable salt form thereof defined herein below in Table 1.

TABLE 1

Entry	n	R	R ³
1	1	H	Phenyl
2	2	H	Phenyl
3	3	H	Phenyl
4	1	Me	Phenyl
5	2	Me	Phenyl
6	3	Me	Phenyl
7	1	CH ₂ Ph	Phenyl
8	2	CH ₂ Ph	Phenyl
9	3	CH ₂ Ph	Phenyl
10	1	COMe	Phenyl
11	2	COMe	Phenyl
12	3	COMe	Phenyl
13	1	CO ₂ Me	Phenyl
14	2	CO ₂ Me	Phenyl
15	3	CO ₂ Me	Phenyl
16	1	CO ₂ tBu	Phenyl
17	2	CO ₂ tBu	Phenyl
18	3	CO ₂ tBu	Phenyl
19	1	CONHMe	Phenyl
20	2	CONHMe	Phenyl
21	3	CONHMe	Phenyl
22	1	SO ₂ Me	Phenyl
23	2	SO ₂ Me	Phenyl
24	3	SO ₂ Me	Phenyl
25	1	SO ₂ NH ₂	Phenyl
26	2	SO ₂ NH ₂	Phenyl
27	3	SO ₂ NH ₂	Phenyl
28	1	H	3-OH-Phenyl
29	2	H	3-OH-Phenyl
30	3	H	3-OH-Phenyl
31	1	Me	3-OH-Phenyl
32	2	Me	3-OH-Phenyl
33	3	Me	3-OH-Phenyl
34	1	CH ₂ Ph	3-OH-Phenyl
35	2	CH ₂ Ph	3-OH-Phenyl
36	3	CH ₂ Ph	3-OH-Phenyl
37	1	COMe	3-OH-Phenyl
38	2	COMe	3-OH-Phenyl
39	3	COMe	3-OH-Phenyl
40	1	CO ₂ Me	3-OH-Phenyl
41	2	CO ₂ Me	3-OH-Phenyl
42	3	CO ₂ Me	3-OH-Phenyl
43	1	CO ₂ tBu	3-OH-Phenyl
44	2	CO ₂ tBu	3-OH-Phenyl
45	3	CO ₂ tBu	3-OH-Phenyl
46	1	CONHMe	3-OH-Phenyl
47	2	CONHMe	3-OH-Phenyl
48	3	CONHMe	3-OH-Phenyl
49	1	SO ₂ Me	3-OH-Phenyl
50	2	SO ₂ Me	3-OH-Phenyl
51	3	SO ₂ Me	3-OH-Phenyl
52	1	SO ₂ NH ₂	3-OH-Phenyl
53	2	SO ₂ NH ₂	3-OH-Phenyl
54	3	SO ₂ NH ₂	3-OH-Phenyl
55	1	H	4-NO ₂ -Phenyl
56	2	H	4-NO ₂ -Phenyl
57	3	H	4-NO ₂ -Phenyl
58	1	Me	4-NO ₂ -Phenyl

TABLE 1-continued

Entry	n	R	R ³
59	2	Me	4-NO ₂ -Phenyl
60	3	Me	4-NO ₂ -Phenyl
61	1	CH ₂ Ph	4-NO ₂ -Phenyl
62	2	CH ₂ Ph	4-NO ₂ -Phenyl
63	3	CH ₂ Ph	4-NO ₂ -Phenyl
64	1	COMe	4-NO ₂ -Phenyl
65	2	COMe	4-NO ₂ -Phenyl
66	3	COMe	4-NO ₂ -Phenyl
67	1	CO ₂ Me	4-NO ₂ -Phenyl
68	2	CO ₂ Me	4-NO ₂ -Phenyl
69	3	CO ₂ Me	4-NO ₂ -Phenyl
70	1	CO ₂ tBu	4-NO ₂ -Phenyl
71	2	CO ₂ tBu	4-NO ₂ -Phenyl
72	3	CO ₂ tBu	4-NO ₂ -Phenyl
73	1	CONHMe	4-NO ₂ -Phenyl
74	2	CONHMe	4-NO ₂ -Phenyl
75	3	CONHMe	4-NO ₂ -Phenyl
76	1	SO ₂ Me	4-NO ₂ -Phenyl
77	2	SO ₂ Me	4-NO ₂ -Phenyl
78	3	SO ₂ Me	4-NO ₂ -Phenyl
79	1	SO ₂ NH ₂	4-NO ₂ -Phenyl
80	2	SO ₂ NH ₂	4-NO ₂ -Phenyl
81	3	SO ₂ NH ₂	4-NO ₂ -Phenyl
82	1	H	3-OMe-Phenyl
83	2	H	3-OMe-Phenyl
84	3	H	3-OMe-Phenyl
85	1	Me	3-OMe-Phenyl
86	2	Me	3-OMe-Phenyl
87	3	Me	3-OMe-Phenyl
88	1	CH ₂ Ph	3-OMe-Phenyl
89	2	CH ₂ Ph	3-OMe-Phenyl
90	3	CH ₂ Ph	3-OMe-Phenyl
91	1	COMe	3-OMe-Phenyl
92	2	COMe	3-OMe-Phenyl
93	3	COMe	3-OMe-Phenyl
94	1	CO ₂ Me	3-OMe-Phenyl
95	2	CO ₂ Me	3-OMe-Phenyl
96	3	CO ₂ Me	3-OMe-Phenyl
97	1	CO ₂ tBu	3-OMe-Phenyl
98	2	CO ₂ tBu	3-OMe-Phenyl
99	3	CO ₂ tBu	3-OMe-Phenyl
100	1	CONHMe	3-OMe-Phenyl
101	2	CONHMe	3-OMe-Phenyl
102	3	CONHMe	3-OMe-Phenyl
103	1	SO ₂ Me	3-OMe-Phenyl
104	2	SO ₂ Me	3-OMe-Phenyl
105	3	SO ₂ Me	3-OMe-Phenyl
106	1	SO ₂ NH ₂	3-OMe-Phenyl
107	2	SO ₂ NH ₂	3-OMe-Phenyl
108	3	SO ₂ NH ₂	3-OMe-Phenyl
109	1	H	4-CN-Phenyl
110	2	H	4-CN-Phenyl
111	3	H	4-CN-Phenyl
112	1	Me	4-CN-Phenyl
113	2	Me	4-CN-Phenyl
114	3	Me	4-CN-Phenyl
115	1	CH ₂ Ph	4-CN-Phenyl
116	2	CH ₂ Ph	4-CN-Phenyl
117	3	CH ₂ Ph	4-CN-Phenyl
118	1	COMe	4-CN-Phenyl
119	2	COMe	4-CN-Phenyl
120	3	COMe	4-CN-Phenyl
121	1	CO ₂ Me	4-CN-Phenyl
122	2	CO ₂ Me	4-CN-Phenyl
123	3	CO ₂ Me	4-CN-Phenyl
124	1	CO ₂ tBu	4-CN-Phenyl
125	2	CO ₂ tBu	4-CN-Phenyl
126	3	CO ₂ tBu	4-CN-Phenyl
127	1	CONHMe	4-CN-Phenyl
128	2	CONHMe	4-CN-Phenyl
129	3	CONHMe	4-CN-Phenyl
130	1	SO ₂ Me	4-CN-Phenyl
131	2	SO ₂ Me	4-CN-Phenyl
132	3	SO ₂ Me	4-CN-Phenyl
133	1	SO ₂ NH ₂	4-CN-Phenyl

TABLE 1-continued

Entry	n	R	R ³
134	2	SO ₂ NH ₂	4-CN-Phenyl
135	3	SO ₂ NH ₂	4-CN-Phenyl
136	1	H	2-CN-Phenyl
137	2	H	2-CN-Phenyl
138	3	H	2-CN-Phenyl
139	1	Me	2-CN-Phenyl
140	2	Me	2-CN-Phenyl
141	3	Me	2-CN-Phenyl
142	1	CH ₂ Ph	2-CN-Phenyl
143	2	CH ₂ Ph	2-CN-Phenyl
144	3	CH ₂ Ph	2-CN-Phenyl
145	1	COMe	2-CN-Phenyl
146	2	COMe	2-CN-Phenyl
147	3	COMe	2-CN-Phenyl
148	1	CO ₂ Me	2-CN-Phenyl
149	2	CO ₂ Me	2-CN-Phenyl
150	3	CO ₂ Me	2-CN-Phenyl
151	1	CO ₂ tBu	2-CN-Phenyl
152	2	CO ₂ tBu	2-CN-Phenyl
153	3	CO ₂ tBu	2-CN-Phenyl
154	1	CONHMe	2-CN-Phenyl
155	2	CONHMe	2-CN-Phenyl
156	3	CONHMe	2-CN-Phenyl
157	1	SO ₂ Me	2-CN-Phenyl
158	2	SO ₂ Me	2-CN-Phenyl
159	3	SO ₂ Me	2-CN-Phenyl
160	1	SO ₂ NH ₂	2-CN-Phenyl
161	2	SO ₂ NH ₂	2-CN-Phenyl
162	3	SO ₂ NH ₂	2-CN-Phenyl
163	1	H	3-Me-Phenyl
164	2	H	3-Me-Phenyl
165	3	H	3-Me-Phenyl
166	1	Me	3-Me-Phenyl
167	2	Me	3-Me-Phenyl
168	3	Me	3-Me-Phenyl
169	1	CH ₂ Ph	3-Me-Phenyl
170	2	CH ₂ Ph	3-Me-Phenyl
171	3	CH ₂ Ph	3-Me-Phenyl
172	1	COMe	3-Me-Phenyl
173	2	COMe	3-Me-Phenyl
174	3	COMe	3-Me-Phenyl
175	1	CO ₂ Me	3-Me-Phenyl
176	2	CO ₂ Me	3-Me-Phenyl
177	3	CO ₂ Me	3-Me-Phenyl
178	1	CO ₂ tBu	3-Me-Phenyl
179	2	CO ₂ tBu	3-Me-Phenyl
180	3	CO ₂ tBu	3-Me-Phenyl
181	1	CONHMe	3-Me-Phenyl
182	2	CONHMe	3-Me-Phenyl
183	3	CONHMe	3-Me-Phenyl
184	1	SO ₂ Me	3-Me-Phenyl
185	2	SO ₂ Me	3-Me-Phenyl
186	3	SO ₂ Me	3-Me-Phenyl
187	1	SO ₂ NH ₂	3-Me-Phenyl
188	2	SO ₂ NH ₂	3-Me-Phenyl
189	3	SO ₂ NH ₂	3-Me-Phenyl
190	1	H	2-F-Phenyl
191	2	H	2-F-Phenyl
192	3	H	2-F-Phenyl
193	1	Me	2-F-Phenyl
194	2	Me	2-F-Phenyl
195	3	Me	2-F-Phenyl
196	1	CH ₂ Ph	2-F-Phenyl
197	2	CH ₂ Ph	2-F-Phenyl
198	3	CH ₂ Ph	2-F-Phenyl
199	1	COMe	2-F-Phenyl
200	2	COMe	2-F-Phenyl
201	3	COMe	2-F-Phenyl
202	1	CO ₂ Me	2-F-Phenyl
203	2	CO ₂ Me	2-F-Phenyl
204	3	CO ₂ Me	2-F-Phenyl
205	1	CO ₂ tBu	2-F-Phenyl
206	2	CO ₂ tBu	2-F-Phenyl
207	3	CO ₂ tBu	2-F-Phenyl
208	1	CONHMe	2-F-Phenyl

TABLE 1-continued

Entry	n	R	R ³
209	2	CONHMe	2-F-Phenyl
210	3	CONHMe	2-F-Phenyl
211	1	SO ₂ Me	2-F-Phenyl
212	2	SO ₂ Me	2-F-Phenyl
213	3	SO ₂ Me	2-F-Phenyl
214	1	SO ₂ NH ₂	2-F-Phenyl
215	2	SO ₂ NH ₂	2-F-Phenyl
216	3	SO ₂ NH ₂	2-F-Phenyl
217	1	H	4-F-Phenyl
218	2	H	4-F-Phenyl
219	3	H	4-F-Phenyl
220	1	Me	4-F-Phenyl
221	2	Me	4-F-Phenyl
222	3	Me	4-F-Phenyl
223	1	CH ₂ Ph	4-F-Phenyl
224	2	CH ₂ Ph	4-F-Phenyl
225	3	CH ₂ Ph	4-F-Phenyl
226	1	COMe	4-F-Phenyl
227	2	COMe	4-F-Phenyl
228	3	COMe	4-F-Phenyl
229	1	CO ₂ Me	4-F-Phenyl
230	2	CO ₂ Me	4-F-Phenyl
231	3	CO ₂ Me	4-F-Phenyl
232	1	CO ₂ tBu	4-F-Phenyl
233	2	CO ₂ tBu	4-F-Phenyl
234	3	CO ₂ tBu	4-F-Phenyl
235	1	CONHMe	4-F-Phenyl
236	2	CONHMe	4-F-Phenyl
237	3	CONHMe	4-F-Phenyl
238	1	SO ₂ Me	4-F-Phenyl
239	2	SO ₂ Me	4-F-Phenyl
240	3	SO ₂ Me	4-F-Phenyl
241	1	SO ₂ NH ₂	4-F-Phenyl
242	2	SO ₂ NH ₂	4-F-Phenyl
243	3	SO ₂ NH ₂	4-F-Phenyl
244	1	H	3-Cl-Phenyl
245	2	H	3-Cl-Phenyl
246	3	H	3-Cl-Phenyl
247	1	Me	3-Cl-Phenyl
248	2	Me	3-Cl-Phenyl
249	3	Me	3-Cl-Phenyl
250	1	CH ₂ Ph	3-Cl-Phenyl
251	2	CH ₂ Ph	3-Cl-Phenyl
252	3	CH ₂ Ph	3-Cl-Phenyl
253	1	COMe	3-Cl-Phenyl
254	2	COMe	3-Cl-Phenyl
255	3	COMe	3-Cl-Phenyl
256	1	CO ₂ Me	3-Cl-Phenyl
257	2	CO ₂ Me	3-Cl-Phenyl
258	3	CO ₂ Me	3-Cl-Phenyl
259	1	CO ₂ tBu	3-Cl-Phenyl
260	2	CO ₂ tBu	3-Cl-Phenyl
261	3	CO ₂ tBu	3-Cl-Phenyl
262	1	CONHMe	3-Cl-Phenyl
263	2	CONHMe	3-Cl-Phenyl
264	3	CONHMe	3-Cl-Phenyl
265	1	SO ₂ Me	3-Cl-Phenyl
266	2	SO ₂ Me	3-Cl-Phenyl
267	3	SO ₂ Me	3-Cl-Phenyl
268	1	SO ₂ NH ₂	3-Cl-Phenyl
269	2	SO ₂ NH ₂	3-Cl-Phenyl
270	3	SO ₂ NH ₂	3-Cl-Phenyl
271	1	H	2-Br-Phenyl
272	2	H	2-Br-Phenyl
273	3	H	2-Br-Phenyl
274	1	Me	2-Br-Phenyl
275	2	Me	2-Br-Phenyl
276	3	Me	2-Br-Phenyl
277	1	CH ₂ Ph	2-Br-Phenyl
278	2	CH ₂ Ph	2-Br-Phenyl
279	3	CH ₂ Ph	2-Br-Phenyl
280	1	COMe	2-Br-Phenyl
281	2	COMe	2-Br-Phenyl
282	3	COMe	2-Br-Phenyl
283	1	CO ₂ Me	2-Br-Phenyl

TABLE 1-continued

Entry	n	R	R ³
284	2	CO ₂ Me	2-Br-Phenyl
285	3	CO ₂ Me	2-Br-Phenyl
286	1	CO ₂ tBu	2-Br-Phenyl
287	2	CO ₂ tBu	2-Br-Phenyl
288	3	CO ₂ tBu	2-Br-Phenyl
289	1	CONHMe	2-Br-Phenyl
290	2	CONHMe	2-Br-Phenyl
291	3	CONHMe	2-Br-Phenyl
292	1	SO ₂ Me	2-Br-Phenyl
293	2	SO ₂ Me	2-Br-Phenyl
294	3	SO ₂ Me	2-Br-Phenyl
295	1	SO ₂ NH ₂	2-Br-Phenyl
296	2	SO ₂ NH ₂	2-Br-Phenyl
297	3	SO ₂ NH ₂	2-Br-Phenyl
298	1	H	4-Br-Phenyl
299	2	H	4-Br-Phenyl
300	3	H	4-Br-Phenyl
301	1	Me	4-Br-Phenyl
302	2	Me	4-Br-Phenyl
303	3	Me	4-Br-Phenyl
304	1	CH ₂ Ph	4-Br-Phenyl
305	2	CH ₂ Ph	4-Br-Phenyl
306	3	CH ₂ Ph	4-Br-Phenyl
307	1	COMe	4-Br-Phenyl
308	2	COMe	4-Br-Phenyl
309	3	COMe	4-Br-Phenyl
310	1	CO ₂ Me	4-Br-Phenyl
311	2	CO ₂ Me	4-Br-Phenyl
312	3	CO ₂ Me	4-Br-Phenyl
313	1	CO ₂ tBu	4-Br-Phenyl
314	2	CO ₂ tBu	4-Br-Phenyl
315	3	CO ₂ tBu	4-Br-Phenyl
316	1	CONHMe	4-Br-Phenyl
317	2	CONHMe	4-Br-Phenyl
318	3	CONHMe	4-Br-Phenyl
319	1	SO ₂ Me	4-Br-Phenyl
320	2	SO ₂ Me	4-Br-Phenyl
321	3	SO ₂ Me	4-Br-Phenyl
322	1	SO ₂ NH ₂	4-Br-Phenyl
323	2	SO ₂ NH ₂	4-Br-Phenyl
324	3	SO ₂ NH ₂	4-Br-Phenyl
325	1	H	3-CF ₃ -Phenyl
326	2	H	3-CF ₃ -Phenyl
327	3	H	3-CF ₃ -Phenyl
328	1	Me	3-CF ₃ -Phenyl
329	2	Me	3-CF ₃ -Phenyl
330	3	Me	3-CF ₃ -Phenyl
331	1	CH ₂ Ph	3-CF ₃ -Phenyl
332	2	CH ₂ Ph	3-CF ₃ -Phenyl
333	3	CH ₂ Ph	3-CF ₃ -Phenyl
334	1	COMe	3-CF ₃ -Phenyl
335	2	COMe	3-CF ₃ -Phenyl
336	3	COMe	3-CF ₃ -Phenyl
337	1	CO ₂ Me	3-CF ₃ -Phenyl
338	2	CO ₂ Me	3-CF ₃ -Phenyl
339	3	CO ₂ Me	3-CF ₃ -Phenyl
340	1	CO ₂ tBu	3-CF ₃ -Phenyl
341	2	CO ₂ tBu	3-CF ₃ -Phenyl
342	3	CO ₂ tBu	3-CF ₃ -Phenyl
343	1	CONHMe	3-CF ₃ -Phenyl
344	2	CONHMe	3-CF ₃ -Phenyl
345	3	CONHMe	3-CF ₃ -Phenyl
346	1	SO ₂ Me	3-CF ₃ -Phenyl
347	2	SO ₂ Me	3-CF ₃ -Phenyl
348	3	SO ₂ Me	3-CF ₃ -Phenyl
349	1	SO ₂ NH ₂	3-CF ₃ -Phenyl
350	2	SO ₂ NH ₂	3-CF ₃ -Phenyl
351	3	SO ₂ NH ₂	3-CF ₃ -Phenyl
352	1	H	2-iPr-Phenyl
353	2	H	2-iPr-Phenyl
354	3	H	2-iPr-Phenyl
355	1	Me	2-iPr-Phenyl
356	2	Me	2-iPr-Phenyl
357	3	Me	2-iPr-Phenyl
358	1	CH ₂ Ph	2-iPr-Phenyl

TABLE 1-continued

Entry	n	R	R ³
359	2	CH ₂ Ph	2-iPr-Phenyl
360	3	CH ₂ Ph	2-iPr-Phenyl
361	1	COMe	2-iPr-Phenyl
362	2	COMe	2-iPr-Phenyl
363	3	COMe	2-iPr-Phenyl
364	1	CO ₂ Me	2-iPr-Phenyl
365	2	CO ₂ Me	2-iPr-Phenyl
366	3	CO ₂ Me	2-iPr-Phenyl
367	1	CO ₂ tBu	2-iPr-Phenyl
368	2	CO ₂ tBu	2-iPr-Phenyl
369	3	CO ₂ tBu	2-iPr-Phenyl
370	1	CONHMe	2-iPr-Phenyl
371	2	CONHMe	2-iPr-Phenyl
372	3	CONHMe	2-iPr-Phenyl
373	1	SO ₂ Me	2-iPr-Phenyl
374	2	SO ₂ Me	2-iPr-Phenyl
375	3	SO ₂ Me	2-iPr-Phenyl
376	1	SO ₂ NH ₂	2-iPr-Phenyl
377	2	SO ₂ NH ₂	2-iPr-Phenyl
378	3	SO ₂ NH ₂	2-iPr-Phenyl
379	1	H	4-iPr-Phenyl
380	2	H	4-iPr-Phenyl
381	3	H	4-iPr-Phenyl
382	1	Me	4-iPr-Phenyl
383	2	Me	4-iPr-Phenyl
384	3	Me	4-iPr-Phenyl
385	1	CH ₂ Ph	4-iPr-Phenyl
386	2	CH ₂ Ph	4-iPr-Phenyl
387	3	CH ₂ Ph	4-iPr-Phenyl
388	1	COMe	4-iPr-Phenyl
389	2	COMe	4-iPr-Phenyl
390	3	COMe	4-iPr-Phenyl
391	1	CO ₂ Me	4-iPr-Phenyl
392	2	CO ₂ Me	4-iPr-Phenyl
393	3	CO ₂ Me	4-iPr-Phenyl
394	1	CO ₂ tBu	4-iPr-Phenyl
395	2	CO ₂ tBu	4-iPr-Phenyl
396	3	CO ₂ tBu	4-iPr-Phenyl
397	1	CONHMe	4-iPr-Phenyl
398	2	CONHMe	4-iPr-Phenyl
399	3	CONHMe	4-iPr-Phenyl
400	1	SO ₂ Me	4-iPr-Phenyl
401	2	SO ₂ Me	4-iPr-Phenyl
402	3	SO ₂ Me	4-iPr-Phenyl
403	1	SO ₂ NH ₂	4-iPr-Phenyl
404	2	SO ₂ NH ₂	4-iPr-Phenyl
405	3	SO ₂ NH ₂	4-iPr-Phenyl
406	1	H	3-NH ₂ -Phenyl
407	2	H	3-NH ₂ -Phenyl
408	3	H	3-NH ₂ -Phenyl
409	1	Me	3-NH ₂ -Phenyl
410	2	Me	3-NH ₂ -Phenyl
411	3	Me	3-NH ₂ -Phenyl
412	1	CH ₂ Ph	3-NH ₂ -Phenyl
413	2	CH ₂ Ph	3-NH ₂ -Phenyl
414	3	CH ₂ Ph	3-NH ₂ -Phenyl
415	1	COMe	3-NH ₂ -Phenyl
416	2	COMe	3-NH ₂ -Phenyl
417	3	COMe	3-NH ₂ -Phenyl
418	1	CO ₂ Me	3-NH ₂ -Phenyl
419	2	CO ₂ Me	3-NH ₂ -Phenyl
420	3	CO ₂ Me	3-NH ₂ -Phenyl
421	1	CO ₂ tBu	3-NH ₂ -Phenyl
422	2	CO ₂ tBu	3-NH ₂ -Phenyl
423	3	CO ₂ tBu	3-NH ₂ -Phenyl
424	1	CONHMe	3-NH ₂ -Phenyl
425	2	CONHMe	3-NH ₂ -Phenyl
426	3	CONHMe	3-NH ₂ -Phenyl
427	1	SO ₂ Me	3-NH ₂ -Phenyl
428	2	SO ₂ Me	3-NH ₂ -Phenyl
429	3	SO ₂ Me	3-NH ₂ -Phenyl
430	1	SO ₂ NH ₂	3-NH ₂ -Phenyl
431	2	SO ₂ NH ₂	3-NH ₂ -Phenyl
432	3	SO ₂ NH ₂	3-NH ₂ -Phenyl
433	1	H	2,4-di-Me-Phenyl

TABLE 1-continued

Entry	n	R	R ³
434	2	H	2,4-di-Me-Phenyl
435	3	H	2,4-di-Me-Phenyl
436	1	Me	2,4-di-Me-Phenyl
437	2	Me	2,4-di-Me-Phenyl
438	3	Me	2,4-di-Me-Phenyl
439	1	CH ₂ Ph	2,4-di-Me-Phenyl
440	2	CH ₂ Ph	2,4-di-Me-Phenyl
441	3	CH ₂ Ph	2,4-di-Me-Phenyl
442	1	COMe	2,4-di-Me-Phenyl
443	2	COMe	2,4-di-Me-Phenyl
444	3	COMe	2,4-di-Me-Phenyl
445	1	CO ₂ Me	2,4-di-Me-Phenyl
446	2	CO ₂ Me	2,4-di-Me-Phenyl
447	3	CO ₂ Me	2,4-di-Me-Phenyl
448	1	CO ₂ tBu	2,4-di-Me-Phenyl
449	2	CO ₂ tBu	2,4-di-Me-Phenyl
450	3	CO ₂ tBu	2,4-di-Me-Phenyl
451	1	CONHMe	2,4-di-Me-Phenyl
452	2	CONHMe	2,4-di-Me-Phenyl
453	3	CONHMe	2,4-di-Me-Phenyl
454	1	SO ₂ Me	2,4-di-Me-Phenyl
455	2	SO ₂ Me	2,4-di-Me-Phenyl
456	3	SO ₂ Me	2,4-di-Me-Phenyl
457	1	SO ₂ NH ₂	2,4-di-Me-Phenyl
458	2	SO ₂ NH ₂	2,4-di-Me-Phenyl
459	3	SO ₂ NH ₂	2,4-di-Me-Phenyl
460	1	H	2,6-di-iPr-Phenyl
461	2	H	2,6-di-iPr-Phenyl
462	3	H	2,6-di-iPr-Phenyl
463	1	Me	2,6-di-iPr-Phenyl
464	2	Me	2,6-di-iPr-Phenyl
465	3	Me	2,6-di-iPr-Phenyl
466	1	CH ₂ Ph	2,6-di-iPr-Phenyl
467	2	CH ₂ Ph	2,6-di-iPr-Phenyl
468	3	CH ₂ Ph	2,6-di-iPr-Phenyl
469	1	COMe	2,6-di-iPr-Phenyl
470	2	COMe	2,6-di-iPr-Phenyl
471	3	COMe	2,6-di-iPr-Phenyl
472	1	CO ₂ Me	2,6-di-iPr-Phenyl
473	2	CO ₂ Me	2,6-di-iPr-Phenyl
474	3	CO ₂ Me	2,6-di-iPr-Phenyl
475	1	CO ₂ tBu	2,6-di-iPr-Phenyl
476	2	CO ₂ tBu	2,6-di-iPr-Phenyl
477	3	CO ₂ tBu	2,6-di-iPr-Phenyl
478	1	CONHMe	2,6-di-iPr-Phenyl
479	2	CONHMe	2,6-di-iPr-Phenyl
480	3	CONHMe	2,6-di-iPr-Phenyl
481	1	SO ₂ Me	2,6-di-iPr-Phenyl
482	2	SO ₂ Me	2,6-di-iPr-Phenyl
483	3	SO ₂ Me	2,6-di-iPr-Phenyl
484	1	SO ₂ NH ₂	2,6-di-iPr-Phenyl
485	2	SO ₂ NH ₂	2,6-di-iPr-Phenyl
486	3	SO ₂ NH ₂	2,6-di-iPr-Phenyl
487	1	H	3-Ph-Phenyl
488	2	H	3-Ph-Phenyl
489	3	H	3-Ph-Phenyl
490	1	Me	3-Ph-Phenyl
491	2	Me	3-Ph-Phenyl
492	3	Me	3-Ph-Phenyl
493	1	CH ₂ Ph	3-Ph-Phenyl
494	2	CH ₂ Ph	3-Ph-Phenyl
495	3	CH ₂ Ph	3-Ph-Phenyl
496	1	COMe	3-Ph-Phenyl
497	2	COMe	3-Ph-Phenyl
498	3	COMe	3-Ph-Phenyl
499	1	CO ₂ Me	3-Ph-Phenyl
500	2	CO ₂ Me	3-Ph-Phenyl
501	3	CO ₂ Me	3-Ph-Phenyl
502	1	CO ₂ tBu	3-Ph-Phenyl
503	2	CO ₂ tBu	3-Ph-Phenyl
504	3	CO ₂ tBu	3-Ph-Phenyl
505	1	CONHMe	3-Ph-Phenyl
506	2	CONHMe	3-Ph-Phenyl
507	3	CONHMe	3-Ph-Phenyl
508	1	SO ₂ Me	3-Ph-Phenyl

TABLE 1-continued

Entry	n	R	R ³
509	2	SO ₂ Me	3-Ph-Phenyl
510	3	SO ₂ Me	3-Ph-Phenyl
511	1	SO ₂ NH ₂	3-Ph-Phenyl
512	2	SO ₂ NH ₂	3-Ph-Phenyl
513	3	SO ₂ NH ₂	3-Ph-Phenyl
514	1	H	2-morpholino-phenyl
515	2	H	2-morpholino-phenyl
516	3	H	2-morpholino-phenyl
517	1	Me	2-morpholino-phenyl
518	2	Me	2-morpholino-phenyl
519	3	Me	2-morpholino-phenyl
520	1	CH ₂ Ph	2-morpholino-phenyl
521	2	CH ₂ Ph	2-morpholino-phenyl
522	3	CH ₂ Ph	2-morpholino-phenyl
523	1	COMe	2-morpholino-phenyl
524	2	COMe	2-morpholino-phenyl
525	3	COMe	2-morpholino-phenyl
526	1	CO ₂ Me	2-morpholino-phenyl
527	2	CO ₂ Me	2-morpholino-phenyl
528	3	CO ₂ Me	2-morpholino-phenyl
529	1	CO ₂ tBu	2-morpholino-phenyl
530	2	CO ₂ tBu	2-morpholino-phenyl
531	3	CO ₂ tBu	2-morpholino-phenyl
532	1	CONHMe	2-morpholino-phenyl
533	2	CONHMe	2-morpholino-phenyl
534	3	CONHMe	2-morpholino-phenyl
535	1	SO ₂ Me	2-morpholino-phenyl
536	2	SO ₂ Me	2-morpholino-phenyl
537	3	SO ₂ Me	2-morpholino-phenyl
538	1	SO ₂ NH ₂	2-morpholino-phenyl
539	2	SO ₂ NH ₂	2-morpholino-phenyl
540	3	SO ₂ NH ₂	2-morpholino-phenyl
541	1	H	4-morpholino-phenyl
542	2	H	4-morpholino-phenyl
543	3	H	4-morpholino-phenyl
544	1	Me	4-morpholino-phenyl
545	2	Me	4-morpholino-phenyl
546	3	Me	4-morpholino-phenyl
547	1	CH ₂ Ph	4-morpholino-phenyl
548	2	CH ₂ Ph	4-morpholino-phenyl
549	3	CH ₂ Ph	4-morpholino-phenyl
550	1	COMe	4-morpholino-phenyl
551	2	COMe	4-morpholino-phenyl
552	3	COMe	4-morpholino-phenyl
553	1	CO ₂ Me	4-morpholino-phenyl
554	2	CO ₂ Me	4-morpholino-phenyl
555	3	CO ₂ Me	4-morpholino-phenyl
556	1	CO ₂ tBu	4-morpholino-phenyl
557	2	CO ₂ tBu	4-morpholino-phenyl
558	3	CO ₂ tBu	4-morpholino-phenyl
559	1	CONHMe	4-morpholino-phenyl
560	2	CONHMe	4-morpholino-phenyl
561	3	CONHMe	4-morpholino-phenyl
562	1	SO ₂ Me	4-morpholino-phenyl
563	2	SO ₂ Me	4-morpholino-phenyl
564	3	SO ₂ Me	4-morpholino-phenyl
565	1	SO ₂ NH ₂	4-morpholino-phenyl
566	2	SO ₂ NH ₂	4-morpholino-phenyl
567	3	SO ₂ NH ₂	4-morpholino-phenyl
568	1	H	2-pyrimidinyl
569	2	H	2-pyrimidinyl
570	3	H	2-pyrimidinyl
571	1	Me	2-pyrimidinyl
572	2	Me	2-pyrimidinyl
573	3	Me	2-pyrimidinyl
574	1	CH ₂ Ph	2-pyrimidinyl
575	2	CH ₂ Ph	2-pyrimidinyl
576	3	CH ₂ Ph	2-pyrimidinyl
577	1	COMe	2-pyrimidinyl
578	2	COMe	2-pyrimidinyl
579	3	COMe	2-pyrimidinyl
580	1	CO ₂ Me	2-pyrimidinyl
581	2	CO ₂ Me	2-pyrimidinyl
582	3	CO ₂ Me	2-pyrimidinyl
583	1	CO ₂ tBu	2-pyrimidinyl

TABLE 1-continued

Entry	n	R	R ³
584	2	CO ₂ tBu	2-pyrimidinyl
585	3	CO ₂ tBu	2-pyrimidinyl
586	1	CONHMe	2-pyrimidinyl
587	2	CONHMe	2-pyrimidinyl
588	3	CONHMe	2-pyrimidinyl
589	1	SO ₂ Me	2-pyrimidinyl
590	2	SO ₂ Me	2-pyrimidinyl
591	3	SO ₂ Me	2-pyrimidinyl
592	1	SO ₂ NH ₂	2-pyrimidinyl
593	2	SO ₂ NH ₂	2-pyrimidinyl
594	3	SO ₂ NH ₂	2-pyrimidinyl
595	1	H	2-methyl-1H- benzo[d]imidazol-4-yl
596	2	H	2-methyl-1H- benzo[d]imidazol-4-yl
597	3	H	2-methyl-1H- benzo[d]imidazol-4-yl
598	1	Me	2-methyl-1H- benzo[d]imidazol-4-yl
599	2	Me	2-methyl-1H- benzo[d]imidazol-4-yl
600	3	Me	2-methyl-1H- benzo[d]imidazol-4-yl
601	1	CH ₂ Ph	2-methyl-1H- benzo[d]imidazol-4-yl
602	2	CH ₂ Ph	2-methyl-1H- benzo[d]imidazol-4-yl
603	3	CH ₂ Ph	2-methyl-1H- benzo[d]imidazol-4-yl
604	1	COMe	2-methyl-1H- benzo[d]imidazol-4-yl
605	2	COMe	2-methyl-1H- benzo[d]imidazol-4-yl
606	3	COMe	2-methyl-1H- benzo[d]imidazol-4-yl
607	1	CO ₂ Me	2-methyl-1H- benzo[d]imidazol-4-yl
608	2	CO ₂ Me	2-methyl-1H- benzo[d]imidazol-4-yl
609	3	CO ₂ Me	2-methyl-1H- benzo[d]imidazol-4-yl
610	1	CO ₂ tBu	2-methyl-1H- benzo[d]imidazol-4-yl
611	2	CO ₂ tBu	2-methyl-1H- benzo[d]imidazol-4-yl
612	3	CO ₂ tBu	2-methyl-1H- benzo[d]imidazol-4-yl
613	1	CONHMe	2-methyl-1H- benzo[d]imidazol-4-yl
614	2	CONHMe	2-methyl-1H- benzo[d]imidazol-4-yl
615	3	CONHMe	2-methyl-1H- benzo[d]imidazol-4-yl
616	1	SO ₂ Me	2-methyl-1H- benzo[d]imidazol-4-yl
617	2	SO ₂ Me	2-methyl-1H- benzo[d]imidazol-4-yl
618	3	SO ₂ Me	2-methyl-1H- benzo[d]imidazol-4-yl
619	1	SO ₂ NH ₂	2-methyl-1H- benzo[d]imidazol-4-yl
620	2	SO ₂ NH ₂	2-methyl-1H- benzo[d]imidazol-4-yl
621	3	SO ₂ NH ₂	2-methyl-1H- benzo[d]imidazol-4-yl
622	1	H	4-OH-Phenyl
623	2	H	4-OH-Phenyl
624	3	H	4-OH-Phenyl
625	1	Me	4-OH-Phenyl
626	2	Me	4-OH-Phenyl
627	3	Me	4-OH-Phenyl
628	1	CH ₂ Ph	4-OH-Phenyl
629	2	CH ₂ Ph	4-OH-Phenyl
630	3	CH ₂ Ph	4-OH-Phenyl
631	1	COMe	4-OH-Phenyl

TABLE 1-continued

Entry	n	R	R ³
632	2	COMe	4-OH-Phenyl
633	3	COMe	4-OH-Phenyl
634	1	CO ₂ Me	4-OH-Phenyl
635	2	CO ₂ Me	4-OH-Phenyl
636	3	CO ₂ Me	4-OH-Phenyl
637	1	CO ₂ tBu	4-OH-Phenyl
638	2	CO ₂ tBu	4-OH-Phenyl
639	3	CO ₂ tBu	4-OH-Phenyl
640	1	CONHMe	4-OH-Phenyl
641	2	CONHMe	4-OH-Phenyl
642	3	CONHMe	4-OH-Phenyl
643	1	SO ₂ Me	4-OH-Phenyl
644	2	SO ₂ Me	4-OH-Phenyl
645	3	SO ₂ Me	4-OH-Phenyl
646	1	SO ₂ NH ₂	4-OH-Phenyl
647	2	SO ₂ NH ₂	4-OH-Phenyl
648	3	SO ₂ NH ₂	4-OH-Phenyl
649	1	H	2-OH-Phenyl
650	2	H	2-OH-Phenyl
651	3	H	2-OH-Phenyl
652	1	Me	2-OH-Phenyl
653	2	Me	2-OH-Phenyl
654	3	Me	2-OH-Phenyl
655	1	CH ₂ Ph	2-OH-Phenyl
656	2	CH ₂ Ph	2-OH-Phenyl
657	3	CH ₂ Ph	2-OH-Phenyl
658	1	COMe	2-OH-Phenyl
659	2	COMe	2-OH-Phenyl
660	3	COMe	2-OH-Phenyl
661	1	CO ₂ Me	2-OH-Phenyl
662	2	CO ₂ Me	2-OH-Phenyl
663	3	CO ₂ Me	2-OH-Phenyl
664	1	CO ₂ tBu	2-OH-Phenyl
665	2	CO ₂ tBu	2-OH-Phenyl
666	3	CO ₂ tBu	2-OH-Phenyl
667	1	CONHMe	2-OH-Phenyl
668	2	CONHMe	2-OH-Phenyl
669	3	CONHMe	2-OH-Phenyl
670	1	SO ₂ Me	2-OH-Phenyl
671	2	SO ₂ Me	2-OH-Phenyl
672	3	SO ₂ Me	2-OH-Phenyl
673	1	SO ₂ NH ₂	2-OH-Phenyl
674	2	SO ₂ NH ₂	2-OH-Phenyl
675	3	SO ₂ NH ₂	2-OH-Phenyl
676	1	H	4-OMe-Phenyl
677	2	H	4-OMe-Phenyl
678	3	H	4-OMe-Phenyl
679	1	Me	4-OMe-Phenyl
680	2	Me	4-OMe-Phenyl
681	3	Me	4-OMe-Phenyl
682	1	CH ₂ Ph	4-OMe-Phenyl
683	2	CH ₂ Ph	4-OMe-Phenyl
684	3	CH ₂ Ph	4-OMe-Phenyl
685	1	COMe	4-OMe-Phenyl
686	2	COMe	4-OMe-Phenyl
687	3	COMe	4-OMe-Phenyl
688	1	CO ₂ Me	4-OMe-Phenyl
689	2	CO ₂ Me	4-OMe-Phenyl
690	3	CO ₂ Me	4-OMe-Phenyl
691	1	CO ₂ tBu	4-OMe-Phenyl
692	2	CO ₂ tBu	4-OMe-Phenyl
693	3	CO ₂ tBu	4-OMe-Phenyl
694	1	CONHMe	4-OMe-Phenyl
695	2	CONHMe	4-OMe-Phenyl
696	3	CONHMe	4-OMe-Phenyl
697	1	SO ₂ Me	4-OMe-Phenyl
698	2	SO ₂ Me	4-OMe-Phenyl
699	3	SO ₂ Me	4-OMe-Phenyl
700	1	SO ₂ NH ₂	4-OMe-Phenyl
701	2	SO ₂ NH ₂	4-OMe-Phenyl
702	3	SO ₂ NH ₂	4-OMe-Phenyl
703	1	H	2-OMe-Phenyl
704	2	H	2-OMe-Phenyl
705	3	H	2-OMe-Phenyl
706	1	Me	2-OMe-Phenyl

TABLE 1-continued

Entry	n	R	R ³
707	2	Me	2-OMe-Phenyl
708	3	Me	2-OMe-Phenyl
709	1	CH ₂ Ph	2-OMe-Phenyl
710	2	CH ₂ Ph	2-OMe-Phenyl
711	3	CH ₂ Ph	2-OMe-Phenyl
712	1	COMe	2-OMe-Phenyl
713	2	COMe	2-OMe-Phenyl
714	3	COMe	2-OMe-Phenyl
715	1	CO ₂ Me	2-OMe-Phenyl
716	2	CO ₂ Me	2-OMe-Phenyl
717	3	CO ₂ Me	2-OMe-Phenyl
718	1	CO ₂ tBu	2-OMe-Phenyl
719	2	CO ₂ tBu	2-OMe-Phenyl
720	3	CO ₂ tBu	2-OMe-Phenyl
721	1	CONHMe	2-OMe-Phenyl
722	2	CONHMe	2-OMe-Phenyl
723	3	CONHMe	2-OMe-Phenyl
724	1	SO ₂ Me	2-OMe-Phenyl
725	2	SO ₂ Me	2-OMe-Phenyl
726	3	SO ₂ Me	2-OMe-Phenyl
727	1	SO ₂ NH ₂	2-OMe-Phenyl
728	2	SO ₂ NH ₂	2-OMe-Phenyl
729	3	SO ₂ NH ₂	2-OMe-Phenyl
730	1	H	3-CN-Phenyl
731	2	H	3-CN-Phenyl
732	3	H	3-CN-Phenyl
733	1	Me	3-CN-Phenyl
734	2	Me	3-CN-Phenyl
735	3	Me	3-CN-Phenyl
736	1	CH ₂ Ph	3-CN-Phenyl
737	2	CH ₂ Ph	3-CN-Phenyl
738	3	CH ₂ Ph	3-CN-Phenyl
739	1	COMe	3-CN-Phenyl
740	2	COMe	3-CN-Phenyl
741	3	COMe	3-CN-Phenyl
742	1	CO ₂ Me	3-CN-Phenyl
743	2	CO ₂ Me	3-CN-Phenyl
744	3	CO ₂ Me	3-CN-Phenyl
745	1	CO ₂ tBu	3-CN-Phenyl
746	2	CO ₂ tBu	3-CN-Phenyl
747	3	CO ₂ tBu	3-CN-Phenyl
748	1	CONHMe	3-CN-Phenyl
749	2	CONHMe	3-CN-Phenyl
750	3	CONHMe	3-CN-Phenyl
751	1	SO ₂ Me	3-CN-Phenyl
752	2	SO ₂ Me	3-CN-Phenyl
753	3	SO ₂ Me	3-CN-Phenyl
754	1	SO ₂ NH ₂	3-CN-Phenyl
755	2	SO ₂ NH ₂	3-CN-Phenyl
756	3	SO ₂ NH ₂	3-CN-Phenyl
757	1	H	2-Me-Phenyl
758	2	H	2-Me-Phenyl
759	3	H	2-Me-Phenyl
760	1	Me	2-Me-Phenyl
761	2	Me	2-Me-Phenyl
762	3	Me	2-Me-Phenyl
763	1	CH ₂ Ph	2-Me-Phenyl
764	2	CH ₂ Ph	2-Me-Phenyl
765	3	CH ₂ Ph	2-Me-Phenyl
766	1	COMe	2-Me-Phenyl
767	2	COMe	2-Me-Phenyl
768	3	COMe	2-Me-Phenyl
769	1	CO ₂ Me	2-Me-Phenyl
770	2	CO ₂ Me	2-Me-Phenyl
771	3	CO ₂ Me	2-Me-Phenyl
772	1	CO ₂ tBu	2-Me-Phenyl
773	2	CO ₂ tBu	2-Me-Phenyl
774	3	CO ₂ tBu	2-Me-Phenyl
775	1	CONHMe	2-Me-Phenyl
776	2	CONHMe	2-Me-Phenyl
777	3	CONHMe	2-Me-Phenyl
778	1	SO ₂ Me	2-Me-Phenyl
779	2	SO ₂ Me	2-Me-Phenyl
780	3	SO ₂ Me	2-Me-Phenyl
781	1	SO ₂ NH ₂	2-Me-Phenyl

TABLE 1-continued

Entry	n	R	R ³
782	2	SO ₂ NH ₂	2-Me-Phenyl
783	3	SO ₂ NH ₂	2-Me-Phenyl
784	1	H	4-Me-Phenyl
785	2	H	4-Me-Phenyl
786	3	H	4-Me-Phenyl
787	1	Me	4-Me-Phenyl
788	2	Me	4-Me-Phenyl
789	3	Me	4-Me-Phenyl
790	1	CH ₂ Ph	4-Me-Phenyl
791	2	CH ₂ Ph	4-Me-Phenyl
792	3	CH ₂ Ph	4-Me-Phenyl
793	1	COMe	4-Me-Phenyl
794	2	COMe	4-Me-Phenyl
795	3	COMe	4-Me-Phenyl
796	1	CO ₂ Me	4-Me-Phenyl
797	2	CO ₂ Me	4-Me-Phenyl
798	3	CO ₂ Me	4-Me-Phenyl
799	1	CO ₂ tBu	4-Me-Phenyl
800	2	CO ₂ tBu	4-Me-Phenyl
801	3	CO ₂ tBu	4-Me-Phenyl
802	1	CONHMe	4-Me-Phenyl
803	2	CONHMe	4-Me-Phenyl
804	3	CONHMe	4-Me-Phenyl
805	1	SO ₂ Me	4-Me-Phenyl
806	2	SO ₂ Me	4-Me-Phenyl
807	3	SO ₂ Me	4-Me-Phenyl
808	1	SO ₂ NH ₂	4-Me-Phenyl
809	2	SO ₂ NH ₂	4-Me-Phenyl
810	3	SO ₂ NH ₂	4-Me-Phenyl
811	1	H	3-F-Phenyl
812	2	H	3-F-Phenyl
813	3	H	3-F-Phenyl
814	1	Me	3-F-Phenyl
815	2	Me	3-F-Phenyl
816	3	Me	3-F-Phenyl
817	1	CH ₂ Ph	3-F-Phenyl
818	2	CH ₂ Ph	3-F-Phenyl
819	3	CH ₂ Ph	3-F-Phenyl
820	1	COMe	3-F-Phenyl
821	2	COMe	3-F-Phenyl
822	3	COMe	3-F-Phenyl
823	1	CO ₂ Me	3-F-Phenyl
824	2	CO ₂ Me	3-F-Phenyl
825	3	CO ₂ Me	3-F-Phenyl
826	1	CO ₂ tBu	3-F-Phenyl
827	2	CO ₂ tBu	3-F-Phenyl
828	3	CO ₂ tBu	3-F-Phenyl
829	1	CONHMe	3-F-Phenyl
830	2	CONHMe	3-F-Phenyl
831	3	CONHMe	3-F-Phenyl
832	1	SO ₂ Me	3-F-Phenyl
833	2	SO ₂ Me	3-F-Phenyl
834	3	SO ₂ Me	3-F-Phenyl
835	1	SO ₂ NH ₂	3-F-Phenyl
836	2	SO ₂ NH ₂	3-F-Phenyl
837	3	SO ₂ NH ₂	3-F-Phenyl
838	1	H	2-Cl-Phenyl
839	2	H	2-Cl-Phenyl
840	3	H	2-Cl-Phenyl
841	1	Me	2-Cl-Phenyl
842	2	Me	2-Cl-Phenyl
843	3	Me	2-Cl-Phenyl
844	1	CH ₂ Ph	2-Cl-Phenyl
845	2	CH ₂ Ph	2-Cl-Phenyl
846	3	CH ₂ Ph	2-Cl-Phenyl
847	1	COMe	2-Cl-Phenyl
848	2	COMe	2-Cl-Phenyl
849	3	COMe	2-Cl-Phenyl
850	1	CO ₂ Me	2-Cl-Phenyl
851	2	CO ₂ Me	2-Cl-Phenyl
852	3	CO ₂ Me	2-Cl-Phenyl
853	1	CO ₂ tBu	2-Cl-Phenyl
854	2	CO ₂ tBu	2-Cl-Phenyl
855	3	CO ₂ tBu	2-Cl-Phenyl
856	1	CONHMe	2-Cl-Phenyl

TABLE 1-continued

Entry	n	R	R ³
857	2	CONHMe	2-Cl-Phenyl
858	3	CONHMe	2-Cl-Phenyl
859	1	SO ₂ Me	2-Cl-Phenyl
860	2	SO ₂ Me	2-Cl-Phenyl
861	3	SO ₂ Me	2-Cl-Phenyl
862	1	SO ₂ NH ₂	2-Cl-Phenyl
863	2	SO ₂ NH ₂	2-Cl-Phenyl
864	3	SO ₂ NH ₂	2-Cl-Phenyl
865	1	H	4-Cl-Phenyl
866	2	H	4-Cl-Phenyl
867	3	H	4-Cl-Phenyl
868	1	Me	4-Cl-Phenyl
869	2	Me	4-Cl-Phenyl
870	3	Me	4-Cl-Phenyl
871	1	CH ₂ Ph	4-Cl-Phenyl
872	2	CH ₂ Ph	4-Cl-Phenyl
873	3	CH ₂ Ph	4-Cl-Phenyl
874	1	COMe	4-Cl-Phenyl
875	2	COMe	4-Cl-Phenyl
876	3	COMe	4-Cl-Phenyl
877	1	CO ₂ Me	4-Cl-Phenyl
878	2	CO ₂ Me	4-Cl-Phenyl
879	3	CO ₂ Me	4-Cl-Phenyl
880	1	CO ₂ tBu	4-Cl-Phenyl
881	2	CO ₂ tBu	4-Cl-Phenyl
882	3	CO ₂ tBu	4-Cl-Phenyl
883	1	CONHMe	4-Cl-Phenyl
884	2	CONHMe	4-Cl-Phenyl
885	3	CONHMe	4-Cl-Phenyl
886	1	SO ₂ Me	4-Cl-Phenyl
887	2	SO ₂ Me	4-Cl-Phenyl
888	3	SO ₂ Me	4-Cl-Phenyl
889	1	SO ₂ NH ₂	4-Cl-Phenyl
890	2	SO ₂ NH ₂	4-Cl-Phenyl
891	3	SO ₂ NH ₂	4-Cl-Phenyl
892	1	H	3-Br-Phenyl
893	2	H	3-Br-Phenyl
894	3	H	3-Br-Phenyl
895	1	Me	3-Br-Phenyl
896	2	Me	3-Br-Phenyl
897	3	Me	3-Br-Phenyl
898	1	CH ₂ Ph	3-Br-Phenyl
899	2	CH ₂ Ph	3-Br-Phenyl
900	3	CH ₂ Ph	3-Br-Phenyl
901	1	COMe	3-Br-Phenyl
902	2	COMe	3-Br-Phenyl
903	3	COMe	3-Br-Phenyl
904	1	CO ₂ Me	3-Br-Phenyl
905	2	CO ₂ Me	3-Br-Phenyl
906	3	CO ₂ Me	3-Br-Phenyl
907	1	CO ₂ tBu	3-Br-Phenyl
908	2	CO ₂ tBu	3-Br-Phenyl
909	3	CO ₂ tBu	3-Br-Phenyl
910	1	CONHMe	3-Br-Phenyl
911	2	CONHMe	3-Br-Phenyl
912	3	CONHMe	3-Br-Phenyl
913	1	SO ₂ Me	3-Br-Phenyl
914	2	SO ₂ Me	3-Br-Phenyl
915	3	SO ₂ Me	3-Br-Phenyl
916	1	SO ₂ NH ₂	3-Br-Phenyl
917	2	SO ₂ NH ₂	3-Br-Phenyl
918	3	SO ₂ NH ₂	3-Br-Phenyl
919	1	H	2-CF ₃ -Phenyl
920	2	H	2-CF ₃ -Phenyl
921	3	H	2-CF ₃ -Phenyl
922	1	Me	2-CF ₃ -Phenyl
923	2	Me	2-CF ₃ -Phenyl
924	3	Me	2-CF ₃ -Phenyl
925	1	CH ₂ Ph	2-CF ₃ -Phenyl
926	2	CH ₂ Ph	2-CF ₃ -Phenyl
927	3	CH ₂ Ph	2-CF ₃ -Phenyl
928	1	COMe	2-CF ₃ -Phenyl
929	2	COMe	2-CF ₃ -Phenyl
930	3	COMe	2-CF ₃ -Phenyl
931	1	CO ₂ Me	2-CF ₃ -Phenyl

TABLE 1-continued

Entry	n	R	R ³
932	2	CO ₂ Me	2-CF ₃ -Phenyl
933	3	CO ₂ Me	2-CF ₃ -Phenyl
934	1	CO ₂ tBu	2-CF ₃ -Phenyl
935	2	CO ₂ tBu	2-CF ₃ -Phenyl
936	3	CO ₂ tBu	2-CF ₃ -Phenyl
937	1	CONHMe	2-CF ₃ -Phenyl
938	2	CONHMe	2-CF ₃ -Phenyl
939	3	CONHMe	2-CF ₃ -Phenyl
940	1	SO ₂ Me	2-CF ₃ -Phenyl
941	2	SO ₂ Me	2-CF ₃ -Phenyl
942	3	SO ₂ Me	2-CF ₃ -Phenyl
943	1	SO ₂ NH ₂	2-CF ₃ -Phenyl
944	2	SO ₂ NH ₂	2-CF ₃ -Phenyl
945	3	SO ₂ NH ₂	2-CF ₃ -Phenyl
946	1	H	4-CF ₃ -Phenyl
947	2	H	4-CF ₃ -Phenyl
948	3	H	4-CF ₃ -Phenyl
949	1	Me	4-CF ₃ -Phenyl
950	2	Me	4-CF ₃ -Phenyl
951	3	Me	4-CF ₃ -Phenyl
952	1	CH ₂ Ph	4-CF ₃ -Phenyl
953	2	CH ₂ Ph	4-CF ₃ -Phenyl
954	3	CH ₂ Ph	4-CF ₃ -Phenyl
955	1	COMe	4-CF ₃ -Phenyl
956	2	COMe	4-CF ₃ -Phenyl
957	3	COMe	4-CF ₃ -Phenyl
958	1	CO ₂ Me	4-CF ₃ -Phenyl
959	2	CO ₂ Me	4-CF ₃ -Phenyl
960	3	CO ₂ Me	4-CF ₃ -Phenyl
961	1	CO ₂ tBu	4-CF ₃ -Phenyl
962	2	CO ₂ tBu	4-CF ₃ -Phenyl
963	3	CO ₂ tBu	4-CF ₃ -Phenyl
964	1	CONHMe	4-CF ₃ -Phenyl
965	2	CONHMe	4-CF ₃ -Phenyl
966	3	CONHMe	4-CF ₃ -Phenyl
967	1	SO ₂ Me	4-CF ₃ -Phenyl
968	2	SO ₂ Me	4-CF ₃ -Phenyl
969	3	SO ₂ Me	4-CF ₃ -Phenyl
970	1	SO ₂ NH ₂	4-CF ₃ -Phenyl
971	2	SO ₂ NH ₂	4-CF ₃ -Phenyl
972	3	SO ₂ NH ₂	4-CF ₃ -Phenyl
973	1	H	3-iPr-Phenyl
974	2	H	3-iPr-Phenyl
975	3	H	3-iPr-Phenyl
976	1	Me	3-iPr-Phenyl
977	2	Me	3-iPr-Phenyl
978	3	Me	3-iPr-Phenyl
979	1	CH ₂ Ph	3-iPr-Phenyl
980	2	CH ₂ Ph	3-iPr-Phenyl
981	3	CH ₂ Ph	3-iPr-Phenyl
982	1	COMe	3-iPr-Phenyl
983	2	COMe	3-iPr-Phenyl
984	3	COMe	3-iPr-Phenyl
985	1	CO ₂ Me	3-iPr-Phenyl
986	2	CO ₂ Me	3-iPr-Phenyl
987	3	CO ₂ Me	3-iPr-Phenyl
988	1	CO ₂ tBu	3-iPr-Phenyl
989	2	CO ₂ tBu	3-iPr-Phenyl
990	3	CO ₂ tBu	3-iPr-Phenyl
991	1	CONHMe	3-iPr-Phenyl
992	2	CONHMe	3-iPr-Phenyl
993	3	CONHMe	3-iPr-Phenyl
994	1	SO ₂ Me	3-iPr-Phenyl
995	2	SO ₂ Me	3-iPr-Phenyl
996	3	SO ₂ Me	3-iPr-Phenyl
997	1	SO ₂ NH ₂	3-iPr-Phenyl
998	2	SO ₂ NH ₂	3-iPr-Phenyl
999	3	SO ₂ NH ₂	3-iPr-Phenyl
1000	1	H	4-NH ₂ -Phenyl
1001	2	H	4-NH ₂ -Phenyl
1002	3	H	4-NH ₂ -Phenyl
1003	1	Me	4-NH ₂ -Phenyl
1004	2	Me	4-NH ₂ -Phenyl
1005	3	Me	4-NH ₂ -Phenyl
1006	1	CH ₂ Ph	4-NH ₂ -Phenyl

TABLE 1-continued

Entry	n	R	R ³
1007	2	CH ₂ Ph	4-NH ₂ -Phenyl
1008	3	CH ₂ Ph	4-NH ₂ -Phenyl
1009	1	COMe	4-NH ₂ -Phenyl
1010	2	COMe	4-NH ₂ -Phenyl
1011	3	COMe	4-NH ₂ -Phenyl
1012	1	CO ₂ Me	4-NH ₂ -Phenyl
1013	2	CO ₂ Me	4-NH ₂ -Phenyl
1014	3	CO ₂ Me	4-NH ₂ -Phenyl
1015	1	CO ₂ tBu	4-NH ₂ -Phenyl
1016	2	CO ₂ tBu	4-NH ₂ -Phenyl
1017	3	CO ₂ tBu	4-NH ₂ -Phenyl
1018	1	CONHMe	4-NH ₂ -Phenyl
1019	2	CONHMe	4-NH ₂ -Phenyl
1020	3	CONHMe	4-NH ₂ -Phenyl
1021	1	SO ₂ Me	4-NH ₂ -Phenyl
1022	2	SO ₂ Me	4-NH ₂ -Phenyl
1023	3	SO ₂ Me	4-NH ₂ -Phenyl
1024	1	SO ₂ NH ₂	4-NH ₂ -Phenyl
1025	2	SO ₂ NH ₂	4-NH ₂ -Phenyl
1026	3	SO ₂ NH ₂	4-NH ₂ -Phenyl
1027	1	H	2-NH ₂ -Phenyl
1028	2	H	2-NH ₂ -Phenyl
1029	3	H	2-NH ₂ -Phenyl
1030	1	Me	2-NH ₂ -Phenyl
1031	2	Me	2-NH ₂ -Phenyl
1032	3	Me	2-NH ₂ -Phenyl
1033	1	CH ₂ Ph	2-NH ₂ -Phenyl
1034	2	CH ₂ Ph	2-NH ₂ -Phenyl
1035	3	CH ₂ Ph	2-NH ₂ -Phenyl
1036	1	COMe	2-NH ₂ -Phenyl
1037	2	COMe	2-NH ₂ -Phenyl
1038	3	COMe	2-NH ₂ -Phenyl
1039	1	CO ₂ Me	2-NH ₂ -Phenyl
1040	2	CO ₂ Me	2-NH ₂ -Phenyl
1041	3	CO ₂ Me	2-NH ₂ -Phenyl
1042	1	CO ₂ tBu	2-NH ₂ -Phenyl
1043	2	CO ₂ tBu	2-NH ₂ -Phenyl
1044	3	CO ₂ tBu	2-NH ₂ -Phenyl
1045	1	CONHMe	2-NH ₂ -Phenyl
1046	2	CONHMe	2-NH ₂ -Phenyl
1047	3	CONHMe	2-NH ₂ -Phenyl
1048	1	SO ₂ Me	2-NH ₂ -Phenyl
1049	2	SO ₂ Me	2-NH ₂ -Phenyl
1050	3	SO ₂ Me	2-NH ₂ -Phenyl
1051	1	SO ₂ NH ₂	2-NH ₂ -Phenyl
1052	2	SO ₂ NH ₂	2-NH ₂ -Phenyl
1053	3	SO ₂ NH ₂	2-NH ₂ -Phenyl
1054	1	H	2,6-di-Me-Phenyl
1055	2	H	2,6-di-Me-Phenyl
1056	3	H	2,6-di-Me-Phenyl
1057	1	Me	2,6-di-Me-Phenyl
1058	2	Me	2,6-di-Me-Phenyl
1059	3	Me	2,6-di-Me-Phenyl
1060	1	CH ₂ Ph	2,6-di-Me-Phenyl
1061	2	CH ₂ Ph	2,6-di-Me-Phenyl
1062	3	CH ₂ Ph	2,6-di-Me-Phenyl
1063	1	COMe	2,6-di-Me-Phenyl
1064	2	COMe	2,6-di-Me-Phenyl
1065	3	COMe	2,6-di-Me-Phenyl
1066	1	CO ₂ Me	2,6-di-Me-Phenyl
1067	2	CO ₂ Me	2,6-di-Me-Phenyl
1068	3	CO ₂ Me	2,6-di-Me-Phenyl
1069	1	CO ₂ tBu	2,6-di-Me-Phenyl
1070	2	CO ₂ tBu	2,6-di-Me-Phenyl
1071	3	CO ₂ tBu	2,6-di-Me-Phenyl
1072	1	CONHMe	2,6-di-Me-Phenyl
1073	2	CONHMe	2,6-di-Me-Phenyl
1074	3	CONHMe	2,6-di-Me-Phenyl
1075	1	SO ₂ Me	2,6-di-Me-Phenyl
1076	2	SO ₂ Me	2,6-di-Me-Phenyl
1077	3	SO ₂ Me	2,6-di-Me-Phenyl
1078	1	SO ₂ NH ₂	2,6-di-Me-Phenyl
1079	2	SO ₂ NH ₂	2,6-di-Me-Phenyl
1080	3	SO ₂ NH ₂	2,6-di-Me-Phenyl
1081	1	H	2-Ph-Phenyl

TABLE 1-continued

Entry	n	R	R ³
1082	2	H	2-Ph-Phenyl
1083	3	H	2-Ph-Phenyl
1084	1	Me	2-Ph-Phenyl
1085	2	Me	2-Ph-Phenyl
1086	3	Me	2-Ph-Phenyl
1087	1	CH ₂ Ph	2-Ph-Phenyl
1088	2	CH ₂ Ph	2-Ph-Phenyl
1089	3	CH ₂ Ph	2-Ph-Phenyl
1090	1	COMe	2-Ph-Phenyl
1091	2	COMe	2-Ph-Phenyl
1092	3	COMe	2-Ph-Phenyl
1093	1	CO ₂ Me	2-Ph-Phenyl
1094	2	CO ₂ Me	2-Ph-Phenyl
1095	3	CO ₂ Me	2-Ph-Phenyl
1096	1	CO ₂ tBu	2-Ph-Phenyl
1097	2	CO ₂ tBu	2-Ph-Phenyl
1098	3	CO ₂ tBu	2-Ph-Phenyl
1099	1	CONHMe	2-Ph-Phenyl
1100	2	CONHMe	2-Ph-Phenyl
1101	3	CONHMe	2-Ph-Phenyl
1102	1	SO ₂ Me	2-Ph-Phenyl
1103	2	SO ₂ Me	2-Ph-Phenyl
1104	3	SO ₂ Me	2-Ph-Phenyl
1105	1	SO ₂ NH ₂	2-Ph-Phenyl
1106	2	SO ₂ NH ₂	2-Ph-Phenyl
1107	3	SO ₂ NH ₂	2-Ph-Phenyl
1108	1	H	4-Ph-Phenyl
1109	2	H	4-Ph-Phenyl
1110	3	H	4-Ph-Phenyl
1111	1	Me	4-Ph-Phenyl
1112	2	Me	4-Ph-Phenyl
1113	3	Me	4-Ph-Phenyl
1114	1	CH ₂ Ph	4-Ph-Phenyl
1115	2	CH ₂ Ph	4-Ph-Phenyl
1116	3	CH ₂ Ph	4-Ph-Phenyl
1117	1	COMe	4-Ph-Phenyl
1118	2	COMe	4-Ph-Phenyl
1119	3	COMe	4-Ph-Phenyl
1120	1	CO ₂ Me	4-Ph-Phenyl
1121	2	CO ₂ Me	4-Ph-Phenyl
1122	3	CO ₂ Me	4-Ph-Phenyl
1123	1	CO ₂ tBu	4-Ph-Phenyl
1124	2	CO ₂ tBu	4-Ph-Phenyl
1125	3	CO ₂ tBu	4-Ph-Phenyl
1126	1	CONHMe	4-Ph-Phenyl
1127	2	CONHMe	4-Ph-Phenyl
1128	3	CONHMe	4-Ph-Phenyl
1129	1	SO ₂ Me	4-Ph-Phenyl
1130	2	SO ₂ Me	4-Ph-Phenyl
1131	3	SO ₂ Me	4-Ph-Phenyl
1132	1	SO ₂ NH ₂	4-Ph-Phenyl
1133	2	SO ₂ NH ₂	4-Ph-Phenyl
1134	3	SO ₂ NH ₂	4-Ph-Phenyl
1135	1	H	3-morpholino-phenyl
1136	2	H	3-morpholino-phenyl
1137	3	H	3-morpholino-phenyl
1138	1	Me	3-morpholino-phenyl
1139	2	Me	3-morpholino-phenyl
1140	3	Me	3-morpholino-phenyl
1141	1	CH ₂ Ph	3-morpholino-phenyl
1142	2	CH ₂ Ph	3-morpholino-phenyl
1143	3	CH ₂ Ph	3-morpholino-phenyl
1144	1	COMe	3-morpholino-phenyl
1145	2	COMe	3-morpholino-phenyl
1146	3	COMe	3-morpholino-phenyl
1147	1	CO ₂ Me	3-morpholino-phenyl
1148	2	CO ₂ Me	3-morpholino-phenyl
1149	3	CO ₂ Me	3-morpholino-phenyl
1150	1	CO ₂ tBu	3-morpholino-phenyl
1151	2	CO ₂ tBu	3-morpholino-phenyl
1152	3	CO ₂ tBu	3-morpholino-phenyl
1153	1	CONHMe	3-morpholino-phenyl
1154	2	CONHMe	3-morpholino-phenyl
1155	3	CONHMe	3-morpholino-phenyl
1156	1	SO ₂ Me	3-morpholino-phenyl

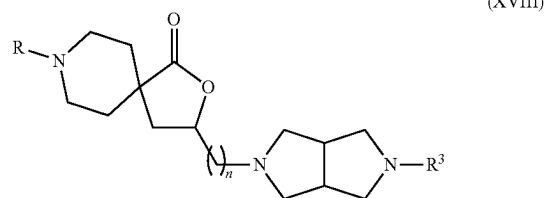
TABLE 1-continued

Entry	n	R	R ³
1157	2	SO ₂ Me	3-morpholino-phenyl
1158	3	SO ₂ Me	3-morpholino-phenyl
1159	1	SO ₂ NH ₂	3-morpholino-phenyl
1160	2	SO ₂ NH ₂	3-morpholino-phenyl
1161	3	SO ₂ NH ₂	3-morpholino-phenyl
1162	1	H	2-pyrazinyl
1163	2	H	2-pyrazinyl
1164	3	H	2-pyrazinyl
1165	1	Me	2-pyrazinyl
1166	2	Me	2-pyrazinyl
1167	3	Me	2-pyrazinyl
1168	1	CH ₂ Ph	2-pyrazinyl
1169	2	CH ₂ Ph	2-pyrazinyl
1170	3	CH ₂ Ph	2-pyrazinyl
1171	1	COMe	2-pyrazinyl
1172	2	COMe	2-pyrazinyl
1173	3	COMe	2-pyrazinyl
1174	1	CO ₂ Me	2-pyrazinyl
1175	2	CO ₂ Me	2-pyrazinyl
1176	3	CO ₂ Me	2-pyrazinyl
1177	1	CO ₂ tBu	2-pyrazinyl
1178	2	CO ₂ tBu	2-pyrazinyl
1179	3	CO ₂ tBu	2-pyrazinyl
1180	1	CONHMe	2-pyrazinyl
1181	2	CONHMe	2-pyrazinyl
1182	3	CONHMe	2-pyrazinyl
1183	1	SO ₂ Me	2-pyrazinyl
1184	2	SO ₂ Me	2-pyrazinyl
1185	3	SO ₂ Me	2-pyrazinyl
1186	1	SO ₂ NH ₂	2-pyrazinyl
1187	2	SO ₂ NH ₂	2-pyrazinyl
1188	3	SO ₂ NH ₂	2-pyrazinyl
1189	1	H	5-indolyl
1190	2	H	5-indolyl
1191	3	H	5-indolyl
1192	1	Me	5-indolyl
1193	2	Me	5-indolyl
1194	3	Me	5-indolyl
1195	1	CH ₂ Ph	5-indolyl
1196	2	CH ₂ Ph	5-indolyl
1197	3	CH ₂ Ph	5-indolyl
1198	1	COMe	5-indolyl
1199	2	COMe	5-indolyl
1200	3	COMe	5-indolyl
1201	1	CO ₂ Me	5-indolyl
1202	2	CO ₂ Me	5-indolyl
1203	3	CO ₂ Me	5-indolyl
1204	1	CO ₂ tBu	5-indolyl
1205	2	CO ₂ tBu	5-indolyl
1206	3	CO ₂ tBu	5-indolyl
1207	1	CONHMe	5-indolyl
1208	2	CONHMe	5-indolyl
1209	3	CONHMe	5-indolyl
1210	1	SO ₂ Me	5-indolyl
1211	2	SO ₂ Me	5-indolyl
1212	3	SO ₂ Me	5-indolyl
1213	1	SO ₂ NH ₂	5-indolyl
1214	2	SO ₂ NH ₂	5-indolyl
1215	3	SO ₂ NH ₂	5-indolyl
1216	1	H	1H-benzo[d]imidazol-4-yl
1217	2	H	1H-benzo[d]imidazol-4-yl
1218	3	H	1H-benzo[d]imidazol-4-yl
1219	1	Me	1H-benzo[d]imidazol-4-yl
1220	2	Me	1H-benzo[d]imidazol-4-yl
1221	3	Me	1H-benzo[d]imidazol-4-yl
1222	1	CH ₂ Ph	1H-benzo[d]imidazol-4-yl
1223	2	CH ₂ Ph	1H-benzo[d]imidazol-4-yl

TABLE 1-continued

Entry	n	R	R ³
1224	3	CH ₂ Ph	1H-benzo[d]imidazol-4-yl
1225	1	COMe	1H-benzo[d]imidazol-4-yl
1226	2	COMe	1H-benzo[d]imidazol-4-yl
1227	3	COMe	1H-benzo[d]imidazol-4-yl
1228	1	CO ₂ Me	1H-benzo[d]imidazol-4-yl
1229	2	CO ₂ Me	1H-benzo[d]imidazol-4-yl
1230	3	CO ₂ Me	1H-benzo[d]imidazol-4-yl
1231	1	CO ₂ tBu	1H-benzo[d]imidazol-4-yl
1232	2	CO ₂ tBu	1H-benzo[d]imidazol-4-yl
1233	3	CO ₂ tBu	1H-benzo[d]imidazol-4-yl
1234	1	CONHMe	1H-benzo[d]imidazol-4-yl
1235	2	CONHMe	1H-benzo[d]imidazol-4-yl
1236	3	CONHMe	1H-benzo[d]imidazol-4-yl
1237	1	SO ₂ Me	1H-benzo[d]imidazol-4-yl
1238	2	SO ₂ Me	1H-benzo[d]imidazol-4-yl
1239	3	SO ₂ Me	1H-benzo[d]imidazol-4-yl
1240	1	SO ₂ NH ₂	1H-benzo[d]imidazol-4-yl
1241	2	SO ₂ NH ₂	1H-benzo[d]imidazol-4-yl
1242	3	SO ₂ NH ₂	1H-benzo[d]imidazol-4-yl

[0401] Exemplary embodiments include compounds having the formula (XVIII)



or a pharmaceutically acceptable salt form thereof defined herein below in Table 2.

TABLE 2

Entry	n	R	R ³
1	1	H	Phenyl
2	2	H	Phenyl
3	3	H	Phenyl
4	1	Me	Phenyl
5	2	Me	Phenyl
6	3	Me	Phenyl
7	1	CH ₂ Ph	Phenyl
8	2	CH ₂ Ph	Phenyl
9	3	CH ₂ Ph	Phenyl
10	1	COMe	Phenyl

TABLE 2-continued

Entry	n	R	R ³
11	2	COMe	Phenyl
12	3	COMe	Phenyl
13	1	CO ₂ Me	Phenyl
14	2	CO ₂ Me	Phenyl
15	3	CO ₂ Me	Phenyl
16	1	CO ₂ tBu	Phenyl
17	2	CO ₂ tBu	Phenyl
18	3	CO ₂ tBu	Phenyl
19	1	CONHMe	Phenyl
20	2	CONHMe	Phenyl
21	3	CONHMe	Phenyl
22	1	SO ₂ Me	Phenyl
23	2	SO ₂ Me	Phenyl
24	3	SO ₂ Me	Phenyl
25	1	SO ₂ NH ₂	Phenyl
26	2	SO ₂ NH ₂	Phenyl
27	3	SO ₂ NH ₂	Phenyl
28	1	H	3-OH-Phenyl
29	2	H	3-OH-Phenyl
30	3	H	3-OH-Phenyl
31	1	Me	3-OH-Phenyl
32	2	Me	3-OH-Phenyl
33	3	Me	3-OH-Phenyl
34	1	CH ₂ Ph	3-OH-Phenyl
35	2	CH ₂ Ph	3-OH-Phenyl
36	3	CH ₂ Ph	3-OH-Phenyl
37	1	COMe	3-OH-Phenyl
38	2	COMe	3-OH-Phenyl
39	3	COMe	3-OH-Phenyl
40	1	CO ₂ Me	3-OH-Phenyl
41	2	CO ₂ Me	3-OH-Phenyl
42	3	CO ₂ Me	3-OH-Phenyl
43	1	CO ₂ tBu	3-OH-Phenyl
44	2	CO ₂ tBu	3-OH-Phenyl
45	3	CO ₂ tBu	3-OH-Phenyl
46	1	CONHMe	3-OH-Phenyl
47	2	CONHMe	3-OH-Phenyl
48	3	CONHMe	3-OH-Phenyl
49	1	SO ₂ Me	3-OH-Phenyl
50	2	SO ₂ Me	3-OH-Phenyl
51	3	SO ₂ Me	3-OH-Phenyl
52	1	SO ₂ NH ₂	3-OH-Phenyl
53	2	SO ₂ NH ₂	3-OH-Phenyl
54	3	SO ₂ NH ₂	3-OH-Phenyl
55	1	H	4-NO ₂ -Phenyl
56	2	H	4-NO ₂ -Phenyl
57	3	H	4-NO ₂ -Phenyl
58	1	Me	4-NO ₂ -Phenyl
59	2	Me	4-NO ₂ -Phenyl
60	3	Me	4-NO ₂ -Phenyl
61	1	CH ₂ Ph	4-NO ₂ -Phenyl
62	2	CH ₂ Ph	4-NO ₂ -Phenyl
63	3	CH ₂ Ph	4-NO ₂ -Phenyl
64	1	COMe	4-NO ₂ -Phenyl
65	2	COMe	4-NO ₂ -Phenyl
66	3	COMe	4-NO ₂ -Phenyl
67	1	CO ₂ Me	4-NO ₂ -Phenyl
68	2	CO ₂ Me	4-NO ₂ -Phenyl
69	3	CO ₂ Me	4-NO ₂ -Phenyl
70	1	CO ₂ tBu	4-NO ₂ -Phenyl
71	2	CO ₂ tBu	4-NO ₂ -Phenyl
72	3	CO ₂ tBu	4-NO ₂ -Phenyl
73	1	CONHMe	4-NO ₂ -Phenyl
74	2	CONHMe	4-NO ₂ -Phenyl
75	3	CONHMe	4-NO ₂ -Phenyl
76	1	SO ₂ Me	4-NO ₂ -Phenyl
77	2	SO ₂ Me	4-NO ₂ -Phenyl
78	3	SO ₂ Me	4-NO ₂ -Phenyl
79	1	SO ₂ NH ₂	4-NO ₂ -Phenyl
80	2	SO ₂ NH ₂	4-NO ₂ -Phenyl
81	3	SO ₂ NH ₂	4-NO ₂ -Phenyl
82	1	H	3-OMe-Phenyl
83	2	H	3-OMe-Phenyl
84	3	H	3-OMe-Phenyl
85	1	Me	3-OMe-Phenyl

TABLE 2-continued

Entry	n	R	R ³
86	2	Me	3-OMe-Phenyl
87	3	Me	3-OMe-Phenyl
88	1	CH ₂ Ph	3-OMe-Phenyl
89	2	CH ₂ Ph	3-OMe-Phenyl
90	3	CH ₂ Ph	3-OMe-Phenyl
91	1	COMe	3-OMe-Phenyl
92	2	COMe	3-OMe-Phenyl
93	3	COMe	3-OMe-Phenyl
94	1	CO ₂ Me	3-OMe-Phenyl
95	2	CO ₂ Me	3-OMe-Phenyl
96	3	CO ₂ Me	3-OMe-Phenyl
97	1	CO ₂ tBu	3-OMe-Phenyl
98	2	CO ₂ tBu	3-OMe-Phenyl
99	3	CO ₂ tBu	3-OMe-Phenyl
100	1	CONHMe	3-OMe-Phenyl
101	2	CONHMe	3-OMe-Phenyl
102	3	CONHMe	3-OMe-Phenyl
103	1	SO ₂ Me	3-OMe-Phenyl
104	2	SO ₂ Me	3-OMe-Phenyl
105	3	SO ₂ Me	3-OMe-Phenyl
106	1	SO ₂ NH ₂	3-OMe-Phenyl
107	2	SO ₂ NH ₂	3-OMe-Phenyl
108	3	SO ₂ NH ₂	3-OMe-Phenyl
109	1	H	4-CN-Phenyl
110	2	H	4-CN-Phenyl
111	3	H	4-CN-Phenyl
112	1	Me	4-CN-Phenyl
113	2	Me	4-CN-Phenyl
114	3	Me	4-CN-Phenyl
115	1	CH ₂ Ph	4-CN-Phenyl
116	2	CH ₂ Ph	4-CN-Phenyl
117	3	CH ₂ Ph	4-CN-Phenyl
118	1	COMe	4-CN-Phenyl
119	2	COMe	4-CN-Phenyl
120	3	COMe	4-CN-Phenyl
121	1	CO ₂ Me	4-CN-Phenyl
122	2	CO ₂ Me	4-CN-Phenyl
123	3	CO ₂ Me	4-CN-Phenyl
124	1	CO ₂ tBu	4-CN-Phenyl
125	2	CO ₂ tBu	4-CN-Phenyl
126	3	CO ₂ tBu	4-CN-Phenyl
127	1	CONHMe	4-CN-Phenyl
128	2	CONHMe	4-CN-Phenyl
129	3	CONHMe	4-CN-Phenyl
130	1	SO ₂ Me	4-CN-Phenyl
131	2	SO ₂ Me	4-CN-Phenyl
132	3	SO ₂ Me	4-CN-Phenyl
133	1	SO ₂ NH ₂	4-CN-Phenyl
134	2	SO ₂ NH ₂	4-CN-Phenyl
135	3	SO ₂ NH ₂	4-CN-Phenyl
136	1	H	2-CN-Phenyl
137	2	H	2-CN-Phenyl
138	3	H	2-CN-Phenyl
139	1	Me	2-CN-Phenyl
140	2	Me	2-CN-Phenyl
141	3	Me	2-CN-Phenyl
142	1	CH ₂ Ph	2-CN-Phenyl
143	2	CH ₂ Ph	2-CN-Phenyl
144	3	CH ₂ Ph	2-CN-Phenyl
145	1	COMe	2-CN-Phenyl
146	2	COMe	2-CN-Phenyl
147	3	COMe	2-CN-Phenyl
148	1	CO ₂ Me	2-CN-Phenyl
149	2	CO ₂ Me	2-CN-Phenyl
150	3	CO ₂ Me	2-CN-Phenyl
151	1	CO ₂ tBu	2-CN-Phenyl
152	2	CO ₂ tBu	2-CN-Phenyl
153	3	CO ₂ tBu	2-CN-Phenyl
154	1	CONHMe	2-CN-Phenyl
155	2	CONHMe	2-CN-Phenyl
156	3	CONHMe	2-CN-Phenyl
157	1	SO ₂ Me	2-CN-Phenyl
158	2	SO ₂ Me	2-CN-Phenyl
159	3	SO ₂ Me	2-CN-Phenyl
160	1	SO ₂ NH ₂	2-CN-Phenyl

TABLE 2-continued

Entry	n	R	R ³
161	2	SO ₂ NH ₂	2-CN-Phenyl
162	3	SO ₂ NH ₂	2-CN-Phenyl
163	1	H	3-Me-Phenyl
164	2	H	3-Me-Phenyl
165	3	H	3-Me-Phenyl
166	1	Me	3-Me-Phenyl
167	2	Me	3-Me-Phenyl
168	3	Me	3-Me-Phenyl
169	1	CH ₂ Ph	3-Me-Phenyl
170	2	CH ₂ Ph	3-Me-Phenyl
171	3	CH ₂ Ph	3-Me-Phenyl
172	1	COMe	3-Me-Phenyl
173	2	COMe	3-Me-Phenyl
174	3	COMe	3-Me-Phenyl
175	1	CO ₂ Me	3-Me-Phenyl
176	2	CO ₂ Me	3-Me-Phenyl
177	3	CO ₂ Me	3-Me-Phenyl
178	1	CO ₂ tBu	3-Me-Phenyl
179	2	CO ₂ tBu	3-Me-Phenyl
180	3	CO ₂ tBu	3-Me-Phenyl
181	1	CONHMe	3-Me-Phenyl
182	2	CONHMe	3-Me-Phenyl
183	3	CONHMe	3-Me-Phenyl
184	1	SO ₂ Me	3-Me-Phenyl
185	2	SO ₂ Me	3-Me-Phenyl
186	3	SO ₂ Me	3-Me-Phenyl
187	1	SO ₂ NH ₂	3-Me-Phenyl
188	2	SO ₂ NH ₂	3-Me-Phenyl
189	3	SO ₂ NH ₂	3-Me-Phenyl
190	1	H	2-F-Phenyl
191	2	H	2-F-Phenyl
192	3	H	2-F-Phenyl
193	1	Me	2-F-Phenyl
194	2	Me	2-F-Phenyl
195	3	Me	2-F-Phenyl
196	1	CH ₂ Ph	2-F-Phenyl
197	2	CH ₂ Ph	2-F-Phenyl
198	3	CH ₂ Ph	2-F-Phenyl
199	1	COMe	2-F-Phenyl
200	2	COMe	2-F-Phenyl
201	3	COMe	2-F-Phenyl
202	1	CO ₂ Me	2-F-Phenyl
203	2	CO ₂ Me	2-F-Phenyl
204	3	CO ₂ Me	2-F-Phenyl
205	1	CO ₂ tBu	2-F-Phenyl
206	2	CO ₂ tBu	2-F-Phenyl
207	3	CO ₂ tBu	2-F-Phenyl
208	1	CONHMe	2-F-Phenyl
209	2	CONHMe	2-F-Phenyl
210	3	CONHMe	2-F-Phenyl
211	1	SO ₂ Me	2-F-Phenyl
212	2	SO ₂ Me	2-F-Phenyl
213	3	SO ₂ Me	2-F-Phenyl
214	1	SO ₂ NH ₂	2-F-Phenyl
215	2	SO ₂ NH ₂	2-F-Phenyl
216	3	SO ₂ NH ₂	2-F-Phenyl
217	1	H	4-F-Phenyl
218	2	H	4-F-Phenyl
219	3	H	4-F-Phenyl
220	1	Me	4-F-Phenyl
221	2	Me	4-F-Phenyl
222	3	Me	4-F-Phenyl
223	1	CH ₂ Ph	4-F-Phenyl
224	2	CH ₂ Ph	4-F-Phenyl
225	3	CH ₂ Ph	4-F-Phenyl
226	1	COMe	4-F-Phenyl
227	2	COMe	4-F-Phenyl
228	3	COMe	4-F-Phenyl
229	1	CO ₂ Me	4-F-Phenyl
230	2	CO ₂ Me	4-F-Phenyl
231	3	CO ₂ Me	4-F-Phenyl
232	1	CO ₂ tBu	4-F-Phenyl
233	2	CO ₂ tBu	4-F-Phenyl
234	3	CO ₂ tBu	4-F-Phenyl
235	1	CONHMe	4-F-Phenyl

TABLE 2-continued

Entry	n	R	R ³
236	2	CONHMe	4-F-Phenyl
237	3	CONHMe	4-F-Phenyl
238	1	SO ₂ Me	4-F-Phenyl
239	2	SO ₂ Me	4-F-Phenyl
240	3	SO ₂ Me	4-F-Phenyl
241	1	SO ₂ NH ₂	4-F-Phenyl
242	2	SO ₂ NH ₂	4-F-Phenyl
243	3	SO ₂ NH ₂	4-F-Phenyl
244	1	H	3-Cl-Phenyl
245	2	H	3-Cl-Phenyl
246	3	H	3-Cl-Phenyl
247	1	Me	3-Cl-Phenyl
248	2	Me	3-Cl-Phenyl
249	3	Me	3-Cl-Phenyl
250	1	CH ₂ Ph	3-Cl-Phenyl
251	2	CH ₂ Ph	3-Cl-Phenyl
252	3	CH ₂ Ph	3-Cl-Phenyl
253	1	COMe	3-Cl-Phenyl
254	2	COMe	3-Cl-Phenyl
255	3	COMe	3-Cl-Phenyl
256	1	CO ₂ Me	3-Cl-Phenyl
257	2	CO ₂ Me	3-Cl-Phenyl
258	3	CO ₂ Me	3-Cl-Phenyl
259	1	CO ₂ tBu	3-Cl-Phenyl
260	2	CO ₂ tBu	3-Cl-Phenyl
261	3	CO ₂ tBu	3-Cl-Phenyl
262	1	CONHMe	3-Cl-Phenyl
263	2	CONHMe	3-Cl-Phenyl
264	3	CONHMe	3-Cl-Phenyl
265	1	SO ₂ Me	3-Cl-Phenyl
266	2	SO ₂ Me	3-Cl-Phenyl
267	3	SO ₂ Me	3-Cl-Phenyl
268	1	SO ₂ NH ₂	3-Cl-Phenyl
269	2	SO ₂ NH ₂	3-Cl-Phenyl
270	3	SO ₂ NH ₂	3-Cl-Phenyl
271	1	H	2-Br-Phenyl
272	2	H	2-Br-Phenyl
273	3	H	2-Br-Phenyl
274	1	Me	2-Br-Phenyl
275	2	Me	2-Br-Phenyl
276	3	Me	2-Br-Phenyl
277	1	CH ₂ Ph	2-Br-Phenyl
278	2	CH ₂ Ph	2-Br-Phenyl
279	3	CH ₂ Ph	2-Br-Phenyl
280	1	COMe	2-Br-Phenyl
281	2	COMe	2-Br-Phenyl
282	3	COMe	2-Br-Phenyl
283	1	CO ₂ Me	2-Br-Phenyl
284	2	CO ₂ Me	2-Br-Phenyl
285	3	CO ₂ Me	2-Br-Phenyl
286	1	CO ₂ tBu	2-Br-Phenyl
287	2	CO ₂ tBu	2-Br-Phenyl
288	3	CO ₂ tBu	2-Br-Phenyl
289	1	CONHMe	2-Br-Phenyl
290	2	CONHMe	2-Br-Phenyl
291	3	CONHMe	2-Br-Phenyl
292	1	SO ₂ Me	2-Br-Phenyl
293	2	SO ₂ Me	2-Br-Phenyl
294	3	SO ₂ Me	2-Br-Phenyl
295	1	SO ₂ NH ₂	2-Br-Phenyl
296	2	SO ₂ NH ₂	2-Br-Phenyl
297	3	SO ₂ NH ₂	2-Br-Phenyl
298	1	H	4-Br-Phenyl
299	2	H	4-Br-Phenyl
300	3	H	4-Br-Phenyl
301	1	Me	4-Br-Phenyl
302	2	Me	4-Br-Phenyl
303	3	Me	4-Br-Phenyl
304	1	CH ₂ Ph	4-Br-Phenyl
305	2	CH ₂ Ph	4-Br-Phenyl
306	3	CH ₂ Ph	4-Br-Phenyl
307	1	COMe	4-Br-Phenyl
308	2	COMe	4-Br-Phenyl
309	3	COMe	4-Br-Phenyl
310	1	CO ₂ Me	4-Br-Phenyl

TABLE 2-continued

Entry	n	R	R ³
311	2	CO ₂ Me	4-Br-Phenyl
312	3	CO ₂ Me	4-Br-Phenyl
313	1	CO ₂ tBu	4-Br-Phenyl
314	2	CO ₂ tBu	4-Br-Phenyl
315	3	CO ₂ tBu	4-Br-Phenyl
316	1	CONHMe	4-Br-Phenyl
317	2	CONHMe	4-Br-Phenyl
318	3	CONHMe	4-Br-Phenyl
319	1	SO ₂ Me	4-Br-Phenyl
320	2	SO ₂ Me	4-Br-Phenyl
321	3	SO ₂ Me	4-Br-Phenyl
322	1	SO ₂ NH ₂	4-Br-Phenyl
323	2	SO ₂ NH ₂	4-Br-Phenyl
324	3	SO ₂ NH ₂	4-Br-Phenyl
325	1	H	3-CF ₃ -Phenyl
326	2	H	3-CF ₃ -Phenyl
327	3	H	3-CF ₃ -Phenyl
328	1	Me	3-CF ₃ -Phenyl
329	2	Me	3-CF ₃ -Phenyl
330	3	Me	3-CF ₃ -Phenyl
331	1	CH ₂ Ph	3-CF ₃ -Phenyl
332	2	CH ₂ Ph	3-CF ₃ -Phenyl
333	3	CH ₂ Ph	3-CF ₃ -Phenyl
334	1	COMe	3-CF ₃ -Phenyl
335	2	COMe	3-CF ₃ -Phenyl
336	3	COMe	3-CF ₃ -Phenyl
337	1	CO ₂ Me	3-CF ₃ -Phenyl
338	2	CO ₂ Me	3-CF ₃ -Phenyl
339	3	CO ₂ Me	3-CF ₃ -Phenyl
340	1	CO ₂ tBu	3-CF ₃ -Phenyl
341	2	CO ₂ tBu	3-CF ₃ -Phenyl
342	3	CO ₂ tBu	3-CF ₃ -Phenyl
343	1	CONHMe	3-CF ₃ -Phenyl
344	2	CONHMe	3-CF ₃ -Phenyl
345	3	CONHMe	3-CF ₃ -Phenyl
346	1	SO ₂ Me	3-CF ₃ -Phenyl
347	2	SO ₂ Me	3-CF ₃ -Phenyl
348	3	SO ₂ Me	3-CF ₃ -Phenyl
349	1	SO ₂ NH ₂	3-CF ₃ -Phenyl
350	2	SO ₂ NH ₂	3-CF ₃ -Phenyl
351	3	SO ₂ NH ₂	3-CF ₃ -Phenyl
352	1	H	2-iPr-Phenyl
353	2	H	2-iPr-Phenyl
354	3	H	2-iPr-Phenyl
355	1	Me	2-iPr-Phenyl
356	2	Me	2-iPr-Phenyl
357	3	Me	2-iPr-Phenyl
358	1	CH ₂ Ph	2-iPr-Phenyl
359	2	CH ₂ Ph	2-iPr-Phenyl
360	3	CH ₂ Ph	2-iPr-Phenyl
361	1	COMe	2-iPr-Phenyl
362	2	COMe	2-iPr-Phenyl
363	3	COMe	2-iPr-Phenyl
364	1	CO ₂ Me	2-iPr-Phenyl
365	2	CO ₂ Me	2-iPr-Phenyl
366	3	CO ₂ Me	2-iPr-Phenyl
367	1	CO ₂ tBu	2-iPr-Phenyl
368	2	CO ₂ tBu	2-iPr-Phenyl
369	3	CO ₂ tBu	2-iPr-Phenyl
370	1	CONHMe	2-iPr-Phenyl
371	2	CONHMe	2-iPr-Phenyl
372	3	CONHMe	2-iPr-Phenyl
373	1	SO ₂ Me	2-iPr-Phenyl
374	2	SO ₂ Me	2-iPr-Phenyl
375	3	SO ₂ Me	2-iPr-Phenyl
376	1	SO ₂ NH ₂	2-iPr-Phenyl
377	2	SO ₂ NH ₂	2-iPr-Phenyl
378	3	SO ₂ NH ₂	2-iPr-Phenyl
379	1	H	4-iPr-Phenyl
380	2	H	4-iPr-Phenyl
381	3	H	4-iPr-Phenyl
382	1	Me	4-iPr-Phenyl
383	2	Me	4-iPr-Phenyl
384	3	Me	4-iPr-Phenyl
385	1	CH ₂ Ph	4-iPr-Phenyl

TABLE 2-continued

Entry	n	R	R ³
386	2	CH ₂ Ph	4-iPr-Phenyl
387	3	CH ₂ Ph	4-iPr-Phenyl
388	1	COMe	4-iPr-Phenyl
389	2	COMe	4-iPr-Phenyl
390	3	COMe	4-iPr-Phenyl
391	1	CO ₂ Me	4-iPr-Phenyl
392	2	CO ₂ Me	4-iPr-Phenyl
393	3	CO ₂ Me	4-iPr-Phenyl
394	1	CO ₂ tBu	4-iPr-Phenyl
395	2	CO ₂ tBu	4-iPr-Phenyl
396	3	CO ₂ tBu	4-iPr-Phenyl
397	1	CONHMe	4-iPr-Phenyl
398	2	CONHMe	4-iPr-Phenyl
399	3	CONHMe	4-iPr-Phenyl
400	1	SO ₂ Me	4-iPr-Phenyl
401	2	SO ₂ Me	4-iPr-Phenyl
402	3	SO ₂ Me	4-iPr-Phenyl
403	1	SO ₂ NH ₂	4-iPr-Phenyl
404	2	SO ₂ NH ₂	4-iPr-Phenyl
405	3	SO ₂ NH ₂	4-iPr-Phenyl
406	1	H	3-NH ₂ -Phenyl
407	2	H	3-NH ₂ -Phenyl
408	3	H	3-NH ₂ -Phenyl
409	1	Me	3-NH ₂ -Phenyl
410	2	Me	3-NH ₂ -Phenyl
411	3	Me	3-NH ₂ -Phenyl
412	1	CH ₂ Ph	3-NH ₂ -Phenyl
413	2	CH ₂ Ph	3-NH ₂ -Phenyl
414	3	CH ₂ Ph	3-NH ₂ -Phenyl
415	1	COMe	3-NH ₂ -Phenyl
416	2	COMe	3-NH ₂ -Phenyl
417	3	COMe	3-NH ₂ -Phenyl
418	1	CO ₂ Me	3-NH ₂ -Phenyl
419	2	CO ₂ Me	3-NH ₂ -Phenyl
420	3	CO ₂ Me	3-NH ₂ -Phenyl
421	1	CO ₂ tBu	3-NH ₂ -Phenyl
422	2	CO ₂ tBu	3-NH ₂ -Phenyl
423	3	CO ₂ tBu	3-NH ₂ -Phenyl
424	1	CONHMe	3-NH ₂ -Phenyl
425	2	CONHMe	3-NH ₂ -Phenyl
426	3	CONHMe	3-NH ₂ -Phenyl
427	1	SO ₂ Me	3-NH ₂ -Phenyl
428	2	SO ₂ Me	3-NH ₂ -Phenyl
429	3	SO ₂ Me	3-NH ₂ -Phenyl
430	1	SO ₂ NH ₂	3-NH ₂ -Phenyl
431	2	SO ₂ NH ₂	3-NH ₂ -Phenyl
432	3	SO ₂ NH ₂	3-NH ₂ -Phenyl
433	1	H	2,4-di-Me-Phenyl
434	2	H	2,4-di-Me-Phenyl
435	3	H	2,4-di-Me-Phenyl
436	1	Me	2,4-di-Me-Phenyl
437	2	Me	2,4-di-Me-Phenyl
438	3	Me	2,4-di-Me-Phenyl
439	1	CH ₂ Ph	2,4-di-Me-Phenyl
440	2	CH ₂ Ph	2,4-di-Me-Phenyl
441	3	CH ₂ Ph	2,4-di-Me-Phenyl
442	1	COMe	2,4-di-Me-Phenyl
443	2	COMe	2,4-di-Me-Phenyl
444	3	COMe	2,4-di-Me-Phenyl
445	1	CO ₂ Me	2,4-di-Me-Phenyl
446	2	CO ₂ Me	2,4-di-Me-Phenyl
447	3	CO ₂ Me	2,4-di-Me-Phenyl
448	1	CO ₂ tBu	2,4-di-Me-Phenyl
449	2	CO ₂ tBu	2,4-di-Me-Phenyl
450	3	CO ₂ tBu	2,4-di-Me-Phenyl
451	1	CONHMe	2,4-di-Me-Phenyl
452	2	CONHMe	2,4-di-Me-Phenyl
453	3	CONHMe	2,4-di-Me-Phenyl
454	1	SO ₂ Me	2,4-di-Me-Phenyl
455	2	SO ₂ Me	2,4-di-Me-Phenyl
456	3	SO ₂ Me	2,4-di-Me-Phenyl
457	1	SO ₂ NH ₂	2,4-di-Me-Phenyl
458	2	SO ₂ NH ₂	2,4-di-Me-Phenyl
459	3	SO ₂ NH ₂	2,4-di-Me-Phenyl
460	1	H	2,6-di-iPr-Phenyl

TABLE 2-continued

Entry	n	R	R ³
461	2	H	2,6-di-iPr-Phenyl
462	3	H	2,6-di-iPr-Phenyl
463	1	Me	2,6-di-iPr-Phenyl
464	2	Me	2,6-di-iPr-Phenyl
465	3	Me	2,6-di-iPr-Phenyl
466	1	CH ₂ Ph	2,6-di-iPr-Phenyl
467	2	CH ₂ Ph	2,6-di-iPr-Phenyl
468	3	CH ₂ Ph	2,6-di-iPr-Phenyl
469	1	COMe	2,6-di-iPr-Phenyl
470	2	COMe	2,6-di-iPr-Phenyl
471	3	COMe	2,6-di-iPr-Phenyl
472	1	CO ₂ Me	2,6-di-iPr-Phenyl
473	2	CO ₂ Me	2,6-di-iPr-Phenyl
474	3	CO ₂ Me	2,6-di-iPr-Phenyl
475	1	CO ₂ tBu	2,6-di-iPr-Phenyl
476	2	CO ₂ tBu	2,6-di-iPr-Phenyl
477	3	CO ₂ tBu	2,6-di-iPr-Phenyl
478	1	CONHMe	2,6-di-iPr-Phenyl
479	2	CONHMe	2,6-di-iPr-Phenyl
480	3	CONHMe	2,6-di-iPr-Phenyl
481	1	SO ₂ Me	2,6-di-iPr-Phenyl
482	2	SO ₂ Me	2,6-di-iPr-Phenyl
483	3	SO ₂ Me	2,6-di-iPr-Phenyl
484	1	SO ₂ NH ₂	2,6-di-iPr-Phenyl
485	2	SO ₂ NH ₂	2,6-di-iPr-Phenyl
486	3	SO ₂ NH ₂	2,6-di-iPr-Phenyl
487	1	H	3-Ph-Phenyl
488	2	H	3-Ph-Phenyl
489	3	H	3-Ph-Phenyl
490	1	Me	3-Ph-Phenyl
491	2	Me	3-Ph-Phenyl
492	3	Me	3-Ph-Phenyl
493	1	CH ₂ Ph	3-Ph-Phenyl
494	2	CH ₂ Ph	3-Ph-Phenyl
495	3	CH ₂ Ph	3-Ph-Phenyl
496	1	COMe	3-Ph-Phenyl
497	2	COMe	3-Ph-Phenyl
498	3	COMe	3-Ph-Phenyl
499	1	CO ₂ Me	3-Ph-Phenyl
500	2	CO ₂ Me	3-Ph-Phenyl
501	3	CO ₂ Me	3-Ph-Phenyl
502	1	CO ₂ tBu	3-Ph-Phenyl
503	2	CO ₂ tBu	3-Ph-Phenyl
504	3	CO ₂ tBu	3-Ph-Phenyl
505	1	CONHMe	3-Ph-Phenyl
506	2	CONHMe	3-Ph-Phenyl
507	3	CONHMe	3-Ph-Phenyl
508	1	SO ₂ Me	3-Ph-Phenyl
509	2	SO ₂ Me	3-Ph-Phenyl
510	3	SO ₂ Me	3-Ph-Phenyl
511	1	SO ₂ NH ₂	3-Ph-Phenyl
512	2	SO ₂ NH ₂	3-Ph-Phenyl
513	3	SO ₂ NH ₂	3-Ph-Phenyl
514	1	H	2-morpholino-phenyl
515	2	H	2-morpholino-phenyl
516	3	H	2-morpholino-phenyl
517	1	Me	2-morpholino-phenyl
518	2	Me	2-morpholino-phenyl
519	3	Me	2-morpholino-phenyl
520	1	CH ₂ Ph	2-morpholino-phenyl
521	2	CH ₂ Ph	2-morpholino-phenyl
522	3	CH ₂ Ph	2-morpholino-phenyl
523	1	COMe	2-morpholino-phenyl
524	2	COMe	2-morpholino-phenyl
525	3	COMe	2-morpholino-phenyl
526	1	CO ₂ Me	2-morpholino-phenyl
527	2	CO ₂ Me	2-morpholino-phenyl
528	3	CO ₂ Me	2-morpholino-phenyl
529	1	CO ₂ tBu	2-morpholino-phenyl
530	2	CO ₂ tBu	2-morpholino-phenyl
531	3	CO ₂ tBu	2-morpholino-phenyl
532	1	CONHMe	2-morpholino-phenyl
533	2	CONHMe	2-morpholino-phenyl
534	3	CONHMe	2-morpholino-phenyl
535	1	SO ₂ Me	2-morpholino-phenyl

TABLE 2-continued

Entry	n	R	R ³
536	2	SO ₂ Me	2-morpholino-phenyl
537	3	SO ₂ Me	2-morpholino-phenyl
538	1	SO ₂ NH ₂	2-morpholino-phenyl
539	2	SO ₂ NH ₂	2-morpholino-phenyl
540	3	SO ₂ NH ₂	2-morpholino-phenyl
541	1	H	4-morpholino-phenyl
542	2	H	4-morpholino-phenyl
543	3	H	4-morpholino-phenyl
544	1	Me	4-morpholino-phenyl
545	2	Me	4-morpholino-phenyl
546	3	Me	4-morpholino-phenyl
547	1	CH ₂ Ph	4-morpholino-phenyl
548	2	CH ₂ Ph	4-morpholino-phenyl
549	3	CH ₂ Ph	4-morpholino-phenyl
550	1	COMe	4-morpholino-phenyl
551	2	COMe	4-morpholino-phenyl
552	3	COMe	4-morpholino-phenyl
553	1	CO ₂ Me	4-morpholino-phenyl
554	2	CO ₂ Me	4-morpholino-phenyl
555	3	CO ₂ Me	4-morpholino-phenyl
556	1	CO ₂ tBu	4-morpholino-phenyl
557	2	CO ₂ tBu	4-morpholino-phenyl
558	3	CO ₂ tBu	4-morpholino-phenyl
559	1	CONHMe	4-morpholino-phenyl
560	2	CONHMe	4-morpholino-phenyl
561	3	CONHMe	4-morpholino-phenyl
562	1	SO ₂ Me	4-morpholino-phenyl
563	2	SO ₂ Me	4-morpholino-phenyl
564	3	SO ₂ Me	4-morpholino-phenyl
565	1	SO ₂ NH ₂	4-morpholino-phenyl
566	2	SO ₂ NH ₂	4-morpholino-phenyl
567	3	SO ₂ NH ₂	4-morpholino-phenyl
568	1	H	2-pyrimidinyl
569	2	H	2-pyrimidinyl
570	3	H	2-pyrimidinyl
571	1	Me	2-pyrimidinyl
572	2	Me	2-pyrimidinyl
573	3	Me	2-pyrimidinyl
574	1	CH ₂ Ph	2-pyrimidinyl
575	2	CH ₂ Ph	2-pyrimidinyl
576	3	CH ₂ Ph	2-pyrimidinyl
577	1	COMe	2-pyrimidinyl
578	2	COMe	2-pyrimidinyl
579	3	COMe	2-pyrimidinyl
580	1	CO ₂ Me	2-pyrimidinyl
581	2	CO ₂ Me	2-pyrimidinyl
582	3	CO ₂ Me	2-pyrimidinyl
583	1	CO ₂ tBu	2-pyrimidinyl
584	2	CO ₂ tBu	2-pyrimidinyl
585	3	CO ₂ tBu	2-pyrimidinyl
586	1	CONHMe	2-pyrimidinyl
587	2	CONHMe	2-pyrimidinyl
588	3	CONHMe	2-pyrimidinyl
589	1	SO ₂ Me	2-pyrimidinyl
590	2	SO ₂ Me	2-pyrimidinyl
591	3	SO ₂ Me	2-pyrimidinyl
592	1	SO ₂ NH ₂	2-pyrimidinyl
593	2	SO ₂ NH ₂	2-pyrimidinyl
594	3	SO ₂ NH ₂	2-pyrimidinyl
595	1	H	2-methyl-1H-benzo[d]imidazol-4-yl
596	2	H	2-methyl-1H-benzo[d]imidazol-4-yl
597	3	H	2-methyl-1H-benzo[d]imidazol-4-yl
598	1	Me	2-methyl-1H-benzo[d]imidazol-4-yl
599	2	Me	2-methyl-1H-benzo[d]imidazol-4-yl
600	3	Me	2-methyl-1H-benzo[d]imidazol-4-yl
601	1	CH ₂ Ph	2-methyl-1H-benzo[d]imidazol-4-yl
602	2	CH ₂ Ph	2-methyl-1H-benzo[d]imidazol-4-yl

TABLE 2-continued

Entry	n	R	R ³
603	3	CH ₂ Ph	2-methyl-1H-benzo[d]imidazol-4-yl
604	1	COMe	2-methyl-1H-benzo[d]imidazol-4-yl
605	2	COMe	2-methyl-1H-benzo[d]imidazol-4-yl
606	3	COMe	2-methyl-1H-benzo[d]imidazol-4-yl
607	1	CO ₂ Me	2-methyl-1H-benzo[d]imidazol-4-yl
608	2	CO ₂ Me	2-methyl-1H-benzo[d]imidazol-4-yl
609	3	CO ₂ Me	2-methyl-1H-benzo[d]imidazol-4-yl
610	1	CO ₂ tBu	2-methyl-1H-benzo[d]imidazol-4-yl
611	2	CO ₂ tBu	2-methyl-1H-benzo[d]imidazol-4-yl
612	3	CO ₂ tBu	2-methyl-1H-benzo[d]imidazol-4-yl
613	1	CONHMe	2-methyl-1H-benzo[d]imidazol-4-yl
614	2	CONHMe	2-methyl-1H-benzo[d]imidazol-4-yl
615	3	CONHMe	2-methyl-1H-benzo[d]imidazol-4-yl
616	1	SO ₂ Me	2-methyl-1H-benzo[d]imidazol-4-yl
617	2	SO ₂ Me	2-methyl-1H-benzo[d]imidazol-4-yl
618	3	SO ₂ Me	2-methyl-1H-benzo[d]imidazol-4-yl
619	1	SO ₂ NH ₂	2-methyl-1H-benzo[d]imidazol-4-yl
620	2	SO ₂ NH ₂	2-methyl-1H-benzo[d]imidazol-4-yl
621	3	SO ₂ NH ₂	2-methyl-1H-benzo[d]imidazol-4-yl
622	1	H	4-OH-Phenyl
623	2	H	4-OH-Phenyl
624	3	H	4-OH-Phenyl
625	1	Me	4-OH-Phenyl
626	2	Me	4-OH-Phenyl
627	3	Me	4-OH-Phenyl
628	1	CH ₂ Ph	4-OH-Phenyl
629	2	CH ₂ Ph	4-OH-Phenyl
630	3	CH ₂ Ph	4-OH-Phenyl
631	1	COMe	4-OH-Phenyl
632	2	COMe	4-OH-Phenyl
633	3	COMe	4-OH-Phenyl
634	1	CO ₂ Me	4-OH-Phenyl
635	2	CO ₂ Me	4-OH-Phenyl
636	3	CO ₂ Me	4-OH-Phenyl
637	1	CO ₂ tBu	4-OH-Phenyl
638	2	CO ₂ tBu	4-OH-Phenyl
639	3	CO ₂ tBu	4-OH-Phenyl
640	1	CONHMe	4-OH-Phenyl
641	2	CONHMe	4-OH-Phenyl
642	3	CONHMe	4-OH-Phenyl
643	1	SO ₂ Me	4-OH-Phenyl
644	2	SO ₂ Me	4-OH-Phenyl
645	3	SO ₂ Me	4-OH-Phenyl
646	1	SO ₂ NH ₂	4-OH-Phenyl
647	2	SO ₂ NH ₂	4-OH-Phenyl
648	3	SO ₂ NH ₂	4-OH-Phenyl
649	1	H	2-OH-Phenyl
650	2	H	2-OH-Phenyl
651	3	H	2-OH-Phenyl
652	1	Me	2-OH-Phenyl
653	2	Me	2-OH-Phenyl
654	3	Me	2-OH-Phenyl
655	1	CH ₂ Ph	2-OH-Phenyl
656	2	CH ₂ Ph	2-OH-Phenyl
657	3	CH ₂ Ph	2-OH-Phenyl
658	1	COMe	2-OH-Phenyl

TABLE 2-continued

Entry	n	R	R ³
659	2	COMe	2-OH-Phenyl
660	3	COMe	2-OH-Phenyl
661	1	CO ₂ Me	2-OH-Phenyl
662	2	CO ₂ Me	2-OH-Phenyl
663	3	CO ₂ Me	2-OH-Phenyl
664	1	CO ₂ tBu	2-OH-Phenyl
665	2	CO ₂ tBu	2-OH-Phenyl
666	3	CO ₂ tBu	2-OH-Phenyl
667	1	CONHMe	2-OH-Phenyl
668	2	CONHMe	2-OH-Phenyl
669	3	CONHMe	2-OH-Phenyl
670	1	SO ₂ Me	2-OH-Phenyl
671	2	SO ₂ Me	2-OH-Phenyl
672	3	SO ₂ Me	2-OH-Phenyl
673	1	SO ₂ NH ₂	2-OH-Phenyl
674	2	SO ₂ NH ₂	2-OH-Phenyl
675	3	SO ₂ NH ₂	2-OH-Phenyl
676	1	H	4-OMe-Phenyl
677	2	H	4-OMe-Phenyl
678	3	H	4-OMe-Phenyl
679	1	Me	4-OMe-Phenyl
680	2	Me	4-OMe-Phenyl
681	3	Me	4-OMe-Phenyl
682	1	CH ₂ Ph	4-OMe-Phenyl
683	2	CH ₂ Ph	4-OMe-Phenyl
684	3	CH ₂ Ph	4-OMe-Phenyl
685	1	COMe	4-OMe-Phenyl
686	2	COMe	4-OMe-Phenyl
687	3	COMe	4-OMe-Phenyl
688	1	CO ₂ Me	4-OMe-Phenyl
689	2	CO ₂ Me	4-OMe-Phenyl
690	3	CO ₂ Me	4-OMe-Phenyl
691	1	CO ₂ tBu	4-OMe-Phenyl
692	2	CO ₂ tBu	4-OMe-Phenyl
693	3	CO ₂ tBu	4-OMe-Phenyl
694	1	CONHMe	4-OMe-Phenyl
695	2	CONHMe	4-OMe-Phenyl
696	3	CONHMe	4-OMe-Phenyl
697	1	SO ₂ Me	4-OMe-Phenyl
698	2	SO ₂ Me	4-OMe-Phenyl
699	3	SO ₂ Me	4-OMe-Phenyl
700	1	SO ₂ NH ₂	4-OMe-Phenyl
701	2	SO ₂ NH ₂	4-OMe-Phenyl
702	3	SO ₂ NH ₂	4-OMe-Phenyl
703	1	H	2-OMe-Phenyl
704	2	H	2-OMe-Phenyl
705	3	H	2-OMe-Phenyl
706	1	Me	2-OMe-Phenyl
707	2	Me	2-OMe-Phenyl
708	3	Me	2-OMe-Phenyl
709	1	CH ₂ Ph	2-OMe-Phenyl
710	2	CH ₂ Ph	2-OMe-Phenyl
711	3	CH ₂ Ph	2-OMe-Phenyl
712	1	COMe	2-OMe-Phenyl
713	2	COMe	2-OMe-Phenyl
714	3	COMe	2-OMe-Phenyl
715	1	CO ₂ Me	2-OMe-Phenyl
716	2	CO ₂ Me	2-OMe-Phenyl
717	3	CO ₂ Me	2-OMe-Phenyl
718	1	CO ₂ tBu	2-OMe-Phenyl
719	2	CO ₂ tBu	2-OMe-Phenyl
720	3	CO ₂ tBu	2-OMe-Phenyl
721	1	CONHMe	2-OMe-Phenyl
722	2	CONHMe	2-OMe-Phenyl
723	3	CONHMe	2-OMe-Phenyl
724	1	SO ₂ Me	2-OMe-Phenyl
725	2	SO ₂ Me	2-OMe-Phenyl
726	3	SO ₂ Me	2-OMe-Phenyl
727	1	SO ₂ NH ₂	2-OMe-Phenyl
728	2	SO ₂ NH ₂	2-OMe-Phenyl
729	3	SO ₂ NH ₂	2-OMe-Phenyl
730	1	H	3-CN-Phenyl
731	2	H	3-CN-Phenyl
732	3	H	3-CN-Phenyl
733	1	Me	3-CN-Phenyl

TABLE 2-continued

Entry	n	R	R ³
734	2	Me	3-CN-Phenyl
735	3	Me	3-CN-Phenyl
736	1	CH ₂ Ph	3-CN-Phenyl
737	2	CH ₂ Ph	3-CN-Phenyl
738	3	CH ₂ Ph	3-CN-Phenyl
739	1	COMe	3-CN-Phenyl
740	2	COMe	3-CN-Phenyl
741	3	COMe	3-CN-Phenyl
742	1	CO ₂ Me	3-CN-Phenyl
743	2	CO ₂ Me	3-CN-Phenyl
744	3	CO ₂ Me	3-CN-Phenyl
745	1	CO ₂ tBu	3-CN-Phenyl
746	2	CO ₂ tBu	3-CN-Phenyl
747	3	CO ₂ tBu	3-CN-Phenyl
748	1	CONHMe	3-CN-Phenyl
749	2	CONHMe	3-CN-Phenyl
750	3	CONHMe	3-CN-Phenyl
751	1	SO ₂ Me	3-CN-Phenyl
752	2	SO ₂ Me	3-CN-Phenyl
753	3	SO ₂ Me	3-CN-Phenyl
754	1	SO ₂ NH ₂	3-CN-Phenyl
755	2	SO ₂ NH ₂	3-CN-Phenyl
756	3	SO ₂ NH ₂	3-CN-Phenyl
757	1	H	2-Me-Phenyl
758	2	H	2-Me-Phenyl
759	3	H	2-Me-Phenyl
760	1	Me	2-Me-Phenyl
761	2	Me	2-Me-Phenyl
762	3	Me	2-Me-Phenyl
763	1	CH ₂ Ph	2-Me-Phenyl
764	2	CH ₂ Ph	2-Me-Phenyl
765	3	CH ₂ Ph	2-Me-Phenyl
766	1	COMe	2-Me-Phenyl
767	2	COMe	2-Me-Phenyl
768	3	COMe	2-Me-Phenyl
769	1	CO ₂ Me	2-Me-Phenyl
770	2	CO ₂ Me	2-Me-Phenyl
771	3	CO ₂ Me	2-Me-Phenyl
772	1	CO ₂ tBu	2-Me-Phenyl
773	2	CO ₂ tBu	2-Me-Phenyl
774	3	CO ₂ tBu	2-Me-Phenyl
775	1	CONHMe	2-Me-Phenyl
776	2	CONHMe	2-Me-Phenyl
777	3	CONHMe	2-Me-Phenyl
778	1	SO ₂ Me	2-Me-Phenyl
779	2	SO ₂ Me	2-Me-Phenyl
780	3	SO ₂ Me	2-Me-Phenyl
781	1	SO ₂ NH ₂	2-Me-Phenyl
782	2	SO ₂ NH ₂	2-Me-Phenyl
783	3	SO ₂ NH ₂	2-Me-Phenyl
784	1	H	4-Me-Phenyl
785	2	H	4-Me-Phenyl
786	3	H	4-Me-Phenyl
787	1	Me	4-Me-Phenyl
788	2	Me	4-Me-Phenyl
789	3	Me	4-Me-Phenyl
790	1	CH ₂ Ph	4-Me-Phenyl
791	2	CH ₂ Ph	4-Me-Phenyl
792	3	CH ₂ Ph	4-Me-Phenyl
793	1	COMe	4-Me-Phenyl
794	2	COMe	4-Me-Phenyl
795	3	COMe	4-Me-Phenyl
796	1	CO ₂ Me	4-Me-Phenyl
797	2	CO ₂ Me	4-Me-Phenyl
798	3	CO ₂ Me	4-Me-Phenyl
799	1	CO ₂ tBu	4-Me-Phenyl
800	2	CO ₂ tBu	4-Me-Phenyl
801	3	CO ₂ tBu	4-Me-Phenyl
802	1	CONHMe	4-Me-Phenyl
803	2	CONHMe	4-Me-Phenyl
804	3	CONHMe	4-Me-Phenyl
805	1	SO ₂ Me	4-Me-Phenyl
806	2	SO ₂ Me	4-Me-Phenyl
807	3	SO ₂ Me	4-Me-Phenyl
808	1	SO ₂ NH ₂	4-Me-Phenyl

TABLE 2-continued

Entry	n	R	R ³
809	2	SO ₂ NH ₂	4-Me-Phenyl
810	3	SO ₂ NH ₂	4-Me-Phenyl
811	1	H	3-F-Phenyl
812	2	H	3-F-Phenyl
813	3	H	3-F-Phenyl
814	1	Me	3-F-Phenyl
815	2	Me	3-F-Phenyl
816	3	Me	3-F-Phenyl
817	1	CH ₂ Ph	3-F-Phenyl
818	2	CH ₂ Ph	3-F-Phenyl
819	3	CH ₂ Ph	3-F-Phenyl
820	1	COMe	3-F-Phenyl
821	2	COMe	3-F-Phenyl
822	3	COMe	3-F-Phenyl
823	1	CO ₂ Me	3-F-Phenyl
824	2	CO ₂ Me	3-F-Phenyl
825	3	CO ₂ Me	3-F-Phenyl
826	1	CO ₂ tBu	3-F-Phenyl
827	2	CO ₂ tBu	3-F-Phenyl
828	3	CO ₂ tBu	3-F-Phenyl
829	1	CONHMe	3-F-Phenyl
830	2	CONHMe	3-F-Phenyl
831	3	CONHMe	3-F-Phenyl
832	1	SO ₂ Me	3-F-Phenyl
833	2	SO ₂ Me	3-F-Phenyl
834	3	SO ₂ Me	3-F-Phenyl
835	1	SO ₂ NH ₂	3-F-Phenyl
836	2	SO ₂ NH ₂	3-F-Phenyl
837	3	SO ₂ NH ₂	3-F-Phenyl
838	1	H	2-Cl-Phenyl
839	2	H	2-Cl-Phenyl
840	3	H	2-Cl-Phenyl
841	1	Me	2-Cl-Phenyl
842	2	Me	2-Cl-Phenyl
843	3	Me	2-Cl-Phenyl
844	1	CH ₂ Ph	2-Cl-Phenyl
845	2	CH ₂ Ph	2-Cl-Phenyl
846	3	CH ₂ Ph	2-Cl-Phenyl
847	1	COMe	2-Cl-Phenyl
848	2	COMe	2-Cl-Phenyl
849	3	COMe	2-Cl-Phenyl
850	1	CO ₂ Me	2-Cl-Phenyl
851	2	CO ₂ Me	2-Cl-Phenyl
852	3	CO ₂ Me	2-Cl-Phenyl
853	1	CO ₂ tBu	2-Cl-Phenyl
854	2	CO ₂ tBu	2-Cl-Phenyl
855	3	CO ₂ tBu	2-Cl-Phenyl
856	1	CONHMe	2-Cl-Phenyl
857	2	CONHMe	2-Cl-Phenyl
858	3	CONHMe	2-Cl-Phenyl
859	1	SO ₂ Me	2-Cl-Phenyl
860	2	SO ₂ Me	2-Cl-Phenyl
861	3	SO ₂ Me	2-Cl-Phenyl
862	1	SO ₂ NH ₂	2-Cl-Phenyl
863	2	SO ₂ NH ₂	2-Cl-Phenyl
864	3	SO ₂ NH ₂	2-Cl-Phenyl
865	1	H	4-Cl-Phenyl
866	2	H	4-Cl-Phenyl
867	3	H	4-Cl-Phenyl
868	1	Me	4-Cl-Phenyl
869	2	Me	4-Cl-Phenyl
870	3	Me	4-Cl-Phenyl
871	1	CH ₂ Ph	4-Cl-Phenyl
872	2	CH ₂ Ph	4-Cl-Phenyl
873	3	CH ₂ Ph	4-Cl-Phenyl
874	1	COMe	4-Cl-Phenyl
875	2	COMe	4-Cl-Phenyl
876	3	COMe	4-Cl-Phenyl
877	1	CO ₂ Me	4-Cl-Phenyl
878	2	CO ₂ Me	4-Cl-Phenyl
879	3	CO ₂ Me	4-Cl-Phenyl
880	1	CO ₂ tBu	4-Cl-Phenyl
881	2	CO ₂ tBu	4-Cl-Phenyl
882	3	CO ₂ tBu	4-Cl-Phenyl
883	1	CONHMe	4-Cl-Phenyl

TABLE 2-continued

Entry	n	R	R ³
884	2	CONHMe	4-Cl-Phenyl
885	3	CONHMe	4-Cl-Phenyl
886	1	SO ₂ Me	4-Cl-Phenyl
887	2	SO ₂ Me	4-Cl-Phenyl
888	3	SO ₂ Me	4-Cl-Phenyl
889	1	SO ₂ NH ₂	4-Cl-Phenyl
890	2	SO ₂ NH ₂	4-Cl-Phenyl
891	3	SO ₂ NH ₂	4-Cl-Phenyl
892	1	H	3-Br-Phenyl
893	2	H	3-Br-Phenyl
894	3	H	3-Br-Phenyl
895	1	Me	3-Br-Phenyl
896	2	Me	3-Br-Phenyl
897	3	Me	3-Br-Phenyl
898	1	CH ₂ Ph	3-Br-Phenyl
899	2	CH ₂ Ph	3-Br-Phenyl
900	3	CH ₂ Ph	3-Br-Phenyl
901	1	COMe	3-Br-Phenyl
902	2	COMe	3-Br-Phenyl
903	3	COMe	3-Br-Phenyl
904	1	CO ₂ Me	3-Br-Phenyl
905	2	CO ₂ Me	3-Br-Phenyl
906	3	CO ₂ Me	3-Br-Phenyl
907	1	CO ₂ tBu	3-Br-Phenyl
908	2	CO ₂ tBu	3-Br-Phenyl
909	3	CO ₂ tBu	3-Br-Phenyl
910	1	CONHMe	3-Br-Phenyl
911	2	CONHMe	3-Br-Phenyl
912	3	CONHMe	3-Br-Phenyl
913	1	SO ₂ Me	3-Br-Phenyl
914	2	SO ₂ Me	3-Br-Phenyl
915	3	SO ₂ Me	3-Br-Phenyl
916	1	SO ₂ NH ₂	3-Br-Phenyl
917	2	SO ₂ NH ₂	3-Br-Phenyl
918	3	SO ₂ NH ₂	3-Br-Phenyl
919	1	H	2-CF ₃ -Phenyl
920	2	H	2-CF ₃ -Phenyl
921	3	H	2-CF ₃ -Phenyl
922	1	Me	2-CF ₃ -Phenyl
923	2	Me	2-CF ₃ -Phenyl
924	3	Me	2-CF ₃ -Phenyl
925	1	CH ₂ Ph	2-CF ₃ -Phenyl
926	2	CH ₂ Ph	2-CF ₃ -Phenyl
927	3	CH ₂ Ph	2-CF ₃ -Phenyl
928	1	COMe	2-CF ₃ -Phenyl
929	2	COMe	2-CF ₃ -Phenyl
930	3	COMe	2-CF ₃ -Phenyl
931	1	CO ₂ Me	2-CF ₃ -Phenyl
932	2	CO ₂ Me	2-CF ₃ -Phenyl
933	3	CO ₂ Me	2-CF ₃ -Phenyl
934	1	CO ₂ tBu	2-CF ₃ -Phenyl
935	2	CO ₂ tBu	2-CF ₃ -Phenyl
936	3	CO ₂ tBu	2-CF ₃ -Phenyl
937	1	CONHMe	2-CF ₃ -Phenyl
938	2	CONHMe	2-CF ₃ -Phenyl
939	3	CONHMe	2-CF ₃ -Phenyl
940	1	SO ₂ Me	2-CF ₃ -Phenyl
941	2	SO ₂ Me	2-CF ₃ -Phenyl
942	3	SO ₂ Me	2-CF ₃ -Phenyl
943	1	SO ₂ NH ₂	2-CF ₃ -Phenyl
944	2	SO ₂ NH ₂	2-CF ₃ -Phenyl
945	3	SO ₂ NH ₂	2-CF ₃ -Phenyl
946	1	H	4-CF ₃ -Phenyl
947	2	H	4-CF ₃ -Phenyl
948	3	H	4-CF ₃ -Phenyl
949	1	Me	4-CF ₃ -Phenyl
950	2	Me	4-CF ₃ -Phenyl
951	3	Me	4-CF ₃ -Phenyl
952	1	CH ₂ Ph	4-CF ₃ -Phenyl
953	2	CH ₂ Ph	4-CF ₃ -Phenyl
954	3	CH ₂ Ph	4-CF ₃ -Phenyl
955	1	COMe	4-CF ₃ -Phenyl
956	2	COMe	4-CF ₃ -Phenyl
957	3	COMe	4-CF ₃ -Phenyl
958	1	CO ₂ Me	4-CF ₃ -Phenyl

TABLE 2-continued

Entry	n	R	R ³
959	2	CO ₂ Me	4-CF ₃ -Phenyl
960	3	CO ₂ Me	4-CF ₃ -Phenyl
961	1	CO ₂ tBu	4-CF ₃ -Phenyl
962	2	CO ₂ tBu	4-CF ₃ -Phenyl
963	3	CO ₂ tBu	4-CF ₃ -Phenyl
964	1	CONHMe	4-CF ₃ -Phenyl
965	2	CONHMe	4-CF ₃ -Phenyl
966	3	CONHMe	4-CF ₃ -Phenyl
967	1	SO ₂ Me	4-CF ₃ -Phenyl
968	2	SO ₂ Me	4-CF ₃ -Phenyl
969	3	SO ₂ Me	4-CF ₃ -Phenyl
970	1	SO ₂ NH ₂	4-CF ₃ -Phenyl
971	2	SO ₂ NH ₂	4-CF ₃ -Phenyl
972	3	SO ₂ NH ₂	4-CF ₃ -Phenyl
973	1	H	3-iPr-Phenyl
974	2	H	3-iPr-Phenyl
975	3	H	3-iPr-Phenyl
976	1	Me	3-iPr-Phenyl
977	2	Me	3-iPr-Phenyl
978	3	Me	3-iPr-Phenyl
979	1	CH ₂ Ph	3-iPr-Phenyl
980	2	CH ₂ Ph	3-iPr-Phenyl
981	3	CH ₂ Ph	3-iPr-Phenyl
982	1	COMe	3-iPr-Phenyl
983	2	COMe	3-iPr-Phenyl
984	3	COMe	3-iPr-Phenyl
985	1	CO ₂ Me	3-iPr-Phenyl
986	2	CO ₂ Me	3-iPr-Phenyl
987	3	CO ₂ Me	3-iPr-Phenyl
988	1	CO ₂ tBu	3-iPr-Phenyl
989	2	CO ₂ tBu	3-iPr-Phenyl
990	3	CO ₂ tBu	3-iPr-Phenyl
991	1	CONHMe	3-iPr-Phenyl
992	2	CONHMe	3-iPr-Phenyl
993	3	CONHMe	3-iPr-Phenyl
994	1	SO ₂ Me	3-iPr-Phenyl
995	2	SO ₂ Me	3-iPr-Phenyl
996	3	SO ₂ Me	3-iPr-Phenyl
997	1	SO ₂ NH ₂	3-iPr-Phenyl
998	2	SO ₂ NH ₂	3-iPr-Phenyl
999	3	SO ₂ NH ₂	3-iPr-Phenyl
1000	1	H	4-NH ₂ -Phenyl
1001	2	H	4-NH ₂ -Phenyl
1002	3	H	4-NH ₂ -Phenyl
1003	1	Me	4-NH ₂ -Phenyl
1004	2	Me	4-NH ₂ -Phenyl
1005	3	Me	4-NH ₂ -Phenyl
1006	1	CH ₂ Ph	4-NH ₂ -Phenyl
1007	2	CH ₂ Ph	4-NH ₂ -Phenyl
1008	3	CH ₂ Ph	4-NH ₂ -Phenyl
1009	1	COMe	4-NH ₂ -Phenyl
1010	2	COMe	4-NH ₂ -Phenyl
1011	3	COMe	4-NH ₂ -Phenyl
1012	1	CO ₂ Me	4-NH ₂ -Phenyl
1013	2	CO ₂ Me	4-NH ₂ -Phenyl
1014	3	CO ₂ Me	4-NH ₂ -Phenyl
1015	1	CO ₂ tBu	4-NH ₂ -Phenyl
1016	2	CO ₂ tBu	4-NH ₂ -Phenyl
1017	3	CO ₂ tBu	4-NH ₂ -Phenyl
1018	1	CONHMe	4-NH ₂ -Phenyl
1019	2	CONHMe	4-NH ₂ -Phenyl
1020	3	CONHMe	4-NH ₂ -Phenyl
1021	1	SO ₂ Me	4-NH ₂ -Phenyl
1022	2	SO ₂ Me	4-NH ₂ -Phenyl
1023	3	SO ₂ Me	4-NH ₂ -Phenyl
1024	1	SO ₂ NH ₂	4-NH ₂ -Phenyl
1025	2	SO ₂ NH ₂	4-NH ₂ -Phenyl
1026	3	SO ₂ NH ₂	4-NH ₂ -Phenyl
1027	1	H	2-NH ₂ -Phenyl
1028	2	H	2-NH ₂ -Phenyl
1029	3	H	2-NH ₂ -Phenyl
1030	1	Me	2-NH ₂ -Phenyl
1031	2	Me	2-NH ₂ -Phenyl
1032	3	Me	2-NH ₂ -Phenyl
1033	1	CH ₂ Ph	2-NH ₂ -Phenyl

TABLE 2-continued

Entry	n	R	R ³
1034	2	CH ₂ Ph	2-NH ₂ -Phenyl
1035	3	CH ₂ Ph	2-NH ₂ -Phenyl
1036	1	COMe	2-NH ₂ -Phenyl
1037	2	COMe	2-NH ₂ -Phenyl
1038	3	COMe	2-NH ₂ -Phenyl
1039	1	CO ₂ Me	2-NH ₂ -Phenyl
1040	2	CO ₂ Me	2-NH ₂ -Phenyl
1041	3	CO ₂ Me	2-NH ₂ -Phenyl
1042	1	CO ₂ tBu	2-NH ₂ -Phenyl
1043	2	CO ₂ tBu	2-NH ₂ -Phenyl
1044	3	CO ₂ tBu	2-NH ₂ -Phenyl
1045	1	CONHMe	2-NH ₂ -Phenyl
1046	2	CONHMe	2-NH ₂ -Phenyl
1047	3	CONHMe	2-NH ₂ -Phenyl
1048	1	SO ₂ Me	2-NH ₂ -Phenyl
1049	2	SO ₂ Me	2-NH ₂ -Phenyl
1050	3	SO ₂ Me	2-NH ₂ -Phenyl
1051	1	SO ₂ NH ₂	2-NH ₂ -Phenyl
1052	2	SO ₂ NH ₂	2-NH ₂ -Phenyl
1053	3	SO ₂ NH ₂	2-NH ₂ -Phenyl
1054	1	H	2,6-di-Me-Phenyl
1055	2	H	2,6-di-Me-Phenyl
1056	3	H	2,6-di-Me-Phenyl
1057	1	Me	2,6-di-Me-Phenyl
1058	2	Me	2,6-di-Me-Phenyl
1059	3	Me	2,6-di-Me-Phenyl
1060	1	CH ₂ Ph	2,6-di-Me-Phenyl
1061	2	CH ₂ Ph	2,6-di-Me-Phenyl
1062	3	CH ₂ Ph	2,6-di-Me-Phenyl
1063	1	COMe	2,6-di-Me-Phenyl
1064	2	COMe	2,6-di-Me-Phenyl
1065	3	COMe	2,6-di-Me-Phenyl
1066	1	CO ₂ Me	2,6-di-Me-Phenyl
1067	2	CO ₂ Me	2,6-di-Me-Phenyl
1068	3	CO ₂ Me	2,6-di-Me-Phenyl
1069	1	CO ₂ tBu	2,6-di-Me-Phenyl
1070	2	CO ₂ tBu	2,6-di-Me-Phenyl
1071	3	CO ₂ tBu	2,6-di-Me-Phenyl
1072	1	CONHMe	2,6-di-Me-Phenyl
1073	2	CONHMe	2,6-di-Me-Phenyl
1074	3	CONHMe	2,6-di-Me-Phenyl
1075	1	SO ₂ Me	2,6-di-Me-Phenyl
1076	2	SO ₂ Me	2,6-di-Me-Phenyl
1077	3	SO ₂ Me	2,6-di-Me-Phenyl
1078	1	SO ₂ NH ₂	2,6-di-Me-Phenyl
1079	2	SO ₂ NH ₂	2,6-di-Me-Phenyl
1080	3	SO ₂ NH ₂	2,6-di-Me-Phenyl
1081	1	H	2-Ph-Phenyl
1082	2	H	2-Ph-Phenyl
1083	3	H	2-Ph-Phenyl
1084	1	Me	2-Ph-Phenyl
1085	2	Me	2-Ph-Phenyl
1086	3	Me	2-Ph-Phenyl
1087	1	CH ₂ Ph	2-Ph-Phenyl
1088	2	CH ₂ Ph	2-Ph-Phenyl
1089	3	CH ₂ Ph	2-Ph-Phenyl
1090	1	COMe	2-Ph-Phenyl
1091	2	COMe	2-Ph-Phenyl
1092	3	COMe	2-Ph-Phenyl
1093	1	CO ₂ Me	2-Ph-Phenyl
1094	2	CO ₂ Me	2-Ph-Phenyl
1095	3	CO ₂ Me	2-Ph-Phenyl
1096	1	CO ₂ tBu	2-Ph-Phenyl
1097	2	CO ₂ tBu	2-Ph-Phenyl
1098	3	CO ₂ tBu	2-Ph-Phenyl
1099	1	CONHMe	2-Ph-Phenyl
1100	2	CONHMe	2-Ph-Phenyl
1101	3	CONHMe	2-Ph-Phenyl
1102	1	SO ₂ Me	2-Ph-Phenyl
1103	2	SO ₂ Me	2-Ph-Phenyl
1104	3	SO ₂ Me	2-Ph-Phenyl
1105	1	SO ₂ NH ₂	2-Ph-Phenyl
1106	2	SO ₂ NH ₂	2-Ph-Phenyl
1107	3	SO ₂ NH ₂	2-Ph-Phenyl
1108	1	H	4-Ph-Phenyl

TABLE 2-continued

Entry	n	R	R ³
1109	2	H	4-Ph-Phenyl
1110	3	H	4-Ph-Phenyl
1111	1	Me	4-Ph-Phenyl
1112	2	Me	4-Ph-Phenyl
1113	3	Me	4-Ph-Phenyl
1114	1	CH ₂ Ph	4-Ph-Phenyl
1115	2	CH ₂ Ph	4-Ph-Phenyl
1116	3	CH ₂ Ph	4-Ph-Phenyl
1117	1	COMe	4-Ph-Phenyl
1118	2	COMe	4-Ph-Phenyl
1119	3	COMe	4-Ph-Phenyl
1120	1	CO ₂ Me	4-Ph-Phenyl
1121	2	CO ₂ Me	4-Ph-Phenyl
1122	3	CO ₂ Me	4-Ph-Phenyl
1123	1	CO ₂ tBu	4-Ph-Phenyl
1124	2	CO ₂ tBu	4-Ph-Phenyl
1125	3	CO ₂ tBu	4-Ph-Phenyl
1126	1	CONHMe	4-Ph-Phenyl
1127	2	CONHMe	4-Ph-Phenyl
1128	3	CONHMe	4-Ph-Phenyl
1129	1	SO ₂ Me	4-Ph-Phenyl
1130	2	SO ₂ Me	4-Ph-Phenyl
1131	3	SO ₂ Me	4-Ph-Phenyl
1132	1	SO ₂ NH ₂	4-Ph-Phenyl
1133	2	SO ₂ NH ₂	4-Ph-Phenyl
1134	3	SO ₂ NH ₂	4-Ph-Phenyl
1135	1	H	3-morpholino-phenyl
1136	2	H	3-morpholino-phenyl
1137	3	H	3-morpholino-phenyl
1138	1	Me	3-morpholino-phenyl
1139	2	Me	3-morpholino-phenyl
1140	3	Me	3-morpholino-phenyl
1141	1	CH ₂ Ph	3-morpholino-phenyl
1142	2	CH ₂ Ph	3-morpholino-phenyl
1143	3	CH ₂ Ph	3-morpholino-phenyl
1144	1	COMe	3-morpholino-phenyl
1145	2	COMe	3-morpholino-phenyl
1146	3	COMe	3-morpholino-phenyl
1147	1	CO ₂ Me	3-morpholino-phenyl
1148	2	CO ₂ Me	3-morpholino-phenyl
1149	3	CO ₂ Me	3-morpholino-phenyl
1150	1	CO ₂ tBu	3-morpholino-phenyl
1151	2	CO ₂ tBu	3-morpholino-phenyl
1152	3	CO ₂ tBu	3-morpholino-phenyl
1153	1	CONHMe	3-morpholino-phenyl
1154	2	CONHMe	3-morpholino-phenyl
1155	3	CONHMe	3-morpholino-phenyl
1156	1	SO ₂ Me	3-morpholino-phenyl
1157	2	SO ₂ Me	3-morpholino-phenyl
1158	3	SO ₂ Me	3-morpholino-phenyl
1159	1	SO ₂ NH ₂	3-morpholino-phenyl
1160	2	SO ₂ NH ₂	3-morpholino-phenyl
1161	3	SO ₂ NH ₂	3-morpholino-phenyl
1162	1	H	2-pyrazinyl
1163	2	H	2-pyrazinyl
1164	3	H	2-pyrazinyl
1165	1	Me	2-pyrazinyl
1166	2	Me	2-pyrazinyl
1167	3	Me	2-pyrazinyl
1168	1	CH ₂ Ph	2-pyrazinyl
1169	2	CH ₂ Ph	2-pyrazinyl
1170	3	CH ₂ Ph	2-pyrazinyl
1171	1	COMe	2-pyrazinyl
1172	2	COMe	2-pyrazinyl
1173	3	COMe	2-pyrazinyl
1174	1	CO ₂ Me	2-pyrazinyl
1175	2	CO ₂ Me	2-pyrazinyl
1176	3	CO ₂ Me	2-pyrazinyl
1177	1	CO ₂ tBu	2-pyrazinyl
1178	2	CO ₂ tBu	2-pyrazinyl
1179	3	CO ₂ tBu	2-pyrazinyl
1180	1	CONHMe	2-pyrazinyl
1181	2	CONHMe	2-pyrazinyl
1182	3	CONHMe	2-pyrazinyl
1183	1	SO ₂ Me	2-pyrazinyl

TABLE 2-continued

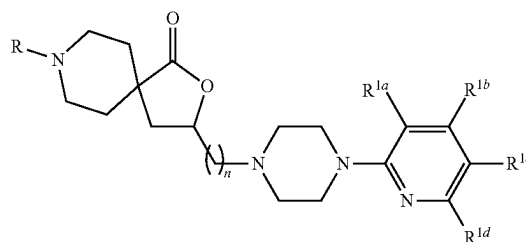
Entry	n	R	R ³
1184	2	SO ₂ Me	2-pyrazinyl
1185	3	SO ₂ Me	2-pyrazinyl
1186	1	SO ₂ NH ₂	2-pyrazinyl
1187	2	SO ₂ NH ₂	2-pyrazinyl
1188	3	SO ₂ NH ₂	2-pyrazinyl
1189	1	H	5-indolyl
1190	2	H	5-indolyl
1191	3	H	5-indolyl
1192	1	Me	5-indolyl
1193	2	Me	5-indolyl
1194	3	Me	5-indolyl
1195	1	CH ₂ Ph	5-indolyl
1196	2	CH ₂ Ph	5-indolyl
1197	3	CH ₂ Ph	5-indolyl
1198	1	COMe	5-indolyl
1199	2	COMe	5-indolyl
1200	3	COMe	5-indolyl
1201	1	CO ₂ Me	5-indolyl
1202	2	CO ₂ Me	5-indolyl
1203	3	CO ₂ Me	5-indolyl
1204	1	CO ₂ tBu	5-indolyl
1205	2	CO ₂ tBu	5-indolyl
1206	3	CO ₂ tBu	5-indolyl
1207	1	CONHMe	5-indolyl
1208	2	CONHMe	5-indolyl
1209	3	CONHMe	5-indolyl
1210	1	SO ₂ Me	5-indolyl
1211	2	SO ₂ Me	5-indolyl
1212	3	SO ₂ Me	5-indolyl
1213	1	SO ₂ NH ₂	5-indolyl
1214	2	SO ₂ NH ₂	5-indolyl
1215	3	SO ₂ NH ₂	5-indolyl
1216	1	H	1H-benzo[d]imidazol-4-yl
1217	2	H	1H-benzo[d]imidazol-4-yl
1218	3	H	1H-benzo[d]imidazol-4-yl
1219	1	Me	1H-benzo[d]imidazol-4-yl
1220	2	Me	1H-benzo[d]imidazol-4-yl
1221	3	Me	1H-benzo[d]imidazol-4-yl
1222	1	CH ₂ Ph	1H-benzo[d]imidazol-4-yl
1223	2	CH ₂ Ph	1H-benzo[d]imidazol-4-yl
1224	3	CH ₂ Ph	1H-benzo[d]imidazol-4-yl
1225	1	COMe	1H-benzo[d]imidazol-4-yl
1226	2	COMe	1H-benzo[d]imidazol-4-yl
1227	3	COMe	1H-benzo[d]imidazol-4-yl
1228	1	CO ₂ Me	1H-benzo[d]imidazol-4-yl
1229	2	CO ₂ Me	1H-benzo[d]imidazol-4-yl
1230	3	CO ₂ Me	1H-benzo[d]imidazol-4-yl
1231	1	CO ₂ tBu	1H-benzo[d]imidazol-4-yl
1232	2	CO ₂ tBu	1H-benzo[d]imidazol-4-yl
1233	3	CO ₂ tBu	1H-benzo[d]imidazol-4-yl
1234	1	CONHMe	1H-benzo[d]imidazol-4-yl
1235	2	CONHMe	1H-benzo[d]imidazol-4-yl
1236	3	CONHMe	1H-benzo[d]imidazol-4-yl

TABLE 2-continued

Entry	n	R	R ³
1237	1	SO ₂ Me	1H-benzo[d]imidazol-4-yl
1238	2	SO ₂ Me	1H-benzo[d]imidazol-4-yl
1239	3	SO ₂ Me	1H-benzo[d]imidazol-4-yl
1240	1	SO ₂ NH ₂	1H-benzo[d]imidazol-4-yl
1241	2	SO ₂ NH ₂	1H-benzo[d]imidazol-4-yl
1242	3	SO ₂ NH ₂	1H-benzo[d]imidazol-4-yl

[0402] Exemplary embodiments include compounds having the formula (XIX)

(XIX)



or a pharmaceutically acceptable salt form thereof defined herein below in Table 3.

TABLE 3

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
1	1	H	H	H	H	H
2	2	H	H	H	H	H
3	3	H	H	H	H	H
4	1	Me	H	H	H	H
5	2	Me	H	H	H	H
6	3	Me	H	H	H	H
7	1	CH ₂ Ph	H	H	H	H
8	2	CH ₂ Ph	H	H	H	H
9	3	CH ₂ Ph	H	H	H	H
10	1	COMe	H	H	H	H
11	2	COMe	H	H	H	H
12	3	COMe	H	H	H	H
13	1	CO ₂ Me	H	H	H	H
14	2	CO ₂ Me	H	H	H	H
15	3	CO ₂ Me	H	H	H	H
16	1	CO ₂ tBu	H	H	H	H
17	2	CO ₂ tBu	H	H	H	H
18	3	CO ₂ tBu	H	H	H	H
19	1	CONHMe	H	H	H	H
20	2	CONHMe	H	H	H	H
21	3	CONHMe	H	H	H	H
22	1	SO ₂ Me	H	H	H	H
23	2	SO ₂ Me	H	H	H	H
24	3	SO ₂ Me	H	H	H	H
25	1	SO ₂ NH ₂	H	H	H	H
26	2	SO ₂ NH ₂	H	H	H	H
27	3	SO ₂ NH ₂	H	H	H	H
28	1	H	H	H	OH	H
29	2	H	H	H	OH	H
30	3	H	H	H	OH	H
31	1	Me	H	H	OH	H
32	2	Me	H	H	OH	H
33	3	Me	H	H	OH	H

TABLE 3-continued

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
34	1	CH ₂ Ph	H	H	OH	H
35	2	CH ₂ Ph	H	H	OH	H
36	3	CH ₂ Ph	H	H	OH	H
37	1	COMe	H	H	OH	H
38	2	COMe	H	H	OH	H
39	3	COMe	H	H	OH	H
40	1	CO ₂ Me	H	H	OH	H
41	2	CO ₂ Me	H	H	OH	H
42	3	CO ₂ Me	H	H	OH	H
43	1	CO ₂ tBu	H	H	OH	H
44	2	CO ₂ tBu	H	H	OH	H
45	3	CO ₂ tBu	H	H	OH	H
46	1	CONHMe	H	H	OH	H
47	2	CONHMe	H	H	OH	H
48	3	CONHMe	H	H	OH	H
49	1	SO ₂ Me	H	H	OH	H
50	2	SO ₂ Me	H	H	OH	H
51	3	SO ₂ Me	H	H	OH	H
52	1	SO ₂ NH ₂	H	H	OH	H
53	2	SO ₂ NH ₂	H	H	OH	H
54	3	SO ₂ NH ₂	H	H	OH	H
55	1	H	H	H	OMe	H
56	2	H	H	H	OMe	H
57	3	H	H	H	OMe	H
58	1	Me	H	H	OMe	H
59	2	Me	H	H	OMe	H
60	3	Me	H	H	OMe	H
61	1	CH ₂ Ph	H	H	OMe	H
62	2	CH ₂ Ph	H	H	OMe	H
63	3	CH ₂ Ph	H	H	OMe	H
64	1	COMe	H	H	OMe	H
65	2	COMe	H	H	OMe	H
66	3	COMe	H	H	OMe	H
67	1	CO ₂ Me	H	H	OMe	H
68	2	CO ₂ Me	H	H	OMe	H
69	3	CO ₂ Me	H	H	OMe	H
70	1	CO ₂ tBu	H	H	OMe	H
71	2	CO ₂ tBu	H	H	OMe	H
72	3	CO ₂ tBu	H	H	OMe	H
73	1	CONHMe	H	H	OMe	H
74	2	CONHMe	H	H	OMe	H
75	3	CONHMe	H	H	OMe	H
76	1	SO ₂ Me	H	H	OMe	H
77	2	SO ₂ Me	H	H	OMe	H
78	3	SO ₂ Me	H	H	OMe	H
79	1	SO ₂ NH ₂	H	H	OMe	H
80	2	SO ₂ NH ₂	H	H	OMe	H
81	3	SO ₂ NH ₂	H	H	OMe	H
82	1	H	H	H	Me	H
83	2	H	H	H	Me	H
84	3	H	H	H	Me	H
85	1	Me	H	H	Me	H
86	2	Me	H	H	Me	H
87	3	Me	H	H	Me	H
88	1	CH ₂ Ph	H	H	Me	H
89	2	CH ₂ Ph	H	H	Me	H
90	3	CH ₂ Ph	H	H	Me	H
91	1	COMe	H	H	Me	H
92	2	COMe	H	H	Me	H
93	3	COMe	H	H	Me	H
94	1	CO ₂ Me	H	H	Me	H
95	2	CO ₂ Me	H	H	Me	H
96	3	CO ₂ Me	H	H	Me	H
97	1	CO ₂ tBu	H	H	Me	H
98	2	CO ₂ tBu	H	H	Me	H
99	3	CO ₂ tBu	H	H	Me	H
100	1	CONHMe	H	H	Me	H
101	2	CONHMe	H	H	Me	H
102	3	CONHMe	H	H	Me	H
103	1	SO ₂ Me	H	H	Me	H
104	2	SO ₂ Me	H	H	Me	H
105	3	SO ₂ Me	H	H	Me	H
106	1	SO ₂ NH ₂	H	H	Me	H
107	2	SO ₂ NH ₂	H	H	Me	H
108	3	SO ₂ NH ₂	H	H	Me	H

TABLE 3-continued

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
109	1	H	H	H	CF ₃	H
110	2	H	H	H	CF ₃	H
111	3	H	H	H	CF ₃	H
112	1	Me	H	H	CF ₃	H
113	2	Me	H	H	CF ₃	H
114	3	Me	H	H	CF ₃	H
115	1	CH ₂ Ph	H	H	CF ₃	H
116	2	CH ₂ Ph	H	H	CF ₃	H
117	3	CH ₂ Ph	H	H	CF ₃	H
118	1	COMe	H	H	CF ₃	H
119	2	COMe	H	H	CF ₃	H
120	3	COMe	H	H	CF ₃	H
121	1	CO ₂ Me	H	H	CF ₃	H
122	2	CO ₂ Me	H	H	CF ₃	H
123	3	CO ₂ Me	H	H	CF ₃	H
124	1	CO ₂ tBu	H	H	CF ₃	H
125	2	CO ₂ tBu	H	H	CF ₃	H
126	3	CO ₂ tBu	H	H	CF ₃	H
127	1	CONHMe	H	H	CF ₃	H
128	2	CONHMe	H	H	CF ₃	H
129	3	CONHMe	H	H	CF ₃	H
130	1	SO ₂ Me	H	H	CF ₃	H
131	2	SO ₂ Me	H	H	CF ₃	H
132	3	SO ₂ Me	H	H	CF ₃	H
133	1	SO ₂ NH ₂	H	H	CF ₃	H
134	2	SO ₂ NH ₂	H	H	CF ₃	H
135	3	SO ₂ NH ₂	H	H	CF ₃	H
136	1	H	H	H	F	H
137	2	H	H	H	F	H
138	3	H	H	H	F	H
139	1	Me	H	H	F	H
140	2	Me	H	H	F	H
141	3	Me	H	H	F	H
142	1	CH ₂ Ph	H	H	F	H
143	2	CH ₂ Ph	H	H	F	H
144	3	CH ₂ Ph	H	H	F	H
145	1	COMe	H	H	F	H
146	2	COMe	H	H	F	H
147	3	COMe	H	H	F	H
148	1	CO ₂ Me	H	H	F	H
149	2	CO ₂ Me	H	H	F	H
150	3	CO ₂ Me	H	H	F	H
151	1	CO ₂ tBu	H	H	F	H
152	2	CO ₂ tBu	H	H	F	H
153	3	CO ₂ tBu	H	H	F	H
154	1	CONHMe	H	H	F	H
155	2	CONHMe	H	H	F	H
156	3	CONHMe	H	H	F	H
157	1	SO ₂ Me	H	H	F	H
158	2	SO ₂ Me	H	H	F	H
159	3	SO ₂ Me	H	H	F	H
160	1	SO ₂ NH ₂	H	H	F	H
161	2	SO ₂ NH ₂	H	H	F	H
162	3	SO ₂ NH ₂	H	H	F	H
163	1	H	H	H	Cl	H
164	2	H	H	H	Cl	H
165	3	H	H	H	Cl	H
166	1	Me	H	H	Cl	H
167	2	Me	H	H	Cl	H
168	3	Me	H	H	Cl	H
169	1	CH ₂ Ph	H	H	Cl	H
170	2	CH ₂ Ph	H	H	Cl	H
171	3	CH ₂ Ph	H	H	Cl	H
172	1	COMe	H	H	Cl	H
173	2	COMe	H	H	Cl	H
174	3	COMe	H	H	Cl	H
175	1	CO ₂ Me	H	H	Cl	H
176	2	CO ₂ Me	H	H	Cl	H
177	3	CO ₂ Me	H	H	Cl	H
178	1	CO ₂ tBu	H	H	Cl	H
179	2	CO ₂ tBu	H	H	Cl	H
180	3	CO ₂ tBu	H	H	Cl	H
181	1	CONHMe	H	H	Cl	H
182	2	CONHMe	H	H	Cl	H
183	3	CONHMe	H	H	Cl	H

TABLE 3-continued

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
184	1	SO ₂ Me	H	H	Cl	H
185	2	SO ₂ Me	H	H	Cl	H
186	3	SO ₂ Me	H	H	Cl	H
187	1	SO ₂ NH ₂	H	H	Cl	H
188	2	SO ₂ NH ₂	H	H	Cl	H
189	3	SO ₂ NH ₂	H	H	Cl	H
190	1	H	H	H	CN	H
191	2	H	H	H	CN	H
192	3	H	H	H	CN	H
193	1	Me	H	H	CN	H
194	2	Me	H	H	CN	H
195	3	Me	H	H	CN	H
196	1	CH ₂ Ph	H	H	CN	H
197	2	CH ₂ Ph	H	H	CN	H
198	3	CH ₂ Ph	H	H	CN	H
199	1	COMe	H	H	CN	H
200	2	COMe	H	H	CN	H
201	3	COMe	H	H	CN	H
202	1	CO ₂ Me	H	H	CN	H
203	2	CO ₂ Me	H	H	CN	H
204	3	CO ₂ Me	H	H	CN	H
205	1	CO ₂ tBu	H	H	CN	H
206	2	CO ₂ tBu	H	H	CN	H
207	3	CO ₂ tBu	H	H	CN	H
208	1	CONHMe	H	H	CN	H
209	2	CONHMe	H	H	CN	H
210	3	CONHMe	H	H	CN	H
211	1	SO ₂ Me	H	H	CN	H
212	2	SO ₂ Me	H	H	CN	H
213	3	SO ₂ Me	H	H	CN	H
214	1	SO ₂ NH ₂	H	H	CN	H
215	2	SO ₂ NH ₂	H	H	CN	H
216	3	SO ₂ NH ₂	H	H	CN	H
217	1	H	H	OH	H	H
218	2	H	H	OH	H	H
219	3	H	H	OH	H	H
220	1	Me	H	OH	H	H
221	2	Me	H	OH	H	H
222	3	Me	H	OH	H	H
223	1	CH ₂ Ph	H	OH	H	H
224	2	CH ₂ Ph	H	OH	H	H
225	3	CH ₂ Ph	H	OH	H	H
226	1	COMe	H	OH	H	H
227	2	COMe	H	OH	H	H
228	3	COMe	H	OH	H	H
229	1	CO ₂ Me	H	OH	H	H
230	2	CO ₂ Me	H	OH	H	H
231	3	CO ₂ Me	H	OH	H	H
232	1	CO ₂ tBu	H	OH	H	H
233	2	CO ₂ tBu	H	OH	H	H
234	3	CO ₂ tBu	H	OH	H	H
235	1	CONHMe	H	OH	H	H
236	2	CONHMe	H	OH	H	H
237	3	CONHMe	H	OH	H	H
238	1	SO ₂ Me	H	OH	H	H
239	2	SO ₂ Me	H	OH	H	H
240	3	SO ₂ Me	H	OH	H	H
241	1	SO ₂ NH ₂	H	OH	H	H
242	2	SO ₂ NH ₂	H	OH	H	H
243	3	SO ₂ NH ₂	H	OH	H	H
244	1	H	H	OMe	H	H
245	2	H	H	OMe	H	H
246	3	H	H	OMe	H	H
247	1	Me	H	OMe	H	H
248	2	Me	H	OMe	H	H
249	3	Me	H	OMe	H	H
250	1	CH ₂ Ph	H	OMe	H	H
251	2	CH ₂ Ph	H	OMe	H	H
252	3	CH ₂ Ph	H	OMe	H	H
253	1	COMe	H	OMe	H	H
254	2	COMe	H	OMe	H	H
255	3	COMe	H	OMe	H	H
256	1	CO ₂ Me	H	OMe	H	H
257	2	CO ₂ Me	H	OMe	H	H
258	3	CO ₂ Me	H	OMe	H	H

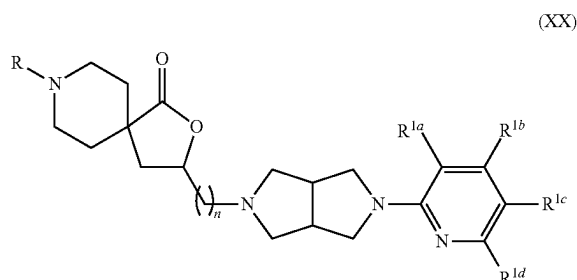
TABLE 3-continued

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
259	1	CO ₂ tBu	H	OMe	H	H
260	2	CO ₂ tBu	H	OMe	H	H
261	3	CO ₂ tBu	H	OMe	H	H
262	1	CONHMe	H	OMe	H	H
263	2	CONHMe	H	OMe	H	H
264	3	CONHMe	H	OMe	H	H
265	1	SO ₂ Me	H	OMe	H	H
266	2	SO ₂ Me	H	OMe	H	H
267	3	SO ₂ Me	H	OMe	H	H
268	1	SO ₂ NH ₂	H	OMe	H	H
269	2	SO ₂ NH ₂	H	OMe	H	H
270	3	SO ₂ NH ₂	H	OMe	H	H
271	1	H	H	Me	H	H
272	2	H	H	Me	H	H
273	3	H	H	Me	H	H
274	1	Me	H	Me	H	H
275	2	Me	H	Me	H	H
276	3	Me	H	Me	H	H
277	1	CH ₂ Ph	H	Me	H	H
278	2	CH ₂ Ph	H	Me	H	H
279	3	CH ₂ Ph	H	Me	H	H
280	1	COMe	H	Me	H	H
281	2	COMe	H	Me	H	H
282	3	COMe	H	Me	H	H
283	1	CO ₂ Me	H	Me	H	H
284	2	CO ₂ Me	H	Me	H	H
285	3	CO ₂ Me	H	Me	H	H
286	1	CO ₂ tBu	H	Me	H	H
287	2	CO ₂ tBu	H	Me	H	H
288	3	CO ₂ tBu	H	Me	H	H
289	1	CONHMe	H	Me	H	H
290	2	CONHMe	H	Me	H	H
291	3	CONHMe	H	Me	H	H
292	1	SO ₂ Me	H	Me	H	H
293	2	SO ₂ Me	H	Me	H	H
294	3	SO ₂ Me	H	Me	H	H
295	1	SO ₂ NH ₂	H	Me	H	H
296	2	SO ₂ NH ₂	H	Me	H	H
297	3	SO ₂ NH ₂	H	Me	H	H
298	1	H	H	CF ₃	H	H
299	2	H	H	CF ₃	H	H
300	3	H	H	CF ₃	H	H
301	1	Me	H	CF ₃	H	H
302	2	Me	H	CF ₃	H	H
303	3	Me	H	CF ₃	H	H
304	1	CH ₂ Ph	H	CF ₃	H	H
305	2	CH ₂ Ph	H	CF ₃	H	H
306	3	CH ₂ Ph	H	CF ₃	H	H
307	1	COMe	H	CF ₃	H	H
308	2	COMe	H	CF ₃	H	H
309	3	COMe	H	CF ₃	H	H
310	1	CO ₂ Me	H	CF ₃	H	H
311	2	CO ₂ Me	H	CF ₃	H	H
312	3	CO ₂ Me	H	CF ₃	H	H
313	1	CO ₂ tBu	H	CF ₃	H	H
314	2	CO ₂ tBu	H	CF ₃	H	H
315	3	CO ₂ tBu	H	CF ₃	H	H
316	1	CONHMe	H	CF ₃	H	H
317	2	CONHMe	H	CF ₃	H	H
318	3	CONHMe	H	CF ₃	H	H
319	1	SO ₂ Me	H	CF ₃	H	H
320	2	SO ₂ Me	H	CF ₃	H	H
321	3	SO ₂ Me	H	CF ₃	H	H
322	1	SO ₂ NH ₂	H	CF ₃	H	H
323	2	SO ₂ NH ₂	H	CF ₃	H	H
324	3	SO ₂ NH ₂	H	CF ₃	H	H
325	1	H	H	F	H	H
326	2	H	H	F	H	H
327	3	H	H	F	H	H
328	1	Me	H	F	H	H
329	2	Me	H	F	H	H
330	3	Me	H	F	H	H
331	1	CH ₂ Ph	H	F	H	H
332	2	CH ₂ Ph	H	F	H	H
333	3	CH ₂ Ph	H	F	H	H

TABLE 3-continued

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
334	1	COMe	H	F	H	H
335	2	COMe	H	F	H	H
336	3	COMe	H	F	H	H
337	1	CO ₂ Me	H	F	H	H
338	2	CO ₂ Me	H	F	H	H
339	3	CO ₂ Me	H	F	H	H
340	1	CO ₂ tBu	H	F	H	H
341	2	CO ₂ tBu	H	F	H	H
342	3	CO ₂ tBu	H	F	H	H
343	1	CONHMe	H	F	H	H
344	2	CONHMe	H	F	H	H
345	3	CONHMe	H	F	H	H
346	1	SO ₂ Me	H	F	H	H
347	2	SO ₂ Me	H	F	H	H
348	3	SO ₂ Me	H	F	H	H
349	1	SO ₂ NH ₂	H	F	H	H
350	2	SO ₂ NH ₂	H	F	H	H
351	3	SO ₂ NH ₂	H	F	H	H
352	1	H	H	Cl	H	H
353	2	H	H	Cl	H	H
354	3	H	H	Cl	H	H
355	1	Me	H	Cl	H	H
356	2	Me	H	Cl	H	H
357	3	Me	H	Cl	H	H
358	1	CH ₂ Ph	H	Cl	H	H
359	2	CH ₂ Ph	H	Cl	H	H
360	3	CH ₂ Ph	H	Cl	H	H
361	1	COMe	H	Cl	H	H
362	2	COMe	H	Cl	H	H
363	3	COMe	H	Cl	H	H
364	1	CO ₂ Me	H	Cl	H	H
365	2	CO ₂ Me	H	Cl	H	H
366	3	CO ₂ Me	H	Cl	H	H
367	1	CO ₂ tBu	H	Cl	H	H
368	2	CO ₂ tBu	H	Cl	H	H
369	3	CO ₂ tBu	H	Cl	H	H
370	1	CONHMe	H	Cl	H	H
371	2	CONHMe	H	Cl	H	H
372	3	CONHMe	H	Cl	H	H
373	1	SO ₂ Me	H	Cl	H	H
374	2	SO ₂ Me	H	Cl	H	H
375	3	SO ₂ Me	H	Cl	H	H
376	1	SO ₂ NH ₂	H	Cl	H	H
377	2	SO ₂ NH ₂	H	Cl	H	H
378	3	SO ₂ NH ₂	H	Cl	H	H
379	1	H	H	CN	H	H
380	2	H	H	CN	H	H
381	3	H	H	CN	H	H
382	1	Me	H	CN	H	H
383	2	Me	H	CN	H	H
384	3	Me	H	CN	H	H
385	1	CH ₂ Ph	H	CN	H	H
386	2	CH ₂ Ph	H	CN	H	H
387	3	CH ₂ Ph	H	CN	H	H
388	1	COMe	H	CN	H	H
389	2	COMe	H	CN	H	H
390	3	COMe	H	CN	H	H
391	1	CO ₂ Me	H	CN	H	H
392	2	CO ₂ Me	H	CN	H	H
393	3	CO ₂ Me	H	CN	H	H
394	1	CO ₂ tBu	H	CN	H	H
395	2	CO ₂ tBu	H	CN	H	H
396	3	CO ₂ tBu	H	CN	H	H
397	1	CONHMe	H	CN	H	H
398	2	CONHMe	H	CN	H	H
399	3	CONHMe	H	CN	H	H
400	1	SO ₂ Me	H	CN	H	H
401	2	SO ₂ Me	H	CN	H	H
402	3	SO ₂ Me	H	CN	H	H
403	1	SO ₂ NH ₂	H	CN	H	H
404	2	SO ₂ NH ₂	H	CN	H	H
405	3	SO ₂ NH ₂	H	CN	H	H

[0403] Exemplary embodiments include compounds having the formula (XX)



or a pharmaceutically acceptable salt form thereof defined herein below in Table 4.

TABLE 4

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
1	1	H	H	H	H	H
2	2	H	H	H	H	H
3	3	H	H	H	H	H
4	1	Me	H	H	H	H
5	2	Me	H	H	H	H
6	3	Me	H	H	H	H
7	1	CH ₂ Ph	H	H	H	H
8	2	CH ₂ Ph	H	H	H	H
9	3	CH ₂ Ph	H	H	H	H
10	1	COMe	H	H	H	H
11	2	COMe	H	H	H	H
12	3	COMe	H	H	H	H
13	1	CO ₂ Me	H	H	H	H
14	2	CO ₂ Me	H	H	H	H
15	3	CO ₂ Me	H	H	H	H
16	1	CO ₂ tBu	H	H	H	H
17	2	CO ₂ tBu	H	H	H	H
18	3	CO ₂ tBu	H	H	H	H
19	1	CONHMe	H	H	H	H
20	2	CONHMe	H	H	H	H
21	3	CONHMe	H	H	H	H
22	1	SO ₂ Me	H	H	H	H
23	2	SO ₂ Me	H	H	H	H
24	3	SO ₂ Me	H	H	H	H
25	1	SO ₂ NH ₂	H	H	H	H
26	2	SO ₂ NH ₂	H	H	H	H
27	3	SO ₂ NH ₂	H	H	H	H
28	1	H	H	H	OH	H
29	2	H	H	H	OH	H
30	3	H	H	H	OH	H
31	1	Me	H	H	OH	H
32	2	Me	H	H	OH	H
33	3	Me	H	H	OH	H
34	1	CH ₂ Ph	H	H	OH	H
35	2	CH ₂ Ph	H	H	OH	H
36	3	CH ₂ Ph	H	H	OH	H
37	1	COMe	H	H	OH	H
38	2	COMe	H	H	OH	H
39	3	COMe	H	H	OH	H
40	1	CO ₂ Me	H	H	OH	H
41	2	CO ₂ Me	H	H	OH	H
42	3	CO ₂ Me	H	H	OH	H
43	1	CO ₂ tBu	H	H	OH	H
44	2	CO ₂ tBu	H	H	OH	H
45	3	CO ₂ tBu	H	H	OH	H
46	1	CONHMe	H	H	OH	H
47	2	CONHMe	H	H	OH	H
48	3	CONHMe	H	H	OH	H
49	1	SO ₂ Me	H	H	OH	H
50	2	SO ₂ Me	H	H	OH	H
51	3	SO ₂ Me	H	H	OH	H
52	1	SO ₂ NH ₂	H	H	OH	H
53	2	SO ₂ NH ₂	H	H	OH	H
54	3	SO ₂ NH ₂	H	H	OH	H
55	1	H	H	H	OMe	H

TABLE 4-continued

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
56	2	H	H	H	OMe	H
57	3	H	H	H	OMe	H
58	1	Me	H	H	OMe	H
59	2	Me	H	H	OMe	H
60	3	Me	H	H	OMe	H
61	1	CH ₂ Ph	H	H	OMe	H
62	2	CH ₂ Ph	H	H	OMe	H
63	3	CH ₂ Ph	H	H	OMe	H
64	1	COMe	H	H	OMe	H
65	2	COMe	H	H	OMe	H
66	3	COMe	H	H	OMe	H
67	1	CO ₂ Me	H	H	OMe	H
68	2	CO ₂ Me	H	H	OMe	H
69	3	CO ₂ Me	H	H	OMe	H
70	1	CO ₂ tBu	H	H	OMe	H
71	2	CO ₂ tBu	H	H	OMe	H
72	3	CO ₂ tBu	H	H	OMe	H
73	1	CONHMe	H	H	OMe	H
74	2	CONHMe	H	H	OMe	H
75	3	CONHMe	H	H	OMe	H
76	1	SO ₂ Me	H	H	OMe	H
77	2	SO ₂ Me	H	H	OMe	H
78	3	SO ₂ Me	H	H	OMe	H
79	1	SO ₂ NH ₂	H	H	OMe	H
80	2	SO ₂ NH ₂	H	H	OMe	H
81	3	SO ₂ NH ₂	H	H	OMe	H
82	1	H	H	H	Me	H
83	2	H	H	H	Me	H
84	3	H	H	H	Me	H
85	1	Me	H	H	Me	H
86	2	Me	H	H	Me	H
87	3	Me	H	H	Me	H
88	1	CH ₂ Ph	H	H	Me	H
89	2	CH ₂ Ph	H	H	Me	H
90	3	CH ₂ Ph	H	H	Me	H
91	1	COMe	H	H	Me	H
92	2	COMe	H	H	Me	H
93	3	COMe	H	H	Me	H
94	1	CO ₂ Me	H	H	Me	H
95	2	CO ₂ Me	H	H	Me	H
96	3	CO ₂ Me	H	H	Me	H
97	1	CO ₂ tBu	H	H	Me	H
98	2	CO ₂ tBu	H	H	Me	H
99	3	CO ₂ tBu	H	H	Me	H
100	1	CONHMe	H	H	Me	H
101	2	CONHMe	H	H	Me	H
102	3	CONHMe	H	H	Me	H
103	1	SO ₂ Me	H	H	Me	H
104	2	SO ₂ Me	H	H	Me	H
105	3	SO ₂ Me	H	H	Me	H
106	1	SO ₂ NH ₂	H	H	Me	H
107	2	SO ₂ NH ₂	H	H	Me	H
108	3	SO ₂ NH ₂	H	H	Me	H
109	1	H	H	H	CF ₃	H
110	2	H	H	H	CF ₃	H
111	3	H	H	H	CF ₃	H
112	1	Me	H	H	CF ₃	H
113	2	Me	H	H	CF ₃	H
114	3	Me	H	H	CF ₃	H
115	1	CH ₂ Ph	H	H	CF ₃	H
116	2	CH ₂ Ph	H	H	CF ₃	H
117	3	CH ₂ Ph	H	H	CF ₃	H
118	1	COMe	H	H	CF ₃	H
119	2	COMe	H	H	CF ₃	H
120	3	COMe	H	H	CF ₃	H
121	1	CO ₂ Me	H	H	CF ₃	H
122	2	CO ₂ Me	H	H	CF ₃	H
123	3	CO ₂ Me	H	H	CF ₃	H
124	1	CO ₂ tBu	H	H	CF ₃	H
125	2	CO ₂ tBu	H	H	CF ₃	H
126	3	CO ₂ tBu	H	H	CF ₃	H
127	1	CONHMe	H	H	CF ₃	H
128	2	CONHMe	H	H	CF ₃	H
129	3	CONHMe	H	H	CF ₃	H
130	1	SO ₂ Me	H	H	CF ₃	H

TABLE 4-continued

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
131	2	SO ₂ Me	H	H	CF ₃	H
132	3	SO ₂ Me	H	H	CF ₃	H
133	1	SO ₂ NH ₂	H	H	CF ₃	H
134	2	SO ₂ NH ₂	H	H	CF ₃	H
135	3	SO ₂ NH ₂	H	H	CF ₃	H
136	1	H	H	H	F	H
137	2	H	H	H	F	H
138	3	H	H	H	F	H
139	1	Me	H	H	F	H
140	2	Me	H	H	F	H
141	3	Me	H	H	F	H
142	1	CH ₂ Ph	H	H	F	H
143	2	CH ₂ Ph	H	H	F	H
144	3	CH ₂ Ph	H	H	F	H
145	1	COMe	H	H	F	H
146	2	COMe	H	H	F	H
147	3	COMe	H	H	F	H
148	1	CO ₂ Me	H	H	F	H
149	2	CO ₂ Me	H	H	F	H
150	3	CO ₂ Me	H	H	F	H
151	1	CO ₂ tBu	H	H	F	H
152	2	CO ₂ tBu	H	H	F	H
153	3	CO ₂ tBu	H	H	F	H
154	1	CONHMe	H	H	F	H
155	2	CONHMe	H	H	F	H
156	3	CONHMe	H	H	F	H
157	1	SO ₂ Me	H	H	F	H
158	2	SO ₂ Me	H	H	F	H
159	3	SO ₂ Me	H	H	F	H
160	1	SO ₂ NH ₂	H	H	F	H
161	2	SO ₂ NH ₂	H	H	F	H
162	3	SO ₂ NH ₂	H	H	F	H
163	1	H	H	H	Cl	H
164	2	H	H	H	Cl	H
165	3	H	H	H	Cl	H
166	1	Me	H	H	Cl	H
167	2	Me	H	H	Cl	H
168	3	Me	H	H	Cl	H
169	1	CH ₂ Ph	H	H	Cl	H
170	2	CH ₂ Ph	H	H	Cl	H
171	3	CH ₂ Ph	H	H	Cl	H
172	1	COMe	H	H	Cl	H
173	2	COMe	H	H	Cl	H
174	3	COMe	H	H	Cl	H
175	1	CO ₂ Me	H	H	Cl	H
176	2	CO ₂ Me	H	H	Cl	H
177	3	CO ₂ Me	H	H	Cl	H
178	1	CO ₂ tBu	H	H	Cl	H
179	2	CO ₂ tBu	H	H	Cl	H
180	3	CO ₂ tBu	H	H	Cl	H
181	1	CONHMe	H	H	Cl	H
182	2	CONHMe	H	H	Cl	H
183	3	CONHMe	H	H	Cl	H
184	1	SO ₂ Me	H	H	Cl	H
185	2	SO ₂ Me	H	H	Cl	H
186	3	SO ₂ Me	H	H	Cl	H
187	1	SO ₂ NH ₂	H	H	Cl	H
188	2	SO ₂ NH ₂	H	H	Cl	H
189	3	SO ₂ NH ₂	H	H	Cl	H
190	1	H	H	H	CN	H
191	2	H	H	H	CN	H
192	3	H	H	H	CN	H
193	1	Me	H	H	CN	H
194	2	Me	H	H	CN	H
195	3	Me	H	H	CN	H
196	1	CH ₂ Ph	H	H	CN	H
197	2	CH ₂ Ph	H	H	CN	H
198	3	CH ₂ Ph	H	H	CN	H
199	1	COMe	H	H	CN	H
200	2	COMe	H	H	CN	H
201	3	COMe	H	H	CN	H
202	1	CO ₂ Me	H	H	CN	H
203	2	CO ₂ Me	H	H	CN	H
204	3	CO ₂ Me	H	H	CN	H
205	1	CO ₂ tBu	H	H	CN	H

TABLE 4-continued

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
206	2	CO ₂ tBu	H	H	CN	H
207	3	CO ₂ tBu	H	H	CN	H
208	1	CONHMe	H	H	CN	H
209	2	CONHMe	H	H	CN	H
210	3	CONHMe	H	H	CN	H
211	1	SO ₂ Me	H	H	CN	H
212	2	SO ₂ Me	H	H	CN	H
213	3	SO ₂ Me	H	H	CN	H
214	1	SO ₂ NH ₂	H	H	CN	H
215	2	SO ₂ NH ₂	H	H	CN	H
216	3	SO ₂ NH ₂	H	H	CN	H
217	1	H	H	OH	H	H
218	2	H	H	OH	H	H
219	3	H	H	OH	H	H
220	1	Me	H	OH	H	H
221	2	Me	H	OH	H	H
222	3	Me	H	OH	H	H
223	1	CH ₂ Ph	H	OH	H	H
224	2	CH ₂ Ph	H	OH	H	H
225	3	CH ₂ Ph	H	OH	H	H
226	1	COMe	H	OH	H	H
227	2	COMe	H	OH	H	H
228	3	COMe	H	OH	H	H
229	1	CO ₂ Me	H	OH	H	H
230	2	CO ₂ Me	H	OH	H	H
231	3	CO ₂ Me	H	OH	H	H
232	1	CO ₂ tBu	H	OH	H	H
233	2	CO ₂ tBu	H	OH	H	H
234	3	CO ₂ tBu	H	OH	H	H
235	1	CONHMe	H	OH	H	H
236	2	CONHMe	H	OH	H	H
237	3	CONHMe	H	OH	H	H
238	1	SO ₂ Me	H	OH	H	H
239	2	SO ₂ Me	H	OH	H	H
240	3	SO ₂ Me	H	OH	H	H
241	1	SO ₂ NH ₂	H	OH	H	H
242	2	SO ₂ NH ₂	H	OH	H	H
243	3	SO ₂ NH ₂	H	OH	H	H
244	1	H	H	OMe	H	H
245	2	H	H	OMe	H	H
246	3	H	H	OMe	H	H
247	1	Me	H	OMe	H	H
248	2	Me	H	OMe	H	H
249	3	Me	H	OMe	H	H
250	1	CH ₂ Ph	H	OMe	H	H
251	2	CH ₂ Ph	H	OMe	H	H
252	3	CH ₂ Ph	H	OMe	H	H
253	1	COMe	H	OMe	H	H
254	2	COMe	H	OMe	H	H
255	3	COMe	H	OMe	H	H
256	1	CO ₂ Me	H	OMe	H	H
257	2	CO ₂ Me	H	OMe	H	H
258	3	CO ₂ Me	H	OMe	H	H
259	1	CO ₂ tBu	H	OMe	H	H
260	2	CO ₂ tBu	H	OMe	H	H
261	3	CO ₂ tBu	H	OMe	H	H
262	1	CONHMe	H	OMe	H	H
263	2	CONHMe	H	OMe	H	H
264	3	CONHMe	H	OMe	H	H
265	1	SO ₂ Me	H	OMe	H	H
266	2	SO ₂ Me	H	OMe	H	H
267	3	SO ₂ Me	H	OMe	H	H
268	1	SO ₂ NH ₂	H	OMe	H	H
269	2	SO ₂ NH ₂	H	OMe	H	H
270	3	SO ₂ NH ₂	H	OMe	H	H
271	1	H	H	Me	H	H
272	2	H	H	Me	H	H
273	3	H	H	Me	H	H
274	1	Me	H	Me	H	H
275	2	Me	H	Me	H	H
276	3	Me	H	Me	H	H
277	1	CH ₂ Ph	H	Me	H	H
278	2	CH ₂ Ph	H	Me	H	H
279	3	CH ₂ Ph	H	Me	H	H
280	1	COMe	H	Me	H	H

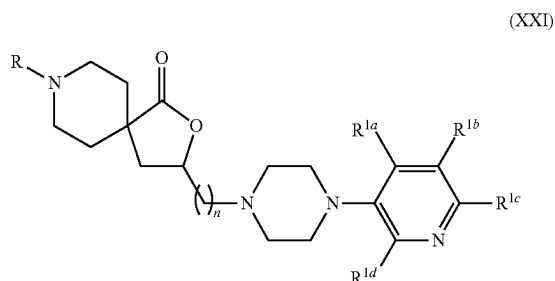
TABLE 4-continued

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
281	2	COMe	H	Me	H	H
282	3	COMe	H	Me	H	H
283	1	CO ₂ Me	H	Me	H	H
284	2	CO ₂ Me	H	Me	H	H
285	3	CO ₂ Me	H	Me	H	H
286	1	CO ₂ tBu	H	Me	H	H
287	2	CO ₂ tBu	H	Me	H	H
288	3	CO ₂ tBu	H	Me	H	H
289	1	CONHMe	H	Me	H	H
290	2	CONHMe	H	Me	H	H
291	3	CONHMe	H	Me	H	H
292	1	SO ₂ Me	H	Me	H	H
293	2	SO ₂ Me	H	Me	H	H
294	3	SO ₂ Me	H	Me	H	H
295	1	SO ₂ NH ₂	H	Me	H	H
296	2	SO ₂ NH ₂	H	Me	H	H
297	3	SO ₂ NH ₂	H	Me	H	H
298	1	H	H	CF ₃	H	H
299	2	H	H	CF ₃	H	H
300	3	H	H	CF ₃	H	H
301	1	Me	H	CF ₃	H	H
302	2	Me	H	CF ₃	H	H
303	3	Me	H	CF ₃	H	H
304	1	CH ₂ Ph	H	CF ₃	H	H
305	2	CH ₂ Ph	H	CF ₃	H	H
306	3	CH ₂ Ph	H	CF ₃	H	H
307	1	COMe	H	CF ₃	H	H
308	2	COMe	H	CF ₃	H	H
309	3	COMe	H	CF ₃	H	H
310	1	CO ₂ Me	H	CF ₃	H	H
311	2	CO ₂ Me	H	CF ₃	H	H
312	3	CO ₂ Me	H	CF ₃	H	H
313	1	CO ₂ tBu	H	CF ₃	H	H
314	2	CO ₂ tBu	H	CF ₃	H	H
315	3	CO ₂ tBu	H	CF ₃	H	H
316	1	CONHMe	H	CF ₃	H	H
317	2	CONHMe	H	CF ₃	H	H
318	3	CONHMe	H	CF ₃	H	H
319	1	SO ₂ Me	H	CF ₃	H	H
320	2	SO ₂ Me	H	CF ₃	H	H
321	3	SO ₂ Me	H	CF ₃	H	H
322	1	SO ₂ NH ₂	H	CF ₃	H	H
323	2	SO ₂ NH ₂	H	CF ₃	H	H
324	3	SO ₂ NH ₂	H	CF ₃	H	H
325	1	H	H	F	H	H
326	2	H	H	F	H	H
327	3	H	H	F	H	H
328	1	Me	H	F	H	H
329	2	Me	H	F	H	H
330	3	Me	H	F	H	H
331	1	CH ₂ Ph	H	F	H	H
332	2	CH ₂ Ph	H	F	H	H
333	3	CH ₂ Ph	H	F	H	H
334	1	COMe	H	F	H	H
335	2	COMe	H	F	H	H
336	3	COMe	H	F	H	H
337	1	CO ₂ Me	H	F	H	H
338	2	CO ₂ Me	H	F	H	H
339	3	CO ₂ Me	H	F	H	H
340	1	CO ₂ tBu	H	F	H	H
341	2	CO ₂ tBu	H	F	H	H
342	3	CO ₂ tBu	H	F	H	H
343	1	CONHMe	H	F	H	H
344	2	CONHMe	H	F	H	H
345	3	CONHMe	H	F	H	H
346	1	SO ₂ Me	H	F	H	H
347	2	SO ₂ Me	H	F	H	H
348	3	SO ₂ Me	H	F	H	H
349	1	SO ₂ NH ₂	H	F	H	H
350	2	SO ₂ NH ₂	H	F	H	H
351	3	SO ₂ NH ₂	H	F	H	H
352	1	H	H	Cl	H	H
353	2	H	H	Cl	H	H
354	3	H	H	Cl	H	H
355	1	Me	H	Cl	H	H

TABLE 4-continued

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
356	2	Me	H	Cl	H	H
357	3	Me	H	Cl	H	H
358	1	CH ₂ Ph	H	Cl	H	H
359	2	CH ₂ Ph	H	Cl	H	H
360	3	CH ₂ Ph	H	Cl	H	H
361	1	COMe	H	Cl	H	H
362	2	COMe	H	Cl	H	H
363	3	COMe	H	Cl	H	H
364	1	CO ₂ Me	H	Cl	H	H
365	2	CO ₂ Me	H	Cl	H	H
366	3	CO ₂ Me	H	Cl	H	H
367	1	CO ₂ tBu	H	Cl	H	H
368	2	CO ₂ tBu	H	Cl	H	H
369	3	CO ₂ tBu	H	Cl	H	H
370	1	CONHMe	H	Cl	H	H
371	2	CONHMe	H	Cl	H	H
372	3	CONHMe	H	Cl	H	H
373	1	SO ₂ Me	H	Cl	H	H
374	2	SO ₂ Me	H	Cl	H	H
375	3	SO ₂ Me	H	Cl	H	H
376	1	SO ₂ NH ₂	H	Cl	H	H
377	2	SO ₂ NH ₂	H	Cl	H	H
378	3	SO ₂ NH ₂	H	Cl	H	H
379	1	H	H	CN	H	H
380	2	H	H	CN	H	H
381	3	H	H	CN	H	H
382	1	Me	H	CN	H	H
383	2	Me	H	CN	H	H
384	3	Me	H	CN	H	H
385	1	CH ₂ Ph	H	CN	H	H
386	2	CH ₂ Ph	H	CN	H	H
387	3	CH ₂ Ph	H	CN	H	H
388	1	COMe	H	CN	H	H
389	2	COMe	H	CN	H	H
390	3	COMe	H	CN	H	H
391	1	CO ₂ Me	H	CN	H	H
392	2	CO ₂ Me	H	CN	H	H
393	3	CO ₂ Me	H	CN	H	H
394	1	CO ₂ tBu	H	CN	H	H
395	2	CO ₂ tBu	H	CN	H	H
396	3	CO ₂ tBu	H	CN	H	H
397	1	CONHMe	H	CN	H	H
398	2	CONHMe	H	CN	H	H
399	3	CONHMe	H	CN	H	H
400	1	SO ₂ Me	H	CN	H	H
401	2	SO ₂ Me	H	CN	H	H
402	3	SO ₂ Me	H	CN	H	H
403	1	SO ₂ NH ₂	H	CN	H	H
404	2	SO ₂ NH ₂	H	CN	H	H
405	3	SO ₂ NH ₂	H	CN	H	H

[0404] Exemplary embodiments include compounds having the formula (XXI)

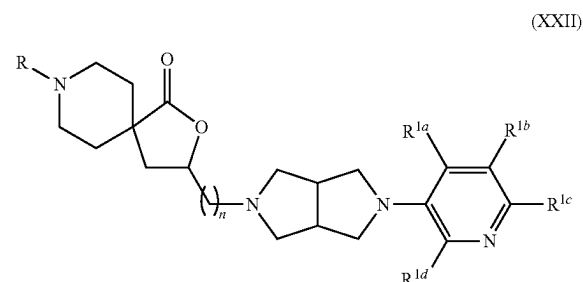


or a pharmaceutically acceptable salt form thereof defined herein below in Table 5.

TABLE 5

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
1	1	H	H	H	H	H
2	2	H	H	H	H	H
3	3	H	H	H	H	H
4	1	Me	H	H	H	H
5	2	Me	H	H	H	H
6	3	Me	H	H	H	H
7	1	CH ₂ Ph	H	H	H	H
8	2	CH ₂ Ph	H	H	H	H
9	3	CH ₂ Ph	H	H	H	H
10	1	COMe	H	H	H	H
11	2	COMe	H	H	H	H
12	3	COMe	H	H	H	H
13	1	CO ₂ Me	H	H	H	H
14	2	CO ₂ Me	H	H	H	H
15	3	CO ₂ Me	H	H	H	H
16	1	CO ₂ tBu	H	H	H	H
17	2	CO ₂ tBu	H	H	H	H
18	3	CO ₂ tBu	H	H	H	H
19	1	CONHMe	H	H	H	H
20	2	CONHMe	H	H	H	H
21	3	CONHMe	H	H	H	H
22	1	SO ₂ Me	H	H	H	H
23	2	SO ₂ Me	H	H	H	H
24	3	SO ₂ Me	H	H	H	H
25	1	SO ₂ NH ₂	H	H	H	H
26	2	SO ₂ NH ₂	H	H	H	H
27	3	SO ₂ NH ₂	H	H	H	H

[0405] Exemplary embodiments include compounds having the formula (XXII)



or a pharmaceutically acceptable salt form thereof defined herein below in Table 6.

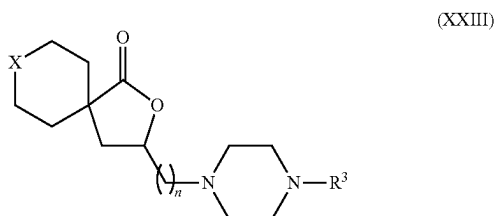
TABLE 6

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
1	1	H	H	H	H	H
2	2	H	H	H	H	H
3	3	H	H	H	H	H
4	1	Me	H	H	H	H
5	2	Me	H	H	H	H
6	3	Me	H	H	H	H
7	1	CH ₂ Ph	H	H	H	H
8	2	CH ₂ Ph	H	H	H	H
9	3	CH ₂ Ph	H	H	H	H
10	1	COMe	H	H	H	H
11	2	COMe	H	H	H	H
12	3	COMe	H	H	H	H
13	1	CO ₂ Me	H	H	H	H
14	2	CO ₂ Me	H	H	H	H
15	3	CO ₂ Me	H	H	H	H
16	1	CO ₂ tBu	H	H	H	H
17	2	CO ₂ tBu	H	H	H	H
18	3	CO ₂ tBu	H	H	H	H

TABLE 6-continued

Entry	n	R	R ^{1a}	R ^{1b}	R ^{1c}	R ^{1d}
19	1	CONHMe	H	H	H	H
20	2	CONHMe	H	H	H	H
21	3	CONHMe	H	H	H	H
22	1	SO ₂ Me	H	H	H	H
23	2	SO ₂ Me	H	H	H	H
24	3	SO ₂ Me	H	H	H	H
25	1	SO ₂ NH ₂	H	H	H	H
26	2	SO ₂ NH ₂	H	H	H	H
27	3	SO ₂ NH ₂	H	H	H	H

[0406] Exemplary embodiments include compounds having the formula (XXIII)



or a pharmaceutically acceptable salt form thereof defined herein below in Table 7.

TABLE 7

Entry	n	X	R ³
1	1	O	Phenyl
2	2	O	Phenyl
3	3	O	Phenyl
4	1	S	Phenyl
5	2	S	Phenyl
6	3	S	Phenyl
7	1	SO	Phenyl
8	2	SO	Phenyl
9	3	SO	Phenyl
10	1	SO ₂	Phenyl
11	2	SO ₂	Phenyl
12	3	SO ₂	Phenyl
13	1	O	3-OH-Phenyl
14	2	O	3-OH-Phenyl
15	3	O	3-OH-Phenyl
16	1	S	3-OH-Phenyl
17	2	S	3-OH-Phenyl
18	3	S	3-OH-Phenyl
19	1	SO	3-OH-Phenyl
20	2	SO	3-OH-Phenyl
21	3	SO	3-OH-Phenyl
22	1	SO ₂	3-OH-Phenyl
23	2	SO ₂	3-OH-Phenyl
24	3	SO ₂	3-OH-Phenyl
25	1	O	4-NO ₂ -Phenyl
26	2	O	4-NO ₂ -Phenyl
27	3	O	4-NO ₂ -Phenyl
28	1	S	4-NO ₂ -Phenyl
29	2	S	4-NO ₂ -Phenyl
30	3	S	4-NO ₂ -Phenyl
31	1	SO	4-NO ₂ -Phenyl
32	2	SO	4-NO ₂ -Phenyl
33	3	SO	4-NO ₂ -Phenyl
34	1	SO ₂	4-NO ₂ -Phenyl
35	2	SO ₂	4-NO ₂ -Phenyl
36	3	SO ₂	4-NO ₂ -Phenyl
37	1	O	3-OMe-Phenyl
38	2	O	3-OMe-Phenyl
39	3	O	3-OMe-Phenyl

TABLE 7-continued

Entry	n	X	R ³
40	1	S	3-OMe-Phenyl
41	2	S	3-OMe-Phenyl
42	3	S	3-OMe-Phenyl
43	1	SO	3-OMe-Phenyl
44	2	SO	3-OMe-Phenyl
45	3	SO	3-OMe-Phenyl
46	1	SO ₂	3-OMe-Phenyl
47	2	SO ₂	3-OMe-Phenyl
48	3	SO ₂	3-OMe-Phenyl
49	1	O	4-CN-Phenyl
50	2	O	4-CN-Phenyl
51	3	O	4-CN-Phenyl
52	1	S	4-CN-Phenyl
53	2	S	4-CN-Phenyl
54	3	S	4-CN-Phenyl
55	1	SO	4-CN-Phenyl
56	2	SO	4-CN-Phenyl
57	3	SO	4-CN-Phenyl
58	1	SO ₂	4-CN-Phenyl
59	2	SO ₂	4-CN-Phenyl
60	3	SO ₂	4-CN-Phenyl
61	1	O	2-CN-Phenyl
62	2	O	2-CN-Phenyl
63	3	O	2-CN-Phenyl
64	1	S	2-CN-Phenyl
65	2	S	2-CN-Phenyl
66	3	S	2-CN-Phenyl
67	1	SO	2-CN-Phenyl
68	2	SO	2-CN-Phenyl
69	3	SO	2-CN-Phenyl
70	1	SO ₂	2-CN-Phenyl
71	2	SO ₂	2-CN-Phenyl
72	3	SO ₂	2-CN-Phenyl
73	1	O	3-Me-Phenyl
74	2	O	3-Me-Phenyl
75	3	O	3-Me-Phenyl
76	1	S	3-Me-Phenyl
77	2	S	3-Me-Phenyl
78	3	S	3-Me-Phenyl
79	1	SO	3-Me-Phenyl
80	2	SO	3-Me-Phenyl
81	3	SO	3-Me-Phenyl
82	1	SO ₂	3-Me-Phenyl
83	2	SO ₂	3-Me-Phenyl
84	3	SO ₂	3-Me-Phenyl
85	1	O	2-F-Phenyl
86	2	O	2-F-Phenyl
87	3	O	2-F-Phenyl
88	1	S	2-F-Phenyl
89	2	S	2-F-Phenyl
90	3	S	2-F-Phenyl
91	1	SO	2-F-Phenyl
92	2	SO	2-F-Phenyl
93	3	SO	2-F-Phenyl
94	1	SO ₂	2-F-Phenyl
95	2	SO ₂	2-F-Phenyl
96	3	SO ₂	2-F-Phenyl
97	1	O	4-F-Phenyl
98	2	O	4-F-Phenyl
99	3	O	4-F-Phenyl
100	1	S	4-F-Phenyl
101	2	S	4-F-Phenyl
102	3	S	4-F-Phenyl
103	1	SO	4-F-Phenyl
104	2	SO	4-F-Phenyl
105	3	SO	4-F-Phenyl
106	1	SO ₂	4-F-Phenyl
107	2	SO ₂	4-F-Phenyl
108	3	SO ₂	4-F-Phenyl
109	1	O	3-Cl-Phenyl
110	2	O	3-Cl-Phenyl
111	3	O	3-Cl-Phenyl
112	1	S	3-Cl-Phenyl
113	2	S	3-Cl-Phenyl
114	3	S	3-Cl-Phenyl

TABLE 7-continued

Entry	n	X	R ³
115	1	SO	3-Cl-Phenyl
116	2	SO	3-Cl-Phenyl
117	3	SO	3-Cl-Phenyl
118	1	SO ₂	3-Cl-Phenyl
119	2	SO ₂	3-Cl-Phenyl
120	3	SO ₂	3-Cl-Phenyl
121	1	O	2-Br-Phenyl
122	2	O	2-Br-Phenyl
123	3	O	2-Br-Phenyl
124	1	S	2-Br-Phenyl
125	2	S	2-Br-Phenyl
126	3	S	2-Br-Phenyl
127	1	SO	2-Br-Phenyl
128	2	SO	2-Br-Phenyl
129	3	SO	2-Br-Phenyl
130	1	SO ₂	2-Br-Phenyl
131	2	SO ₂	2-Br-Phenyl
132	3	SO ₂	2-Br-Phenyl
133	1	O	4-Br-Phenyl
134	2	O	4-Br-Phenyl
135	3	O	4-Br-Phenyl
136	1	S	4-Br-Phenyl
137	2	S	4-Br-Phenyl
138	3	S	4-Br-Phenyl
139	1	SO	4-Br-Phenyl
140	2	SO	4-Br-Phenyl
141	3	SO	4-Br-Phenyl
142	1	SO ₂	4-Br-Phenyl
143	2	SO ₂	4-Br-Phenyl
144	3	SO ₂	4-Br-Phenyl
145	1	O	3-CF ₃ -Phenyl
146	2	O	3-CF ₃ -Phenyl
147	3	O	3-CF ₃ -Phenyl
148	1	S	3-CF ₃ -Phenyl
149	2	S	3-CF ₃ -Phenyl
150	3	S	3-CF ₃ -Phenyl
151	1	SO	3-CF ₃ -Phenyl
152	2	SO	3-CF ₃ -Phenyl
153	3	SO	3-CF ₃ -Phenyl
154	1	SO ₂	3-CF ₃ -Phenyl
155	2	SO ₂	3-CF ₃ -Phenyl
156	3	SO ₂	3-CF ₃ -Phenyl
157	1	O	2-iPr-Phenyl
158	2	O	2-iPr-Phenyl
159	3	O	2-iPr-Phenyl
160	1	S	2-iPr-Phenyl
161	2	S	2-iPr-Phenyl
162	3	S	2-iPr-Phenyl
163	1	SO	2-iPr-Phenyl
164	2	SO	2-iPr-Phenyl
165	3	SO	2-iPr-Phenyl
166	1	SO ₂	2-iPr-Phenyl
167	2	SO ₂	2-iPr-Phenyl
168	3	SO ₂	2-iPr-Phenyl
169	1	O	4-iPr-Phenyl
170	2	O	4-iPr-Phenyl
171	3	O	4-iPr-Phenyl
172	1	S	4-iPr-Phenyl
173	2	S	4-iPr-Phenyl
174	3	S	4-iPr-Phenyl
175	1	SO	4-iPr-Phenyl
176	2	SO	4-iPr-Phenyl
177	3	SO	4-iPr-Phenyl
178	1	SO ₂	4-iPr-Phenyl
179	2	SO ₂	4-iPr-Phenyl
180	3	SO ₂	4-iPr-Phenyl
181	1	O	3-NH ₂ -Phenyl
182	2	O	3-NH ₂ -Phenyl
183	3	O	3-NH ₂ -Phenyl
184	1	S	3-NH ₂ -Phenyl
185	2	S	3-NH ₂ -Phenyl
186	3	S	3-NH ₂ -Phenyl
187	1	SO	3-NH ₂ -Phenyl
188	2	SO	3-NH ₂ -Phenyl
189	3	SO	3-NH ₂ -Phenyl

TABLE 7-continued

Entry	n	X	R ³
190	1	SO ₂	3-NH ₂ -Phenyl
191	2	SO ₂	3-NH ₂ -Phenyl
192	3	SO ₂	3-NH ₂ -Phenyl
193	1	O	2,4-di-Me-Phenyl
194	2	O	2,4-di-Me-Phenyl
195	3	O	2,4-di-Me-Phenyl
196	1	S	2,4-di-Me-Phenyl
197	2	S	2,4-di-Me-Phenyl
198	3	S	2,4-di-Me-Phenyl
199	1	SO	2,4-di-Me-Phenyl
200	2	SO	2,4-di-Me-Phenyl
201	3	SO	2,4-di-Me-Phenyl
202	1	SO ₂	2,4-di-Me-Phenyl
203	2	SO ₂	2,4-di-Me-Phenyl
204	3	SO ₂	2,4-di-Me-Phenyl
205	1	O	2,6-di-iPr-Phenyl
206	2	O	2,6-di-iPr-Phenyl
207	3	O	2,6-di-iPr-Phenyl
208	1	S	2,6-di-iPr-Phenyl
209	2	S	2,6-di-iPr-Phenyl
210	3	S	2,6-di-iPr-Phenyl
211	1	SO	2,6-di-iPr-Phenyl
212	2	SO	2,6-di-iPr-Phenyl
213	3	SO	2,6-di-iPr-Phenyl
214	1	SO ₂	2,6-di-iPr-Phenyl
215	2	SO ₂	2,6-di-iPr-Phenyl
216	3	SO ₂	2,6-di-iPr-Phenyl
217	1	O	3-Ph-Phenyl
218	2	O	3-Ph-Phenyl
219	3	O	3-Ph-Phenyl
220	1	S	3-Ph-Phenyl
221	2	S	3-Ph-Phenyl
222	3	S	3-Ph-Phenyl
223	1	SO	3-Ph-Phenyl
224	2	SO	3-Ph-Phenyl
225	3	SO	3-Ph-Phenyl
226	1	SO ₂	3-Ph-Phenyl
227	2	SO ₂	3-Ph-Phenyl
228	3	SO ₂	3-Ph-Phenyl
229	1	O	2-morpholino-phenyl
230	2	O	2-morpholino-phenyl
231	3	O	2-morpholino-phenyl
232	1	S	2-morpholino-phenyl
233	2	S	2-morpholino-phenyl
234	3	S	2-morpholino-phenyl
235	1	SO	2-morpholino-phenyl
236	2	SO	2-morpholino-phenyl
237	3	SO	2-morpholino-phenyl
238	1	SO ₂	2-morpholino-phenyl
239	2	SO ₂	2-morpholino-phenyl
240	3	SO ₂	2-morpholino-phenyl
241	1	O	4-morpholino-phenyl
242	2	O	4-morpholino-phenyl
243	3	O	4-morpholino-phenyl
244	1	S	4-morpholino-phenyl
245	2	S	4-morpholino-phenyl
246	3	S	4-morpholino-phenyl
247	1	SO	4-morpholino-phenyl
248	2	SO	4-morpholino-phenyl
249	3	SO	4-morpholino-phenyl
250	1	SO ₂	4-morpholino-phenyl
251	2	SO ₂	4-morpholino-phenyl
252	3	SO ₂	4-morpholino-phenyl
253	1	O	2-pyrimidinyl
254	2	O	2-pyrimidinyl
255	3	O	2-pyrimidinyl
256	1	S	2-pyrimidinyl
257	2	S	2-pyrimidinyl
258	3	S	2-pyrimidinyl
259	1	SO	2-pyrimidinyl
260	2	SO	2-pyrimidinyl
261	3	SO	2-pyrimidinyl
262	1	SO ₂	2-pyrimidinyl
263	2	SO ₂	2-pyrimidinyl
264	3	SO ₂	2-pyrimidinyl

TABLE 7-continued

Entry	n	X	R ³
265	1	O	2-methyl-1H-benzo[d]imidazol-4-yl
266	2	O	2-methyl-1H-benzo[d]imidazol-4-yl
267	3	O	2-methyl-1H-benzo[d]imidazol-4-yl
268	1	S	2-methyl-1H-benzo[d]imidazol-4-yl
269	2	S	2-methyl-1H-benzo[d]imidazol-4-yl
270	3	S	2-methyl-1H-benzo[d]imidazol-4-yl
271	1	SO	2-methyl-1H-benzo[d]imidazol-4-yl
272	2	SO	2-methyl-1H-benzo[d]imidazol-4-yl
273	3	SO	2-methyl-1H-benzo[d]imidazol-4-yl
274	1	SO ₂	2-methyl-1H-benzo[d]imidazol-4-yl
275	2	SO ₂	2-methyl-1H-benzo[d]imidazol-4-yl
276	3	SO ₂	2-methyl-1H-benzo[d]imidazol-4-yl
277	1	O	4-OH-Phenyl
278	2	O	4-OH-Phenyl
279	3	O	4-OH-Phenyl
280	1	S	4-OH-Phenyl
281	2	S	4-OH-Phenyl
282	3	S	4-OH-Phenyl
283	1	SO	4-OH-Phenyl
284	2	SO	4-OH-Phenyl
285	3	SO	4-OH-Phenyl
286	1	SO ₂	4-OH-Phenyl
287	2	SO ₂	4-OH-Phenyl
288	3	SO ₂	4-OH-Phenyl
289	1	O	2-OH-Phenyl
290	2	O	2-OH-Phenyl
291	3	O	2-OH-Phenyl
292	1	S	2-OH-Phenyl
293	2	S	2-OH-Phenyl
294	3	S	2-OH-Phenyl
295	1	SO	2-OH-Phenyl
296	2	SO	2-OH-Phenyl
297	3	SO	2-OH-Phenyl
298	1	SO ₂	2-OH-Phenyl
299	2	SO ₂	2-OH-Phenyl
300	3	SO ₂	2-OH-Phenyl
301	1	O	4-OMe-Phenyl
302	2	O	4-OMe-Phenyl
303	3	O	4-OMe-Phenyl
304	1	S	4-OMe-Phenyl
305	2	S	4-OMe-Phenyl
306	3	S	4-OMe-Phenyl
307	1	SO	4-OMe-Phenyl
308	2	SO	4-OMe-Phenyl
309	3	SO	4-OMe-Phenyl
310	1	SO ₂	4-OMe-Phenyl
311	2	SO ₂	4-OMe-Phenyl
312	3	SO ₂	4-OMe-Phenyl
313	1	O	2-OMe-Phenyl
314	2	O	2-OMe-Phenyl
315	3	O	2-OMe-Phenyl
316	1	S	2-OMe-Phenyl
317	2	S	2-OMe-Phenyl
318	3	S	2-OMe-Phenyl
319	1	SO	2-OMe-Phenyl
320	2	SO	2-OMe-Phenyl
321	3	SO	2-OMe-Phenyl
322	1	SO ₂	2-OMe-Phenyl
323	2	SO ₂	2-OMe-Phenyl
324	3	SO ₂	2-OMe-Phenyl
325	1	O	3-CN-Phenyl
326	2	O	3-CN-Phenyl
327	3	O	3-CN-Phenyl
328	1	S	3-CN-Phenyl
329	2	S	3-CN-Phenyl
330	3	S	3-CN-Phenyl
331	1	SO	3-CN-Phenyl
332	2	SO	3-CN-Phenyl
333	3	SO	3-CN-Phenyl
334	1	SO ₂	3-CN-Phenyl
335	2	SO ₂	3-CN-Phenyl
336	3	SO ₂	3-CN-Phenyl
337	1	O	2-Me-Phenyl
338	2	O	2-Me-Phenyl
339	3	O	2-Me-Phenyl

TABLE 7-continued

Entry	n	X	R ³
340	1	S	2-Me-Phenyl
341	2	S	2-Me-Phenyl
342	3	S	2-Me-Phenyl
343	1	SO	2-Me-Phenyl
344	2	SO	2-Me-Phenyl
345	3	SO	2-Me-Phenyl
346	1	SO ₂	2-Me-Phenyl
347	2	SO ₂	2-Me-Phenyl
348	3	SO ₂	2-Me-Phenyl
349	1	O	4-Me-Phenyl
350	2	O	4-Me-Phenyl
351	3	O	4-Me-Phenyl
352	1	S	4-Me-Phenyl
353	2	S	4-Me-Phenyl
354	3	S	4-Me-Phenyl
355	1	SO	4-Me-Phenyl
356	2	SO	4-Me-Phenyl
357	3	SO	4-Me-Phenyl
358	1	SO ₂	4-Me-Phenyl
359	2	SO ₂	4-Me-Phenyl
360	3	SO ₂	4-Me-Phenyl
361	1	O	3-F-Phenyl
362	2	O	3-F-Phenyl
363	3	O	3-F-Phenyl
364	1	S	3-F-Phenyl
365	2	S	3-F-Phenyl
366	3	S	3-F-Phenyl
367	1	SO	3-F-Phenyl
368	2	SO	3-F-Phenyl
369	3	SO	3-F-Phenyl
370	1	SO ₂	3-F-Phenyl
371	2	SO ₂	3-F-Phenyl
372	3	SO ₂	3-F-Phenyl
373	1	O	2-Cl-Phenyl
374	2	O	2-Cl-Phenyl
375	3	O	2-Cl-Phenyl
376	1	S	2-Cl-Phenyl
377	2	S	2-Cl-Phenyl
378	3	S	2-Cl-Phenyl
379	1	SO	2-Cl-Phenyl
380	2	SO	2-Cl-Phenyl
381	3	SO	2-Cl-Phenyl
382	1	SO ₂	2-Cl-Phenyl
383	2	SO ₂	2-Cl-Phenyl
384	3	SO ₂	2-Cl-Phenyl
385	1	O	4-Cl-Phenyl
386	2	O	4-Cl-Phenyl
387	3	O	4-Cl-Phenyl
388	1	S	4-Cl-Phenyl
389	2	S	4-Cl-Phenyl
390	3	S	4-Cl-Phenyl
391	1	SO	4-Cl-Phenyl
392	2	SO	4-Cl-Phenyl
393	3	SO	4-Cl-Phenyl
394	1	SO ₂	4-Cl-Phenyl
395	2	SO ₂	4-Cl-Phenyl
396	3	SO ₂	4-Cl-Phenyl
397	1	O	3-Br-Phenyl
398	2	O	3-Br-Phenyl
399	3	O	3-Br-Phenyl
400	1	S	3-Br-Phenyl
401	2	S	3-Br-Phenyl
402	3	S	3-Br-Phenyl
403	1	SO	3-Br-Phenyl
404	2	SO	3-Br-Phenyl
405	3	SO	3-Br-Phenyl
406	1	SO ₂	3-Br-Phenyl
407	2	SO ₂	3-Br-Phenyl
408	3	SO ₂	3-Br-Phenyl
409	1	O	2-CF ₃ -Phenyl
410	2	O	2-CF ₃ -Phenyl
411	3	O	2-CF ₃ -Phenyl
412	1	S	2-CF ₃ -Phenyl
413	2	S	2-CF ₃ -Phenyl
414	3	S	2-CF ₃ -Phenyl

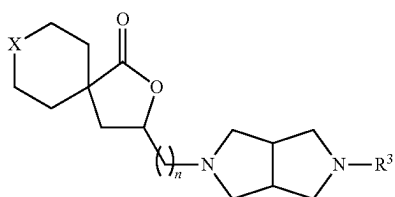
TABLE 7-continued

Entry	n	X	R ³
415	1	SO	2-CF ₃ -Phenyl
416	2	SO	2-CF ₃ -Phenyl
417	3	SO	2-CF ₃ -Phenyl
418	1	SO ₂	2-CF ₃ -Phenyl
419	2	SO ₂	2-CF ₃ -Phenyl
420	3	SO ₂	2-CF ₃ -Phenyl
421	1	O	4-CF ₃ -Phenyl
422	2	O	4-CF ₃ -Phenyl
423	3	O	4-CF ₃ -Phenyl
424	1	S	4-CF ₃ -Phenyl
425	2	S	4-CF ₃ -Phenyl
426	3	S	4-CF ₃ -Phenyl
427	1	SO	4-CF ₃ -Phenyl
428	2	SO	4-CF ₃ -Phenyl
429	3	SO	4-CF ₃ -Phenyl
430	1	SO ₂	4-CF ₃ -Phenyl
431	2	SO ₂	4-CF ₃ -Phenyl
432	3	SO ₂	4-CF ₃ -Phenyl
433	1	O	3-iPr-Phenyl
434	2	O	3-iPr-Phenyl
435	3	O	3-iPr-Phenyl
436	1	S	3-iPr-Phenyl
437	2	S	3-iPr-Phenyl
438	3	S	3-iPr-Phenyl
439	1	SO	3-iPr-Phenyl
440	2	SO	3-iPr-Phenyl
441	3	SO	3-iPr-Phenyl
442	1	SO ₂	3-iPr-Phenyl
443	2	SO ₂	3-iPr-Phenyl
444	3	SO ₂	3-iPr-Phenyl
445	1	O	4-NH ₂ -Phenyl
446	2	O	4-NH ₂ -Phenyl
447	3	O	4-NH ₂ -Phenyl
448	1	S	4-NH ₂ -Phenyl
449	2	S	4-NH ₂ -Phenyl
450	3	S	4-NH ₂ -Phenyl
451	1	SO	4-NH ₂ -Phenyl
452	2	SO	4-NH ₂ -Phenyl
453	3	SO	4-NH ₂ -Phenyl
454	1	SO ₂	4-NH ₂ -Phenyl
455	2	SO ₂	4-NH ₂ -Phenyl
456	3	SO ₂	4-NH ₂ -Phenyl
457	1	O	2-NH ₂ -Phenyl
458	2	O	2-NH ₂ -Phenyl
459	3	O	2-NH ₂ -Phenyl
460	1	S	2-NH ₂ -Phenyl
461	2	S	2-NH ₂ -Phenyl
462	3	S	2-NH ₂ -Phenyl
463	1	SO	2-NH ₂ -Phenyl
464	2	SO	2-NH ₂ -Phenyl
465	3	SO	2-NH ₂ -Phenyl
466	1	SO ₂	2-NH ₂ -Phenyl
467	2	SO ₂	2-NH ₂ -Phenyl
468	3	SO ₂	2-NH ₂ -Phenyl
469	1	O	2,6-di-Me-Phenyl
470	2	O	2,6-di-Me-Phenyl
471	3	O	2,6-di-Me-Phenyl
472	1	S	2,6-di-Me-Phenyl
473	2	S	2,6-di-Me-Phenyl
474	3	S	2,6-di-Me-Phenyl
475	1	SO	2,6-di-Me-Phenyl
476	2	SO	2,6-di-Me-Phenyl
477	3	SO	2,6-di-Me-Phenyl
478	1	SO ₂	2,6-di-Me-Phenyl
479	2	SO ₂	2,6-di-Me-Phenyl
480	3	SO ₂	2,6-di-Me-Phenyl
481	1	O	2-Ph-Phenyl
482	2	O	2-Ph-Phenyl
483	3	O	2-Ph-Phenyl
484	1	S	2-Ph-Phenyl
485	2	S	2-Ph-Phenyl
486	3	S	2-Ph-Phenyl
487	1	SO	2-Ph-Phenyl
488	2	SO	2-Ph-Phenyl
489	3	SO	2-Ph-Phenyl

TABLE 7-continued

Entry	n	X	R ³
490	1	SO ₂	2-Ph-Phenyl
491	2	SO ₂	2-Ph-Phenyl
492	3	SO ₂	2-Ph-Phenyl
493	1	O	4-Ph-Phenyl
494	2	O	4-Ph-Phenyl
495	3	O	4-Ph-Phenyl
496	1	S	4-Ph-Phenyl
497	2	S	4-Ph-Phenyl
498	3	S	4-Ph-Phenyl
499	1	SO	4-Ph-Phenyl
500	2	SO	4-Ph-Phenyl
501	3	SO	4-Ph-Phenyl
502	1	SO ₂	4-Ph-Phenyl
503	2	SO ₂	4-Ph-Phenyl
504	3	SO ₂	4-Ph-Phenyl
505	1	O	3-morpholino-phenyl
506	2	O	3-morpholino-phenyl
507	3	O	3-morpholino-phenyl
508	1	S	3-morpholino-phenyl
509	2	S	3-morpholino-phenyl
510	3	S	3-morpholino-phenyl
511	1	SO	3-morpholino-phenyl
512	2	SO	3-morpholino-phenyl
513	3	SO	3-morpholino-phenyl
514	1	SO ₂	3-morpholino-phenyl
515	2	SO ₂	3-morpholino-phenyl
516	3	SO ₂	3-morpholino-phenyl
517	1	O	2-pyrazinyl
518	2	O	2-pyrazinyl
519	3	O	2-pyrazinyl
520	1	S	2-pyrazinyl
521	2	S	2-pyrazinyl
522	3	S	2-pyrazinyl
523	1	SO	2-pyrazinyl
524	2	SO	2-pyrazinyl
525	3	SO	2-pyrazinyl
526	1	SO ₂	2-pyrazinyl
527	2	SO ₂	2-pyrazinyl
528	3	SO ₂	2-pyrazinyl
529	1	O	5-indolyl
530	2	O	5-indolyl
531	3	O	5-indolyl
532	1	S	5-indolyl
533	2	S	5-indolyl
534	3	S	5-indolyl
535	1	SO	5-indolyl
536	2	SO	5-indolyl
537	3	SO	5-indolyl
538	1	SO ₂	5-indolyl
539	2	SO ₂	5-indolyl
540	3	SO ₂	5-indolyl
541	1	O	1H-benzo[d]imidazol-4-yl
542	2	O	1H-benzo[d]imidazol-4-yl
543	3	O	1H-benzo[d]imidazol-4-yl
544	1	S	1H-benzo[d]imidazol-4-yl
545	2	S	1H-benzo[d]imidazol-4-yl
546	3	S	1H-benzo[d]imidazol-4-yl
547	1	SO	1H-benzo[d]imidazol-4-yl
548	2	SO	1H-benzo[d]imidazol-4-yl
549	3	SO	1H-benzo[d]imidazol-4-yl
550	1	SO ₂	1H-benzo[d]imidazol-4-yl
551	2	SO ₂	1H-benzo[d]imidazol-4-yl
552	3	SO ₂	1H-benzo[d]imidazol-4-yl

[0407] Exemplary embodiments include compounds having the formula (XXIV)



or a pharmaceutically acceptable salt form thereof defined herein below in Table 8.

TABLE 8

Entry	n	X	R ³
1	1	O	Phenyl
2	2	O	Phenyl
3	3	O	Phenyl
4	1	S	Phenyl
5	2	S	Phenyl
6	3	S	Phenyl
7	1	SO	Phenyl
8	2	SO	Phenyl
9	3	SO	Phenyl
10	1	SO ₂	Phenyl
11	2	SO ₂	Phenyl
12	3	SO ₂	Phenyl
13	1	O	3-OH-Phenyl
14	2	O	3-OH-Phenyl
15	3	O	3-OH-Phenyl
16	1	S	3-OH-Phenyl
17	2	S	3-OH-Phenyl
18	3	S	3-OH-Phenyl
19	1	SO	3-OH-Phenyl
20	2	SO	3-OH-Phenyl
21	3	SO	3-OH-Phenyl
22	1	SO ₂	3-OH-Phenyl
23	2	SO ₂	3-OH-Phenyl
24	3	SO ₂	3-OH-Phenyl
25	1	O	4-NO ₂ -Phenyl
26	2	O	4-NO ₂ -Phenyl
27	3	O	4-NO ₂ -Phenyl
28	1	S	4-NO ₂ -Phenyl
29	2	S	4-NO ₂ -Phenyl
30	3	S	4-NO ₂ -Phenyl
31	1	SO	4-NO ₂ -Phenyl
32	2	SO	4-NO ₂ -Phenyl
33	3	SO	4-NO ₂ -Phenyl
34	1	SO ₂	4-NO ₂ -Phenyl
35	2	SO ₂	4-NO ₂ -Phenyl
36	3	SO ₂	4-NO ₂ -Phenyl
37	1	O	3-OMe-Phenyl
38	2	O	3-OMe-Phenyl
39	3	O	3-OMe-Phenyl
40	1	S	3-OMe-Phenyl
41	2	S	3-OMe-Phenyl
42	3	S	3-OMe-Phenyl
43	1	SO	3-OMe-Phenyl
44	2	SO	3-OMe-Phenyl
45	3	SO	3-OMe-Phenyl
46	1	SO ₂	3-OMe-Phenyl
47	2	SO ₂	3-OMe-Phenyl
48	3	SO ₂	3-OMe-Phenyl
49	1	O	4-CN-Phenyl
50	2	O	4-CN-Phenyl
51	3	O	4-CN-Phenyl
52	1	S	4-CN-Phenyl
53	2	S	4-CN-Phenyl
54	3	S	4-CN-Phenyl
55	1	SO	4-CN-Phenyl
56	2	SO	4-CN-Phenyl
57	3	SO	4-CN-Phenyl
58	1	SO ₂	4-CN-Phenyl

TABLE 8-continued

Entry	n	X	R ³
59	2	SO ₂	4-CN-Phenyl
60	3	SO ₂	4-CN-Phenyl
61	1	O	2-CN-Phenyl
62	2	O	2-CN-Phenyl
63	3	O	2-CN-Phenyl
64	1	S	2-CN-Phenyl
65	2	S	2-CN-Phenyl
66	3	S	2-CN-Phenyl
67	1	SO	2-CN-Phenyl
68	2	SO	2-CN-Phenyl
69	3	SO	2-CN-Phenyl
70	1	SO ₂	2-CN-Phenyl
71	2	SO ₂	2-CN-Phenyl
72	3	SO ₂	2-CN-Phenyl
73	1	O	3-Me-Phenyl
74	2	O	3-Me-Phenyl
75	3	O	3-Me-Phenyl
76	1	S	3-Me-Phenyl
77	2	S	3-Me-Phenyl
78	3	S	3-Me-Phenyl
79	1	SO	3-Me-Phenyl
80	2	SO	3-Me-Phenyl
81	3	SO	3-Me-Phenyl
82	1	SO ₂	3-Me-Phenyl
83	2	SO ₂	3-Me-Phenyl
84	3	SO ₂	3-Me-Phenyl
85	1	O	2-F-Phenyl
86	2	O	2-F-Phenyl
87	3	O	2-F-Phenyl
88	1	S	2-F-Phenyl
89	2	S	2-F-Phenyl
90	3	S	2-F-Phenyl
91	1	SO	2-F-Phenyl
92	2	SO	2-F-Phenyl
93	3	SO	2-F-Phenyl
94	1	SO ₂	2-F-Phenyl
95	2	SO ₂	2-F-Phenyl
96	3	SO ₂	2-F-Phenyl
97	1	O	4-F-Phenyl
98	2	O	4-F-Phenyl
99	3	O	4-F-Phenyl
100	1	S	4-F-Phenyl
101	2	S	4-F-Phenyl
102	3	S	4-F-Phenyl
103	1	SO	4-F-Phenyl
104	2	SO	4-F-Phenyl
105	3	SO	4-F-Phenyl
106	1	SO ₂	4-F-Phenyl
107	2	SO ₂	4-F-Phenyl
108	3	SO ₂	4-F-Phenyl
109	1	O	3-Cl-Phenyl
110	2	O	3-Cl-Phenyl
111	3	O	3-Cl-Phenyl
112	1	S	3-Cl-Phenyl
113	2	S	3-Cl-Phenyl
114	3	S	3-Cl-Phenyl
115	1	SO	3-Cl-Phenyl
116	2	SO	3-Cl-Phenyl
117	3	SO	3-Cl-Phenyl
118	1	SO ₂	3-Cl-Phenyl
119	2	SO ₂	3-Cl-Phenyl
120	3	SO ₂	3-Cl-Phenyl
121	1	O	2-Br-Phenyl
122	2	O	2-Br-Phenyl
123	3	O	2-Br-Phenyl
124	1	S	2-Br-Phenyl
125	2	S	2-Br-Phenyl
126	3	S	2-Br-Phenyl
127	1	SO	2-Br-Phenyl
128	2	SO	2-Br-Phenyl
129	3	SO	2-Br-Phenyl
130	1	SO ₂	2-Br-Phenyl
131	2	SO ₂	2-Br-Phenyl
132	3	SO ₂	2-Br-Phenyl
133	1	O	4-Br-Phenyl

TABLE 8-continued

Entry	n	X	R ³
134	2	O	4-Br-Phenyl
135	3	O	4-Br-Phenyl
136	1	S	4-Br-Phenyl
137	2	S	4-Br-Phenyl
138	3	S	4-Br-Phenyl
139	1	SO	4-Br-Phenyl
140	2	SO	4-Br-Phenyl
141	3	SO	4-Br-Phenyl
142	1	SO ₂	4-Br-Phenyl
143	2	SO ₂	4-Br-Phenyl
144	3	SO ₂	4-Br-Phenyl
145	1	O	3-CF ₃ -Phenyl
146	2	O	3-CF ₃ -Phenyl
147	3	O	3-CF ₃ -Phenyl
148	1	S	3-CF ₃ -Phenyl
149	2	S	3-CF ₃ -Phenyl
150	3	S	3-CF ₃ -Phenyl
151	1	SO	3-CF ₃ -Phenyl
152	2	SO	3-CF ₃ -Phenyl
153	3	SO	3-CF ₃ -Phenyl
154	1	SO ₂	3-CF ₃ -Phenyl
155	2	SO ₂	3-CF ₃ -Phenyl
156	3	SO ₂	3-CF ₃ -Phenyl
157	1	O	2-iPr-Phenyl
158	2	O	2-iPr-Phenyl
159	3	O	2-iPr-Phenyl
160	1	S	2-iPr-Phenyl
161	2	S	2-iPr-Phenyl
162	3	S	2-iPr-Phenyl
163	1	SO	2-iPr-Phenyl
164	2	SO	2-iPr-Phenyl
165	3	SO	2-iPr-Phenyl
166	1	SO ₂	2-iPr-Phenyl
167	2	SO ₂	2-iPr-Phenyl
168	3	SO ₂	2-iPr-Phenyl
169	1	O	4-iPr-Phenyl
170	2	O	4-iPr-Phenyl
171	3	O	4-iPr-Phenyl
172	1	S	4-iPr-Phenyl
173	2	S	4-iPr-Phenyl
174	3	S	4-iPr-Phenyl
175	1	SO	4-iPr-Phenyl
176	2	SO	4-iPr-Phenyl
177	3	SO	4-iPr-Phenyl
178	1	SO ₂	4-iPr-Phenyl
179	2	SO ₂	4-iPr-Phenyl
180	3	SO ₂	4-iPr-Phenyl
181	1	O	3-NH ₂ -Phenyl
182	2	O	3-NH ₂ -Phenyl
183	3	O	3-NH ₂ -Phenyl
184	1	S	3-NH ₂ -Phenyl
185	2	S	3-NH ₂ -Phenyl
186	3	S	3-NH ₂ -Phenyl
187	1	SO	3-NH ₂ -Phenyl
188	2	SO	3-NH ₂ -Phenyl
189	3	SO	3-NH ₂ -Phenyl
190	1	SO ₂	3-NH ₂ -Phenyl
191	2	SO ₂	3-NH ₂ -Phenyl
192	3	SO ₂	3-NH ₂ -Phenyl
193	1	O	2,4-di-Me-Phenyl
194	2	O	2,4-di-Me-Phenyl
195	3	O	2,4-di-Me-Phenyl
196	1	S	2,4-di-Me-Phenyl
197	2	S	2,4-di-Me-Phenyl
198	3	S	2,4-di-Me-Phenyl
199	1	SO	2,4-di-Me-Phenyl
200	2	SO	2,4-di-Me-Phenyl
201	3	SO	2,4-di-Me-Phenyl
202	1	SO ₂	2,4-di-Me-Phenyl
203	2	SO ₂	2,4-di-Me-Phenyl
204	3	SO ₂	2,4-di-Me-Phenyl
205	1	O	2,6-di-iPr-Phenyl
206	2	O	2,6-di-iPr-Phenyl
207	3	O	2,6-di-iPr-Phenyl
208	1	S	2,6-di-iPr-Phenyl

TABLE 8-continued

Entry	n	X	R ³
209	2	S	2,6-di-iPr-Phenyl
210	3	S	2,6-di-iPr-Phenyl
211	1	SO	2,6-di-iPr-Phenyl
212	2	SO	2,6-di-iPr-Phenyl
213	3	SO	2,6-di-iPr-Phenyl
214	1	SO ₂	2,6-di-iPr-Phenyl
215	2	SO ₂	2,6-di-iPr-Phenyl
216	3	SO ₂	2,6-di-iPr-Phenyl
217	1	O	3-Ph-Phenyl
218	2	O	3-Ph-Phenyl
219	3	O	3-Ph-Phenyl
220	1	S	3-Ph-Phenyl
221	2	S	3-Ph-Phenyl
222	3	S	3-Ph-Phenyl
223	1	SO	3-Ph-Phenyl
224	2	SO	3-Ph-Phenyl
225	3	SO	3-Ph-Phenyl
226	1	SO ₂	3-Ph-Phenyl
227	2	SO ₂	3-Ph-Phenyl
228	3	SO ₂	3-Ph-Phenyl
229	1	O	2-morpholino-phenyl
230	2	O	2-morpholino-phenyl
231	3	O	2-morpholino-phenyl
232	1	S	2-morpholino-phenyl
233	2	S	2-morpholino-phenyl
234	3	S	2-morpholino-phenyl
235	1	SO	2-morpholino-phenyl
236	2	SO	2-morpholino-phenyl
237	3	SO	2-morpholino-phenyl
238	1	SO ₂	2-morpholino-phenyl
239	2	SO ₂	2-morpholino-phenyl
240	3	SO ₂	2-morpholino-phenyl
241	1	O	4-morpholino-phenyl
242	2	O	4-morpholino-phenyl
243	3	O	4-morpholino-phenyl
244	1	S	4-morpholino-phenyl
245	2	S	4-morpholino-phenyl
246	3	S	4-morpholino-phenyl
247	1	SO	4-morpholino-phenyl
248	2	SO	4-morpholino-phenyl
249	3	SO	4-morpholino-phenyl
250	1	SO ₂	4-morpholino-phenyl
251	2	SO ₂	4-morpholino-phenyl
252	3	SO ₂	4-morpholino-phenyl
253	1	O	2-pyrimidinyl
254	2	O	2-pyrimidinyl
255	3	O	2-pyrimidinyl
256	1	S	2-pyrimidinyl
257	2	S	2-pyrimidinyl
258	3	S	2-pyrimidinyl
259	1	SO	2-pyrimidinyl
260	2	SO	2-pyrimidinyl
261	3	SO	2-pyrimidinyl
262	1	SO ₂	2-pyrimidinyl
263	2	SO ₂	2-pyrimidinyl
264	3	SO ₂	2-pyrimidinyl
265	1	O	2-methyl-1H-benzo[d]imidazol-4-yl
266	2	O	2-methyl-1H-benzo[d]imidazol-4-yl
267	3	O	2-methyl-1H-benzo[d]imidazol-4-yl
268	1	S	2-methyl-1H-benzo[d]imidazol-4-yl
269	2	S	2-methyl-1H-benzo[d]imidazol-4-yl
270	3	S	2-methyl-1H-benzo[d]imidazol-4-yl
271	1	SO	2-methyl-1H-benzo[d]imidazol-4-yl
272	2	SO	2-methyl-1H-benzo[d]imidazol-4-yl
273	3	SO	2-methyl-1H-benzo[d]imidazol-4-yl

TABLE 8-continued

Entry	n	X	R ³
274	1	SO ₂	2-methyl-1H-benzof[d]imidazol-4-yl
275	2	SO ₂	2-methyl-1H-benzof[d]imidazol-4-yl
276	3	SO ₂	2-methyl-1H-benzof[d]imidazol-4-yl
277	1	O	4-OH-Phenyl
278	2	O	4-OH-Phenyl
279	3	O	4-OH-Phenyl
280	1	S	4-OH-Phenyl
281	2	S	4-OH-Phenyl
282	3	S	4-OH-Phenyl
283	1	SO	4-OH-Phenyl
284	2	SO	4-OH-Phenyl
285	3	SO	4-OH-Phenyl
286	1	SO ₂	4-OH-Phenyl
287	2	SO ₂	4-OH-Phenyl
288	3	SO ₂	4-OH-Phenyl
289	1	O	2-OH-Phenyl
290	2	O	2-OH-Phenyl
291	3	O	2-OH-Phenyl
292	1	S	2-OH-Phenyl
293	2	S	2-OH-Phenyl
294	3	S	2-OH-Phenyl
295	1	SO	2-OH-Phenyl
296	2	SO	2-OH-Phenyl
297	3	SO	2-OH-Phenyl
298	1	SO ₂	2-OH-Phenyl
299	2	SO ₂	2-OH-Phenyl
300	3	SO ₂	2-OH-Phenyl
301	1	O	4-OMe-Phenyl
302	2	O	4-OMe-Phenyl
303	3	O	4-OMe-Phenyl
304	1	S	4-OMe-Phenyl
305	2	S	4-OMe-Phenyl
306	3	S	4-OMe-Phenyl
307	1	SO	4-OMe-Phenyl
308	2	SO	4-OMe-Phenyl
309	3	SO	4-OMe-Phenyl
310	1	SO ₂	4-OMe-Phenyl
311	2	SO ₂	4-OMe-Phenyl
312	3	SO ₂	4-OMe-Phenyl
313	1	O	2-OMe-Phenyl
314	2	O	2-OMe-Phenyl
315	3	O	2-OMe-Phenyl
316	1	S	2-OMe-Phenyl
317	2	S	2-OMe-Phenyl
318	3	S	2-OMe-Phenyl
319	1	SO	2-OMe-Phenyl
320	2	SO	2-OMe-Phenyl
321	3	SO	2-OMe-Phenyl
322	1	SO ₂	2-OMe-Phenyl
323	2	SO ₂	2-OMe-Phenyl
324	3	SO ₂	2-OMe-Phenyl
325	1	O	3-CN-Phenyl
326	2	O	3-CN-Phenyl
327	3	O	3-CN-Phenyl
328	1	S	3-CN-Phenyl
329	2	S	3-CN-Phenyl
330	3	S	3-CN-Phenyl
331	1	SO	3-CN-Phenyl
332	2	SO	3-CN-Phenyl
333	3	SO	3-CN-Phenyl
334	1	SO ₂	3-CN-Phenyl
335	2	SO ₂	3-CN-Phenyl
336	3	SO ₂	3-CN-Phenyl
337	1	O	2-Me-Phenyl
338	2	O	2-Me-Phenyl
339	3	O	2-Me-Phenyl
340	1	S	2-Me-Phenyl
341	2	S	2-Me-Phenyl
342	3	S	2-Me-Phenyl
343	1	SO	2-Me-Phenyl
344	2	SO	2-Me-Phenyl
345	3	SO	2-Me-Phenyl

TABLE 8-continued

Entry	n	X	R ³
346	1	SO ₂	2-Me-Phenyl
347	2	SO ₂	2-Me-Phenyl
348	3	SO ₂	2-Me-Phenyl
349	1	O	4-Me-Phenyl
350	2	O	4-Me-Phenyl
351	3	O	4-Me-Phenyl
352	1	S	4-Me-Phenyl
353	2	S	4-Me-Phenyl
354	3	S	4-Me-Phenyl
355	1	SO	4-Me-Phenyl
356	2	SO	4-Me-Phenyl
357	3	SO	4-Me-Phenyl
358	1	SO ₂	4-Me-Phenyl
359	2	SO ₂	4-Me-Phenyl
360	3	SO ₂	4-Me-Phenyl
361	1	O	3-F-Phenyl
362	2	O	3-F-Phenyl
363	3	O	3-F-Phenyl
364	1	S	3-F-Phenyl
365	2	S	3-F-Phenyl
366	3	S	3-F-Phenyl
367	1	SO	3-F-Phenyl
368	2	SO	3-F-Phenyl
369	3	SO	3-F-Phenyl
370	1	SO ₂	3-F-Phenyl
371	2	SO ₂	3-F-Phenyl
372	3	SO ₂	3-F-Phenyl
373	1	O	2-Cl-Phenyl
374	2	O	2-Cl-Phenyl
375	3	O	2-Cl-Phenyl
376	1	S	2-Cl-Phenyl
377	2	S	2-Cl-Phenyl
378	3	S	2-Cl-Phenyl
379	1	SO	2-Cl-Phenyl
380	2	SO	2-Cl-Phenyl
381	3	SO	2-Cl-Phenyl
382	1	SO ₂	2-Cl-Phenyl
383	2	SO ₂	2-Cl-Phenyl
384	3	SO ₂	2-Cl-Phenyl
385	1	O	4-Cl-Phenyl
386	2	O	4-Cl-Phenyl
387	3	O	4-Cl-Phenyl
388	1	S	4-Cl-Phenyl
389	2	S	4-Cl-Phenyl
390	3	S	4-Cl-Phenyl
391	1	SO	4-Cl-Phenyl
392	2	SO	4-Cl-Phenyl
393	3	SO	4-Cl-Phenyl
394	1	SO ₂	4-Cl-Phenyl
395	2	SO ₂	4-Cl-Phenyl
396	3	SO ₂	4-Cl-Phenyl
397	1	O	3-Br-Phenyl
398	2	O	3-Br-Phenyl
399	3	O	3-Br-Phenyl
400	1	S	3-Br-Phenyl
401	2	S	3-Br-Phenyl
402	3	S	3-Br-Phenyl
403	1	SO	3-Br-Phenyl
404	2	SO	3-Br-Phenyl
405	3	SO	3-Br-Phenyl
406	1	SO ₂	3-Br-Phenyl
407	2	SO ₂	3-Br-Phenyl
408	3	SO ₂	3-Br-Phenyl
409	1	O	2-CF ₃ -Phenyl
410	2	O	2-CF ₃ -Phenyl
411	3	O	2-CF ₃ -Phenyl
412	1	S	2-CF ₃ -Phenyl
413	2	S	2-CF ₃ -Phenyl
414	3	S	2-CF ₃ -Phenyl
415	1	SO	2-CF ₃ -Phenyl
416	2	SO	2-CF ₃ -Phenyl
417	3	SO	2-CF ₃ -Phenyl
418	1	SO ₂	2-CF ₃ -Phenyl
419	2	SO ₂	2-CF ₃ -Phenyl
420	3	SO ₂	2-CF ₃ -Phenyl

TABLE 8-continued

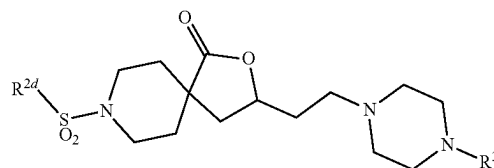
Entry	n	X	R ³
421	1	O	4-CF ₃ -Phenyl
422	2	O	4-CF ₃ -Phenyl
423	3	O	4-CF ₃ -Phenyl
424	1	S	4-CF ₃ -Phenyl
425	2	S	4-CF ₃ -Phenyl
426	3	S	4-CF ₃ -Phenyl
427	1	SO	4-CF ₃ -Phenyl
428	2	SO	4-CF ₃ -Phenyl
429	3	SO	4-CF ₃ -Phenyl
430	1	SO ₂	4-CF ₃ -Phenyl
431	2	SO ₂	4-CF ₃ -Phenyl
432	3	SO ₂	4-CF ₃ -Phenyl
433	1	O	3-iPr-Phenyl
434	2	O	3-iPr-Phenyl
435	3	O	3-iPr-Phenyl
436	1	S	3-iPr-Phenyl
437	2	S	3-iPr-Phenyl
438	3	S	3-iPr-Phenyl
439	1	SO	3-iPr-Phenyl
440	2	SO	3-iPr-Phenyl
441	3	SO	3-iPr-Phenyl
442	1	SO ₂	3-iPr-Phenyl
443	2	SO ₂	3-iPr-Phenyl
444	3	SO ₂	3-iPr-Phenyl
445	1	O	4-NH ₂ -Phenyl
446	2	O	4-NH ₂ -Phenyl
447	3	O	4-NH ₂ -Phenyl
448	1	S	4-NH ₂ -Phenyl
449	2	S	4-NH ₂ -Phenyl
450	3	S	4-NH ₂ -Phenyl
451	1	SO	4-NH ₂ -Phenyl
452	2	SO	4-NH ₂ -Phenyl
453	3	SO	4-NH ₂ -Phenyl
454	1	SO ₂	4-NH ₂ -Phenyl
455	2	SO ₂	4-NH ₂ -Phenyl
456	3	SO ₂	4-NH ₂ -Phenyl
457	1	O	2-NH ₂ -Phenyl
458	2	O	2-NH ₂ -Phenyl
459	3	O	2-NH ₂ -Phenyl
460	1	S	2-NH ₂ -Phenyl
461	2	S	2-NH ₂ -Phenyl
462	3	S	2-NH ₂ -Phenyl
463	1	SO	2-NH ₂ -Phenyl
464	2	SO	2-NH ₂ -Phenyl
465	3	SO	2-NH ₂ -Phenyl
466	1	SO ₂	2-NH ₂ -Phenyl
467	2	SO ₂	2-NH ₂ -Phenyl
468	3	SO ₂	2-NH ₂ -Phenyl
469	1	O	2,6-di-Me-Phenyl
470	2	O	2,6-di-Me-Phenyl
471	3	O	2,6-di-Me-Phenyl
472	1	S	2,6-di-Me-Phenyl
473	2	S	2,6-di-Me-Phenyl
474	3	S	2,6-di-Me-Phenyl
475	1	SO	2,6-di-Me-Phenyl
476	2	SO	2,6-di-Me-Phenyl
477	3	SO	2,6-di-Me-Phenyl
478	1	SO ₂	2,6-di-Me-Phenyl
479	2	SO ₂	2,6-di-Me-Phenyl
480	3	SO ₂	2,6-di-Me-Phenyl
481	1	O	2-Ph-Phenyl
482	2	O	2-Ph-Phenyl
483	3	O	2-Ph-Phenyl
484	1	S	2-Ph-Phenyl
485	2	S	2-Ph-Phenyl
486	3	S	2-Ph-Phenyl
487	1	SO	2-Ph-Phenyl
488	2	SO	2-Ph-Phenyl
489	3	SO	2-Ph-Phenyl
490	1	SO ₂	2-Ph-Phenyl
491	2	SO ₂	2-Ph-Phenyl
492	3	SO ₂	2-Ph-Phenyl
493	1	O	4-Ph-Phenyl
494	2	O	4-Ph-Phenyl
495	3	O	4-Ph-Phenyl

TABLE 8-continued

Entry	n	X	R ³
496	1	S	4-Ph-Phenyl
497	2	S	4-Ph-Phenyl
498	3	S	4-Ph-Phenyl
499	1	SO	4-Ph-Phenyl
500	2	SO	4-Ph-Phenyl
501	3	SO	4-Ph-Phenyl
502	1	SO ₂	4-Ph-Phenyl
503	2	SO ₂	4-Ph-Phenyl
504	3	SO ₂	4-Ph-Phenyl
505	1	O	3-morpholino-phenyl
506	2	O	3-morpholino-phenyl
507	3	O	3-morpholino-phenyl
508	1	S	3-morpholino-phenyl
509	2	S	3-morpholino-phenyl
510	3	S	3-morpholino-phenyl
511	1	SO	3-morpholino-phenyl
512	2	SO	3-morpholino-phenyl
513	3	SO	3-morpholino-phenyl
514	1	SO ₂	3-morpholino-phenyl
515	2	SO ₂	3-morpholino-phenyl
516	3	SO ₂	3-morpholino-phenyl
517	1	O	2-pyrazinyl
518	2	O	2-pyrazinyl
519	3	O	2-pyrazinyl
520	1	S	2-pyrazinyl
521	2	S	2-pyrazinyl
522	3	S	2-pyrazinyl
523	1	SO	2-pyrazinyl
524	2	SO	2-pyrazinyl
525	3	SO	2-pyrazinyl
526	1	SO ₂	2-pyrazinyl
527	2	SO ₂	2-pyrazinyl
528	3	SO ₂	2-pyrazinyl
529	1	O	5-indolyl
530	2	O	5-indolyl
531	3	O	5-indolyl
532	1	S	5-indolyl
533	2	S	5-indolyl
534	3	S	5-indolyl
535	1	SO	5-indolyl
536	2	SO	5-indolyl
537	3	SO	5-indolyl
538	1	SO ₂	5-indolyl
539	2	SO ₂	5-indolyl
540	3	SO ₂	5-indolyl
541	1	O	1H-benzo[d]imidazol-4-yl
542	2	O	1H-benzo[d]imidazol-4-yl
543	3	O	1H-benzo[d]imidazol-4-yl
544	1	S	1H-benzo[d]imidazol-4-yl
545	2	S	1H-benzo[d]imidazol-4-yl
546	3	S	1H-benzo[d]imidazol-4-yl
547	1	SO	1H-benzo[d]imidazol-4-yl
548	2	SO	1H-benzo[d]imidazol-4-yl
549	3	SO	1H-benzo[d]imidazol-4-yl
550	1	SO ₂	1H-benzo[d]imidazol-4-yl
551	2	SO ₂	1H-benzo[d]imidazol-4-yl
552	3	SO ₂	1H-benzo[d]imidazol-4-yl

[0408] Exemplary embodiments include compounds having the formula (XXV)

(XXV)



or a pharmaceutically acceptable salt form thereof defined herein below in Table 9

TABLE 9

Entry	R ^{2d}	R ³
1	ethyl	4-CH ₃ -phenyl
2	n-propyl	4-CH ₃ -phenyl
3	isopropyl	4-CH ₃ -phenyl
4	—CH ₂ CH(CH ₃) ₂	4-CH ₃ -phenyl
5	CF ₃	4-CH ₃ -phenyl
6	—CH ₂ CF ₃	4-CH ₃ -phenyl
7	—CH ₂ CH ₂ CF ₃	4-CH ₃ -phenyl
8	cyclopropyl	4-CH ₃ -phenyl
9	Cyclobutyl	4-CH ₃ -phenyl
10	cyclopentyl	4-CH ₃ -phenyl
11	cyclohexyl	4-CH ₃ -phenyl
12	3-pyridyl	4-CH ₃ -phenyl
13	1-methyl-1H-pyrazol-4-yl	4-CH ₃ -phenyl
14	1H-imidazol-4-yl	4-CH ₃ -phenyl
15	2-furanyl	4-CH ₃ -phenyl
16	ethyl	3-CH ₃ -phenyl
17	n-propyl	3-CH ₃ -phenyl
18	isopropyl	3-CH ₃ -phenyl
19	—CH ₂ CH(CH ₃) ₂	3-CH ₃ -phenyl
20	—CF ₃	3-CH ₃ -phenyl
21	—CH ₂ CF ₃	3-CH ₃ -phenyl
22	—CH ₂ CH ₂ CF ₃	3-CH ₃ -phenyl
23	cyclopropyl	3-CH ₃ -phenyl
24	Cyclobutyl	3-CH ₃ -phenyl
25	cyclopentyl	3-CH ₃ -phenyl
26	cyclohexyl	3-CH ₃ -phenyl
27	3-pyridyl	3-CH ₃ -phenyl
28	1-methyl-1H-pyrazol-4-yl	3-CH ₃ -phenyl
29	1H-imidazol-4-yl	3-CH ₃ -phenyl
30	2-furanyl	3-CH ₃ -phenyl
31	ethyl	3-OH-Phenyl
32	n-propyl	3-OH-Phenyl
33	isopropyl	3-OH-Phenyl
34	—CH ₂ CH(CH ₃) ₂	3-OH-Phenyl
35	—CF ₃	3-OH-Phenyl
36	—CH ₂ CF ₃	3-OH-Phenyl
37	—CH ₂ CH ₂ CF ₃	3-OH-Phenyl
38	cyclopropyl	3-OH-Phenyl
39	Cyclobutyl	3-OH-Phenyl
40	cyclopentyl	3-OH-Phenyl
41	cyclohexyl	3-OH-Phenyl
42	3-pyridyl	3-OH-Phenyl
43	1-methyl-1H-pyrazol-4-yl	3-OH-Phenyl
44	1H-imidazol-4-yl	3-OH-Phenyl
45	2-furanyl	3-OH-Phenyl
46	ethyl	4-OMe-Phenyl
47	n-propyl	4-OMe-Phenyl
48	isopropyl	4-OMe-Phenyl
49	—CH ₂ CH(CH ₃) ₂	4-OMe-Phenyl
50	—CF ₃	4-OMe-Phenyl
51	—CH ₂ CF ₃	4-OMe-Phenyl
52	—CH ₂ CH ₂ CF ₃	4-OMe-Phenyl
53	cyclopropyl	4-OMe-Phenyl
54	Cyclobutyl	4-OMe-Phenyl
55	cyclopentyl	4-OMe-Phenyl
56	cyclohexyl	4-OMe-Phenyl
57	3-pyridyl	4-OMe-Phenyl
58	1-methyl-1H-pyrazol-4-yl	4-OMe-Phenyl
59	1H-imidazol-4-yl	4-OMe-Phenyl
60	2-furanyl	4-OMe-Phenyl
61	ethyl	2-OMe-Phenyl
62	n-propyl	2-OMe-Phenyl
63	isopropyl	2-OMe-Phenyl
64	—CH ₂ CH(CH ₃) ₂	2-OMe-Phenyl
65	—CF ₃	2-OMe-Phenyl
66	—CH ₂ CF ₃	2-OMe-Phenyl
67	—CH ₂ CH ₂ CF ₃	2-OMe-Phenyl

TABLE 9-continued

Entry	R ^{2d}	R ³
68	cyclopropyl	2-OMe-Phenyl
69	Cyclobutyl	2-OMe-Phenyl
70	cyclopentyl	2-OMe-Phenyl
71	cyclohexyl	2-OMe-Phenyl
72	3-pyridyl	2-OMe-Phenyl
73	1-methyl-1H-pyrazol-4-yl	2-OMe-Phenyl
74	1H-imidazol-4-yl	2-OMe-Phenyl
75	2-furanyl	2-OMe-Phenyl
76	ethyl	3-CN-Phenyl
77	n-propyl	3-CN-Phenyl
78	isopropyl	3-CN-Phenyl
79	—CH ₂ CH(CH ₃) ₂	3-CN-Phenyl
80	—CF ₃	3-CN-Phenyl
81	—CH ₂ CF ₃	3-CN-Phenyl
82	—CH ₂ CH ₂ CF ₃	3-CN-Phenyl
83	cyclopropyl	3-CN-Phenyl
84	Cyclobutyl	3-CN-Phenyl
85	cyclopentyl	3-CN-Phenyl
86	cyclohexyl	3-CN-Phenyl
87	3-pyridyl	3-CN-Phenyl
88	1-methyl-1H-pyrazol-4-yl	3-CN-Phenyl
89	1H-imidazol-4-yl	3-CN-Phenyl
90	2-furanyl	3-CN-Phenyl
91	ethyl	2-F-Phenyl
92	n-propyl	2-F-Phenyl
93	isopropyl	2-F-Phenyl
94	—CH ₂ CH(CH ₃) ₂	2-F-Phenyl
95	—CF ₃	2-F-Phenyl
96	—CH ₂ CF ₃	2-F-Phenyl
97	—CH ₂ CH ₂ CF ₃	2-F-Phenyl
98	cyclopropyl	2-F-Phenyl
99	Cyclobutyl	2-F-Phenyl
100	cyclopentyl	2-F-Phenyl
101	cyclohexyl	2-F-Phenyl
102	3-pyridyl	2-F-Phenyl
103	1-methyl-1H-pyrazol-4-yl	2-F-Phenyl
104	1H-imidazol-4-yl	2-F-Phenyl
105	2-furanyl	2-F-Phenyl
106	ethyl	4-F-Phenyl
107	n-propyl	4-F-Phenyl
108	isopropyl	4-F-Phenyl
109	—CH ₂ CH(CH ₃) ₂	4-F-Phenyl
110	—CF ₃	4-F-Phenyl
111	—CH ₂ CF ₃	4-F-Phenyl
112	—CH ₂ CH ₂ CF ₃	4-F-Phenyl
113	cyclopropyl	4-F-Phenyl
114	Cyclobutyl	4-F-Phenyl
115	cyclopentyl	4-F-Phenyl
116	cyclohexyl	4-F-Phenyl
117	3-pyridyl	4-F-Phenyl
118	1-methyl-1H-pyrazol-4-yl	4-F-Phenyl
119	1H-imidazol-4-yl	4-F-Phenyl
120	2-furanyl	4-F-Phenyl
121	ethyl	3-Cl-Phenyl
122	n-propyl	3-Cl-Phenyl
123	isopropyl	3-Cl-Phenyl
124	—CH ₂ CH(CH ₃) ₂	3-Cl-Phenyl
125	—CF ₃	3-Cl-Phenyl
126	—CH ₂ CF ₃	3-Cl-Phenyl
127	—CH ₂ CH ₂ CF ₃	3-Cl-Phenyl
128	cyclopropyl	3-Cl-Phenyl
129	Cyclobutyl	3-Cl-Phenyl
130	cyclopentyl	3-Cl-Phenyl
131	cyclohexyl	3-Cl-Phenyl
132	3-pyridyl	3-Cl-Phenyl
133	1-methyl-1H-pyrazol-4-yl	3-Cl-Phenyl
134	1H-imidazol-4-yl	3-Cl-Phenyl
135	2-furanyl	3-Cl-Phenyl
136	ethyl	2-Br-Phenyl
137	n-propyl	2-Br-Phenyl

TABLE 9-continued

Entry	R ^{2d}	R ³
138	isopropyl	2-Br-Phenyl
139	—CH ₂ CH(CH ₃) ₂	2-Br-Phenyl
140	—CF ₃	2-Br-Phenyl
141	—CH ₂ CF ₃	2-Br-Phenyl
142	—CH ₂ CH ₂ CF ₃	2-Br-Phenyl
143	cyclopropyl	2-Br-Phenyl
144	Cyclobutyl	2-Br-Phenyl
145	cyclopentyl	2-Br-Phenyl
146	cyclohexyl	2-Br-Phenyl
147	3-pyridyl	2-Br-Phenyl
148	1-methyl-1H-pyrazol-4-yl	2-Br-Phenyl
149	1H-imidazol-4-yl	2-Br-Phenyl
150	2-furanyl	2-Br-Phenyl
151	ethyl	4-Br-Phenyl
152	n-propyl	4-Br-Phenyl
153	isopropyl	4-Br-Phenyl
154	—CH ₂ CH(CH ₃) ₂	4-Br-Phenyl
155	—CF ₃	4-Br-Phenyl
156	—CH ₂ CF ₃	4-Br-Phenyl
157	—CH ₂ CH ₂ CF ₃	4-Br-Phenyl
158	cyclopropyl	4-Br-Phenyl
159	Cyclobutyl	4-Br-Phenyl
160	cyclopentyl	4-Br-Phenyl
161	cyclohexyl	4-Br-Phenyl
162	3-pyridyl	4-Br-Phenyl
163	1-methyl-1H-pyrazol-4-yl	4-Br-Phenyl
164	1H-imidazol-4-yl	4-Br-Phenyl
165	2-furanyl	4-Br-Phenyl
166	ethyl	3-CF ₃ -Phenyl
167	n-propyl	3-CF ₃ -Phenyl
168	isopropyl	3-CF ₃ -Phenyl
169	—CH ₂ CH(CH ₃) ₂	3-CF ₃ -Phenyl
170	—CF ₃	3-CF ₃ -Phenyl
171	—CH ₂ CF ₃	3-CF ₃ -Phenyl
172	—CH ₂ CH ₂ CF ₃	3-CF ₃ -Phenyl
173	cyclopropyl	3-CF ₃ -Phenyl
174	Cyclobutyl	3-CF ₃ -Phenyl
175	cyclopentyl	3-CF ₃ -Phenyl
176	cyclohexyl	3-CF ₃ -Phenyl
177	3-pyridyl	3-CF ₃ -Phenyl
178	1-methyl-1H-pyrazol-4-yl	3-CF ₃ -Phenyl
179	1H-imidazol-4-yl	3-CF ₃ -Phenyl
180	2-furanyl	3-CF ₃ -Phenyl
181	ethyl	2-iPr-Phenyl
182	n-propyl	2-iPr-Phenyl
183	isopropyl	2-iPr-Phenyl
184	—CH ₂ CH(CH ₃) ₂	2-iPr-Phenyl
185	—CF ₃	2-iPr-Phenyl
186	—CH ₂ CF ₃	2-iPr-Phenyl
187	—CH ₂ CH ₂ CF ₃	2-iPr-Phenyl
188	cyclopropyl	2-iPr-Phenyl
189	Cyclobutyl	2-iPr-Phenyl
190	cyclopentyl	2-iPr-Phenyl
191	cyclohexyl	2-iPr-Phenyl
192	3-pyridyl	2-iPr-Phenyl
193	1-methyl-1H-pyrazol-4-yl	2-iPr-Phenyl
194	1H-imidazol-4-yl	2-iPr-Phenyl
195	2-furanyl	2-iPr-Phenyl
196	ethyl	4-iPr-Phenyl
197	n-propyl	4-iPr-Phenyl
198	isopropyl	4-iPr-Phenyl
199	—CH ₂ CH(CH ₃) ₂	4-iPr-Phenyl
200	—CF ₃	4-iPr-Phenyl
201	—CH ₂ CF ₃	4-iPr-Phenyl
202	—CH ₂ CH ₂ CF ₃	4-iPr-Phenyl
203	cyclopropyl	4-iPr-Phenyl
204	Cyclobutyl	4-iPr-Phenyl
205	cyclopentyl	4-iPr-Phenyl
206	cyclohexyl	4-iPr-Phenyl
207	3-pyridyl	4-iPr-Phenyl

TABLE 9-continued

Entry	R ^{2d}	R ³
208	1-methyl-1H-pyrazol-4-yl	4-iPr-Phenyl
209	1H-imidazol-4-yl	4-iPr-Phenyl
210	2-furanyl	4-iPr-Phenyl
211	ethyl	3-morpholino-phenyl
212	n-propyl	3-morpholino-phenyl
213	isopropyl	3-morpholino-phenyl
214	—CH ₂ CH(CH ₃) ₂	3-morpholino-phenyl
215	—CF ₃	3-morpholino-phenyl
216	—CH ₂ CF ₃	3-morpholino-phenyl
217	—CH ₂ CH ₂ CF ₃	3-morpholino-phenyl
218	cyclopropyl	3-morpholino-phenyl
219	Cyclobutyl	3-morpholino-phenyl
220	cyclopentyl	3-morpholino-phenyl
221	cyclohexyl	3-morpholino-phenyl
222	3-pyridyl	3-morpholino-phenyl
223	1-methyl-1H-pyrazol-4-yl	3-morpholino-phenyl
224	1H-imidazol-4-yl	3-morpholino-phenyl
225	2-furanyl	3-morpholino-phenyl
226	ethyl	4-cyano-2-morpholino-phenyl
227	n-propyl	4-cyano-2-morpholino-phenyl
228	isopropyl	4-cyano-2-morpholino-phenyl
229	—CH ₂ CH(CH ₃) ₂	4-cyano-2-morpholino-phenyl
230	—CF ₃	4-cyano-2-morpholino-phenyl
231	—CH ₂ CF ₃	4-cyano-2-morpholino-phenyl
232	—CH ₂ CH ₂ CF ₃	4-cyano-2-morpholino-phenyl
233	cyclopropyl	4-cyano-2-morpholino-phenyl
234	Cyclobutyl	4-cyano-2-morpholino-phenyl
235	cyclopentyl	4-cyano-2-morpholino-phenyl
236	cyclohexyl	4-cyano-2-morpholino-phenyl
237	3-pyridyl	4-cyano-2-morpholino-phenyl
238	1-methyl-1H-pyrazol-4-yl	4-cyano-2-morpholino-phenyl
239	1H-imidazol-4-yl	4-cyano-2-morpholino-phenyl
240	2-furanyl	4-cyano-2-morpholino-phenyl
241	ethyl	4-hydroxy-2-morpholino-phenyl
242	n-propyl	4-hydroxy-2-morpholino-phenyl
243	isopropyl	4-hydroxy-2-morpholino-phenyl
244	—CH ₂ CH(CH ₃) ₂	4-hydroxy-2-morpholino-phenyl
245	—CF ₃	4-hydroxy-2-morpholino-phenyl
246	—CH ₂ CF ₃	4-hydroxy-2-morpholino-phenyl
247	—CH ₂ CH ₂ CF ₃	4-hydroxy-2-morpholino-phenyl
248	cyclopropyl	4-hydroxy-2-morpholino-phenyl
249	Cyclobutyl	4-hydroxy-2-morpholino-phenyl
250	cyclopentyl	4-hydroxy-2-morpholino-phenyl
251	cyclohexyl	4-hydroxy-2-morpholino-phenyl
252	3-pyridyl	4-hydroxy-2-morpholino-phenyl
253	1-methyl-1H-pyrazol-4-yl	4-hydroxy-2-morpholino-phenyl

TABLE 9-continued

Entry	R ^{2d}	R ³
254	1H-imidazol-4-yl	4-hydroxy-2-morpholino-phenyl
255	2-furanyl	4-hydroxy-2-morpholino-phenyl
256	ethyl	2-CH ₃ -phenyl
257	n-propyl	2-CH ₃ -phenyl
258	isopropyl	2-CH ₃ -phenyl
259	—CH ₂ CH(CH ₃) ₂	2-CH ₃ -phenyl
260	CF ₃	2-CH ₃ -phenyl
261	—CH ₂ CF ₃	2-CH ₃ -phenyl
262	—CH ₂ CH ₂ CF ₃	2-CH ₃ -phenyl
263	cyclopropyl	2-CH ₃ -phenyl
264	Cyclobutyl	2-CH ₃ -phenyl
265	cyclopentyl	2-CH ₃ -phenyl
266	cyclohexyl	2-CH ₃ -phenyl
267	3-pyridyl	2-CH ₃ -phenyl
268	1-methyl-1H-pyrazol-4-yl	2-CH ₃ -phenyl
269	1H-imidazol-4-yl	2-CH ₃ -phenyl
270	2-furanyl	2-CH ₃ -phenyl
271	ethyl	4-OH-Phenyl
272	n-propyl	4-OH-Phenyl
273	isopropyl	4-OH-Phenyl
274	—CH ₂ CH(CH ₃) ₂	4-OH-Phenyl
275	—CF ₃	4-OH-Phenyl
276	—CH ₂ CF ₃	4-OH-Phenyl
277	—CH ₂ CH ₂ CF ₃	4-OH-Phenyl
278	cyclopropyl	4-OH-Phenyl
279	Cyclobutyl	4-OH-Phenyl
280	cyclopentyl	4-OH-Phenyl
281	cyclohexyl	4-OH-Phenyl
282	3-pyridyl	4-OH-Phenyl
283	1-methyl-1H-pyrazol-4-yl	4-OH-Phenyl
284	1H-imidazol-4-yl	4-OH-Phenyl
285	2-furanyl	4-OH-Phenyl
286	ethyl	2-OH-Phenyl
287	n-propyl	2-OH-Phenyl
288	isopropyl	2-OH-Phenyl
289	—CH ₂ CH(CH ₃) ₂	2-OH-Phenyl
290	—CF ₃	2-OH-Phenyl
291	—CH ₂ CF ₃	2-OH-Phenyl
292	—CH ₂ CH ₂ CF ₃	2-OH-Phenyl
293	cyclopropyl	2-OH-Phenyl
294	Cyclobutyl	2-OH-Phenyl
295	cyclopentyl	2-OH-Phenyl
296	cyclohexyl	2-OH-Phenyl
297	3-pyridyl	2-OH-Phenyl
298	1-methyl-1H-pyrazol-4-yl	2-OH-Phenyl
299	1H-imidazol-4-yl	2-OH-Phenyl
300	2-furanyl	2-OH-Phenyl
301	ethyl	3-OMe-Phenyl
302	n-propyl	3-OMe-Phenyl
303	isopropyl	3-OMe-Phenyl
304	—CH ₂ CH(CH ₃) ₂	3-OMe-Phenyl
305	—CF ₃	3-OMe-Phenyl
306	—CH ₂ CF ₃	3-OMe-Phenyl
307	—CH ₂ CH ₂ CF ₃	3-OMe-Phenyl
308	cyclopropyl	3-OMe-Phenyl
309	Cyclobutyl	3-OMe-Phenyl
310	cyclopentyl	3-OMe-Phenyl
311	cyclohexyl	3-OMe-Phenyl
312	3-pyridyl	3-OMe-Phenyl
313	1-methyl-1H-pyrazol-4-yl	3-OMe-Phenyl
314	1H-imidazol-4-yl	3-OMe-Phenyl
315	2-furanyl	3-OMe-Phenyl
316	ethyl	4-CN-Phenyl
317	n-propyl	4-CN-Phenyl
318	isopropyl	4-CN-Phenyl
319	—CH ₂ CH(CH ₃) ₂	4-CN-Phenyl
320	—CF ₃	4-CN-Phenyl
321	—CH ₂ CF ₃	4-CN-Phenyl
322	—CH ₂ CH ₂ CF ₃	4-CN-Phenyl

TABLE 9-continued

Entry	R ^{2d}	R ³
323	cyclopropyl	4-CN-Phenyl
324	Cyclobutyl	4-CN-Phenyl
325	cyclopentyl	4-CN-Phenyl
326	cyclohexyl	4-CN-Phenyl
327	3-pyridyl	4-CN-Phenyl
328	1-methyl-1H-pyrazol-4-yl	4-CN-Phenyl
329	1H-imidazol-4-yl	4-CN-Phenyl
330	2-furanyl	4-CN-Phenyl
331	ethyl	2-CN-Phenyl
332	n-propyl	2-CN-Phenyl
333	isopropyl	2-CN-Phenyl
334	—CH ₂ CH(CH ₃) ₂	2-CN-Phenyl
335	—CF ₃	2-CN-Phenyl
336	—CH ₂ CF ₃	2-CN-Phenyl
337	—CH ₂ CH ₂ CF ₃	2-CN-Phenyl
338	cyclopropyl	2-CN-Phenyl
339	Cyclobutyl	2-CN-Phenyl
340	cyclopentyl	2-CN-Phenyl
341	cyclohexyl	2-CN-Phenyl
342	3-pyridyl	2-CN-Phenyl
343	1-methyl-1H-pyrazol-4-yl	2-CN-Phenyl
344	1H-imidazol-4-yl	2-CN-Phenyl
345	2-furanyl	2-CN-Phenyl
346	ethyl	3-F-Phenyl
347	n-propyl	3-F-Phenyl
348	isopropyl	3-F-Phenyl
349	—CH ₂ CH(CH ₃) ₂	3-F-Phenyl
350	—CF ₃	3-F-Phenyl
351	—CH ₂ CF ₃	3-F-Phenyl
352	—CH ₂ CH ₂ CF ₃	3-F-Phenyl
353	cyclopropyl	3-F-Phenyl
354	Cyclobutyl	3-F-Phenyl
355	cyclopentyl	3-F-Phenyl
356	cyclohexyl	3-F-Phenyl
357	3-pyridyl	3-F-Phenyl
358	1-methyl-1H-pyrazol-4-yl	3-F-Phenyl
359	1H-imidazol-4-yl	3-F-Phenyl
360	2-furanyl	3-F-Phenyl
361	ethyl	2-Cl-Phenyl
362	n-propyl	2-Cl-Phenyl
363	isopropyl	2-Cl-Phenyl
364	—CH ₂ CH(CH ₃) ₂	2-Cl-Phenyl
365	—CF ₃	2-Cl-Phenyl
366	—CH ₂ CF ₃	2-Cl-Phenyl
367	—CH ₂ CH ₂ CF ₃	2-Cl-Phenyl
368	cyclopropyl	2-Cl-Phenyl
369	Cyclobutyl	2-Cl-Phenyl
370	cyclopentyl	2-Cl-Phenyl
371	cyclohexyl	2-Cl-Phenyl
372	3-pyridyl	2-Cl-Phenyl
373	1-methyl-1H-pyrazol-4-yl	2-Cl-Phenyl
374	1H-imidazol-4-yl	2-Cl-Phenyl
375	2-furanyl	2-Cl-Phenyl
376	ethyl	4-Cl-Phenyl
377	n-propyl	4-Cl-Phenyl
378	isopropyl	4-Cl-Phenyl
379	—CH ₂ CH(CH ₃) ₂	4-Cl-Phenyl
380	—CF ₃	4-Cl-Phenyl
381	—CH ₂ CF ₃	4-Cl-Phenyl
382	—CH ₂ CH ₂ CF ₃	4-Cl-Phenyl
383	cyclopropyl	4-Cl-Phenyl
384	Cyclobutyl	4-Cl-Phenyl
385	cyclopentyl	4-Cl-Phenyl
386	cyclohexyl	4-Cl-Phenyl
387	3-pyridyl	4-Cl-Phenyl
388	1-methyl-1H-pyrazol-4-yl	4-Cl-Phenyl
389	1H-imidazol-4-yl	4-Cl-Phenyl
390	2-furanyl	4-Cl-Phenyl
391	ethyl	3-Br-Phenyl
392	n-propyl	3-Br-Phenyl

TABLE 9-continued

Entry	R ^{2d}	R ³
393	isopropyl	3-Br-Phenyl
394	—CH ₂ CH(CH ₃) ₂	3-Br-Phenyl
395	—CF ₃	3-Br-Phenyl
396	—CH ₂ CF ₃	3-Br-Phenyl
397	—CH ₂ CH ₂ CF ₃	3-Br-Phenyl
398	cyclopropyl	3-Br-Phenyl
399	Cyclobutyl	3-Br-Phenyl
400	cyclopentyl	3-Br-Phenyl
401	cyclohexyl	3-Br-Phenyl
402	3-pyridyl	3-Br-Phenyl
403	1-methyl-1H-pyrazol-4-yl	3-Br-Phenyl
404	1H-imidazol-4-yl	3-Br-Phenyl
405	2-furanyl	3-Br-Phenyl
406	ethyl	2-CF ₃ -Phenyl
407	n-propyl	2-CF ₃ -Phenyl
408	isopropyl	2-CF ₃ -Phenyl
409	—CH ₂ CH(CH ₃) ₂	2-CF ₃ -Phenyl
410	—CF ₃	2-CF ₃ -Phenyl
411	—CH ₂ CF ₃	2-CF ₃ -Phenyl
412	—CH ₂ CH ₂ CF ₃	2-CF ₃ -Phenyl
413	cyclopropyl	2-CF ₃ -Phenyl
414	Cyclobutyl	2-CF ₃ -Phenyl
415	cyclopentyl	2-CF ₃ -Phenyl
416	cyclohexyl	2-CF ₃ -Phenyl
417	3-pyridyl	2-CF ₃ -Phenyl
418	1-methyl-1H-pyrazol-4-yl	2-CF ₃ -Phenyl
419	1H-imidazol-4-yl	2-CF ₃ -Phenyl
420	2-furanyl	2-CF ₃ -Phenyl
421	ethyl	4-CF ₃ -Phenyl
422	n-propyl	4-CF ₃ -Phenyl
423	isopropyl	4-CF ₃ -Phenyl
424	—CH ₂ CH(CH ₃) ₂	4-CF ₃ -Phenyl
425	—CF ₃	4-CF ₃ -Phenyl
426	—CH ₂ CF ₃	4-CF ₃ -Phenyl
427	—CH ₂ CH ₂ CF ₃	4-CF ₃ -Phenyl
428	cyclopropyl	4-CF ₃ -Phenyl
429	Cyclobutyl	4-CF ₃ -Phenyl
430	cyclopentyl	4-CF ₃ -Phenyl
431	cyclohexyl	4-CF ₃ -Phenyl
432	3-pyridyl	4-CF ₃ -Phenyl
433	1-methyl-1H-pyrazol-4-yl	4-CF ₃ -Phenyl
434	1H-imidazol-4-yl	4-CF ₃ -Phenyl
435	2-furanyl	4-CF ₃ -Phenyl
436	ethyl	3-iPr-Phenyl
437	n-propyl	3-iPr-Phenyl
438	isopropyl	3-iPr-Phenyl
439	—CH ₂ CH(CH ₃) ₂	3-iPr-Phenyl
440	—CF ₃	3-iPr-Phenyl
441	—CH ₂ CF ₃	3-iPr-Phenyl
442	—CH ₂ CH ₂ CF ₃	3-iPr-Phenyl
443	cyclopropyl	3-iPr-Phenyl
444	Cyclobutyl	3-iPr-Phenyl
445	cyclopentyl	3-iPr-Phenyl
446	cyclohexyl	3-iPr-Phenyl
447	3-pyridyl	3-iPr-Phenyl
448	1-methyl-1H-pyrazol-4-yl	3-iPr-Phenyl
449	1H-imidazol-4-yl	3-iPr-Phenyl
450	2-furanyl	3-iPr-Phenyl
451	ethyl	2-morpholino-phenyl
452	n-propyl	2-morpholino-phenyl
453	isopropyl	2-morpholino-phenyl
454	—CH ₂ CH(CH ₃) ₂	2-morpholino-phenyl
455	—CF ₃	2-morpholino-phenyl
456	—CH ₂ CF ₃	2-morpholino-phenyl
457	—CH ₂ CH ₂ CF ₃	2-morpholino-phenyl

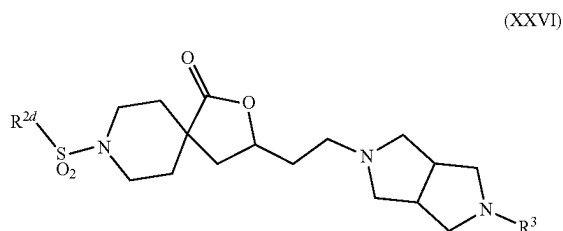
TABLE 9-continued

Entry	R ^{2d}	R ³
458	cyclopropyl	2-morpholino-phenyl
459	Cyclobutyl	2-morpholino-phenyl
460	cyclopentyl	2-morpholino-phenyl
461	cyclohexyl	2-morpholino-phenyl
462	3-pyridyl	2-morpholino-phenyl
463	1-methyl-1H-pyrazol-4-yl	2-morpholino-phenyl
464	1H-imidazol-4-yl	2-morpholino-phenyl
465	2-furanyl	2-morpholino-phenyl
466	ethyl	4-morpholino-phenyl
467	n-propyl	4-morpholino-phenyl
468	isopropyl	4-morpholino-phenyl
469	—CH ₂ CH(CH ₃) ₂	4-morpholino-phenyl
470	—CF ₃	4-morpholino-phenyl
471	—CH ₂ CF ₃	4-morpholino-phenyl
472	—CH ₂ CH ₂ CF ₃	4-morpholino-phenyl
473	cyclopropyl	4-morpholino-phenyl
474	Cyclobutyl	4-morpholino-phenyl
475	cyclopentyl	4-morpholino-phenyl
476	cyclohexyl	4-morpholino-phenyl
477	3-pyridyl	4-morpholino-phenyl
478	1-methyl-1H-pyrazol-4-yl	4-morpholino-phenyl
479	1H-imidazol-4-yl	4-morpholino-phenyl
480	2-furanyl	4-morpholino-phenyl
481	ethyl	4-methyl-2-morpholino-phenyl
482	n-propyl	4-methyl-2-morpholino-phenyl
483	isopropyl	4-methyl-2-morpholino-phenyl
484	—CH ₂ CH(CH ₃) ₂	4-methyl-2-morpholino-phenyl
485	—CF ₃	4-methyl-2-morpholino-phenyl
486	—CH ₂ CF ₃	4-methyl-2-morpholino-phenyl
487	—CH ₂ CH ₂ CF ₃	4-methyl-2-morpholino-phenyl
488	cyclopropyl	4-methyl-2-morpholino-phenyl
489	Cyclobutyl	4-methyl-2-morpholino-phenyl
490	cyclopentyl	4-methyl-2-morpholino-phenyl
491	cyclohexyl	4-methyl-2-morpholino-phenyl
492	3-pyridyl	4-methyl-2-morpholino-phenyl
493	1-methyl-1H-pyrazol-4-yl	4-methyl-2-morpholino-phenyl
494	1H-imidazol-4-yl	4-methyl-2-morpholino-phenyl

TABLE 9-continued

Entry	R ^{2d}	R ³
495	2-furanyl	4-methyl-2-morpholino-phenyl

[0409] Exemplary embodiments include compounds having the formula (XXVI)



or a pharmaceutically acceptable salt form thereof defined herein below in Table 10.

TABLE 10

Entry	R ^{2d}	R ³
1	ethyl	4-CH ₃ -phenyl
2	n-propyl	4-CH ₃ -phenyl
3	isopropyl	4-CH ₃ -phenyl
4	—CH ₂ CH(CH ₃) ₂	4-CH ₃ -phenyl
5	CF ₃	4-CH ₃ -phenyl
6	—CH ₂ CF ₃	4-CH ₃ -phenyl
7	—CH ₂ CH ₂ CF ₃	4-CH ₃ -phenyl
8	cyclopropyl	4-CH ₃ -phenyl
9	Cyclobutyl	4-CH ₃ -phenyl
10	cyclopentyl	4-CH ₃ -phenyl
11	cyclohexyl	4-CH ₃ -phenyl
12	3-pyridyl	4-CH ₃ -phenyl
13	1-methyl-1H-pyrazol-4-yl	4-CH ₃ -phenyl
14	1H-imidazol-4-yl	4-CH ₃ -phenyl
15	2-furanyl	4-CH ₃ -phenyl
16	ethyl	3-CH ₃ -phenyl
17	n-propyl	3-CH ₃ -phenyl
18	isopropyl	3-CH ₃ -phenyl
19	—CH ₂ CH(CH ₃) ₂	3-CH ₃ -phenyl
20	—CF ₃	3-CH ₃ -phenyl
21	—CH ₂ CF ₃	3-CH ₃ -phenyl
22	—CH ₂ CH ₂ CF ₃	3-CH ₃ -phenyl
23	cyclopropyl	3-CH ₃ -phenyl
24	Cyclobutyl	3-CH ₃ -phenyl
25	cyclopentyl	3-CH ₃ -phenyl
26	cyclohexyl	3-CH ₃ -phenyl
27	3-pyridyl	3-CH ₃ -phenyl
28	1-methyl-1H-pyrazol-4-yl	3-CH ₃ -phenyl
29	1H-imidazol-4-yl	3-CH ₃ -phenyl
30	2-furanyl	3-CH ₃ -phenyl
31	ethyl	3-OH-Phenyl
32	n-propyl	3-OH-Phenyl
33	isopropyl	3-OH-Phenyl
34	—CH ₂ CH(CH ₃) ₂	3-OH-Phenyl
35	—CF ₃	3-OH-Phenyl
36	—CH ₂ CF ₃	3-OH-Phenyl
37	—CH ₂ CH ₂ CF ₃	3-OH-Phenyl
38	cyclopropyl	3-OH-Phenyl
39	Cyclobutyl	3-OH-Phenyl
40	cyclopentyl	3-OH-Phenyl
41	cyclohexyl	3-OH-Phenyl
42	3-pyridyl	3-OH-Phenyl

TABLE 10-continued

Entry	R ^{2d}	R ³
43	1-methyl-1H-pyrazol-4-yl	3-OH-Phenyl
44	1H-imidazol-4-yl	3-OH-Phenyl
45	2-furanyl	3-OH-Phenyl
46	ethyl	4-OMe-Phenyl
47	n-propyl	4-OMe-Phenyl
48	isopropyl	4-OMe-Phenyl
49	—CH ₂ CH(CH ₃) ₂	4-OMe-Phenyl
50	—CF ₃	4-OMe-Phenyl
51	—CH ₂ CF ₃	4-OMe-Phenyl
52	—CH ₂ CH ₂ CF ₃	4-OMe-Phenyl
53	cyclopropyl	4-OMe-Phenyl
54	Cyclobutyl	4-OMe-Phenyl
55	cyclopentyl	4-OMe-Phenyl
56	cyclohexyl	4-OMe-Phenyl
57	3-pyridyl	4-OMe-Phenyl
58	1-methyl-1H-pyrazol-4-yl	4-OMe-Phenyl
59	1H-imidazol-4-yl	4-OMe-Phenyl
60	2-furanyl	4-OMe-Phenyl
61	ethyl	2-OMe-Phenyl
62	n-propyl	2-OMe-Phenyl
63	isopropyl	2-OMe-Phenyl
64	—CH ₂ CH(CH ₃) ₂	2-OMe-Phenyl
65	—CF ₃	2-OMe-Phenyl
66	—CH ₂ CF ₃	2-OMe-Phenyl
67	—CH ₂ CH ₂ CF ₃	2-OMe-Phenyl
68	cyclopropyl	2-OMe-Phenyl
69	Cyclobutyl	2-OMe-Phenyl
70	cyclopentyl	2-OMe-Phenyl
71	cyclohexyl	2-OMe-Phenyl
72	3-pyridyl	2-OMe-Phenyl
73	1-methyl-1H-pyrazol-4-yl	2-OMe-Phenyl
74	1H-imidazol-4-yl	2-OMe-Phenyl
75	2-furanyl	2-OMe-Phenyl
76	ethyl	3-CN-Phenyl
77	n-propyl	3-CN-Phenyl
78	isopropyl	3-CN-Phenyl
79	—CH ₂ CH(CH ₃) ₂	3-CN-Phenyl
80	—CF ₃	3-CN-Phenyl
81	—CH ₂ CF ₃	3-CN-Phenyl
82	—CH ₂ CH ₂ CF ₃	3-CN-Phenyl
83	cyclopropyl	3-CN-Phenyl
84	Cyclobutyl	3-CN-Phenyl
85	cyclopentyl	3-CN-Phenyl
86	cyclohexyl	3-CN-Phenyl
87	3-pyridyl	3-CN-Phenyl
88	1-methyl-1H-pyrazol-4-yl	3-CN-Phenyl
89	1H-imidazol-4-yl	3-CN-Phenyl
90	2-furanyl	3-CN-Phenyl
91	ethyl	2-F-Phenyl
92	n-propyl	2-F-Phenyl
93	isopropyl	2-F-Phenyl
94	—CH ₂ CH(CH ₃) ₂	2-F-Phenyl
95	—CF ₃	2-F-Phenyl
96	—CH ₂ CF ₃	2-F-Phenyl
97	—CH ₂ CH ₂ CF ₃	2-F-Phenyl
98	cyclopropyl	2-F-Phenyl
99	Cyclobutyl	2-F-Phenyl
100	cyclopentyl	2-F-Phenyl
101	cyclohexyl	2-F-Phenyl
102	3-pyridyl	2-F-Phenyl
103	1-methyl-1H-pyrazol-4-yl	2-F-Phenyl
104	1H-imidazol-4-yl	2-F-Phenyl
105	2-furanyl	2-F-Phenyl
106	ethyl	4-F-Phenyl
107	n-propyl	4-F-Phenyl
108	isopropyl	4-F-Phenyl
109	—CH ₂ CH(CH ₃) ₂	4-F-Phenyl
110	—CF ₃	4-F-Phenyl
111	—CH ₂ CF ₃	4-F-Phenyl
112	—CH ₂ CH ₂ CF ₃	4-F-Phenyl

TABLE 10-continued

Entry	R ^{2d}	R ³
113	cyclopropyl	4-F-Phenyl
114	Cyclobutyl	4-F-Phenyl
115	cyclopentyl	4-F-Phenyl
116	cyclohexyl	4-F-Phenyl
117	3-pyridyl	4-F-Phenyl
118	1-methyl-1H-pyrazol-4-yl	4-F-Phenyl
119	1H-imidazol-4-yl	4-F-Phenyl
120	2-furanyl	4-F-Phenyl
121	ethyl	3-Cl-Phenyl
122	n-propyl	3-Cl-Phenyl
123	isopropyl	3-Cl-Phenyl
124	—CH ₂ CH(CH ₃) ₂	3-Cl-Phenyl
125	—CF ₃	3-Cl-Phenyl
126	—CH ₂ CF ₃	3-Cl-Phenyl
127	—CH ₂ CH ₂ CF ₃	3-Cl-Phenyl
128	cyclopropyl	3-Cl-Phenyl
129	Cyclobutyl	3-Cl-Phenyl
130	cyclopentyl	3-Cl-Phenyl
131	cyclohexyl	3-Cl-Phenyl
132	3-pyridyl	3-Cl-Phenyl
133	1-methyl-1H-pyrazol-4-yl	3-Cl-Phenyl
134	1H-imidazol-4-yl	3-Cl-Phenyl
135	2-furanyl	3-Cl-Phenyl
136	ethyl	2-Br-Phenyl
137	n-propyl	2-Br-Phenyl
138	isopropyl	2-Br-Phenyl
139	—CH ₂ CH(CH ₃) ₂	2-Br-Phenyl
140	—CF ₃	2-Br-Phenyl
141	—CH ₂ CF ₃	2-Br-Phenyl
142	—CH ₂ CH ₂ CF ₃	2-Br-Phenyl
143	cyclopropyl	2-Br-Phenyl
144	Cyclobutyl	2-Br-Phenyl
145	cyclopentyl	2-Br-Phenyl
146	cyclohexyl	2-Br-Phenyl
147	3-pyridyl	2-Br-Phenyl
148	1-methyl-1H-pyrazol-4-yl	2-Br-Phenyl
149	1H-imidazol-4-yl	2-Br-Phenyl
150	2-furanyl	2-Br-Phenyl
151	ethyl	4-Br-Phenyl
152	n-propyl	4-Br-Phenyl
153	isopropyl	4-Br-Phenyl
154	—CH ₂ CH(CH ₃) ₂	4-Br-Phenyl
155	—CF ₃	4-Br-Phenyl
156	—CH ₂ CF ₃	4-Br-Phenyl
157	—CH ₂ CH ₂ CF ₃	4-Br-Phenyl
158	cyclopropyl	4-Br-Phenyl
159	Cyclobutyl	4-Br-Phenyl
160	cyclopentyl	4-Br-Phenyl
161	cyclohexyl	4-Br-Phenyl
162	3-pyridyl	4-Br-Phenyl
163	1-methyl-1H-pyrazol-4-yl	4-Br-Phenyl
164	1H-imidazol-4-yl	4-Br-Phenyl
165	2-furanyl	4-Br-Phenyl
166	ethyl	3-CF ₃ -Phenyl
167	n-propyl	3-CF ₃ -Phenyl
168	isopropyl	3-CF ₃ -Phenyl
169	—CH ₂ CH(CH ₃) ₂	3-CF ₃ -Phenyl
170	—CF ₃	3-CF ₃ -Phenyl
171	—CH ₂ CF ₃	3-CF ₃ -Phenyl
172	—CH ₂ CH ₂ CF ₃	3-CF ₃ -Phenyl
173	cyclopropyl	3-CF ₃ -Phenyl
174	Cyclobutyl	3-CF ₃ -Phenyl
175	cyclopentyl	3-CF ₃ -Phenyl
176	cyclohexyl	3-CF ₃ -Phenyl
177	3-pyridyl	3-CF ₃ -Phenyl
178	1-methyl-1H-pyrazol-4-yl	3-CF ₃ -Phenyl
179	1H-imidazol-4-yl	3-CF ₃ -Phenyl
180	2-furanyl	3-CF ₃ -Phenyl
181	ethyl	2-iPr-Phenyl
182	n-propyl	2-iPr-Phenyl

TABLE 10-continued

Entry	R ^{2d}	R ³
183	isopropyl	2-iPr-Phenyl
184	—CH ₂ CH(CH ₃) ₂	2-iPr-Phenyl
185	—CF ₃	2-iPr-Phenyl
186	—CH ₂ CF ₃	2-iPr-Phenyl
187	—CH ₂ CH ₂ CF ₃	2-iPr-Phenyl
188	cyclopropyl	2-iPr-Phenyl
189	Cyclobutyl	2-iPr-Phenyl
190	cyclopentyl	2-iPr-Phenyl
191	cyclohexyl	2-iPr-Phenyl
192	3-pyridyl	2-iPr-Phenyl
193	1-methyl-1H-pyrazol-4-yl	2-iPr-Phenyl
194	1H-imidazol-4-yl	2-iPr-Phenyl
195	2-furanyl	2-iPr-Phenyl
196	ethyl	4-iPr-Phenyl
197	n-propyl	4-iPr-Phenyl
198	isopropyl	4-iPr-Phenyl
199	—CH ₂ CH(CH ₃) ₂	4-iPr-Phenyl
200	—CF ₃	4-iPr-Phenyl
201	—CH ₂ CF ₃	4-iPr-Phenyl
202	—CH ₂ CH ₂ CF ₃	4-iPr-Phenyl
203	cyclopropyl	4-iPr-Phenyl
204	Cyclobutyl	4-iPr-Phenyl
205	cyclopentyl	4-iPr-Phenyl
206	cyclohexyl	4-iPr-Phenyl
207	3-pyridyl	4-iPr-Phenyl
208	1-methyl-1H-pyrazol-4-yl	4-iPr-Phenyl
209	1H-imidazol-4-yl	4-iPr-Phenyl
210	2-furanyl	4-iPr-Phenyl
211	ethyl	3-morpholino-phenyl
212	n-propyl	3-morpholino-phenyl
213	isopropyl	3-morpholino-phenyl
214	—CH ₂ CH(CH ₃) ₂	3-morpholino-phenyl
215	—CF ₃	3-morpholino-phenyl
216	—CH ₂ CF ₃	3-morpholino-phenyl
217	—CH ₂ CH ₂ CF ₃	3-morpholino-phenyl
218	cyclopropyl	3-morpholino-phenyl
219	Cyclobutyl	3-morpholino-phenyl
220	cyclopentyl	3-morpholino-phenyl
221	cyclohexyl	3-morpholino-phenyl
222	3-pyridyl	3-morpholino-phenyl
223	1-methyl-1H-pyrazol-4-yl	3-morpholino-phenyl
224	1H-imidazol-4-yl	3-morpholino-phenyl
225	2-furanyl	3-morpholino-phenyl
226	ethyl	4-cyano-2-morpholino-phenyl
227	n-propyl	4-cyano-2-morpholino-phenyl
228	isopropyl	4-cyano-2-morpholino-phenyl
229	—CH ₂ CH(CH ₃) ₂	4-cyano-2-morpholino-phenyl
230	—CF ₃	4-cyano-2-morpholino-phenyl
231	—CH ₂ CF ₃	4-cyano-2-morpholino-phenyl
232	—CH ₂ CH ₂ CF ₃	4-cyano-2-morpholino-phenyl

TABLE 10-continued

Entry	R ^{2d}	R ³
233	cyclopropyl	4-cyano-2-morpholino-phenyl
234	Cyclobutyl	4-cyano-2-morpholino-phenyl
235	cyclopentyl	4-cyano-2-morpholino-phenyl
236	cyclohexyl	4-cyano-2-morpholino-phenyl
237	3-pyridyl	4-cyano-2-morpholino-phenyl
238	1-methyl-1H-pyrazol-4-yl	4-cyano-2-morpholino-phenyl
239	1H-imidazol-4-yl	4-cyano-2-morpholino-phenyl
240	2-furanyl	4-cyano-2-morpholino-phenyl
241	ethyl	4-hydroxy-2-morpholino-phenyl
242	n-propyl	4-hydroxy-2-morpholino-phenyl
243	isopropyl	4-hydroxy-2-morpholino-phenyl
244	—CH ₂ CH(CH ₃) ₂	4-hydroxy-2-morpholino-phenyl
245	—CF ₃	4-hydroxy-2-morpholino-phenyl
246	—CH ₂ CF ₃	4-hydroxy-2-morpholino-phenyl
247	—CH ₂ CH ₂ CF ₃	4-hydroxy-2-morpholino-phenyl
248	cyclopropyl	4-hydroxy-2-morpholino-phenyl
249	Cyclobutyl	4-hydroxy-2-morpholino-phenyl
250	cyclopentyl	4-hydroxy-2-morpholino-phenyl
251	cyclohexyl	4-hydroxy-2-morpholino-phenyl
252	3-pyridyl	4-hydroxy-2-morpholino-phenyl
253	1-methyl-1H-pyrazol-4-yl	4-hydroxy-2-morpholino-phenyl
254	1H-imidazol-4-yl	4-hydroxy-2-morpholino-phenyl
255	2-furanyl	4-hydroxy-2-morpholino-phenyl
256	ethyl	2-CH ₃ -phenyl
257	n-propyl	2-CH ₃ -phenyl
258	isopropyl	2-CH ₃ -phenyl
259	—CH ₂ CH(CH ₃) ₂	2-CH ₃ -phenyl
260	CF ₃	2-CH ₃ -phenyl
261	—CH ₂ CF ₃	2-CH ₃ -phenyl
262	—CH ₂ CH ₂ CF ₃	2-CH ₃ -phenyl
263	cyclopropyl	2-CH ₃ -phenyl
264	Cyclobutyl	2-CH ₃ -phenyl
265	cyclopentyl	2-CH ₃ -phenyl
266	cyclohexyl	2-CH ₃ -phenyl
267	3-pyridyl	2-CH ₃ -phenyl
268	1-methyl-1H-pyrazol-4-yl	2-CH ₃ -phenyl
269	1H-imidazol-4-yl	2-CH ₃ -phenyl
270	2-furanyl	2-CH ₃ -phenyl
271	ethyl	4-OH-Phenyl
272	n-propyl	4-OH-Phenyl
273	isopropyl	4-OH-Phenyl
274	—CH ₂ CH(CH ₃) ₂	4-OH-Phenyl
275	—CF ₃	4-OH-Phenyl
276	—CH ₂ CF ₃	4-OH-Phenyl
277	—CH ₂ CH ₂ CF ₃	4-OH-Phenyl
278	cyclopropyl	4-OH-Phenyl
279	Cyclobutyl	4-OH-Phenyl
280	cyclopentyl	4-OH-Phenyl
281	cyclohexyl	4-OH-Phenyl
282	3-pyridyl	4-OH-Phenyl

TABLE 10-continued

Entry	R ^{2d}	R ³
283	1-methyl-1H-pyrazol-4-yl	4-OH-Phenyl
284	1H-imidazol-4-yl	4-OH-Phenyl
285	2-furanyl	4-OH-Phenyl
286	ethyl	2-OH-Phenyl
287	n-propyl	2-OH-Phenyl
288	isopropyl	2-OH-Phenyl
289	—CH ₂ CH(CH ₃) ₂	2-OH-Phenyl
290	—CF ₃	2-OH-Phenyl
291	—CH ₂ CF ₃	2-OH-Phenyl
292	—CH ₂ CH ₂ CF ₃	2-OH-Phenyl
293	cyclopropyl	2-OH-Phenyl
294	Cyclobutyl	2-OH-Phenyl
295	cyclopentyl	2-OH-Phenyl
296	cyclohexyl	2-OH-Phenyl
297	3-pyridyl	2-OH-Phenyl
298	1-methyl-1H-pyrazol-4-yl	2-OH-Phenyl
299	1H-imidazol-4-yl	2-OH-Phenyl
300	2-furanyl	2-OH-Phenyl
301	ethyl	3-OMe-Phenyl
302	n-propyl	3-OMe-Phenyl
303	isopropyl	3-OMe-Phenyl
304	—CH ₂ CH(CH ₃) ₂	3-OMe-Phenyl
305	—CF ₃	3-OMe-Phenyl
306	—CH ₂ CF ₃	3-OMe-Phenyl
307	—CH ₂ CH ₂ CF ₃	3-OMe-Phenyl
308	cyclopropyl	3-OMe-Phenyl
309	Cyclobutyl	3-OMe-Phenyl
310	cyclopentyl	3-OMe-Phenyl
311	cyclohexyl	3-OMe-Phenyl
312	3-pyridyl	3-OMe-Phenyl
313	1-methyl-1H-pyrazol-4-yl	3-OMe-Phenyl
314	1H-imidazol-4-yl	3-OMe-Phenyl
315	2-furanyl	3-OMe-Phenyl
316	ethyl	4-CN-Phenyl
317	n-propyl	4-CN-Phenyl
318	isopropyl	4-CN-Phenyl
319	—CH ₂ CH(CH ₃) ₂	4-CN-Phenyl
320	—CF ₃	4-CN-Phenyl
321	—CH ₂ CF ₃	4-CN-Phenyl
322	—CH ₂ CH ₂ CF ₃	4-CN-Phenyl
323	cyclopropyl	4-CN-Phenyl
324	Cyclobutyl	4-CN-Phenyl
325	cyclopentyl	4-CN-Phenyl
326	cyclohexyl	4-CN-Phenyl
327	3-pyridyl	4-CN-Phenyl
328	1-methyl-1H-pyrazol-4-yl	4-CN-Phenyl
329	1H-imidazol-4-yl	4-CN-Phenyl
330	2-furanyl	4-CN-Phenyl
331	ethyl	2-CN-Phenyl
332	n-propyl	2-CN-Phenyl
333	isopropyl	2-CN-Phenyl
334	—CH ₂ CH(CH ₃) ₂	2-CN-Phenyl
335	—CF ₃	2-CN-Phenyl
336	—CH ₂ CF ₃	2-CN-Phenyl
337	—CH ₂ CH ₂ CF ₃	2-CN-Phenyl
338	cyclopropyl	2-CN-Phenyl
339	Cyclobutyl	2-CN-Phenyl
340	cyclopentyl	2-CN-Phenyl
341	cyclohexyl	2-CN-Phenyl
342	3-pyridyl	2-CN-Phenyl
343	1-methyl-1H-pyrazol-4-yl	2-CN-Phenyl
344	1H-imidazol-4-yl	2-CN-Phenyl
345	2-furanyl	2-CN-Phenyl
346	ethyl	3-F-Phenyl
347	n-propyl	3-F-Phenyl
348	isopropyl	3-F-Phenyl
349	—CH ₂ CH(CH ₃) ₂	3-F-Phenyl
350	—CF ₃	3-F-Phenyl
351	—CH ₂ CF ₃	3-F-Phenyl
352	—CH ₂ CH ₂ CF ₃	3-F-Phenyl

TABLE 10-continued

Entry	R ^{2d}	R ³
353	cyclopropyl	3-F-Phenyl
354	Cyclobutyl	3-F-Phenyl
355	cyclopentyl	3-F-Phenyl
356	cyclohexyl	3-F-Phenyl
357	3-pyridyl	3-F-Phenyl
358	1-methyl-1H-pyrazol-4-yl	3-F-Phenyl
359	1H-imidazol-4-yl	3-F-Phenyl
360	2-furanyl	3-F-Phenyl
361	ethyl	2-Cl-Phenyl
362	n-propyl	2-Cl-Phenyl
363	isopropyl	2-Cl-Phenyl
364	—CH ₂ CH(CH ₃) ₂	2-Cl-Phenyl
365	—CF ₃	2-Cl-Phenyl
366	—CH ₂ CF ₃	2-Cl-Phenyl
367	—CH ₂ CH ₂ CF ₃	2-Cl-Phenyl
368	cyclopropyl	2-Cl-Phenyl
369	Cyclobutyl	2-Cl-Phenyl
370	cyclopentyl	2-Cl-Phenyl
371	cyclohexyl	2-Cl-Phenyl
372	3-pyridyl	2-Cl-Phenyl
373	1-methyl-1H-pyrazol-4-yl	2-Cl-Phenyl
374	1H-imidazol-4-yl	2-Cl-Phenyl
375	2-furanyl	2-Cl-Phenyl
376	ethyl	4-Cl-Phenyl
377	n-propyl	4-Cl-Phenyl
378	isopropyl	4-Cl-Phenyl
379	—CH ₂ CH(CH ₃) ₂	4-Cl-Phenyl
380	—CF ₃	4-Cl-Phenyl
381	—CH ₂ CF ₃	4-Cl-Phenyl
382	—CH ₂ CH ₂ CF ₃	4-Cl-Phenyl
383	cyclopropyl	4-Cl-Phenyl
384	Cyclobutyl	4-Cl-Phenyl
385	cyclopentyl	4-Cl-Phenyl
386	cyclohexyl	4-Cl-Phenyl
387	3-pyridyl	4-Cl-Phenyl
388	1-methyl-1H-pyrazol-4-yl	4-Cl-Phenyl
389	1H-imidazol-4-yl	4-Cl-Phenyl
390	2-furanyl	4-Cl-Phenyl
391	ethyl	3-Br-Phenyl
392	n-propyl	3-Br-Phenyl
393	isopropyl	3-Br-Phenyl
394	—CH ₂ CH(CH ₃) ₂	3-Br-Phenyl
395	—CF ₃	3-Br-Phenyl
396	—CH ₂ CF ₃	3-Br-Phenyl
397	—CH ₂ CH ₂ CF ₃	3-Br-Phenyl
398	cyclopropyl	3-Br-Phenyl
399	Cyclobutyl	3-Br-Phenyl
400	cyclopentyl	3-Br-Phenyl
401	cyclohexyl	3-Br-Phenyl
402	3-pyridyl	3-Br-Phenyl
403	1-methyl-1H-pyrazol-4-yl	3-Br-Phenyl
404	1H-imidazol-4-yl	3-Br-Phenyl
405	2-furanyl	3-Br-Phenyl
406	ethyl	2-CF ₃ -Phenyl
407	n-propyl	2-CF ₃ -Phenyl
408	isopropyl	2-CF ₃ -Phenyl
409	—CH ₂ CH(CH ₃) ₂	2-CF ₃ -Phenyl
410	—CF ₃	2-CF ₃ -Phenyl
411	—CH ₂ CF ₃	2-CF ₃ -Phenyl
412	—CH ₂ CH ₂ CF ₃	2-CF ₃ -Phenyl
413	cyclopropyl	2-CF ₃ -Phenyl
414	Cyclobutyl	2-CF ₃ -Phenyl
415	cyclopentyl	2-CF ₃ -Phenyl
416	cyclohexyl	2-CF ₃ -Phenyl
417	3-pyridyl	2-CF ₃ -Phenyl
418	1-methyl-1H-pyrazol-4-yl	2-CF ₃ -Phenyl
419	1H-imidazol-4-yl	2-CF ₃ -Phenyl
420	2-furanyl	2-CF ₃ -Phenyl
421	ethyl	4-CF ₃ -Phenyl
422	n-propyl	4-CF ₃ -Phenyl

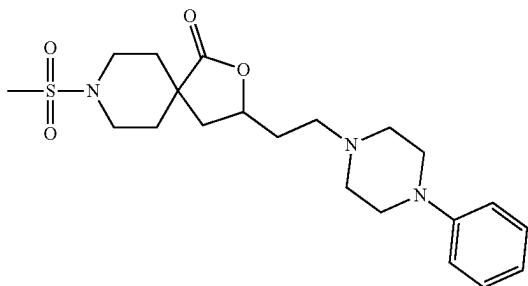
TABLE 10-continued

Entry	R ^{2d}	R ³
423	isopropyl	4-CF ₃ -Phenyl
424	—CH ₂ CH(CH ₃) ₂	4-CF ₃ -Phenyl
425	—CF ₃	4-CF ₃ -Phenyl
426	—CH ₂ CF ₃	4-CF ₃ -Phenyl
427	—CH ₂ CH ₂ CF ₃	4-CF ₃ -Phenyl
428	cyclopropyl	4-CF ₃ -Phenyl
429	Cyclobutyl	4-CF ₃ -Phenyl
430	cyclopentyl	4-CF ₃ -Phenyl
431	cyclohexyl	4-CF ₃ -Phenyl
432	3-pyridyl	4-CF ₃ -Phenyl
433	1-methyl-1H-pyrazol-4-yl	4-CF ₃ -Phenyl
434	1H-imidazol-4-yl	4-CF ₃ -Phenyl
435	2-furanyl	4-CF ₃ -Phenyl
436	ethyl	3-IPr-Phenyl
437	n-propyl	3-IPr-Phenyl
438	isopropyl	3-IPr-Phenyl
439	—CH ₂ CH(CH ₃) ₂	3-IPr-Phenyl
440	—CF ₃	3-IPr-Phenyl
441	—CH ₂ CF ₃	3-IPr-Phenyl
442	—CH ₂ CH ₂ CF ₃	3-IPr-Phenyl
443	cyclopropyl	3-IPr-Phenyl
444	Cyclobutyl	3-IPr-Phenyl
445	cyclopentyl	3-IPr-Phenyl
446	cyclohexyl	3-IPr-Phenyl
447	3-pyridyl	3-IPr-Phenyl
448	1-methyl-1H-pyrazol-4-yl	3-IPr-Phenyl
449	1H-imidazol-4-yl	3-IPr-Phenyl
450	2-furanyl	3-IPr-Phenyl
451	ethyl	2-morpholino-phenyl
452	n-propyl	2-morpholino-phenyl
453	isopropyl	2-morpholino-phenyl
454	—CH ₂ CH(CH ₃) ₂	2-morpholino-phenyl
455	—CF ₃	2-morpholino-phenyl
456	—CH ₂ CF ₃	2-morpholino-phenyl
457	—CH ₂ CH ₂ CF ₃	2-morpholino-phenyl
458	cyclopropyl	2-morpholino-phenyl
459	Cyclobutyl	2-morpholino-phenyl
460	cyclopentyl	2-morpholino-phenyl
461	cyclohexyl	2-morpholino-phenyl
462	3-pyridyl	2-morpholino-phenyl
463	1-methyl-1H-pyrazol-4-yl	2-morpholino-phenyl
464	1H-imidazol-4-yl	2-morpholino-phenyl
465	2-furanyl	2-morpholino-phenyl
466	ethyl	4-morpholino-phenyl
467	n-propyl	4-morpholino-phenyl
468	isopropyl	4-morpholino-phenyl
469	—CH ₂ CH(CH ₃) ₂	4-morpholino-phenyl
470	—CF ₃	4-morpholino-phenyl
471	—CH ₂ CF ₃	4-morpholino-phenyl
472	—CH ₂ CH ₂ CF ₃	4-morpholino-phenyl

TABLE 10-continued

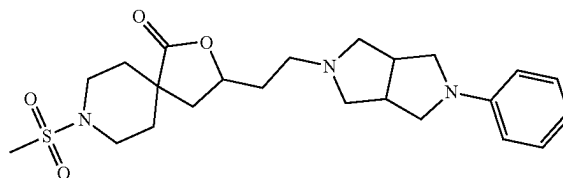
Entry	R ^{2d}	R ³
473	cyclopropyl	4-morpholino-phenyl
474	Cyclobutyl	4-morpholino-phenyl
475	cyclopentyl	4-morpholino-phenyl
476	cyclohexyl	4-morpholino-phenyl
477	3-pyridyl	4-morpholino-phenyl
478	1-methyl-1H-pyrazol-4-yl	4-morpholino-phenyl
479	1H-imidazol-4-yl	4-morpholino-phenyl
480	2-furanyl	4-morpholino-phenyl
481	ethyl	4-methyl-2-morpholino-phenyl
482	n-propyl	4-methyl-2-morpholino-phenyl
483	isopropyl	4-methyl-2-morpholino-phenyl
484	—CH ₂ CH(CH ₃) ₂	4-methyl-2-morpholino-phenyl
485	—CF ₃	4-methyl-2-morpholino-phenyl
486	—CH ₂ CF ₃	4-methyl-2-morpholino-phenyl
487	—CH ₂ CH ₂ CF ₃	4-methyl-2-morpholino-phenyl
488	cyclopropyl	4-methyl-2-morpholino-phenyl
489	Cyclobutyl	4-methyl-2-morpholino-phenyl
490	cyclopentyl	4-methyl-2-morpholino-phenyl
491	cyclohexyl	4-methyl-2-morpholino-phenyl
492	3-pyridyl	4-methyl-2-morpholino-phenyl
493	1-methyl-1H-pyrazol-4-yl	4-methyl-2-morpholino-phenyl
494	1H-imidazol-4-yl	4-methyl-2-morpholino-phenyl
495	2-furanyl	4-methyl-2-morpholino-phenyl

[0410] For the purposes of demonstrating the manner in which the compounds of the present invention are named and referred to herein, the compound having the formula:



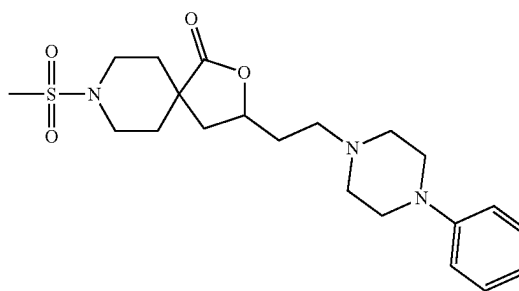
has the chemical name 8-(methylsulfonyl)-3-(2-(4-phenylpiperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one.

[0411] For the purposes of demonstrating the manner in which the compounds of the present invention are named and referred to herein, the compound having the formula:

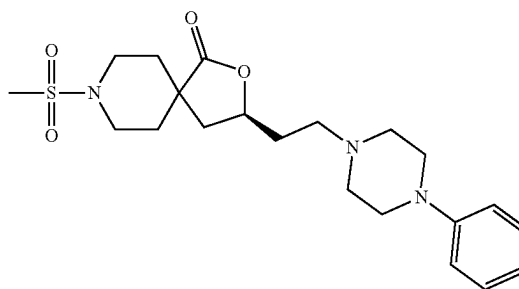


has the chemical name 8-(methylsulfonyl)-3-(2-(5-phenylhexahydropyrrolo[3,4-c]pyrrol-2(1H)-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one.

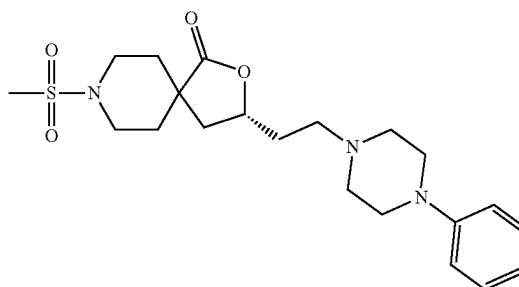
[0412] For the purposes of the present invention, a compound depicted by the racemic formula, for example:



will stand equally well for either of the two enantiomers having the formula:



or the formula:



or mixtures thereof, or in the case where a second chiral center is present, all diastereomers.

[0413] In all of the embodiments provided herein, examples of suitable optional substituents are not intended to limit the scope of the claimed invention. The compounds

of the invention may contain any of the substituents, or combinations of substituents, provided herein. Process for preparing the 5-hydroxytryptamine receptor 7 activity modulators of the invention

[0414] The present invention further relates to a process for preparing the 5-hydroxytryptamine receptor 7 activity modulators of the present invention.

[0415] Compounds of the present teachings can be prepared in accordance with the procedures outlined herein, from commercially available starting materials, compounds known in the literature, or readily prepared intermediates, by employing standard synthetic methods and procedures known to those skilled in the art. Standard synthetic methods and procedures for the preparation of organic molecules and functional group transformations and manipulations can be readily obtained from the relevant scientific literature or from standard textbooks in the field. It will be appreciated that where typical or preferred process conditions (i.e., reaction temperatures, times, mole ratios of reactants, solvents, pressures, etc.) are given, other process conditions can also be used unless otherwise stated. Optimum reaction conditions can vary with the particular reactants or solvent used, but such conditions can be determined by one skilled in the art by routine optimization procedures. Those skilled in the art of organic synthesis will recognize that the nature and order of the synthetic steps presented can be varied for the purpose of optimizing the formation of the compounds described herein.

[0416] The processes described herein can be monitored according to any suitable method known in the art. For example, product formation can be monitored by spectroscopic means, such as nuclear magnetic resonance spectroscopy (e.g., ^1H or ^{13}C), infrared spectroscopy, spectrophotometry (e.g., UV-visible), mass spectrometry, or by chromatography such as high pressure liquid chromatography (HPLC), gas chromatography (GC), gel-permeation chromatography (GPC), or thin layer chromatography (TLC).

[0417] Preparation of the compounds can involve protection and deprotection of various chemical groups. The need for protection and deprotection and the selection of appropriate protecting groups can be readily determined by one skilled in the art. The chemistry of protecting groups can be found, for example, in Greene et al., *Protective Groups in Organic Synthesis*, 2d. Ed. (Wiley & Sons, 1991), the entire disclosure of which is incorporated by reference herein for all purposes.

[0418] The reactions or the processes described herein can be carried out in suitable solvents which can be readily selected by one skilled in the art of organic synthesis. Suitable solvents typically are substantially nonreactive with the reactants, intermediates, and/or products at the temperatures at which the reactions are carried out, i.e., temperatures that can range from the solvent's freezing temperature to the solvent's boiling temperature. A given reaction can be carried out in one solvent or a mixture of more than one solvent. Depending on the particular reaction step, suitable solvents for a particular reaction step can be selected.

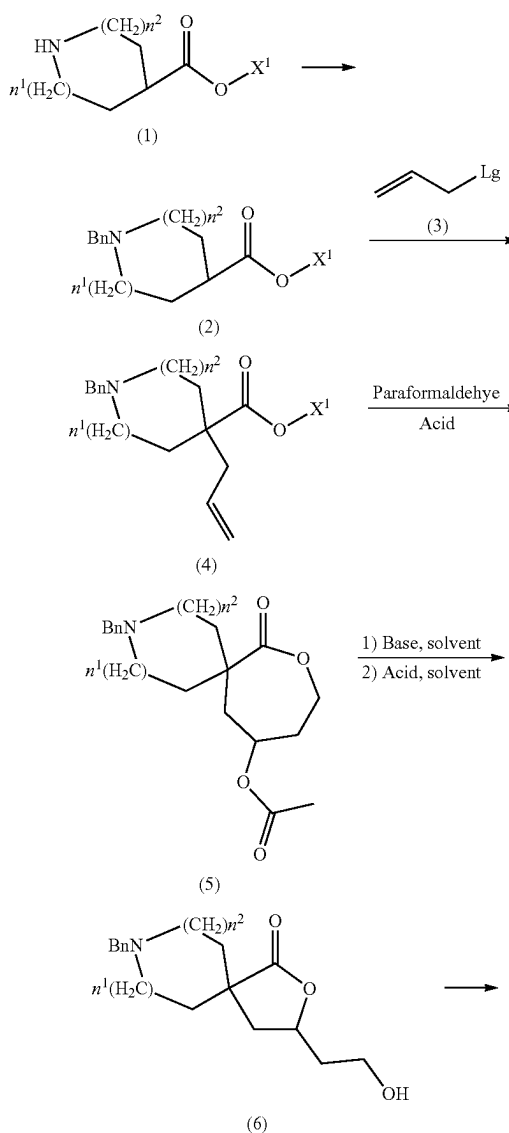
[0419] The compounds of these teachings can be prepared by methods known in the art of organic chemistry. The reagents used in the preparation of the compounds of these teachings can be either commercially obtained or can be prepared by standard procedures described in the literature.

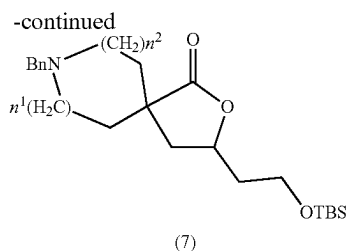
For example, compounds of the present invention can be prepared according to the method illustrated in the General Synthetic Schemes:

General Synthetic Schemes for Preparation of Compounds

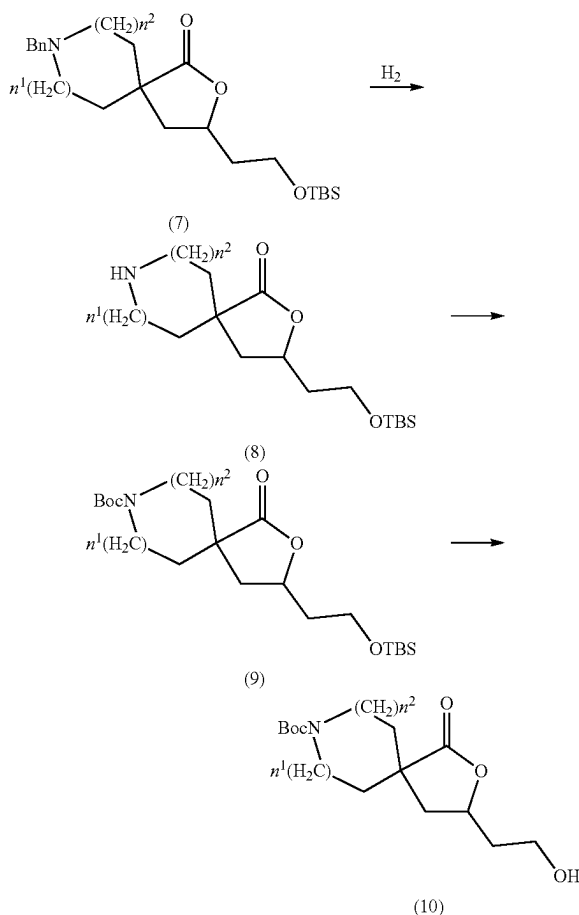
[0420] The reagents used in the preparation of the compounds of this invention can be either commercially obtained or can be prepared by standard procedures described in the literature. In accordance with this invention, compounds in the genus may be produced by one of the following reaction schemes.

[0421] Compounds of the disclosure may be prepared according to the process outlined in Scheme 1-x.

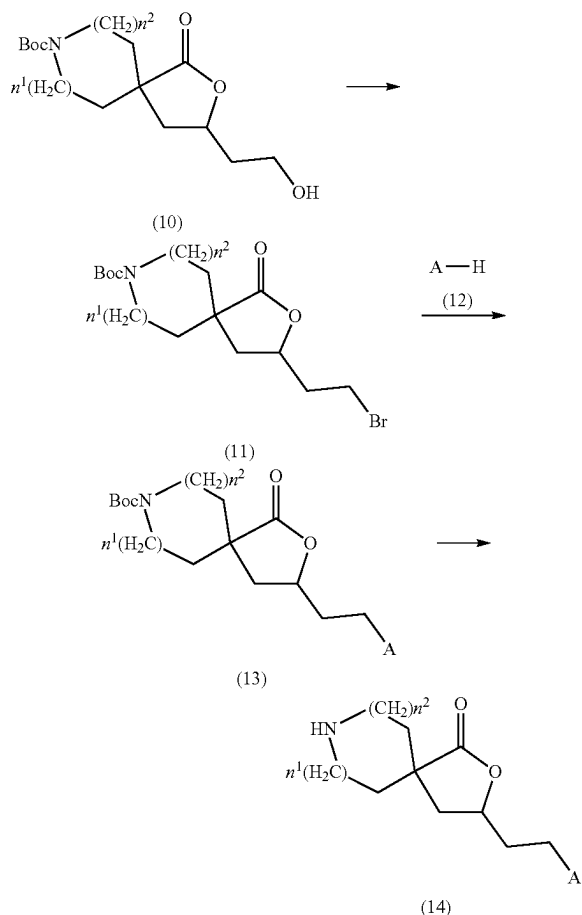




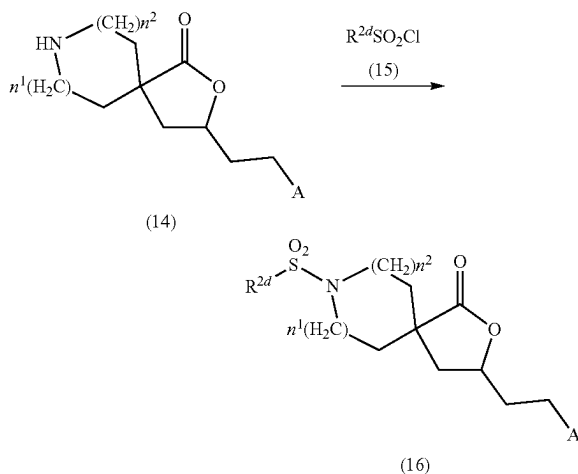
[0422] A suitably substituted compound of formula (1), a known compound or compound prepared by known methods wherein X^1 is an C_{1-6} alkyl, is reacted with benzyl bromide in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6-lutidine, and the like in the presence of a solvent such as ethanol, methanol, isopropanol, tetrahydrofuran, 1,4-dioxane, methylene chloride, N,N-dimethyl formamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (2). A compound of the formula (2) is reacted with a compound of the formula (3), a known compound or a compound prepared by known methods in which LG is a leaving group such as chlorine, bromine, iodine, mesylate, tosylate, and the like, in the presence of a base such as lithium diisopropylamide, sodium diisopropylamide, potassium diisopropylamide, lithium bis(trimethylsilyl)amide, sodium bis(trimethylsilyl)amide, potassium bis(trimethylsilyl)amide, sodium hydride, and the like in an organic solvent such as tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, dimethylformamide, dimethylacetamide, and the like, to provide a compound of the formula (4). A compound of the formula (4) is then treated with paraformaldehyde in the presence of an acid such as sulfuric acid, hydrochloric acid, and the like, in the presence of acetic acid, and optionally in an organic solvent such as methanol, ethanol, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (5). A compound of the formula (5) is then treated with a base such as sodium hydroxide, potassium hydroxide, lithium hydroxide, and the like, in a solvent such as water, methanol, ethanol, isopropanol, and the like, optionally with heating, and then treated with an acid such as sulfuric acid, hydrochloric acid, and the like, in a solvent such as water, methanol, ethanol, isopropanol, and the like, to provide a compound of the formula (6). A compound of the formula (6) is reacted with tert-butyl dimethylchlorosilane in the presence of imidazole, in the presence of a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (7). Alternatively, a compound of the formula (6) is reacted with tert-butyl dimethylsilyl trifluoromethanesulfonate in the presence of a base such as pyridine, 2,6-lutidine, triethylamine, diisopropylethylamine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (7).



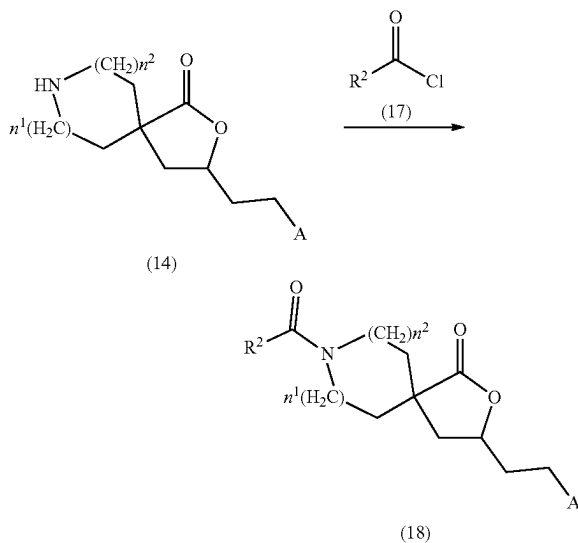
[0423] A compound of the formula (7) is reacted with hydrogen gas in the presence of a palladium catalyst such as palladium on carbon, palladium on barium sulfate, palladium (II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis (triphenylphosphine)palladium(II), palladium on carbon, bis(acetonitrile)dichloropalladium(II), and the like, in an organic solvent such as methanol, ethanol, ethyl acetate, tetrahydrofuran, 1,4-dioxane, dichloromethane, chloroform, 1,2-dichloroethane, N,N-dimethylformamide, and the like, to provide a compound of the formula (8). A compound of the formula (8) is reacted with Di-tert-butyl dicarbonate in the presence of a base such as pyridine, 2,6-lutidine, triethylamine, diisopropylethylamine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (9). A compound of the formula (9) is reacted with Tetra-n-butylammonium fluoride in the presence of solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (10).



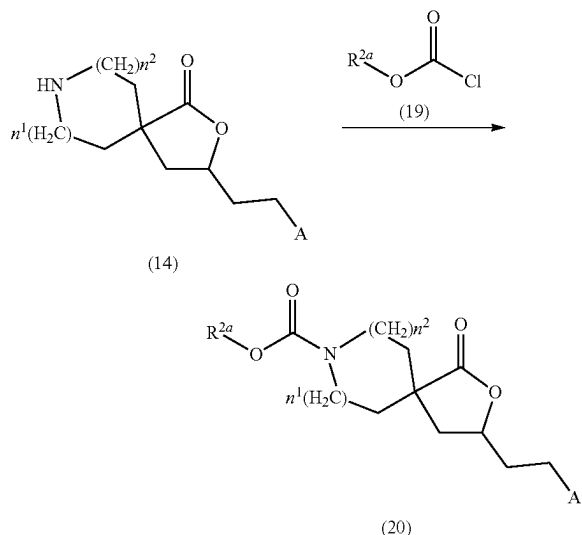
[0424] A compound of the formula (10) is treated with carbon tetrabromide in the presence of triphenylphosphine, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (11). A compound of the formula (11) is reacted with a compound of the formula (12), a known compound or a compound prepared by known methods, in the presence of a base such as sodium carbonate, potassium carbonate, lithium carbonate, sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (13). A compound of the formula (13) is reacted with an acid such as trifluoroacetic acid, formic acid, acetic acid, hydrochloric acid, sulfuric acid, and the like, optionally in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (14).



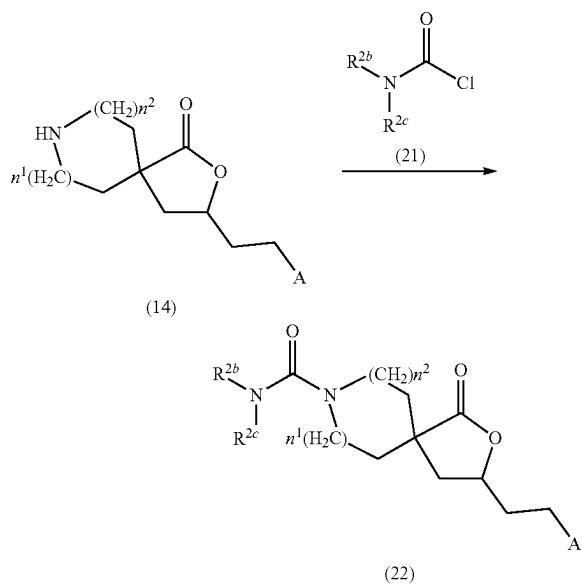
[0425] A compound of the formula (14) is reacted with a compound of the formula (15), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (16).



[0426] A compound of the formula (14) is reacted with a compound of the formula (17), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (18).

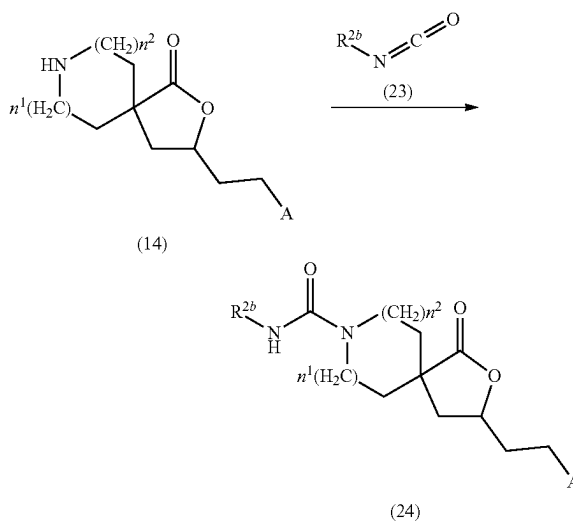


[0427] A compound of the formula (14) is reacted with a compound of the formula (19), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (20).

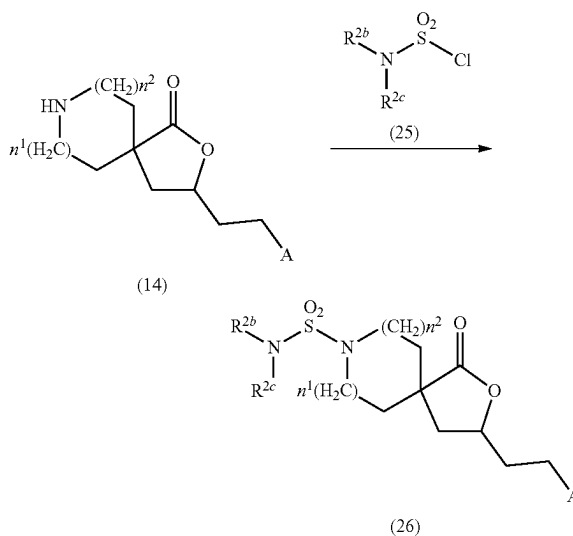


[0428] A compound of the formula (14) is reacted with a compound of the formula (21), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dime-

thoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (22).

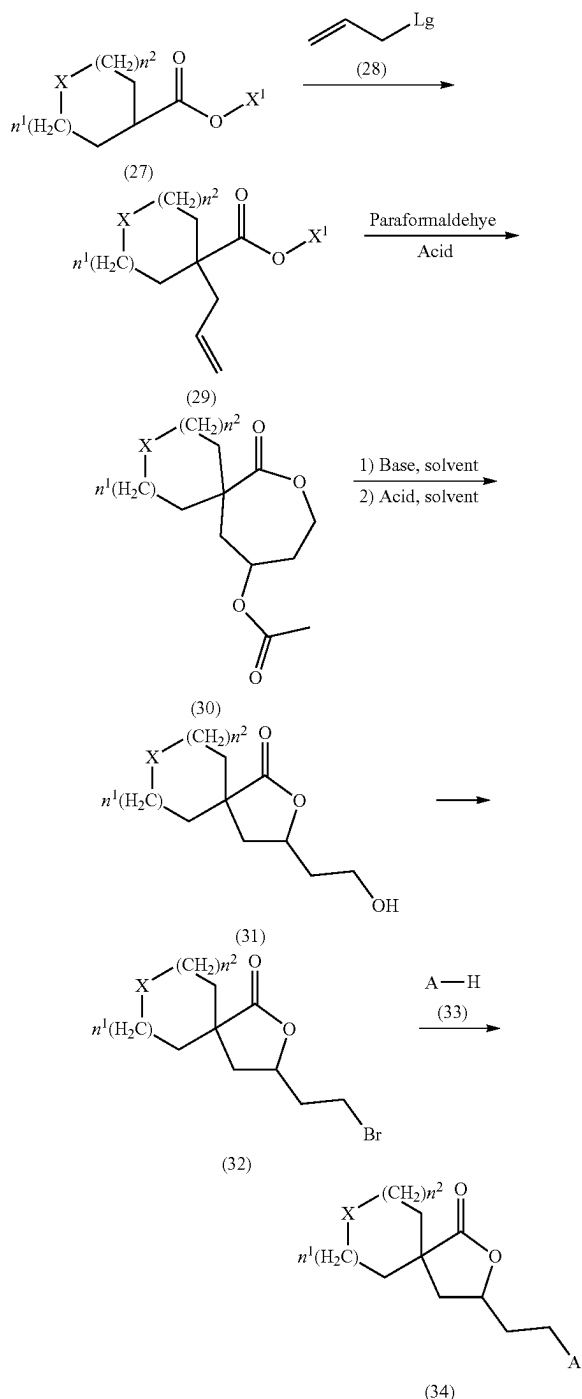


[0429] A compound of the formula (14) is reacted with a compound of the formula (23), a known compound or a compound prepared by known methods, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (24).

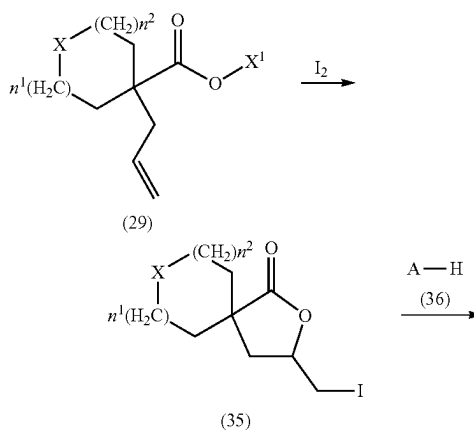


[0430] A compound of the formula (14) is reacted with a compound of the formula (25), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dime-

thoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (26).

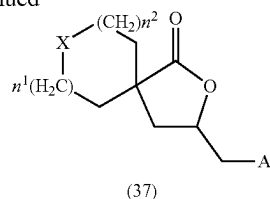


prepared by known methods in which LG is a leaving group such as chlorine, bromine, iodine, mesylate, tosylate, and the like, in the presence of a base such as lithium diisopropylamide, sodium diisopropylamide, potassium diisopropylamide, lithium bis(trimethylsilyl)amide, sodium bis(trimethylsilyl)amide, potassium bis(trimethylsilyl)amide, sodium hydride, and the like in an organic solvent such as tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, dimethylformamide, dimethylacetamide, and the like, to provide a compound of the formula (29). A compound of the formula (29) is then treated with paraformaldehyde in the presence of an acid such as sulfuric acid, hydrochloric acid, and the like, in the presence of acetic acid, and optionally in an organic solvent such as methanol, ethanol, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (30). A compound of the formula (30) is then treated with a base such as sodium hydroxide, potassium hydroxide, lithium hydroxide, and the like, in a solvent such as water, methanol, ethanol, isopropanol, and the like, optionally with heating, and then treated with an acid such as sulfuric acid, hydrochloric acid, and the like, in a solvent such as water, methanol, ethanol, isopropanol, and the like, to provide a compound of the formula (31). A compound of the formula (31) is treated with carbon tetrabromide in the presence of triphenylphosphine, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (32). A compound of the formula (32) is reacted with a compound of the formula (33), a known compound or a compound prepared by known methods, in the presence of a base such as sodium carbonate, potassium carbonate, lithium carbonate, sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (34).

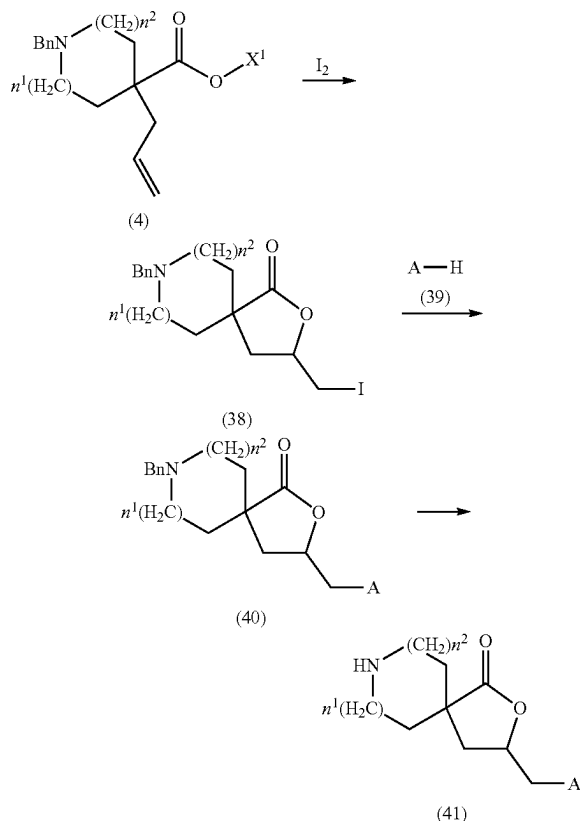


[0431] A suitably substituted compound of formula (27), a known compound or compound prepared by known methods wherein X^1 is an C_{1-6} alkyl, is reacted with a compound of the formula (28), a known compound or a compound

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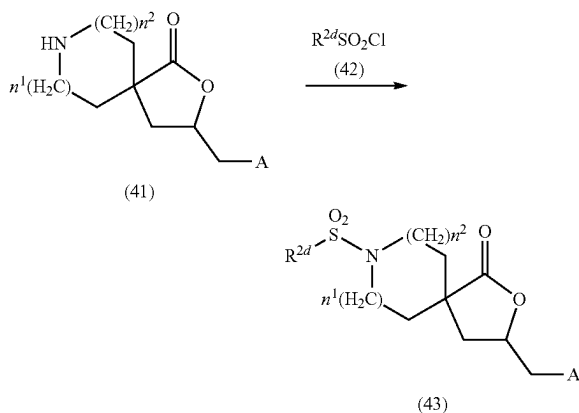


[0432] A compound of the formula (29) is then reacted with iodine in the presence of a base such as sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, sodium carbonate, potassium carbonate, lithium bicarbonate, sodium hydroxide, potassium hydroxide, lithium hydroxide, and the like, in the presence of a solvent such as tetrahydrofuran, ethyl ether, 1,4-dioxane, and the like to provide a compound of the formula (35). A compound of the formula (35) is reacted with a compound of the formula (36), a known compound or compound prepared by known methods, in an organic solvent such as tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, dimethylformamide, dimethylacetamide, and the like, optionally in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6 lutidine, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (37).

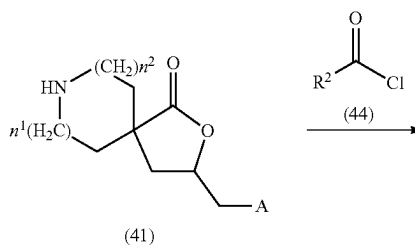


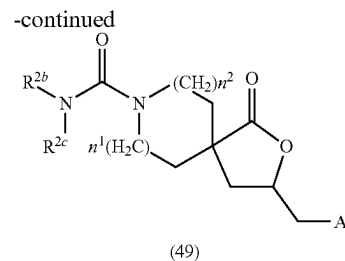
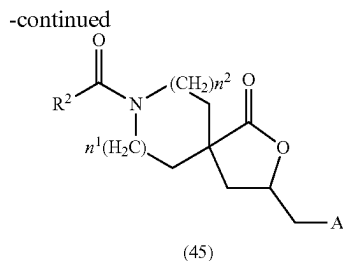
[0433] A compound of the formula (4) is then reacted with iodine in the presence of a base such as sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, sodium carbon-

ate, potassium carbonate, lithium bicarbonate, sodium hydroxide, potassium hydroxide, lithium hydroxide, and the like, in the presence of a solvent such as tetrahydrofuran, ethyl ether, 1,4-dioxane, and the like to provide a compound of the formula (38). A compound of the formula (38) is reacted with a compound of the formula (39), a known compound or compound prepared by known methods, in an organic solvent such as tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, dimethylformamide, dimethylacetamide, and the like, optionally in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6 lutidine, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (40). A compound of the formula (40) is reacted with hydrogen gas in the presence of a palladium catalyst such as palladium on carbon, palladium on barium sulfate, palladium (II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis(triphenylphosphine)palladium(II), palladium on carbon, bis(acetonitrile)dichloropalladium(II), and the like, in an organic solvent such as methanol, ethanol, ethyl acetate, tetrahydrofuran, 1,4-dioxane, dichloromethane, chloroform, 1,2-dichloroethane, N,N-dimethylformamide, and the like, to provide a compound of the formula (41).

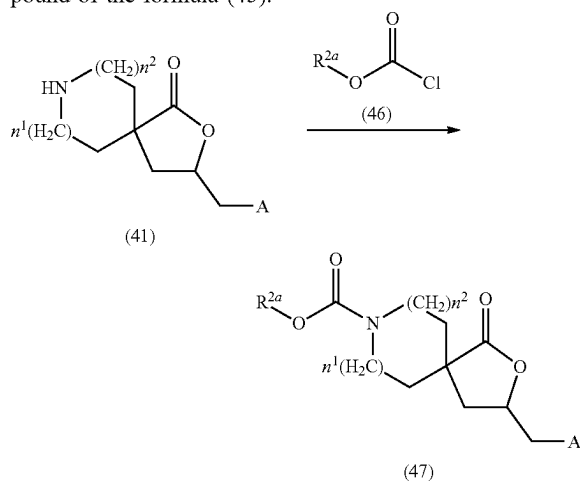


[0434] A compound of the formula (41) is reacted with a compound of the formula (42), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (43).

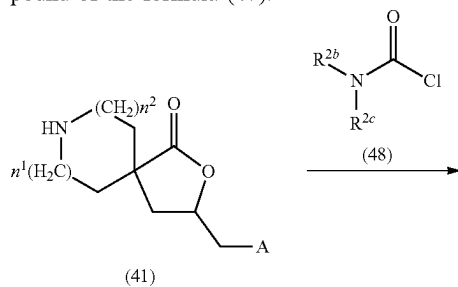




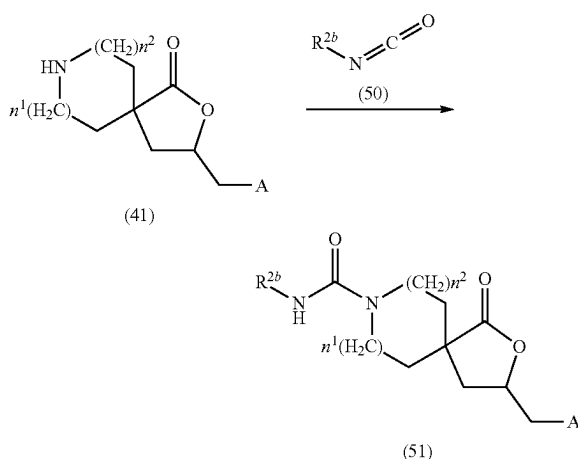
[0435] A compound of the formula (41) is reacted with a compound of the formula (44), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (45).



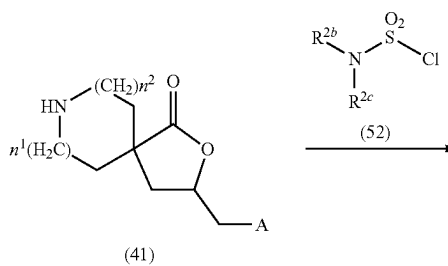
[0436] A compound of the formula (41) is reacted with a compound of the formula (46), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (47).

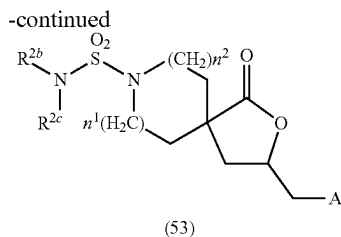


[0437] A compound of the formula (41) is reacted with a compound of the formula (48), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (49).

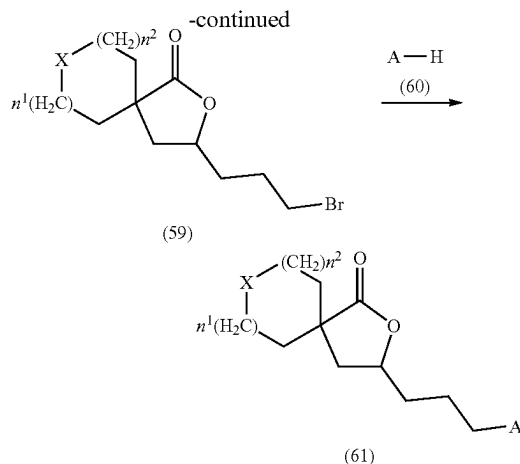
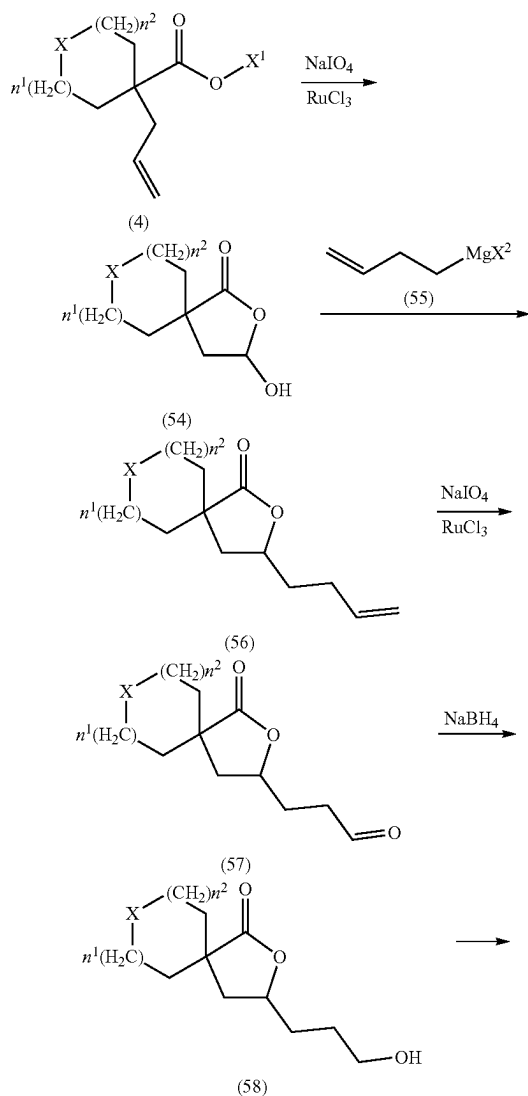


[0438] A compound of the formula (41) is reacted with a compound of the formula (50), a known compound or a compound prepared by known methods, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (51).

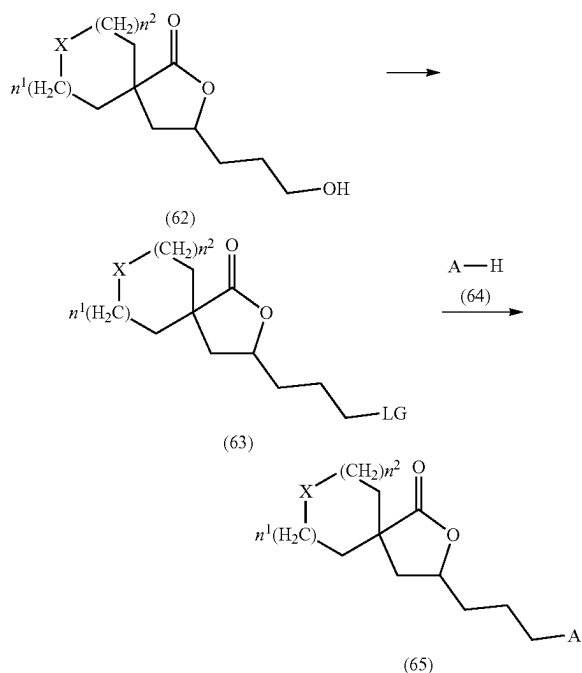




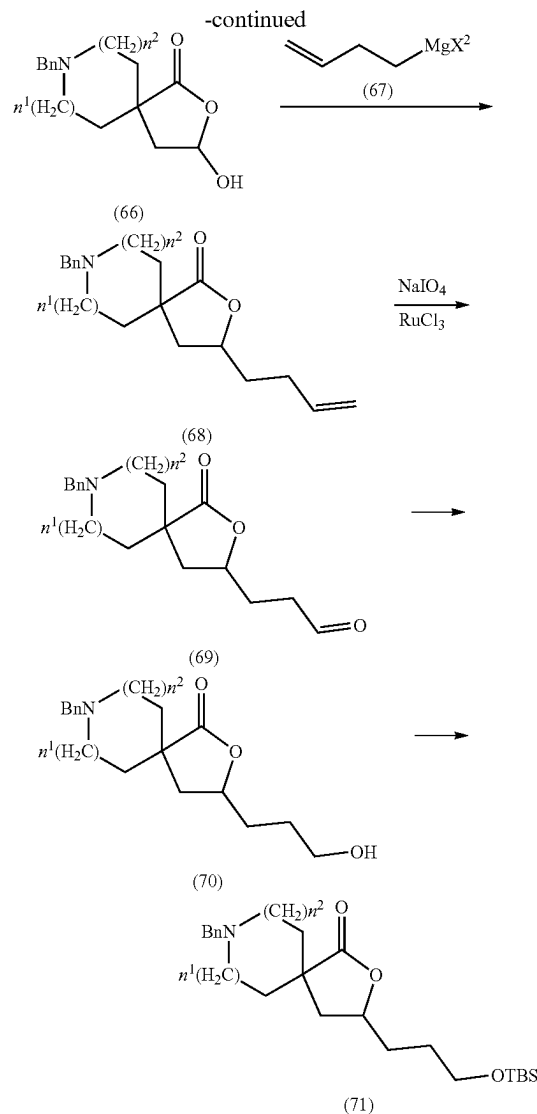
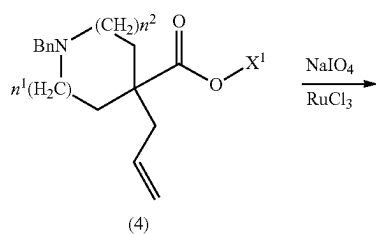
[0439] A compound of the formula (41) is reacted with a compound of the formula (52), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (53).



[0440] A compound of the formula (4) is reacted with ruthenium chloride in the presence of sodium periodate in a solvent such as acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (54). A compound of the formula (54) is reacted with a compound of the formula (55), a known compound or compound prepared by known methods, wherein X² is a halogen, in the presence of a solvent such as ethyl ether, tetrahydrofuran, 1,4-dioxane and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (56). A compound of the formula (56) is reacted with ruthenium chloride in the presence of sodium periodate in a solvent such as acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (57). A compound of the formula (57) is reacted with a reducing agent such as lithium borohydride, sodium borohydride, sodium cyanoborohydride and the like, in a solvent such as methanol, ethanol, isopropanol, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (59). A compound of the formula (59) is reacted with a compound of the formula (60), a known compound or a compound prepared by known methods, in the presence of a base such as sodium carbonate, potassium carbonate, lithium carbonate, sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (61).

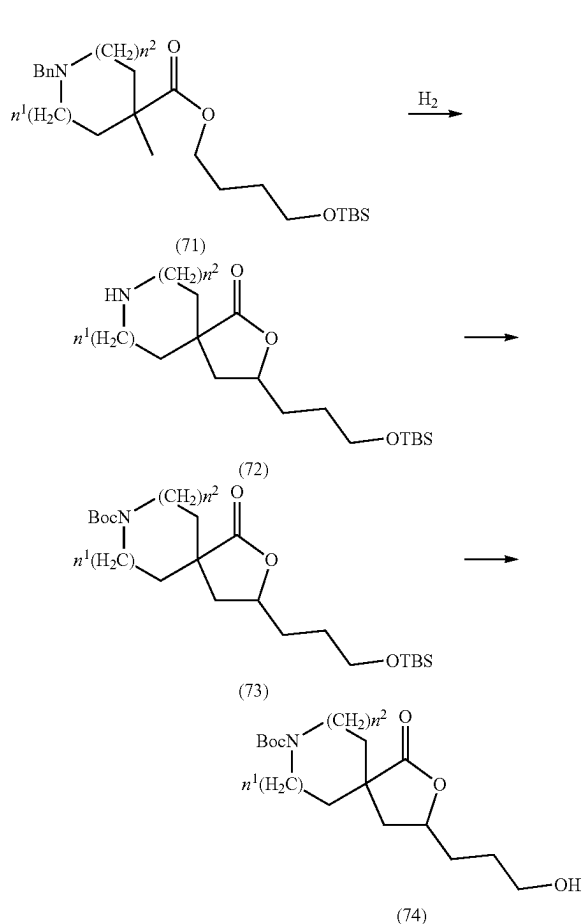


[0441] A compound of the formula (62) is then converted to a compound of the formula (63), wherein LG is a mesylate, tosylate, nosylate, and the like, using methods that are known to one skilled in the art. Thus, a compound of the formula (62) is treated with a sulfonyl chloride such as methanesulfonyl chloride, toluenesulfonyl chloride p-nitrophenyl sulfonyl chloride, and the like, in the presence of a base such as triethylamine, diisopropyl amine, pyridine, 2,6-lutidine, and the like, in an organic solvent such as methylene chloride, dichloroethane, tetrahydrofuran, 1,4-dioxane, N, N-dimethylformamide, tetrahydrofuran, 1,4-dioxane and the like to provide a compound of the formula (63). A compound of the formula (63) is reacted with a compound of the formula (64), a known compound or a compound prepared by known methods, in the presence of a base such as sodium carbonate, potassium carbonate, lithium carbonate, sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (65).



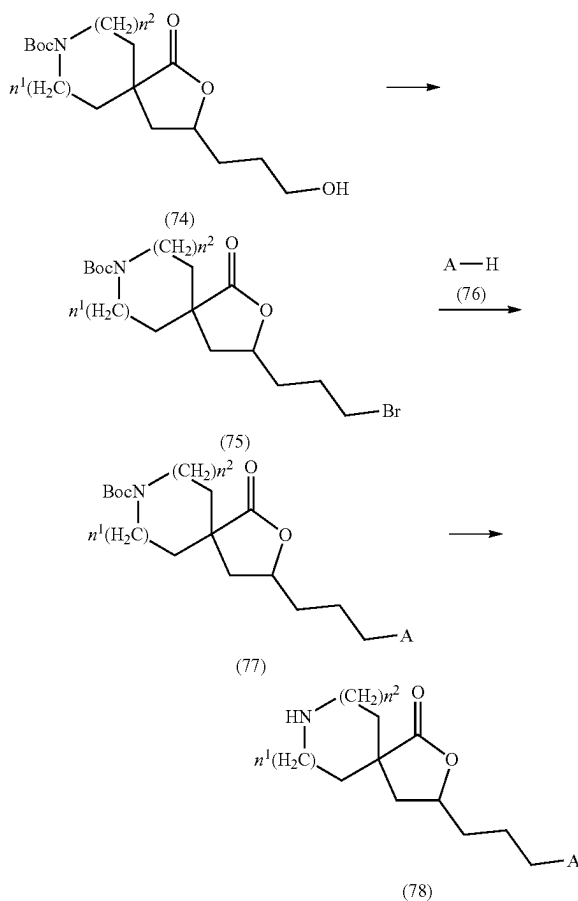
[0442] A compound of the formula (4) is reacted with ruthenium chloride in the presence of sodium periodate in a solvent such as acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (66). A compound of the formula (66) is reacted with a compound of the formula (67), a known compound or compound prepared by known methods, wherein X² is a halogen, in the presence of a solvent such as ethyl ether, tetrahydrofuran, 1,4-dioxane and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (68). A compound of the formula (68) is reacted with ruthenium chloride in the presence of sodium periodate in a solvent such as acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (69). A compound of the formula (69) is reacted with a reducing agent such as lithium borohydride, sodium borohydride, sodium cyanoborohydride and the like, in a solvent such as methanol, ethanol, isopropanol, acetonitrile, and the like, optionally with heating, optionally with microwave

irradiation, to provide a compound of the formula (70). A compound of the formula (70) is reacted with tert-butyldimethylchlorosilane in the presence of imidazole, in the presence of a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (71). Alternatively, a compound of the formula (70) is reacted with tert-butyldimethylsilyl trifluoromethanesulfonate in the presence of a base such as pyridine, 2,6-lutidine, triethylamine, diisopropylethylamine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (71).



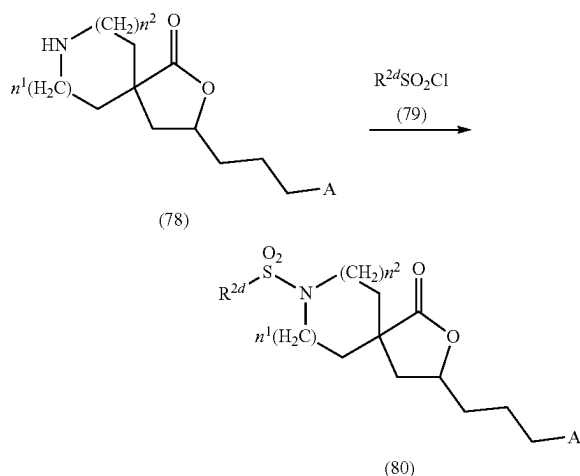
[0443] A compound of the formula (71) is reacted with hydrogen gas in the presence of a palladium catalyst such as palladium on carbon, palladium on barium sulfate, palladium (II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis (triphenylphosphine)palladium(II), palladium on carbon, bis(acetonitrile)dichloropalladium(II), and the like, in an organic solvent such as methanol, ethanol, ethyl acetate, tetrahydrofuran, 1,4-dioxane, dichloromethane, chloroform, 1,2-dichloroethane, N,N-dimethylformamide, and the like, to provide a compound of the formula (72). A

compound of the formula (72) is reacted with Di-tert-butyl dicarbonate in the presence of a base such as pyridine, 2,6-lutidine, triethylamine, diisopropylethylamine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (73). A compound of the formula (73) is reacted with Tetra-n-butylammonium fluoride in the presence of solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (74).

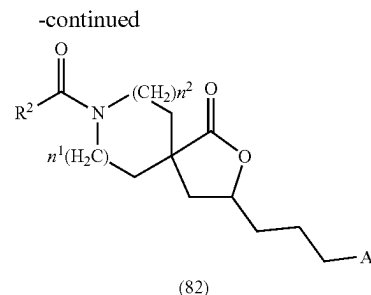
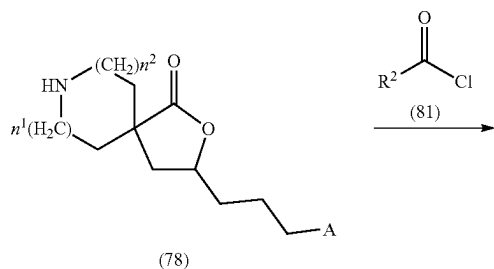


[0444] A compound of the formula (74) is treated with carbon tetrabromide in the presence of triphenylphosphine, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (75). A compound of the formula (75) is reacted with a compound of the formula (76), a known compound or a compound prepared by known methods, in the presence of a base such as sodium carbonate, potassium carbonate, lithium carbonate, sodium

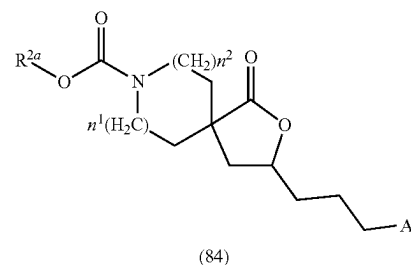
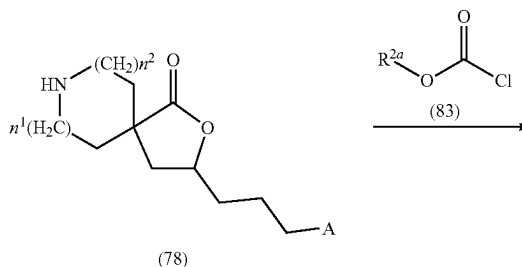
bicarbonate, potassium bicarbonate, lithium bicarbonate, triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (77). A compound of the formula (77) is reacted with an acid such as trifluoroacetic acid, formic acid, acetic acid, hydrochloric acid, sulfuric acid, and the like, optionally in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (78).



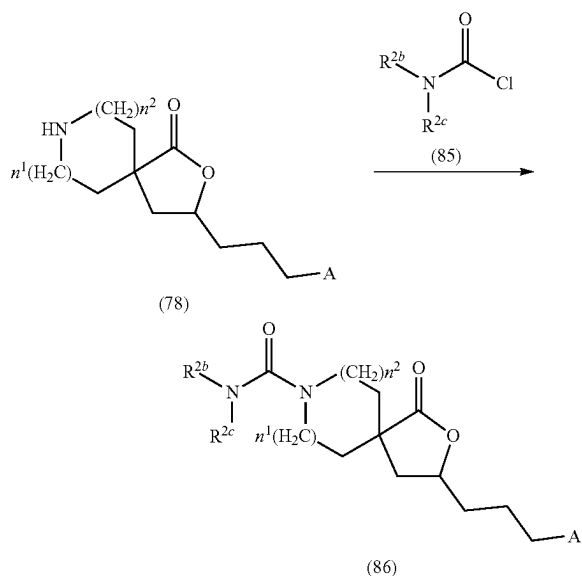
[0445] A compound of the formula (78) is reacted with a compound of the formula (79), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (80).



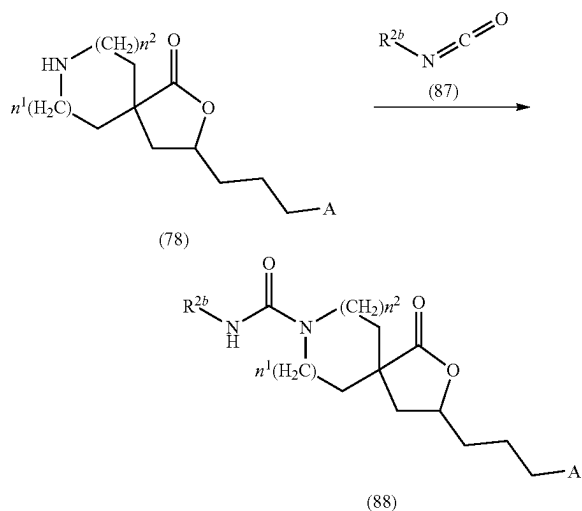
[0446] A compound of the formula (78) is reacted with a compound of the formula (81), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (82).



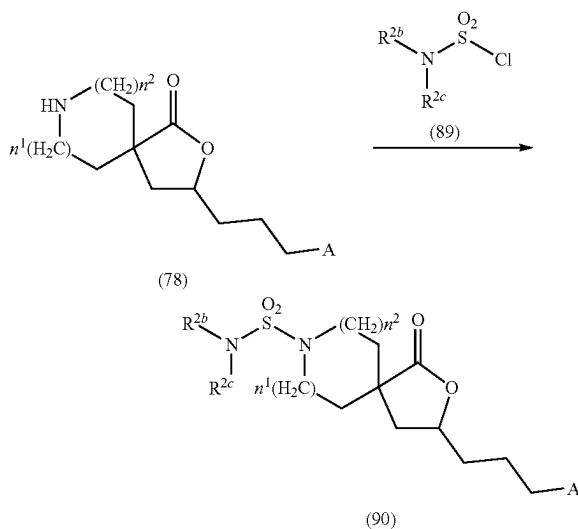
[0447] A compound of the formula (78) is reacted with a compound of the formula (83), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (84).



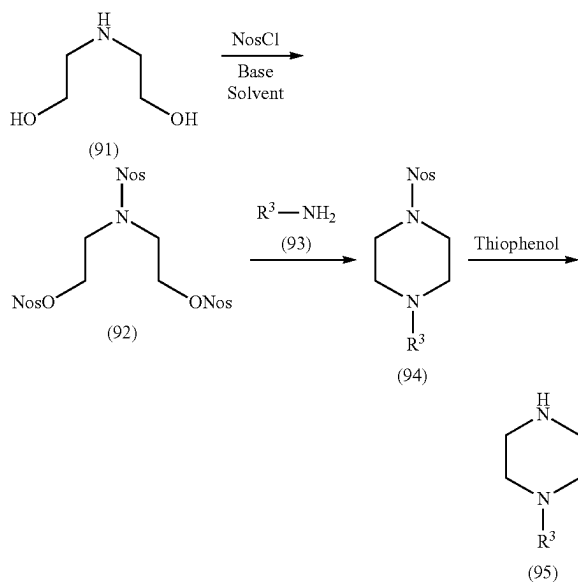
[0448] A compound of the formula (78) is reacted with a compound of the formula (85), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (86).



[0449] A compound of the formula (78) is reacted with a compound of the formula (87), a known compound or a compound prepared by known methods, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (88).

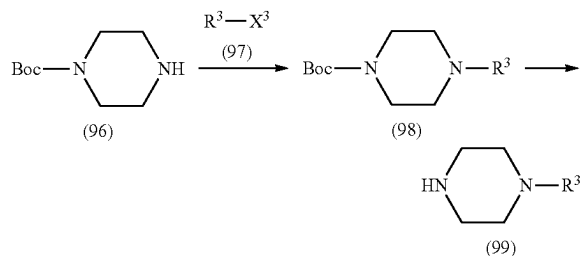


[0450] A compound of the formula (78) is reacted with a compound of the formula (89), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (90).

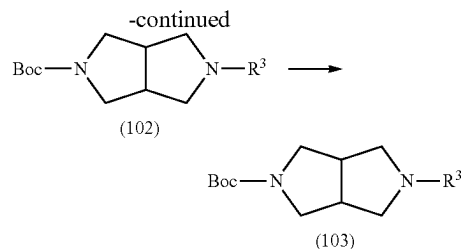
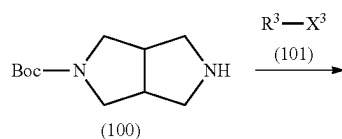


[0451] Diethanolamine (91) is reacted with 4-nitrobenzenesulfonyl chloride (NosCl) in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6-lutidine, and the like in a solvent such as tetrahydrofuran, 1,4-dioxane, methylene chloride and the like to provide a compound of the formula (92). A compound of the formula (92) is then reacted with a compound of the formula (93), a

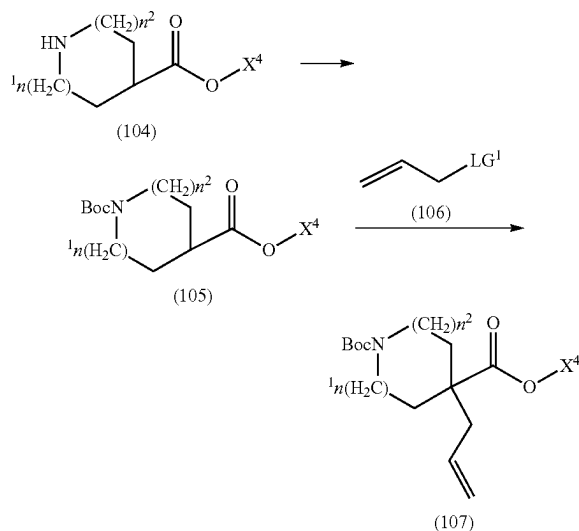
known compound or one prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6-lutidine, and the like, in a solvent such as acetonitrile, methanol, ethanol, dimethyl formamide, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (94). A compound of the formula (94) is reacted with a thiophenol in the presence of a base such as sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, sodium carbonate, potassium carbonate, lithium bicarbonate, sodium hydroxide, potassium hydroxide, lithium hydroxide, and the like, in the presence of a solvent such as tetrahydrofuran, ethyl ether, 1,4-dioxane, acetonitrile and the like, optionally in the presence of dimethylsulfoxide, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (95).



[0452] A compound of the formula (96), a known compound or a compound prepared by known methods, is reacted with a compound of the formula (97), a known compound or a compound prepared by known methods in which X^3 is selected from the group consisting of chlorine, bromine, iodine, and methanetrifluorosulfonate, in the presence of a base such as sodium tert-butoxide, lithium tert-butoxide, potassium tert-butoxide, and the like, optionally in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6-lutidine, and the like, in the presence of a palladium catalyst such as palladium (II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis(triphenylphosphine)palladium(II), palladium on carbon, bis(acetonitrile)dichloropalladium(II), tris(dibenzylideneacetone)dipalladium(0), and the like, in the presence of a solvent such as toluene, benzene, methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, N,N-dimethylformamide, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (98). A compound of the formula (98) is reacted with an acid such as trifluoroacetic acid, formic acid, acetic acid, hydrochloric acid, sulfuric acid, and the like, optionally in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (99).

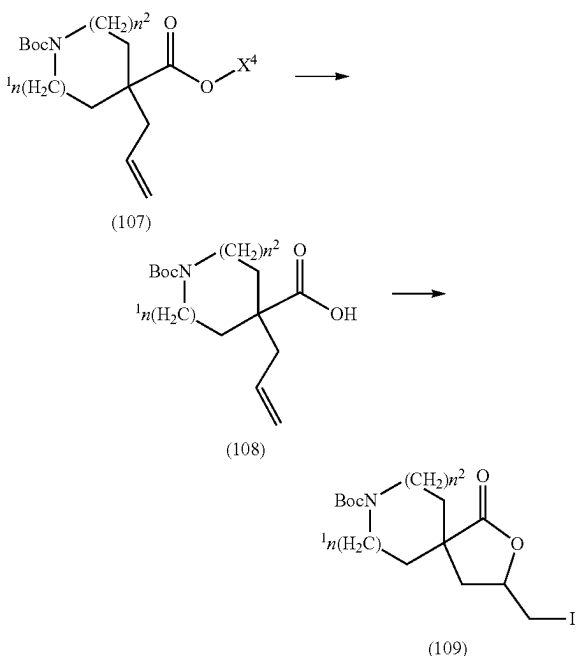


[0453] A compound of the formula (100), a known compound or a compound prepared by known methods, is reacted with a compound of the formula (101), a known compound or a compound prepared by known methods in which X^3 is selected from the group consisting of chlorine, bromine, iodine, and methanetrifluorosulfonate, in the presence of a base such as sodium tert-butoxide, lithium tert-butoxide, potassium tert-butoxide, and the like, optionally in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6-lutidine, and the like, in the presence of a palladium catalyst such as palladium (II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis(triphenylphosphine)palladium(II), palladium on carbon, bis(acetonitrile)dichloropalladium(II), tris(dibenzylideneacetone)dipalladium(0), and the like, in the presence of a solvent such as toluene, benzene, methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, N,N-dimethylformamide, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (102). A compound of the formula (102) is reacted with an acid such as trifluoroacetic acid, formic acid, acetic acid, hydrochloric acid, sulfuric acid, and the like, optionally in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (103).

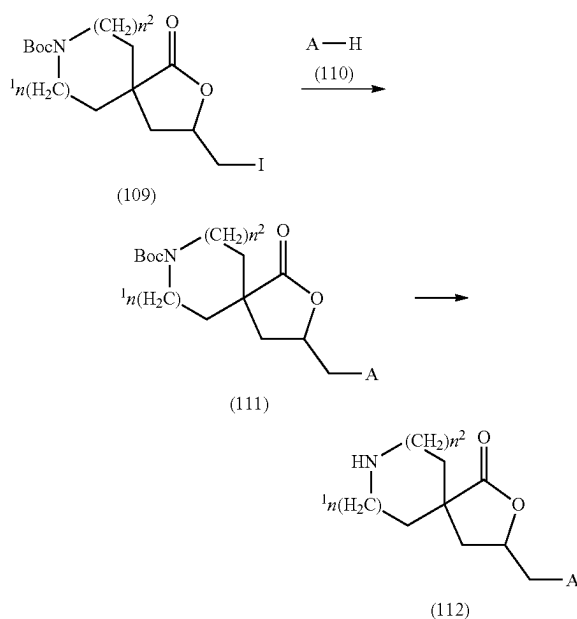


[0454] A compound of the formula (104), a known compound or a compound prepared by known methods in which

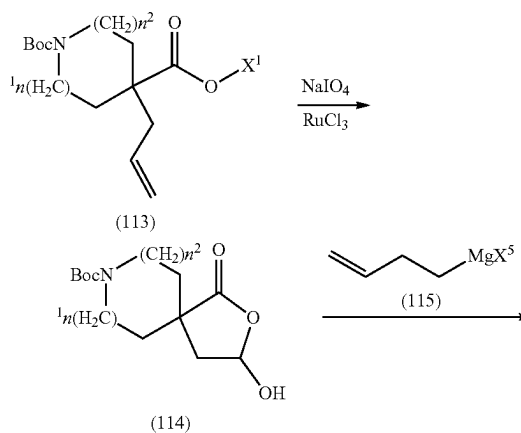
X^4 is an C_{1-6} alkyl, is reacted with di-tert-butyl dicarbonate in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6-lutidine, and the like, in a solvent such as methanol, ethanol, methylene chloride, tetrahydrofuran, 1,4-dioxane, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (105). A compound of the formula (105) is reacted with a compound of the formula (106), a known compound or a compound prepared by known methods where in LG^1 is a leaving group such as chlorine, bromine, iodine, mesylate, tosylate, and the like, in the presence of a base such as lithium diisopropylamide, sodium diisopropylamide, potassium diisopropylamide, lithium bis(trimethylsilyl)amide, sodium bis(trimethylsilyl)amide, potassium bis(trimethylsilyl)amide, sodium hydride, and the like in an organic solvent such as tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, dimethylformamide, dimethylacetamide, and the like, to provide a compound of the formula (107).

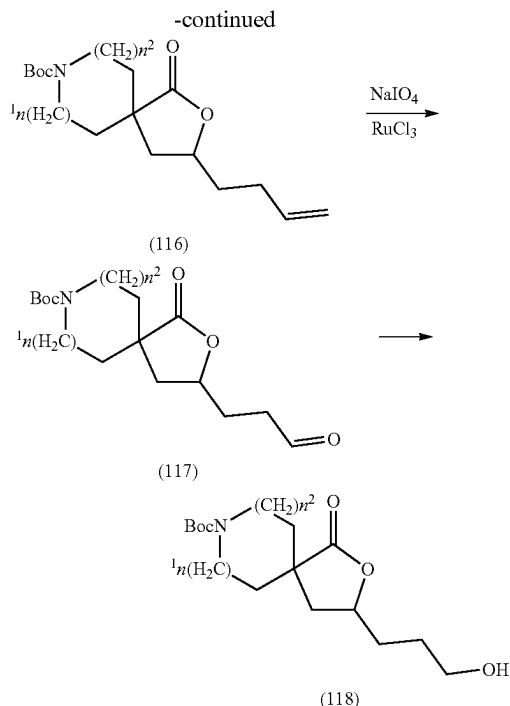


[0455] A compound of the formula (107) is reacted with a base such as sodium hydroxide, lithium hydroxide, potassium hydroxide, sodium carbonate, lithium carbonate, potassium carbonate, and the like in a solvent such as methanol, ethanol, isopropanol, tetrahydrofuran, 1,4-dioxane, N,N-dimethylformamide, and the like, optionally in the presence of water, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (108). A compound of the formula (108) is then reacted with iodine in the presence of a base such as sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, sodium carbonate, potassium carbonate, lithium carbonate, sodium hydroxide, potassium hydroxide, lithium hydroxide, and the like, in the presence of a solvent such as tetrahydrofuran, ethyl ether, 1,4-dioxane, and the like to provide a compound of the formula (109).

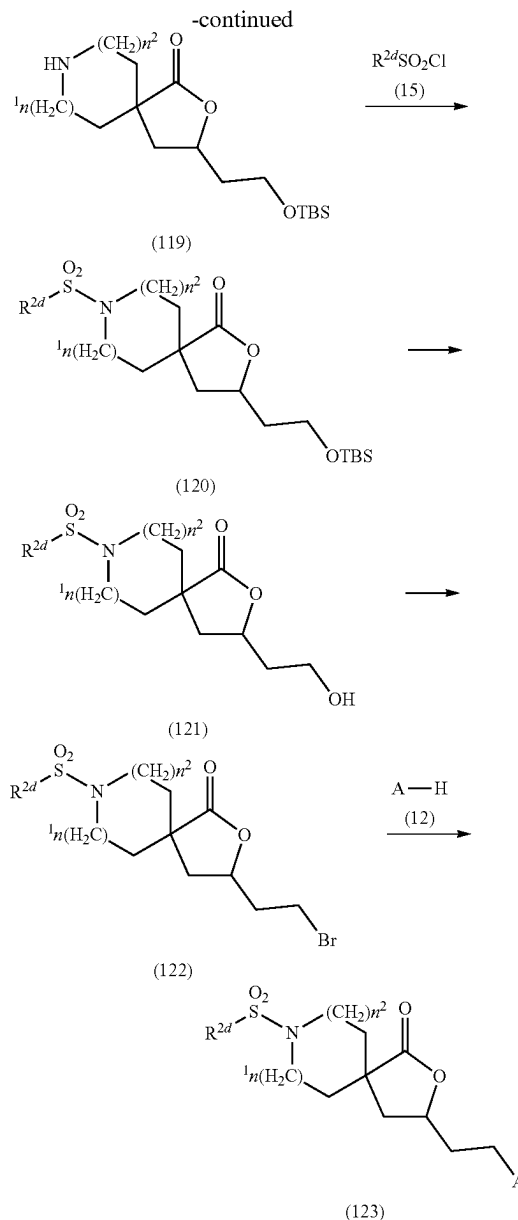
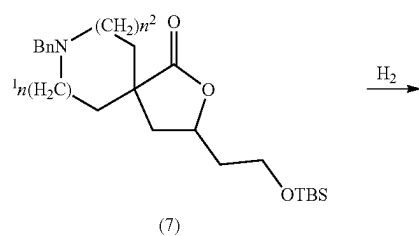


[0456] A compound of the formula (109) is reacted with a compound of the formula (110), a known compound or compound prepared by known methods, in an organic solvent such as tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, dimethylformamide, dimethylacetamide, and the like, optionally in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, 2,6 lutidine, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (111). A compound of the formula (111) is reacted with an acid such as trifluoroacetic acid, formic acid, acetic acid, hydrochloric acid, sulfuric acid, and the like, optionally in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (112).





[0457] A compound of the formula (113) is reacted with ruthenium chloride in the presence of sodium periodate in a solvent such as acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (114). A compound of the formula (114) is reacted with a compound of the formula (115), a known compound or compound prepared by known methods, wherein X^5 is a halogen, in the presence of a solvent such as ethyl ether, tetrahydrofuran, 1,4-dioxane and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (116). A compound of the formula (116) is reacted with ruthenium chloride in the presence of sodium periodate in a solvent such as acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation, to provide a compound of the formula (117). A compound of the formula (117) is reacted with a reducing agent such as lithium borohydride, sodium borohydride, sodium cyanoborohydride and the like, in a solvent such as methanol, ethanol, isopropanol, acetonitrile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (118).



[0458] A compound of the formula (7) is reacted with hydrogen gas in the presence of a palladium catalyst such as palladium on carbon, palladium on barium sulfate, palladium (II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis (triphenylphosphine)palladium(II), palladium on carbon, bis(acetonitrile)dichloropalladium(II), and the like, in an organic solvent such as methanol, ethanol, ethyl acetate, tetrahydrofuran, 1,4-dioxane, dichloromethane, chloroform, 1,2-dichloroethane, N,N-dimethylformamide, and the like, to provide a compound of the formula (119). A compound of the formula (119) is reacted with a compound of the formula (15), a known compound or a compound prepared by known methods, in the presence of a base such as triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile,

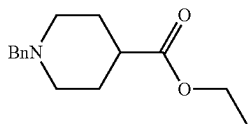
trile, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (120). A compound of the formula (120) is reacted with tetra-n-butylammonium fluoride in the presence of solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (121). A compound of the formula (121) is treated with carbon tetrabromide in the presence of triphenylphosphine, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (122). A compound of the formula (122) is reacted with a compound of the formula (12), a known compound or a compound prepared by known methods, in the presence of a base such as sodium carbonate, potassium carbonate, lithium carbonate, sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, triethylamine, diisopropylethylamine, pyridine, and the like, in a solvent such as methylene chloride, 1,2-dichloroethane, tetrahydrofuran, 1,4-dioxane, 1,2-dimethoxyethane, N,N-dimethylformamide, N,N-dimethylacetamide, acetonitrile, methanol, ethanol, isopropanol, and the like, optionally with heating, optionally with microwave irradiation to provide a compound of the formula (123).

[0459] The Examples provided below provide representative methods for preparing exemplary compounds of the present invention. The skilled practitioner will know how to substitute the appropriate reagents, starting materials and purification methods known to those skilled in the art, in order to prepare the compounds of the present invention.

EXAMPLES

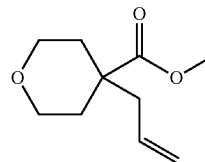
[0460] The practice of the invention is illustrated by the following non-limiting examples. The Examples provided below provide representative methods for preparing exemplary compounds of the present invention. The skilled practitioner will know how to substitute the appropriate reagents, starting materials and purification methods known to those skilled in the art, in order to prepare the compounds of the present invention.

[0461] In the examples that follow, ¹H-NMR spectra were obtained on a Varian Mercury 300-MHz NMR. Purity (%) and mass spectral data were determined with a Waters Alliance 2695 HPLC/MS (Waters Symmetry C18, 4.6×75 mm, 3.5 m) with a 2996 diode array detector from 210-400 nm.

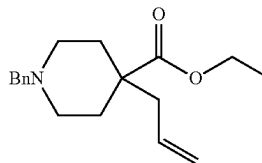


[0462] Example 1: Preparation of ethyl 1-benzylpiperidine-4-carboxylate: To a solution of ethyl piperidine-4-carboxylate (5.0 g, 31.8 mmol, 1.0 eq) and ethanol (15.0 mL), benzyl bromide (7.07 g, mmol, 1.3 eq) was added dropwise at 0° C. Following, triethylamine (1.06 g, 10.5

mmol, 1.5 eq) was added in one portion while at 0° C. The resulting mixture was allowed to warm to RT and stir overnight. The reaction was concentrated in vacuo to remove the presence of ethanol. The resulting residue was suspended in a mixture of ethyl acetate: D.I. water (20 mL:20 mL). The organic layer was separated and the aqueous layer was extracted with ethyl acetate (2×10 mL). The combined extract was dried over Na₂SO₄, then filtered through a plug of silica gel and washed with ethyl acetate. The filtrate was concentrated in vacuo to give product that was used in the next step without further purification. ¹H NMR (400 MHz, CDCl₃) δ 7.41-7.20 (m, 5H), 4.14 (q, J=7.2 Hz, 2H), 3.51 (s, 2H), 2.87 (dt, J=3.5, 11.8 Hz, 2H), 2.29 (m, 1H), 2.04 (td, J=2.5, 11.4 Hz, 2H), 1.95-1.85 (m, 2H), 1.85-1.70 (m, 2H), 1.26 (t, J=7.1 Hz, 3H).

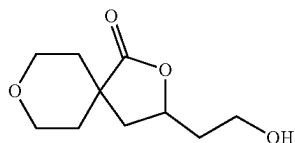


[0463] Example 2: Preparation of methyl 4-allyltetrahydro-2H-pyran-4-carboxylate: This reaction was performed in oven-dried glassware under a nitrogen atmosphere. To a solution of lithium diisopropylamide (1M, 1.20 equiv) in dry tetrahydrofuran (4.16 mL), methyl tetrahydro-2H-pyran-4-carboxylate (0.5 g, 3.47 mmol, 1.0 equiv), in 5 mL dry THF, was added dropwise during 0.5 hours at -78° C. The mixture was allowed to stir at this temperature for 1 hr followed by the addition of allyl bromide (0.457 g, 3.78 mmol, 1.1 eq) dropwise. The reaction mixture was allowed to warm to RT over a 1 hr period. The reaction was quenched with 10% HCl (while cooling in ice bath) until acidic (pH=2). The organic layer was separated and the aqueous layer was extracted with ethyl acetate (3×10 mL). The extract was dried over Na₂SO₄ and then concentrated in vacuo to give product that was used in the next step without further purification. ¹H NMR (400 MHz, CDCl₃) δ 5.68-5.52 (m, 1H), 5.03-4.91 (m, 2H), 3.75 (dt, J=3.7, 11.8 Hz, 2H), 3.63 (s, 3H), 3.37 (td, J=2.1, 11.6 Hz, 2H), 2.21 (d, J=7.4 Hz, 2H), 2.03-1.95 (m, 2H), 1.53-1.40 (m, 2H).



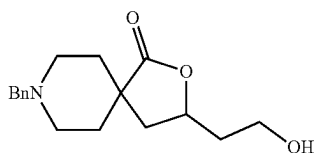
[0464] Example 3: Preparation of ethyl 4-allyl-1-benzylpiperidine-4-carboxylate: This reaction was performed in oven-dried glassware under a nitrogen atmosphere. To a solution of ethyl 1-benzylpiperidine-4-carboxylate (6.24 g, 26.7 mmol, 1.0 eq) and dry THF (50 mL), lithium diisopropylamide (1M, 1.10 equiv) in dry tetrahydrofuran (29.3 mL) was added dropwise during 0.5 hours at -78° C. The mixture was allowed to stir at this temperature for 1 hr followed by the addition of allyl iodine (6.73 g, 3.78 mmol, 1.5 eq) dropwise. The reaction mixture was allowed to warm to RT

and stir for 2 hr. The reaction was quenched with sat. aq. NH_4Cl until neutral pH (while cooling in ice bath). The organic layer was separated and the aqueous layer was extracted with ethyl acetate (2x50 mL). The combined extract was dried over Na_2SO_4 , then filtered through a plug of silica gel and washed with ethyl acetate. The filtrate was concentrated in vacuo to give product that was used in the next step without further purification. ^1H NMR (400 MHz, CDCl_3) δ 7.37-7.20 (m, 5H), 5.78-5.62 (m, 1H), 5.10-4.97 (m, 2H), 4.17 (q, $J=7.1$ Hz, 2H), 3.47 (s, 2H), 2.78-2.64 (m, 2H), 2.28 (d, $J=7.4$ Hz, 2H), 2.18-2.03 (m, 4H), 1.61-1.46 (m, 2H), 1.26 (t, $J=7.1$ Hz, 3H).



[0465] Example 4: Preparation of 3-(2-hydroxyethyl)-2,8-dioxaspiro[4.5]decan-1-one: A mixture of glacial acetic acid (10.9 g, 180 mmol, 53.6 eq), paraformaldehyde (0.309 g, 10.3 mmol, 3.0 eq) and H_2SO_4 (0.191 g, 1.95 mmol, 0.57 eq) was stirred for 30 min at 70°C . before methyl 4-allyltetrahydro-2H-pyran-4-carboxylate (0.632 g, 3.43 mmol, 1.0 equiv) was added dropwise during 10 min. The reaction mixture was then maintained at $70\text{--}80^\circ\text{C}$. and allowed to stir overnight. Acetic acid was removed under reduced pressure and the reaction was quenched with 10% NaHCO_3 solution. The mixture was then extracted with ethyl acetate (3x10 mL) and the combined organic phase was concentrated in vacuo to give a crude oil. The crude oil was used for next step without further purification.

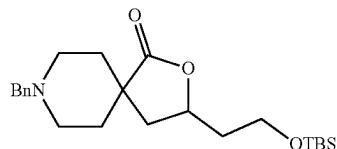
[0466] A mixture of the crude oil (715 mg) and 30% NaOH (2.86 g NaOH, 4x crude oil) aqueous solution was refluxed for 2 hours. The mixture was cooled in an ice bath and excess 30% H_2SO_4 was added until acidic ($\text{pH}<2$). The resulting mixture was extracted with ethyl acetate (3x25 mL), the combined organic phase was washed with 10% NaHCO_3 , (50 mL), brine (50 mL), dried over Na_2SO_4 and concentrated in vacuo to give a crude product which was used in the next step without further purification. ^1H NMR (400 MHz, CDCl_3) δ 4.57 (m, 1H), 3.91 (dt, $J=4.5$, 11.8 Hz, 1H), 3.79 (dt, $J=4.5$, 12.0 Hz, 1H), 3.66 (t, $J=6.0$ Hz, 2H), 3.54-3.44 (m, 1H), 3.43-3.34 (m, 1H), 3.13 (b, 1H), 2.41 (dd, $J=6.1$, 13.2 Hz, 1H), 2.01-1.91 (m, 1H), 1.89-1.64 (m, 4H), 1.54-1.44 (m, 1H), 1.42-1.33 (m, 1H).



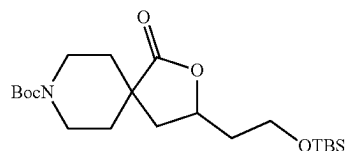
[0467] Example 5: Preparation of 8-benzyl-3-(2-hydroxyethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: A mixture of glacial acetic acid (78.1 g, 1.3 mol, 53.6 eq), paraformaldehyde (2.21 g, 73.5 mmol, 3.0 eq) and H_2SO_4 (3.63 g, 37 mmol, 1.5 eq) was stirred for 30 min at 70°C . before ethyl 4-allyl-1-benzylpiperidine-4-carboxylate (7.03 g, 24.5 mmol, 1.0

equiv) was added dropwise during 10 min. The reaction mixture was then maintained at $70\text{--}80^\circ\text{C}$. and allowed to stir overnight. Acetic acid was removed under reduced pressure and the reaction was quenched with 10% NaHCO_3 solution. The mixture was then extracted with ethyl acetate (3x40 mL) and the combined organic phase was concentrated in vacuo to give a crude oil. The crude oil was used for next step without further purification.

[0468] A mixture of the crude oil (7.07 mg) and 30% NaOH (28 g NaOH, 4x crude oil) aqueous solution was refluxed for 2 hours. The mixture was cooled in an ice bath and excess 30% H_2SO_4 was added until acidic ($\text{pH}<2$). The resulting mixture was the neutralized ($\text{pH}=8\text{--}9$) with sat. aq. NaHCO_3 solution and then extracted with ethyl acetate (3x100 mL), the combined organic phase was dried over Na_2SO_4 and concentrated in vacuo to give a crude product which was used in the next step without further purification. ^1H NMR (400 MHz, CDCl_3) δ 7.39-7.22 (m, 5H), 4.65 (m, 1H), 3.83 (t, $J=5.6$ Hz, 2H), 3.54 (s, 2H), 2.95-2.84 (m, 1H), 2.83-2.73 (m, 1H), 2.42 (dd, $J=6.1$, 13.0 Hz, 1H), 2.30-2.07 (m, 4H), 2.00-1.84 (m, 3H), 1.75-1.59 (m, 2H), 1.58-1.48 (m, 1H).

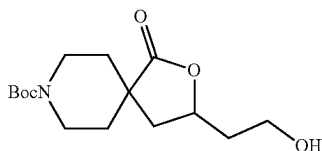


[0469] Example 6: Preparation of 8-benzyl-3-(2-((tert-butyldimethylsilyloxy)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: To a solution of 8-benzyl-3-(2-hydroxyethyl)-2-oxa-8-azaspiro[4.5]decan-1-one (10.0 g, 34.6 mmol, 1.0 eq.), imidazole (2.47 g, 36.3 mmol, 1.05 eq.) and dichloromethane (70 mL), was added a solution of tert-Butyldimethylsilyl chloride (1M, 5.47 g, 36.3 mmol, 1.05 eq.) in dichloromethane (36.3 mL). The reaction was allowed to stir at RT for 2 hr, before being quenched with D.I. water (50 mL). The organic layer was separated and the aqueous layer was extracted with dichloromethane (2x50 mL). The combined organic phase was dried over Na_2SO_4 and concentrated in vacuo to give a crude product which was further purified by column chromatography (Ethyl acetate/Hexanes, 0%~20%). ^1H NMR (400 MHz, CDCl_3) δ 7.32-7.11 (m, 5H), 4.52 (m, 1H), 3.73-3.65 (m, 2H), 3.46 (s, 2H), 2.87-2.76 (m, 1H), 2.72 (dt, $J=4.5$, 11.8 Hz, 1H), 2.31 (dd, $J=6.2$, 12.9 Hz, 1H), 2.22-2.08 (m, 1H), 2.08-1.97 (m, 2H), 1.91-1.70 (m, 3H), 1.62 (dd, $J=9.8$, 12.8 Hz, 1H), 1.59-1.50 (m, 1H), 1.49-1.38 (m, 1H), 0.83 (s, 9H), 0.00 (s, 6H).

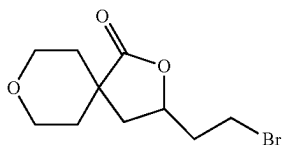


[0470] Example 7: Preparation of tert-butyl 3-(2-((tert-butyldimethylsilyloxy)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate: A mixture of 8-benzyl-3-(2-((tert-butyldimethylsilyloxy)ethyl)-2-oxa-8-azaspiro[4.5]decan-

1-one (4.77 g, 11.8 mmol, 1 eq.), Pd/C (954 mg, 20% wt) and MeOH (79 mL) was stirred at RT under 1 atm of H₂ (filled balloon) overnight. The mixture was filtered through a plug of Celite, washed with MeOH (50 mL) and concentrated in vacuo to give a crude oil. The crude oil (3.78 g) was dissolved in dichloromethane (79 mL) and cooled to 0° C. before the addition of Di-tert-butyl dicarbonate (2.83 g, 13.0 mmol, 1.1 eq.) and trimethylamine (1.8 g, 17.7 mmol, 1.5 eq.). The reaction was allowed to warm to RT and stir for 45 min. At this point the reaction was diluted with sat. aq. NaHCO₃ solution and then extracted with ethyl acetate (3×50 mL), the combined organic phase was dried over Na₂SO₄ and concentrated in vacuo to give a crude product which was used in the next step without further purification. ¹H NMR (400 MHz, CDCl₃) δ 4.57 (m, 1H), 3.91 (b, 1H), 3.77 (b, 1H), 3.73-3.66 (m, 2H), 3.17-3.05 (m, 1H), 3.04-2.93 (m, 1H), 2.31 (dd, J=6.2, 13.0 Hz, 1H), 1.96-1.81 (m, 2H), 1.81-1.64 (m, 3H), 1.59-1.48 (m, 1H), 1.48-1.32 (m, 10H), 0.83 (s, 9H), 0.00 (s, 6H).

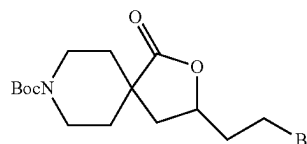


[0471] Example 8: Preparation of tert-butyl 3-(2-hydroxyethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate: To a solution of tert-butyl 3-((tert-butyldimethylsilyl)oxy)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate (4.88 g, 11.8 mmol, 1 eq.) and THF (70 mL) was added tetra-n-butylammonium fluoride (3.24 g, 12.4 mmol, 1.05 eq.), using THF (10 mL) to complete transfer. The resulting solution was allowed to stir at RT for 30 min before being concentrated in vacuo to give a crude product which was further purified by column chromatography (MeOH/DCM, 0%~10%). ¹H NMR (400 MHz, CDCl₃) δ 4.67 (m, 1H), 3.95 (dt, J=5.0, 13.6 Hz, 1H), 3.87-3.73 (m, 3H), 3.23-3.10 (m, 1H), 3.09-2.98 (m, 1H), 2.39 (dd, J=6.0, 13.0 Hz, 1H), 1.99-1.84 (m, 4H), 1.83-1.68 (m, 2H), 1.63-1.53 (m, 1H), 1.53-1.36 (m, 10H).

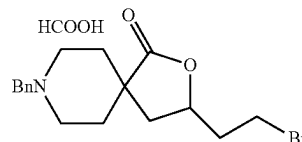


[0472] Example 9: Preparation of 3-(2-bromoethyl)-2,8-dioxaspiro[4.5]decan-1-one: A solution of 3-(2-hydroxyethyl)-2,8-dioxaspiro[4.5]decan-1-one (0.320 g, 1.60 mmol, 1 eq.) and THF (15 mL) was cooled to 0° C. before triphenylphosphine (0.630 g, 2.4 mmol, 1.5 eq.) and carbon tetrabromide (0.795 g, 2.4 mmol, 1.5 eq.) were sequentially added to the solution. The reaction solution was allowed to warm to RT and stir overnight. The resulting mixture was then filtered and concentrated in vacuo to give a crude mixture. This mixture was suspended in diethyl ether (50 mL) and filtered 2× using diethyl ether to wash the filter cakes. The final filtrate was loaded onto Celite in vacuo and

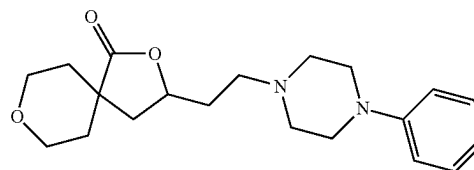
further purified by column chromatography (ethyl acetate/hexanes, 0%~40%). ¹H NMR (400 MHz, CDCl₃) δ 4.67 (m, 1H), 4.04 (dt, J=4.6, 11.8 Hz, 1H), 3.91 (dt, J=4.6, 12.1 Hz, 1H), 3.60 (m, 1H), 3.56-3.45 (m, 3H), 2.50 (dd, J=6.1, 12.9 Hz, 1H), 2.30-2.02 (m, 3H), 1.91 (m, 1H), 1.76 (dd, J=9.8, 13.0 Hz, 1H), 1.64-1.55 (m, 1H), 1.52-1.44 (m, 1H).



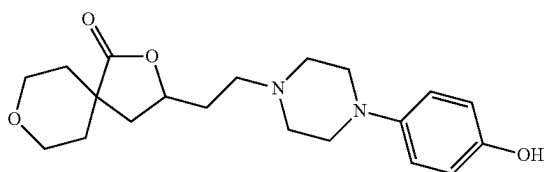
[0473] Example 10: Preparation of tert-butyl 3-(2-bromoethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate: The title compound was prepared according to the procedure for 3-(2-bromoethyl)-2,8-dioxaspiro[4.5]decan-1-one, except tert-butyl 3-(2-hydroxyethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate was substituted for 3-(2-hydroxyethyl)-2,8-dioxaspiro[4.5]decan-1-one: ¹H NMR (400 MHz, CDCl₃) δ 4.68 (m, 1H), 3.97 (dt, J=5.0, 13.5 Hz, 1H), 3.83 (dt, J=5.0, 13.7 Hz, 1H), 3.54 (dd, J=5.3, 7.5 Hz, 2H), 3.27-3.14 (m, 1H), 3.13-3.01 (m, 1H), 2.42 (dd, J=6.0, 13.0 Hz, 1H), 2.31-2.20 (m, 1H), 2.20-2.09 (m, 1H), 2.01-1.90 (m, 1H), 1.89-1.78 (m, 1H), 1.74 (dd, J=9.8, 12.8 Hz, 1H), 1.66-1.56 (m, 1H), 1.54-1.36 (m, 10H).



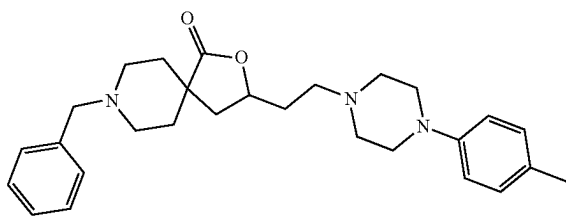
[0474] Example 11: Preparation of 8-benzyl-3-(2-bromoethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate: A solution of 3-(2-hydroxyethyl)-2,8-dioxaspiro[4.5]decan-1-one (2.07 g, 7.16 mmol, 1 eq.) and THF (70 mL) was cooled to 0° C. before triphenylphosphine (2.83 g, 10.8 mmol, 1.5 eq.) and carbon tetrabromide (3.58 g, 10.8 mmol, 1.5 eq.) were sequentially added to the solution. The reaction solution was allowed to warm to RT and stir overnight. The resulting mixture was then filtered and concentrated in vacuo to give a crude mixture. This mixture was suspended in diethyl ether (50 mL) and filtered 2× using diethyl ether to wash the filter cakes. The final filtrate was loaded onto Celite in vacuo and further purified by column chromatography on a C18 column. (ACN/H₂O, 0%~100%, w/0.1% formic acid). ¹H NMR (400 MHz, MeOD) δ 7.53 (b, 2H), 7.47 (b, 3H), 4.75 (m, 1H), 4.27 (s, 2H), 3.64-3.47 (m, 3H), 3.34 (m, 1H), 3.19 (b, 1H), 3.08 (b, 1H), 2.52 (m, 1H), 2.35-2.15 (m, 3H), 2.15-1.97 (m, 2H), 1.96-1.81 (m, 2H).



[0475] Example 12: Preparation of 3-(2-(4-phenylpiperazin-1-yl)ethyl)-2,8-dioxaspiro[4.5]decan-1-one: A solution of 3-(2-bromoethyl)-2,8-dioxaspiro[4.5]decan-1-one (0.050 g, 0.190 mmol, 1 eq.), THF (4 mL) and 1-phenylpiperazine (0.065 g, 0.399 mmol, 2.1 eq.) was heated and stirred at 60° C. for 3 days. The resulting mixture was then filtered and concentrated in vacuo to give a crude residue that was further purified by column chromatography (methanol/dichloromethane, 0%~10%). ¹H NMR (400 MHz, CDCl₃) δ 7.27 (m, 2H), 6.93 (d, J=8.3 Hz, 2H), 6.86 (t, J=7.3 Hz, 1H), 4.58 (m, 1H), 4.06 (dt, J=4.6, 11.9 Hz, 1H), 3.93 (dt, J=4.6, 12.0 Hz, 1H), 3.61 (m, 1H), 3.51 (m, 1H), 3.21 (t, J=5.0 Hz, 4H), 2.70-2.52 (m, 6H), 2.47 (dd, J=6.0, 12.8 Hz, 1H), 2.11 (m, 1H), 2.01-1.83 (m, 3H), 1.79 (dd, J=9.7, 13.1 Hz, 1H), 1.65-1.54 (m, 1H), 1.54-1.45 (m, 1H); MS (LC/MS, M+H⁺): 344.8.

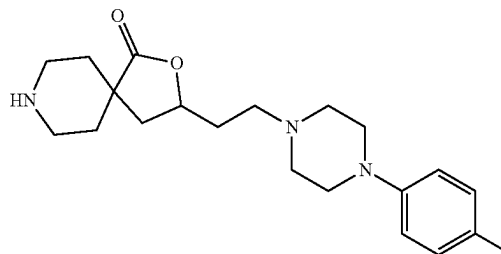


[0476] Example 13: Preparation of 3-(2-(4-(4-hydroxyphenyl)piperazin-1-yl)ethyl)-2,8-dioxaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 3-(2-(4-phenylpiperazin-1-yl)ethyl)-2,8-dioxaspiro[4.5]decan-1-one, except 4-(piperazin-1-yl)phenol was substituted for 1-phenylpiperazine: ¹H NMR (400 MHz, CDCl₃) δ 6.74 (d, J=8.9 Hz, 2H), 6.63 (d, J=8.9 Hz, 2H), 4.45 (m, 1H), 3.97 (dt, J=4.6, 11.8 Hz, 1H), 3.83 (dt, J=4.5, 12.3 Hz, 1H), 3.51 (m, 1H), 3.42 (m, 1H), 3.00 (t, J=4.7 Hz, 4H), 2.67-2.42 (m, 6H), 2.35 (dd, J=6.1, 12.1 Hz, 1H), 2.00 (m, 1H), 1.92-1.74 (m, 3H), 1.67 (dd, J=9.6, 12.9 Hz, 1H), 1.52-1.43 (m, 1H), 1.43-1.34 (m, 1H); MS (LC/MS, M+H⁺): 360.8.

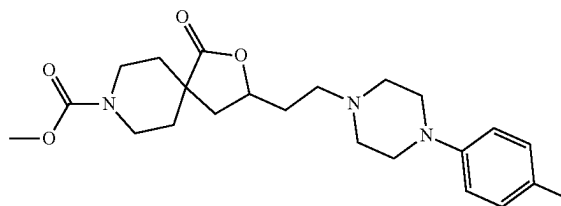


[0477] Example 14: Preparation of 8-benzyl-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: A solution of 8-benzyl-3-(2-bromoethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate (0.545 g, 1.37 mmol, 1 eq.), THF (13.7 mL), 1-(p-tolyl)piperazine (0.507 g, 2.88 mmol, 2.1 eq.) and triethylamine (0.107 g, 1.5 mmol, 1.1 eq.) was heated and stirred at 60° C. for 3 days. The resulting mixture was then filtered and concentrated in vacuo to give a crude residue that was further purified by column chromatography (methanol/dichloromethane, 0%~10%). ¹H NMR (400 MHz, CDCl₃) δ 7.39-7.23 (m, 5H), 7.10 (d, J=8.3 Hz, 2H), 6.87 (d, J=8.6 Hz, 2H), 4.53 (m, 1H), 3.54 (s, 2H), 3.17 (t, J=5.0 Hz, 4H), 2.95-2.85 (m, 1H), 2.84-2.75 (m, 1H), 2.70-2.49 (m, 6H), 2.40 (dd, J=6.2, 12.8 Hz, 1H), 2.30

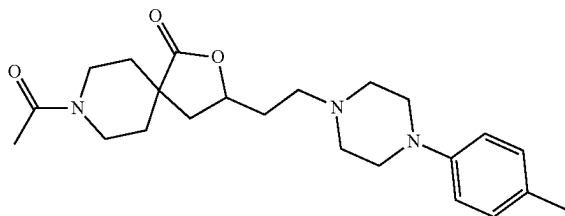
(s, 3H), 2.27-2.05 (m, 3H), 2.01-1.79 (m, 3H), 1.76-1.58 (m, 2H), 1.58-1.46 (m, 1H); MS (LC/MS, M+H⁺): 447.8.



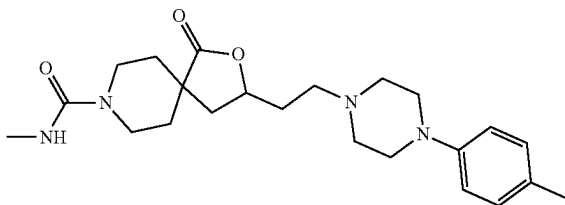
[0478] Example 15: Preparation of 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: A mixture of 8-benzyl-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one (445 mg, 0.993 mmol, 1 eq.), Pd/C (90 mg, 20% wt) and EtOH (6.6 mL) was stirred at RT under 1 atm of H₂ (filled balloon) for 48 hrs. The mixture was filtered through a plug of Celite, washed with MeOH (50 mL) and concentrated in vacuo to give a crude oil. ¹H NMR (400 MHz, MeOD) δ 6.95 (d, J=8.2 Hz, 2H), 6.77 (d, J=8.5 Hz, 2H), 4.56 (m, 1H), 3.47 (m, 1H), 3.28-3.18 (m, 1H), 3.17-3.09 (m, 1H), 3.08-2.96 (m, 5H), 2.66-2.46 (m, 6H), 2.42 (dd, J=6.0, 13.0 Hz, 1H), 2.14 (s, 3H), 2.10-2.00 (m, 1H), 2.00-1.91 (m, 1H), 1.91-1.80 (m, 4H), 1.80-1.70 (m, 1H); MS (LC/MS, M+H⁺): 357.8.



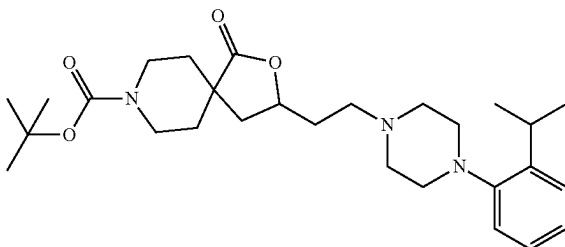
[0479] Example 16: Preparation of methyl 1-oxo-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: A solution of 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one (0.05 g, 0.14 mmol, 1 eq.), dichloromethane (2 mL) and triethylamine (0.44 g, 0.41 mmol, 3 eq.) was cooled to 0° C. before methyl chloroformate (0.027 g, 0.28 mmol, 2 eq.) was added to the solution. The reaction solution was allowed to warm to RT and stir for 3 hours. The reaction was diluted with dichloromethane and loaded onto Celite in vacuo and further purified by column chromatography on a C18 column. (ACN/H₂O, 0%~100%, w/0.1% formic acid). The resulting formate acid salt was dissolved in MeOH (2 mL) and Amberlite IRA-400(OH) resin was added. This mixture was allowed to stir at RT for 30 min and then filtered and concentrated in vacuo to afford pure free based product. ¹H NMR (400 MHz, CDCl₃) δ 7.08 (d, J=8.4 Hz, 2H), 6.85 (d, J=8.6 Hz, 2H), 4.59 (m, 1H), 4.02 (b, 1H), 3.85 (b, 1H), 3.72 (s, 3H), 3.35-3.24 (m, 1H), 3.23-3.10 (m, 5H), 2.71-2.50 (m, 6H), 2.38 (dd, J=6.0, 12.8 Hz, 1H), 2.28 (s, 3H), 2.07-1.72 (m, 5H), 1.69-1.47 (m, 2H); MS (LC/MS, M+H⁺): 415.8.



[0480] Example 17: Preparation of 8-acetyl-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for methyl 1-oxo-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-8-carboxylate, except acetic anhydride was substituted for methyl chloroformate: ¹H NMR (400 MHz, CDCl₃) δ 7.08 (d, J=8.3 Hz, 2H), 6.85 (d, J=8.5 Hz, 2H), 4.60 (m, 1H), 4.19 (m, 0.5 H), 4.04-3.84 (m, 1H), 3.72 (m, 0.5H), 3.45-3.22 (m, 2H), 3.15 (t, J=4.8 Hz, 4H), 2.70-2.49 (m, 6H), 2.43-2.32 (m, 1H), 2.27 (s, 3H), 2.10 (s, 3H), 2.05-1.72 (m, 5H), 1.69-1.49 (m, 2H); MS (LC/MS, M+H⁺): 399.8.

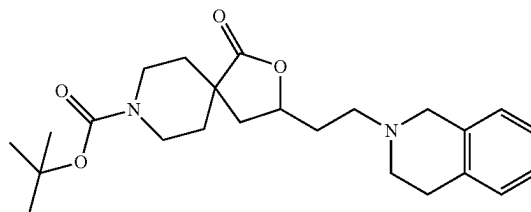


[0481] Example 18: Preparation of N-methyl-1-oxo-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-8-carboxamide: The title compound was prepared according to the procedure for methyl 1-oxo-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-8-carboxylate, except N-methyl-1H-imidazole-1-carboxamide was substituted for methyl chloroformate: ¹H NMR (400 MHz, MeOD) δ 6.96 (d, J=8.3 Hz, 2H), 6.78 (d, J=8.2 Hz, 2H), 4.52 (m, 1H), 3.79 (dt, J=4.7, 13.8 Hz, 1H), 3.68 (dt, J=4.5, 13.8 Hz, 1H), 3.22 (m, 1H), 3.16-2.86 (m, 6H), 2.63 (s, 3H), 2.59-2.36 (m, 7H), 2.15 (s, 3H), 1.91-1.67 (m, 3H), 1.66-1.50 (m, 2H), 1.50-1.38 (m, 1H); MS (LC/MS, M+H⁺): 414.8.

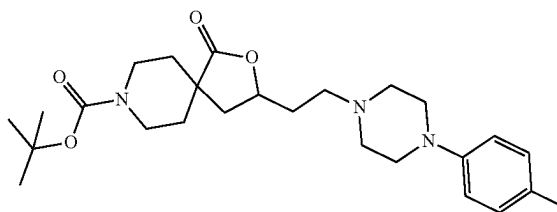


[0482] Example 20: Preparation of tert-butyl 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate: A mixture of tert-butyl 3-(2-bromoethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate (0.500 g, 1.38 mmol, 1 eq.), ACN (7 mL),

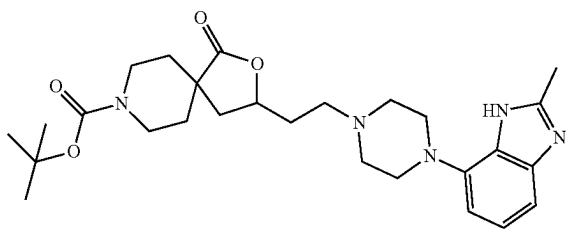
1-(2-isopropylphenyl)piperazine (0.337 g, 1.65 mmol, 1.2 eq.) and K₂CO₃ (0.954 g, 6.9 mmol, 5 eq.) was heated and stirred at 80° C. for 3 days. The resulting mixture was then filtered and concentrated in vacuo to give a crude residue that was further purified by column chromatography (methanol/dichloromethane, 0%~10%). ¹H NMR (400 MHz, CDCl₃) δ 7.16 (dd, J=1.6, 7.4 Hz, 1H), 7.09-6.94 (m, 3H), 4.49 (m, 1H), 3.89 (b, 1H), 3.75 (b, 1H), 3.40 (sep, J=6.9 Hz, 1H), 3.09 (m, 1H), 2.98 (m, 1H), 2.81 (t, J=4.6 Hz, 4H), 2.65-2.39 (m, 5H), 2.30 (dd, J=6.1, 12.8 Hz, 1H), 1.99-1.60 (m, 5H), 1.51 (m, 1H), 1.46-1.29 (m, 11H), 1.12 (s, 3H), 1.10 (s, 6H); MS (LC/MS, M+H⁺): 485.8



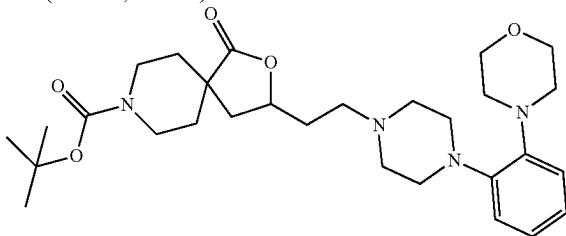
[0483] Example 21: Preparation of tert-butyl 3-(2-(3,4-dihydroisoquinolin-2(1H)-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate: The title compound was prepared according to the procedure for tert-butyl 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate, except 1,2,3,4-tetrahydroisoquinoline hydrochloride was substituted for 1-(2-isopropylphenyl)piperazine: ¹H NMR (400 MHz, CDCl₃) δ 7.07-6.96 (m, 3H), 6.95-6.87 (m, 1H), 4.53 (m, 1H), 3.87 (b, 1H), 3.72 (b, 1H), 3.60-3.46 (m, 2H), 3.06 (m, 1H), 2.95 (m, 1H), 2.80 (t, J=5.8 Hz, 2H), 2.64 (t, J=6.0 Hz, 2H), 2.58 (t, J=7.3 Hz, 2H), 2.28 (dd, J=6.1, 12.9 Hz, 1H), 1.97-1.75 (m, 3H), 1.74-1.58 (m, 2H), 1.54-1.26 (m, 11H); MS (LC/MS, M+H⁺): 414.8



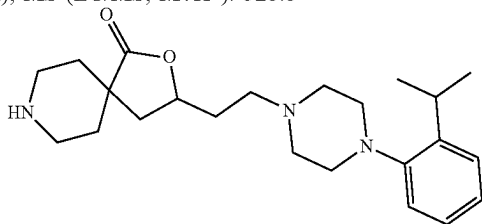
[0484] Example 22: Preparation of tert-butyl 1-oxo-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-8-carboxylate: A solution of tert-butyl 3-(2-bromoethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate (1.5 g, 4.11 mmol, 1.1 eq.), THF (36 mL), 1-(p-tolyl)piperazine (0.660 g, 3.74 mmol, 1 eq.) and triethylamine (0.416 g, 4.11 mmol, 1.1 eq.) was heated and stirred at 70° C. for 3 days. The resulting mixture was then filtered and concentrated in vacuo to give a crude residue that was further purified by column chromatography (methanol/dichloromethane, 0%~10%). ¹H NMR (400 MHz, CDCl₃) δ 7.05 (d, J=8.3 Hz, 2H), 6.82 (d, J=8.5 Hz, 2H), 4.55 (m, 1H), 3.96 (m, 1H), 3.81 (m, 1H), 3.22-2.98 (m, 6H), 2.67-2.45 (m, 6H), 2.36 (dd, J=6.2, 12.9 Hz, 1H), 2.25 (s, 3H), 2.00-1.66 (m, 5H), 1.57 (m, 1H), 1.53-1.34 (m, 10H); MS (LC/MS, M+H⁺): 457.8



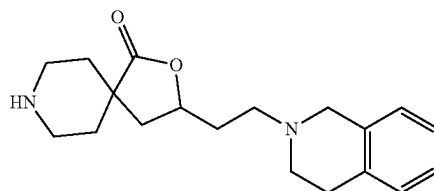
[0485] Example 23: Preparation of tert-butyl 3-(2-(4-(2-methyl-1H-benzo[d]imidazol-7-yl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: A solution of tert-butyl 3-(2-bromoethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate (0.5 g, 1.38 mmol, 1. eq.), THF (12 mL), 2-methyl-7-(piperazin-1-yl)-1H-benzo[d]imidazole (0.549 g, 1.65 mmol, 1.2 eq.) and triethylamine (0.500 g, 4.95 mmol, 3.5 eq.) was heated and stirred at 70° C. for 3 days. The resulting mixture was then filtered and concentrated in vacuo to give a crude residue that was first purified by column chromatography (methanol/dichloromethane, 0%~10%). The resulting fractions were further purified by column chromatography on a C18 column. (ACN/H₂O, 0%~100%, w/0.1% NH₄OH) ¹H NMR (400 MHz, CDCl₃) δ 7.16-6.97 (m, 2H), 6.65 (m, 1H), 4.54 (m, 1H), 3.98 (m, 1H), 3.83 (m, 1H), 3.40 (b, 4H), 3.17 (t, J=11.1 Hz, 1H), 3.06 (t, J=11.5 Hz, 1H), 2.68-2.42 (m, 9H), 2.36 (dd, J=6.2, 13.0 Hz, 1H), 1.99-1.66 (m, 5H), 1.58 (m, 1H), 1.54-1.33 (m, 10H); MS (LC/MS, M+H⁺): 497.8



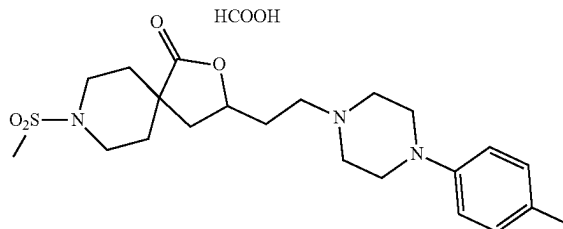
[0486] Example 24: Preparation of tert-butyl 3-(2-(4-(2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: The title compound was prepared according to the procedure for tert-butyl 3-(2-(4-(2-methyl-1H-benzo[d]imidazol-7-yl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate, except 4-(2-(piperazin-1-yl)phenyl)morpholine was substituted for 2-methyl-7-(piperazin-1-yl)-1H-benzo[d]imidazole: ¹H NMR (400 MHz, CDCl₃) δ 6.94-6.87 (m, 2H), 6.87-6.78 (m, 2H), 4.51 (m, 1H), 3.90 (m, 1H), 3.81-3.66 (m, 5H), 3.34-2.84 (m, 10H), 2.67-2.37 (m, 6H), 2.31 (dd, J=6.2, 12.9 Hz, 1H), 1.93-1.59 (m, 5H), 1.52 (m, 1H), 1.47-1.30 (m, 10H); MS (LC/MS, M+H⁺): 528.8



[0487] Example 25: Preparation of 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decane-1-one: To a solution of tert-butyl 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate (0.450 g, 0.930 mmol, 1 eq.) in dichloromethane (3 mL) at 0° C. was added trifluoroacetic acid (3 mL). The reaction was allowed to stir at RT for 30 min before being diluted with MeOH and concentrated in vacuo to afford the product as a TFA salt. The resulting TFA salt was dissolved in MeOH (2 mL) and Amberlite IRA-400(OH) resin was added. This mixture was allowed to stir at RT for 30 min and then filtered and concentrated in vacuo to afford pure free based product. ¹H NMR (400 MHz, MeOD) δ 7.16 (d, J=7.5 Hz, 1H), 7.07-6.93 (m, 3H), 4.50 (m, 1H), 3.44 (sep, J=6.9 Hz, 1H), 2.99 (dt, J=4.3, 12.6 Hz, 1H), 2.89 (dt, J=4.3, 13.2 Hz, 1H), 2.82 (t, J=4.7 Hz, 4H), 2.74-2.37 (m, 9H), 1.91-1.78 (m, 3H), 1.77-1.50 (m, 3H), 1.42 (m, 1H), 1.12 (s, 3H), 1.10 (s, 3H); MS (LC/MS, M+H⁺): 385.8

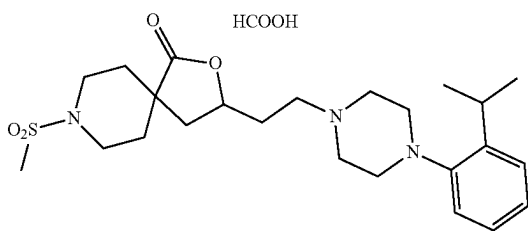


[0488] Example 26: Preparation of 3-(2-(3,4-dihydroisoquinolin-2(1H)-yl)ethyl)-2-oxa-8-azaspiro[4.5]decane-1-one: The title compound was prepared according to the procedure for 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decane-1-one, except tert-butyl 3-(2-(3,4-dihydroisoquinolin-2(1H)-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate was substituted for tert-butyl 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: ¹H NMR (400 MHz, MeOD) δ 7.18-7.09 (m, 3H), 7.09-7.03 (m, 1H), 4.64 (m, 1H), 3.68 (s, 2H), 3.25 (dt, J=5.0, 13.1 Hz, 1H), 3.10 (dt, J=5.0, 13.4 Hz, 1H), 2.99-2.88 (m, 3H), 2.85-2.61 (m, 5H), 2.55 (dd, J=6.1, 12.9 Hz, 1H), 2.09-1.95 (m, 3H), 1.90-1.70 (m, 3H), 1.63 (m, 1H); MS (LC/MS, M+H⁺): 314.8

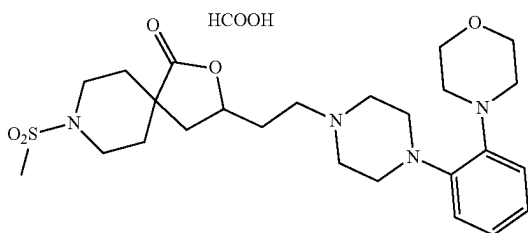


[0489] Example 27: Preparation of 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decane-1-one formate: A solution of 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decane-1-one (0.05 g, 0.14 mmol, 1 eq.), dichloromethane (2 mL) and triethylamine (0.44 g, 0.41 mmol, 3 eq.) was cooled to 0° C. before methanesulfonyl chloride (0.032 g, 0.28 mmol, 2 eq.) was added to the solution. The reaction solution was allowed to

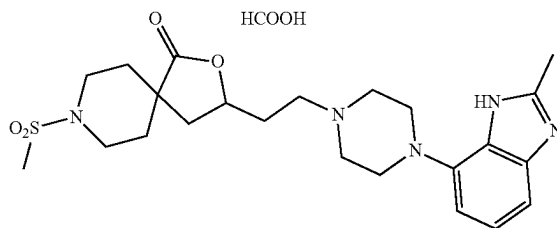
warm to RT and stir for 3 hours. The reaction was diluted with dichloromethane and loaded onto Celite in vacuo and further purified by column chromatography on a C18 column. (ACN/H₂O, 0%~100%, w/0.1% formic acid). ¹H NMR (400 MHz, DMSO) δ 7.01 (d, J=8.5 Hz, 2H), 6.82 (d, J=8.5 Hz, 2H), 4.59 (m, 1H), 3.58-3.45 (m, 1H), 3.44-3.33 (m, 1H), 3.13-2.95 (m, 5H), 2.88 (s, 3H), 2.86-2.78 (m, 1H), 2.62-2.31 (m, 7H), 2.19 (s, 3H), 1.97-1.57 (m, 7H); MS (LC/MS, M+H⁺): 435.8



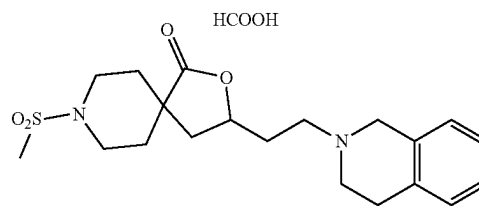
[0490] Example 28: Preparation of 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 3-(2-(4-(2-isopropylphenyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one was substituted for 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: ¹H NMR (400 MHz, MeOD) δ 7.34-7.27 (m, 1H), 7.23-7.08 (m, 3H), 4.67 (m, 1H), 3.70 (dt, J=4.8, 12.3 Hz, 1H), 3.61-3.47 (m, 2H), 3.40-3.09 (m, 11H), 3.04 (m, 1H), 2.88 (s, 3H), 2.56 (dd, J=6.4, 12.8 Hz, 1H), 2.29-2.10 (m, 2H), 2.05 (m, 1H), 1.96-1.80 (m, 3H), 1.75 (m, 1H), 1.24 (s, 3H), 1.22 (s, 3H); MS (LC/MS, M+H⁺): 463.7



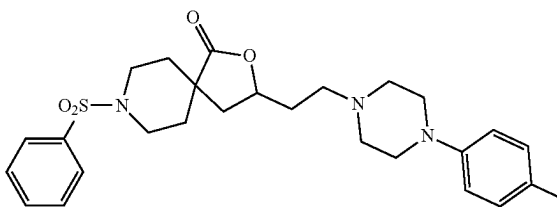
[0491] Example 29: Preparation of 8-(methylsulfonyl)-3-(2-(4-(2-morpholinophenyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 3-(2-(4-(2-morpholinophenyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one was substituted for 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: ¹H NMR (400 MHz, MeOD) δ 7.13-6.90 (m, 4H), 4.67 (m, 1H), 3.86 (t, J=4.6 Hz, 4H), 3.70 (dt, J=4.7, 12.3 Hz, 1H), 3.61-3.08 (m, 16H), 3.04 (m, 1H), 2.88 (s, 3H), 2.56 (dd, J=5.9, 13.0 Hz, 1H), 2.29-2.11 (m, 2H), 2.05 (m, 1H), 1.97-1.81 (m, 3H), 1.75 (m, 1H); MS (LC/MS, M+H⁺): 507.2



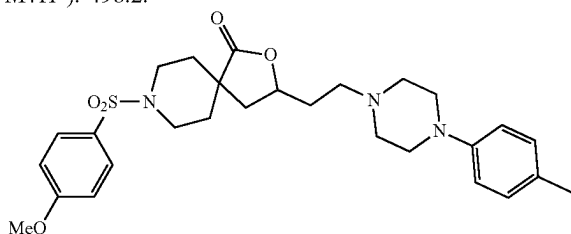
[0492] Example 30: Preparation of 3-(2-(4-(2-methyl-1H-benzo[d]imidazol-7-yl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 3-(2-(4-(2-methyl-1H-benzo[d]imidazol-7-yl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one was substituted for 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: ¹H NMR (400 MHz, MeOD) δ 7.47-7.26 (m, 2H), 7.04 (d, J=6.7 Hz, 1H), 4.69 (m, 1H), 3.79-3.28 (m, 12H), 3.14 (m, 1H), 3.01 (m, 1H), 2.87 (s, 3H), 2.79 (s, 3H), 2.58 (dd, J=5.9, 12.9 Hz, 1H), 2.38-2.14 (m, 2H), 2.12-1.98 (m, 1H), 1.96-1.80 (m, 3H), 1.75 (m, 1H); MS (LC/MS, M+H⁺): 476.2



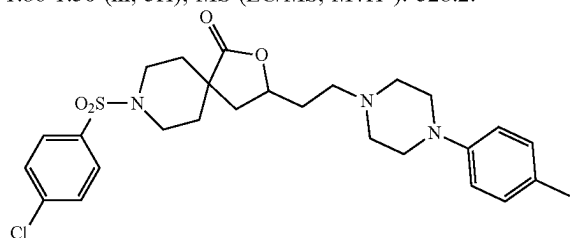
[0493] Example 31: Preparation of 3-(2-(3,4-dihydroisoquinolin-2(1H)-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 3-(2-(3,4-dihydroisoquinolin-2(1H)-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one was substituted for 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: ¹H NMR (400 MHz, MeOD) δ 7.31-7.21 (m, 3H), 7.20-7.15 (m, 1H), 4.67 (m, 1H), 4.26 (s, 2H), 3.70 (dt, J=4.9, 12.3 Hz, 1H), 3.53 (dt, J=5.2, 12.5 Hz, 1H), 3.39 (t, J=6.2 Hz, 2H), 3.31-3.10 (m, 5H), 3.03 (m, 1H), 2.88 (s, 3H), 2.56 (dd, J=6.0, 12.9 Hz, 1H), 2.30-2.12 (m, 2H), 2.05 (m, 1H), 1.97-1.80 (m, 3H), 1.75 (m, 1H); MS (LC/MS, M+H⁺): 392.7.



[0494] Example 32: Preparation of 8-(phenylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except benzenesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.69 (m, 2H), 7.53 (m, 1H), 7.46 (m, 2H), 7.00 (d, $J=8.3$ Hz, 2H), 6.75 (d, $J=8.5$ Hz, 2H), 4.44 (m, 1H), 3.46 (m, 1H), 3.22 (m, 1H), 3.10 (t, $J=4.7$ Hz, 4H), 2.97 (m, 1H), 2.87 (m, 1H), 2.75-2.44 (m, 6H), 2.19 (s, 3H), 2.12 (dd, $J=6.1$, 13.0 Hz, 1H), 2.02-1.77 (m, 4H), 1.72-1.53 (m, 3H); MS (LC/MS, $\text{M}+\text{H}^+$): 498.2.

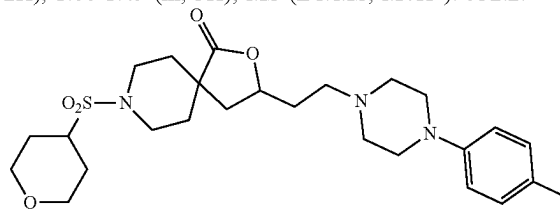


[0495] Example 33: Preparation of 8-((4-methoxyphenyl)sulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 4-methoxybenzenesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.62 (d, $J=9.0$ Hz, 2H), 6.99 (d, $J=8.3$ Hz, 2H), 6.91 (d, $J=8.9$ Hz, 2H), 6.75 (d, $J=8.6$ Hz, 2H), 4.45 (m, 1H), 3.80 (s, 3H), 3.42 (m, 1H), 3.18 (m, 1H), 3.04 (t, $J=4.9$ Hz, 4H), 2.95 (m, 1H), 2.86 (m, 1H), 2.57-2.39 (m, 6H), 2.19 (s, 3H), 2.11 (dd, $J=6.1$, 12.9 Hz, 1H), 2.01-1.86 (m, 2H), 1.86-1.50 (m, 5H); MS (LC/MS, $\text{M}+\text{H}^+$): 528.2.

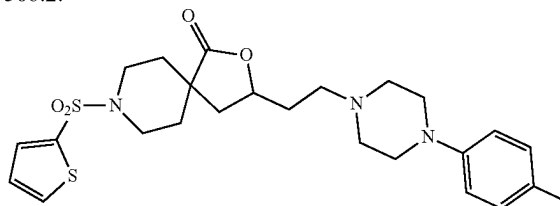


[0496] Example 34: Preparation of 8-((4-chlorophenyl)sulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 4-chlorobenzenesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.56 (d, $J=8.6$ Hz, 2H), 7.37 (d, $J=8.5$ Hz, 2H), 6.93 (d, $J=8.4$ Hz, 2H), 6.69 (d, $J=8.6$ Hz, 2H), 4.40 (m, 1H), 3.37 (m, 1H), 3.10 (m, 1H), 3.02 (t, $J=4.7$ Hz, 4H), 2.96 (m, 1H), 2.87 (m, 1H), 2.59-2.34 (m, 6H), 2.13 (s, 3H), 2.06

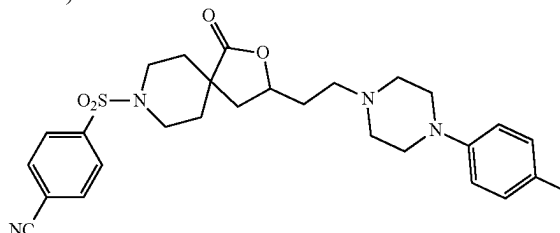
(dd, $J=5.9$, 12.9 Hz, 1H), 1.95-1.81 (m, 2H), 1.81-1.67 (m, 2H), 1.66-1.49 (m, 3H); MS (LC/MS, $\text{M}+\text{H}^+$): 532.2.



[0497] Example 35: Preparation of 8-((tetrahydro-2H-pyran-4-yl)sulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except tetrahydro-2H-pyran-4-sulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.08 (d, $J=8.3$ Hz, 2H), 6.84 (d, $J=8.5$ Hz, 2H), 4.60 (m, 1H), 4.08 (dd, $J=3.6$, 11.5 Hz, 2H), 3.80 (m, 1H), 3.57-3.42 (m, 4H), 3.41-3.30 (m, 3H), 3.22-3.06 (m, 5H), 2.69-2.47 (m, 6H), 2.30 (dd, $J=6.1$, 12.9 Hz, 1H), 2.27 (s, 3H), 2.03-1.65 (m, 9H); MS (LC/MS, $\text{M}+\text{H}^+$): 506.2.

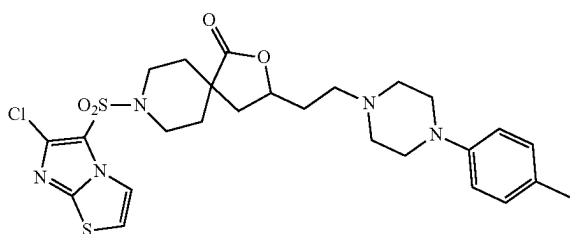


[0498] Example 36: Preparation of 8-(thiophen-2-ylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except thiophene-2-sulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.53 (dd, $J=1.2$, 5.0 Hz, 1H), 7.46 (dd, 1.3, 3.8 Hz, 1H), 7.06 (dd, $J=3.8$, 5.0 Hz, 1H), 6.99 (d, $J=8.4$ Hz, 2H), 6.75 (d, $J=8.5$ Hz, 2H), 4.46 (m, 1H), 3.49 (m, 1H), 3.24 (m, 1H), 3.12-2.97 (m, 5H), 2.92 (m, 1H), 2.61-2.41 (m, 6H), 2.19 (s, 3H), 2.14 (dd, $J=6.0$, 13.0 Hz, 1H), 2.04-1.88 (m, 2H), 1.88-1.73 (m, 2H), 1.73-1.57 (m, 3H); MS (LC/MS, $\text{M}+\text{H}^+$): 504.1.

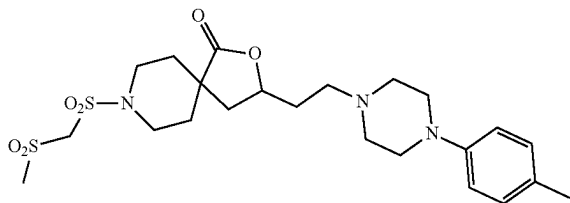


[0499] Example 37: Preparation of 4-((1-oxo-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-8-

yl)sulfonyl)benzonitrile: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 4-cyanobenzenesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.89 (d, $J=8.1$ Hz, 2H), 7.85 (d, 7.8 Hz, 2H), 7.08 (d, $J=8.2$ Hz, 2H), 6.84 (d, $J=8.5$ Hz, 2H), 4.57 (m, 1H), 3.53 (m, 1H), 3.24 (m, 2H), 3.20-3.07 (m, 5H), 2.68-2.48 (m, 6H), 2.28 (s, 3H), 2.21 (dd, $J=6.0$, 13.0 Hz, 1H), 2.08-1.97 (m, 2H), 1.97-1.65 (m, 5H); MS (LC/MS, $\text{M}+\text{H}^+$): 523.2.

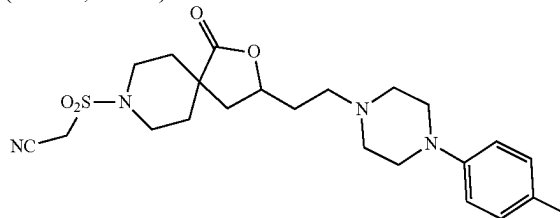


[0500] Example 38: Preparation of 8-((6-chloroimidazo[2,1-b]thiazol-5-yl)sulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 6-chloroimidazo[2,1-b]thiazole-5-sulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.90 (d, $J=4.5$ Hz, 1H), 7.11-7.02 (m, 3H), 6.84 (d, $J=8.6$ Hz, 2H), 4.57 (m, 1H), 3.74 (m, 1H), 3.52 (m, 1H), 3.26 (m, 1H), 3.20-3.06 (m, 5H), 2.67-2.47 (m, 6H), 2.32-2.21 (m, 4H), 2.13-1.97 (m, 2H), 1.97-1.82 (m, 2H), 1.82-1.65 (m, 3H); MS (LC/MS, $\text{M}+\text{H}^+$): 578.1.

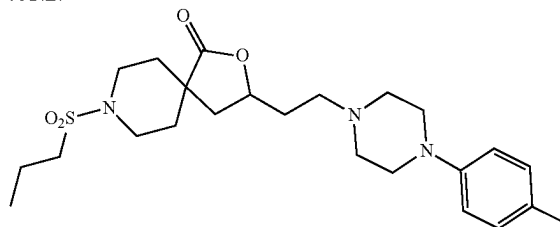


[0501] Example 39: Preparation of 8-(((methylsulfonyl)methyl)sulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except (methylsulfonyl)methanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.09 (d, $J=8.3$ Hz, 2H), 6.86 (d, $J=8.6$ Hz, 2H), 4.61 (m, 1H), 4.45 (s, 2H), 3.89 (m, 1H), 3.66 (m, 1H), 3.45 (m, 1H), 3.34 (m, 1H), 3.23 (s, 3H), 3.17 (t,

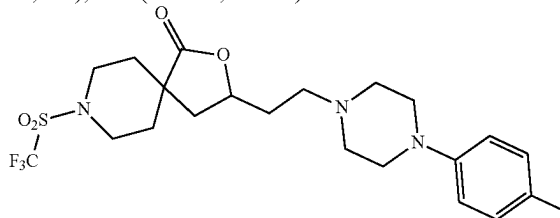
$J=4.9$ Hz, 4H), 2.73-2.52 (m, 6H), 2.37 (dd, $J=6.1$, 12.9 Hz, 1H), 2.28 (s, 3H), 2.15-1.85 (m, 4H), 1.85-1.65 (m, 3H); MS (LC/MS, $\text{M}+\text{H}^+$): 514.2.



[0502] Example 40: Preparation of 2-(((4-(p-tolyl)piperazin-1-yl)ethyl)(2-oxa-8-azaspiro[4.5]decan-8-yl)sulfonyl)acetonitrile: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except cyanomethanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.09 (d, $J=8.3$ Hz, 2H), 6.86 (d, $J=8.5$ Hz, 2H), 4.64 (m, 1H), 4.02-3.91 (m, 3H), 3.69 (m, 1H), 3.58 (m, 1H), 3.48 (m, 1H), 3.18 (t, $J=4.9$ Hz, 4H), 2.73-2.54 (m, 6H), 2.34 (dd, $J=6.1$, 13.0 Hz, 1H), 2.29 (s, 3H), 2.13-2.02 (m, 2H), 2.02-1.89 (m, 2H), 1.89-1.71 (m, 3H); MS (LC/MS, $\text{M}+\text{H}^+$): 461.2.

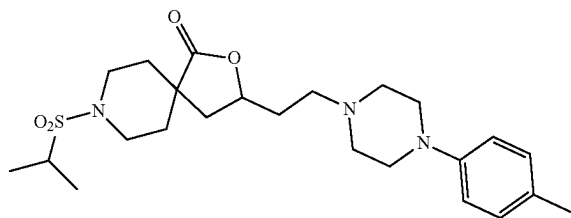


[0503] Example 41: Preparation of 8-(propylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 1-propanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.09 (d, $J=8.3$ Hz, 2H), 6.85 (d, $J=8.4$ Hz, 2H), 4.61 (m, 1H), 3.73 (m, 1H), 3.48-3.35 (m, 2H), 3.34-3.25 (m, 1H), 3.16 (t, $J=4.8$ Hz, 4H), 2.92 (m, 2H), 2.69-2.50 (m, 6H), 2.30 (dd, $J=6.0$, 12.9 Hz, 1H), 2.28 (s, 3H), 2.07-1.65 (m, 9H), 1.07 (t, $J=7.4$ Hz, 3H); MS (LC/MS, $\text{M}+\text{H}^+$): 464.

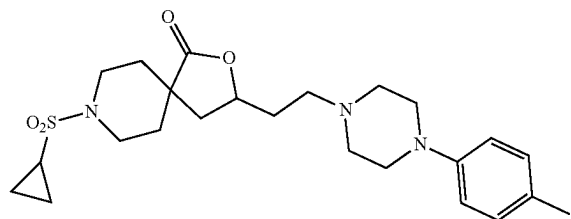


[0504] Example 42: Preparation of 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-8-((trifluoromethyl)sulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared

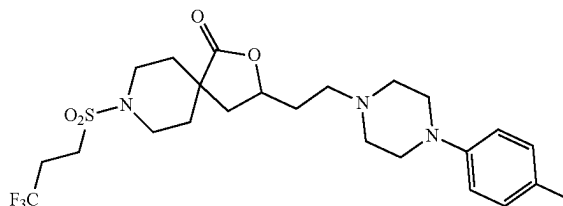
according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except trifluoromethanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 6.96 (d, $J=8.3$ Hz, 2H), 6.72 (d, $J=8.5$ Hz, 2H), 4.50 (m, 1H), 3.87 (m, 1H), 3.70-3.23 (b, 3H), 3.03 (t, $J=4.9$ Hz, 4H), 2.56-2.38 (m, 6H), 2.19 (dd, $J=6.2, 12.9$ Hz, 1H), 2.15 (s, 3H), 1.98-1.86 (m, 2H), 1.86-1.52 (m, 5H); MS (LC/MS, $\text{M}+\text{H}^+$): 490



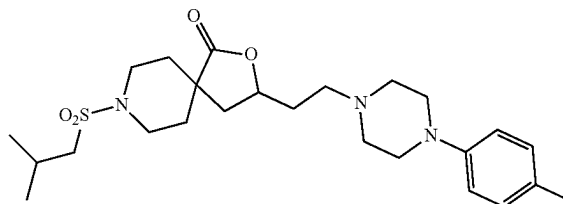
[0505] Example 43: Preparation of 8-(isopropylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 2-propanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 6.95 (d, $J=8.2$ Hz, 2H), 6.72 (d, $J=8.6$ Hz, 2H), 4.47 (m, 1H), 3.67 (m, 1H), 3.44 (m, 1H), 3.35 (m, 1H), 3.25 (m, 1H), 3.12-2.94 (m, 5H), 2.58-2.36 (m, 6H), 2.18 (dd, $J=6.0, 12.9$ Hz, 1H), 2.14 (s, 3H), 1.92-1.71 (m, 4H), 1.71-1.45 (m, 3H); MS (LC/MS, $\text{M}+\text{H}^+$): 464



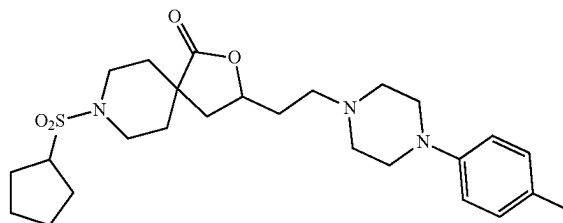
[0506] Example 44: Preparation of 8-(cyclopropylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except cyclopropanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.08 (d, $J=8.3$ Hz, 2H), 6.85 (d, $J=8.5$ Hz, 2H), 4.61 (m, 1H), 3.78 (m, 1H), 3.50 (m, 1H), 3.37 (m, 1H), 3.28 (m, 1H), 3.16 (t, $J=4.9$ Hz, 4H), 2.70-2.50 (m, 6H), 2.36-2.25 (m, 5H), 2.10-1.98 (m, 2H), 1.98-1.65 (m, 5H), 1.23-1.11 (m, 2H), 1.07-0.95 (m, 2H); MS (LC/MS, $\text{M}+\text{H}^+$): 462



[0507] Example 45: Preparation of 3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-8-((3,3,3-trifluoropropyl)sulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 3,3,3-trifluoropropane-1-sulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 6.95 (d, $J=8.3$ Hz, 2H), 6.72 (d, $J=8.5$ Hz, 2H), 4.49 (m, 1H), 3.64 (m, 1H), 3.38-3.27 (m, 2H), 3.23 (m, 1H), 3.09-2.93 (m, 6H), 2.59-2.36 (m, 8H), 2.17 (dd, $J=5.9, 13.0$ Hz, 1H), 2.15 (s, 3H), 1.96-1.85 (m, 2H), 1.85-1.56 (m, 5H); MS (LC/MS, $\text{M}+\text{H}^+$): 518

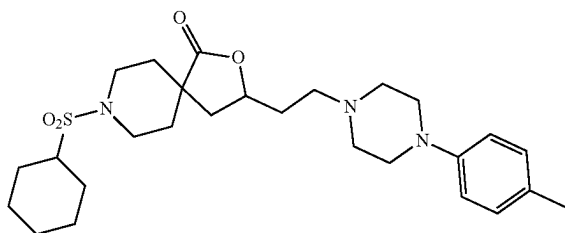


[0508] Example 46: Preparation of 8-(isobutylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except isobutanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.08 (d, $J=8.3$ Hz, 2H), 6.85 (d, $J=8.5$ Hz, 2H), 4.61 (m, 1H), 3.70 (m, 1H), 3.45-3.33 (m, 2H), 3.29 (m, 1H), 3.16 (t, $J=4.8$ Hz, 4H), 2.79 (dd, $J=2.2, 6.6$ Hz, 2H), 2.70-2.49 (m, 6H), 2.37-2.20 (m, 5H), 2.07-1.97 (m, 2H), 1.97-1.67 (m, 5H), 1.12 (d, $J=6.7$ Hz, 6H); MS (LC/MS, $\text{M}+\text{H}^+$): 478

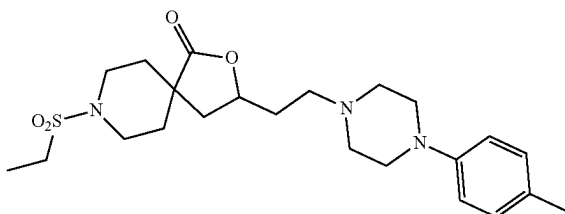


[0509] Example 47: Preparation of 8-(cyclopentylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except cyclopentanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.08 (d, $J=8.3$ Hz, 2H), 6.85 (d, $J=8.5$ Hz, 2H), 4.61 (m, 1H), 3.78 (m, 1H), 3.50 (m, 1H), 3.37 (m, 1H), 3.28 (m, 1H), 3.16 (t, $J=4.9$ Hz, 4H), 2.70-2.50 (m, 6H), 2.36-2.25 (m, 5H), 2.10-1.98 (m, 2H), 1.98-1.65 (m, 5H), 1.23-1.11 (m, 2H), 1.07-0.95 (m, 2H); MS (LC/MS, $\text{M}+\text{H}^+$): 462

[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except cyclopentanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 6.95 (d, $J=8.3$ Hz, 2H), 6.72 (d, $J=8.6$ Hz, 2H), 4.47 (m, 1H), 3.64 (m, 1H), 3.42-3.22 (m, 3H), 3.18 (m, 1H), 3.03 (t, $J=4.9$ Hz, 4H), 2.57-2.37 (m, 6H), 2.18 (dd, $J=6.1$, 12.9 Hz, 1H), 2.15 (s, 3H), 1.95-1.42 (m, 15H); MS (LC/MS, $\text{M}+\text{H}^+$): 490

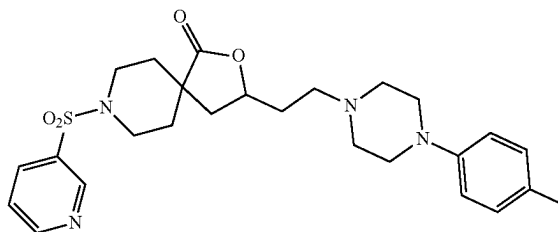


[0510] Example 48: Preparation of 8-(cyclohexylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except cyclohexanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 6.95 (d, $J=8.3$ Hz, 2H), 6.72 (d, $J=8.4$ Hz, 2H), 4.47 (m, 1H), 3.66 (m, 1H), 3.38 (m, 1H), 3.27 (m, 1H), 3.18 (m, 1H), 3.03 (t, $J=4.9$ Hz, 4H), 2.77 (tt, $J=3.4$, 12.0 Hz, 1H), 2.59-2.36 (m, 6H), 2.19 (dd, $J=6.0$, 12.9 Hz, 1H), 2.15 (s, 3H), 2.06-1.94 (b, 2H), 1.91-1.72 (m, 6H), 1.72-1.49 (m, 4H), 1.37 (qd, $J=3.3$, 12.3 Hz, 2H), 1.23-0.99 (m, 3H); MS (LC/MS, $\text{M}+\text{H}^+$): 504

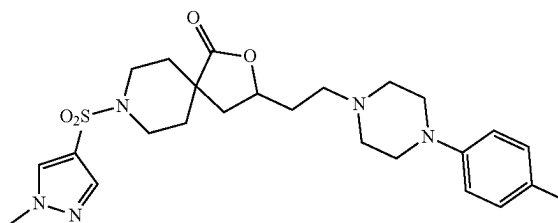


[0511] Example 49: Preparation of 8-(ethylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except ethanesulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/

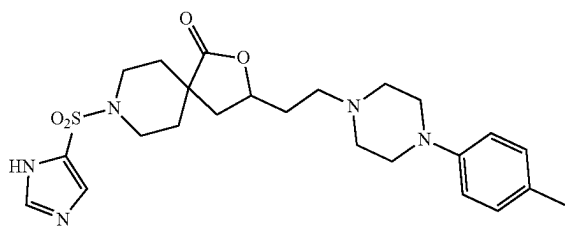
DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.08 (d, $J=8.2$ Hz, 2H), 6.85 (d, $J=8.6$ Hz, 2H), 4.61 (m, 1H), 3.75 (m, 1H), 3.51-3.35 (m, 2H), 3.30 (m, 1H), 3.15 (t, $J=4.9$ Hz, 4H), 2.98 (q, $J=7.4$ Hz, 2H), 2.70-2.48 (m, 6H), 2.31 (dd, $J=6.2$, 13.0 Hz, 1H), 2.28 (s, 3H), 2.07-1.96 (m, 2H), 1.96-1.66 (m, 5), 1.38 (t, $J=7.4$ Hz, 3H); MS (LC/MS, $\text{M}+\text{H}^+$): 450



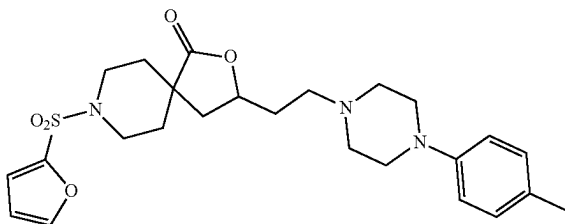
[0512] Example 50: Preparation of 8-(pyridin-3-ylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except pyridine-3-sulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0% ~10%): ^1H NMR (400 MHz, CDCl_3) δ 9.00 (d, $J=2.3$ Hz, 1H), 8.84 (dd, $J=1.5$, 4.8 Hz, 1H), 8.06 (dt, $J=1.9$, 8.0 Hz, 1H), 7.5 (dd, $J=4.9$, 7.9 Hz, 1H), 7.08 (d, $J=8.3$ Hz, 2H), 6.84 (d, $J=8.5$ Hz, 2H), 4.55 (m, 1H), 3.56 (m, 1H), 3.29 (m, 1H), 3.24-3.05 (m, 6H), 2.66-2.47 (m, 6H), 2.27 (s, 3H), 2.09 (dd, $J=6.0$, 13.0 Hz, 1H), 2.09-1.96 (m, 2H), 1.96-1.64 (m, 5H); MS (LC/MS, $\text{M}+\text{H}^+$): 499



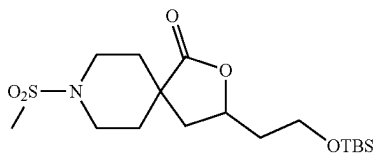
[0513] Example 51: Preparation of 8-((1-methyl-1H-pyrazol-4-yl)sulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 1-methyl-1H-pyrazole-4-sulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0%~10%): ^1H NMR (400 MHz, CDCl_3) δ 7.75 (s, 1H), 7.71 (s, 1H), 7.07 (d, $J=8.4$ Hz, 2H), 6.84 (d, $J=8.6$ Hz, 2H), 4.55 (m, 1H), 3.95 (s, 3H), 3.48 (m, 1H), 3.23 (m, 1H), 3.13 (t, $J=4.9$ Hz, 4H), 3.02 (m, 1H), 2.93 (m, 1H), 2.67-2.48 (m, 6H), 2.27 (s, 3H), 2.22 (dd, $J=6.1$, 12.9 Hz, 1H), 2.11-1.96 (m, 2H), 1.96-1.64 (m, 5H); MS (LC/MS, $\text{M}+\text{H}^+$): 502



[0514] Example 52: Preparation of 8-((1H-imidazol-4-yl)sulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except 1H-imidazole-4-sulfonyl chloride was substituted for methanesulfonyl chloride: ¹H NMR (400 MHz, MeOD) δ 7.88 (s, 1H), 7.74 (s, 1H), 7.09 (d, J=8.4 Hz, 2H), 6.90 (d, J=8.4 Hz, 2H), 4.58 (b, 1H), 3.67 (m, 1H), 3.53 (m, 1H), 3.40-3.05 (m, 10H), 2.99 (m, 1H), 2.85 (m, 1H), 2.38 (dd, J=5.7, 12.9 Hz, 1H), 2.26 (s, 3H), 2.19-1.91 (m, 3H), 1.90-1.72 (m, 3H), 1.72-1.58 (m, 1H); MS (LC/MS, M+H⁺): 488

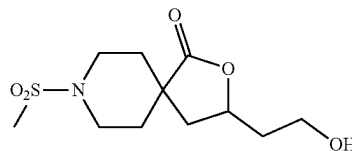


[0515] Example 53: Preparation of 8-(furan-2-ylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for 8-(methylsulfonyl)-3-(2-(4-(p-tolyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one formate, except furan-2-sulfonyl chloride was substituted for methanesulfonyl chloride and the title compound was purified by column chromatography on a silica gel column. (MeOH/DCM, 0% ~10%): ¹H NMR (400 MHz, CDCl₃) δ 7.59 (s, 1H), 7.07 (d, J=8.4 Hz, 2H), 7.02 (d, J=3.4 Hz, 1H), 6.84 (d, J=8.4 Hz, 2H), 6.52 (dd, J=1.7, 3.3 Hz, 1H), 4.56 (m, 1H), 3.70 (m, 1H), 3.47 (m, 1H), 3.24 (m, 1H), 3.19-3.07 (m, 5H), 2.68-2.46 (m, 6H), 2.33-2.18 (m, 4H), 2.07-1.79 (m, 4H), 1.79-1.60 (m, 3H); MS (LC/MS, M+H⁺): 488

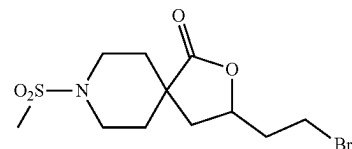


[0516] Example 54: Preparation of 3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: A mixture of 8-benzyl-3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one (3.25 g, 8.04 mmol, 1 eq.), Pd/C (0.65 g, 20% wt) and

MeOH (54 mL) was stirred at RT under 1 atm of H₂ (filled balloon) overnight. The mixture was filtered through a plug of Celite, washed with MeOH (50 mL) and concentrated in vacuo to give a crude oil. The crude oil (2.43 g) was dissolved in dichloromethane (50 mL), followed by addition of triethylamine (5.55 mL, 38.8 mmol, 5 eq.). Cooled to 0° C. before the addition of methanesulfonyl chloride (2.83 g, 13.0 mmol, 1.1 eq.). The reaction was allowed to warm to RT and stir for 30 min. At this point the reaction was diluted with D.I. water and then extracted with dichloromethane (3x50 mL), the combined organic phase was dried over Na₂SO₄ and concentrated in vacuo to give a crude product which was further purified by column chromatography (EtOAc/DCM, 0%~10%). ¹H NMR (400 MHz, CDCl₃) δ 4.61 (m, 1H), 3.75-3.65 (m, 2H), 3.60 (m, 1H), 3.37-3.18 (m, 3H), 2.75 (s, 3H), 2.21 (dd, J=6.0, 13.0 Hz, 1H), 2.02-1.91 (m, 2H), 1.91-1.62 (m, 5H), 0.83 (s, 9H), 0.00 (s, 6H).

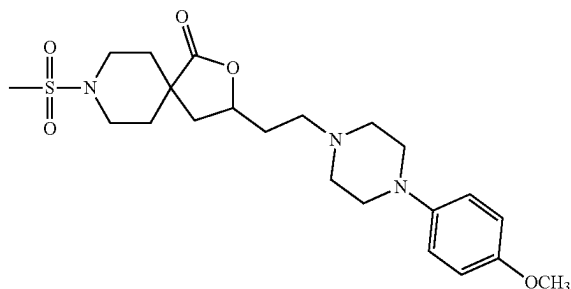


[0517] Example 55: Preparation of 3-(2-hydroxyethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for tert-butyl 3-(2-hydroxyethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate, except 3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one was substituted for tert-butyl 3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate and the product was initially purified by column chromatography on a C18 column. (ACN/H₂O, 0%~100%, w/0.1% formic acid), followed by purification on a silica gel column (MeOH/DCM, 0%~10%). ¹H NMR (400 MHz, CDCl₃) δ 4.73 (m, 1H), 3.84 (t, J=5.5 Hz, 2H), 3.68 (m, 1H), 3.41-3.32 (m, 2H), 3.29 (m, 1H), 2.83 (s, 3H), 2.33 (dd, J=6.0, 13.0 Hz, 1H), 2.10-2.00 (m, 2H), 2.10-1.71 (m, 6H).

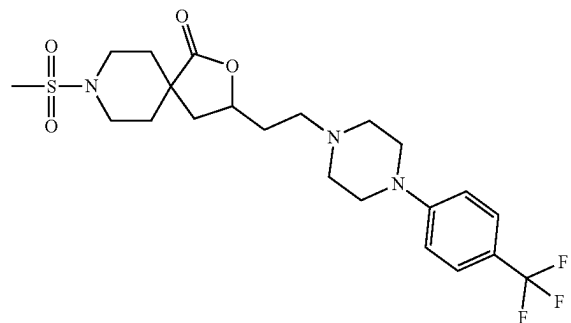


[0518] Example 56: Preparation of 3-(2-bromoethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: A solution of 3-(2-hydroxyethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one (0.890 g, 3.21 mmol, 1 eq.) and dichloromethane (12 mL) was cooled to 0° C. before triphenylphosphine (1.26 g, 4.81 mmol, 1.5 eq.) and carbon tetrabromide (1.6 g, 4.81 mmol, 1.5 eq.) were sequentially added to the solution. The reaction solution was allowed to warm to RT and for 4 hrs. The resulting mixture was then filtered and concentrated in vacuo to give a crude mixture. This mixture was suspended in diethyl ether (50 mL) and filtered 2x using diethyl ether to wash the filter cakes. The

final filtrate was loaded onto Celite in vacuo and further purified by column chromatography (EtOAc/DCM, 0%~40%). ¹H NMR (400 MHz, MeOD) δ 4.72 (m, 1H), 3.67 (m, 1H), 3.54 (dd, $J=5.3, 7.6$ Hz, 2H), 3.44-3.25 (m, 3H), 2.82 (s, 3H), 2.34 (dd, $J=6.0, 12.9$ Hz, 1H), 2.31-2.21 (m, 1H), 2.21-2.10 (m, 1H), 2.10-1.98 (m, 2H), 1.88-1.71 (m, 3H).

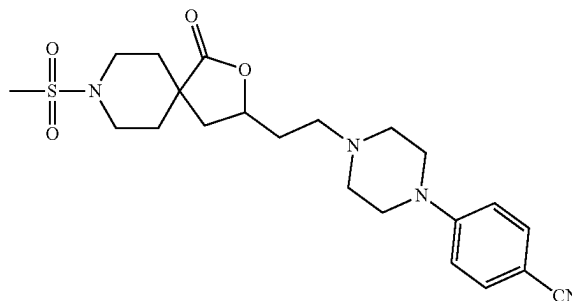


[0519] Example 57: Preparation of 3-(2-(4-(4-methoxyphenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: A solution of 3-(2-bromoethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one (50 mg, 0.147 mmol, 1 eq.), 1-(4-methoxyphenyl)-piperazine (59.33 mg, 0.308 mmol, 2.1 eq.) and Acetonitrile (2 mL) was microwaved for 1 hour at 120° C. The solvent was then evaporated in vacuo and the product was suspended in 15 mL of saturated NaHCO₃ and extracted in dichloromethane (3×15 mL). The combined organic phase was dried over Na₂SO₄ and concentrated in vacuo to give a crude mixture that was then dissolved in dichloromethane and purified by column chromatography (methanol/dichloromethane, 0%~10%). ¹H NMR (400 MHz, CDCl₃) δ 7.4 (d, $J=9.2$ Hz, 2H), 6.5 (d, $J=9.2$ Hz, 2H), 4.61 (m, 1H), 3.77 (s, 3H), 3.67 (m, 1H), 3.36 (m, 2H), 3.29 (m, 1H), 3.1 (t, $J=7.1$ Hz, 4H), 2.8 (s, 3H), 2.62 (m, 4H), 2.56 (t, $J=7.1$ Hz, 2H), 2.29 (dd, $J=7.2, 6$ Hz, 1H), 2.05 (m, 2H), 1.99-1.71 (m, 6H); MS (LC/MS, M+H⁺): 452.

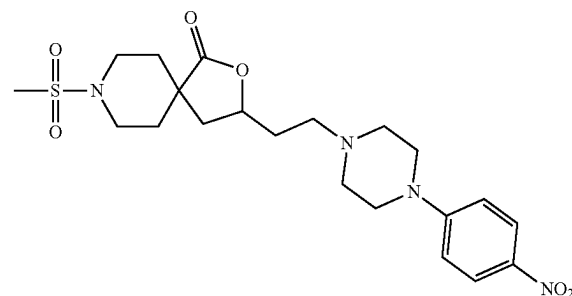


[0520] Example 58: Preparation of 3-(2-(4-(4-methoxyphenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared and purified according to the procedure for 3-(2-(4-(4-methoxyphenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one except that 1-(4-trifluoromethylphenyl)-piperazine was substituted for 1-(4-methoxyphenyl)-piperazine. ¹H NMR (400 MHz,

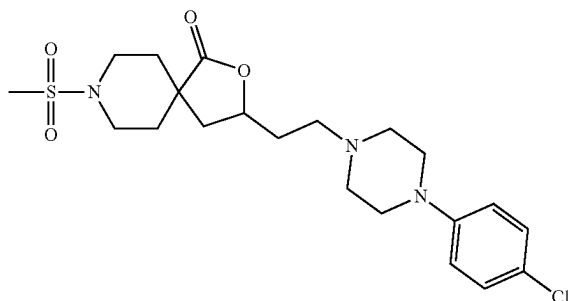
CDCl₃) δ 7.5 (d, $J=8.76$ Hz, 2H), 6.94 (d, $J=8.7$ Hz, 2H), 4.64 (m, 1H), 3.68 (m, 1H), 3.46-3.28 (m, 7H), 2.83 (s, 3H), 2.67-2.55 (m, 6H), 2.3 (dd, $J=7.1, 6.1$ Hz, 1H), 2.11-2.01 (m, 2H), 2.0-1.74 (m, 5H); MS (LC/MS, M+H⁺): 490



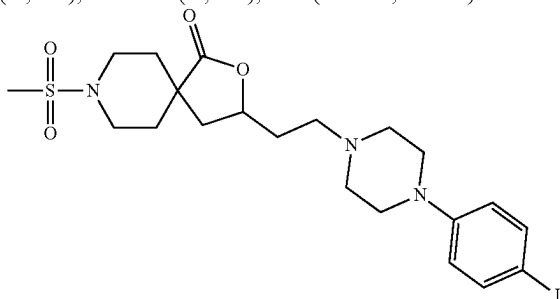
[0521] Example 59: Preparation of 4-(2-(8-(methylsulfonyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-3-yl)ethyl)piperazin-1-ylbenzonitrile: The title compound was prepared and purified according to the procedure for 3-(2-(4-(4-methoxyphenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one except that triethylamine (0.06 mL, 0.44 mmol, 3 eq.) was added in the microwave mixture and 1-(4-cyanophenyl)-piperazine was substituted for 1-(4-methoxyphenyl)-piperazine. ¹H NMR (400 MHz, CDCl₃) δ 7.49 (d, $J=8.4$ Hz, 2H), 6.85 (d, $J=8.4$ Hz, 2H), 4.61 (m, 1H), 3.65 (m, 1H), 3.43-3.26 (m, 7H), 2.81 (s, 3H), 2.64-2.52 (m, 6H), 2.28 (dd, $J=6.8, 5.9$ Hz, 1H), 2.02 (m, 2H), 1.96-1.74 (m, 5H); MS (LC/MS, M+H⁺): 447



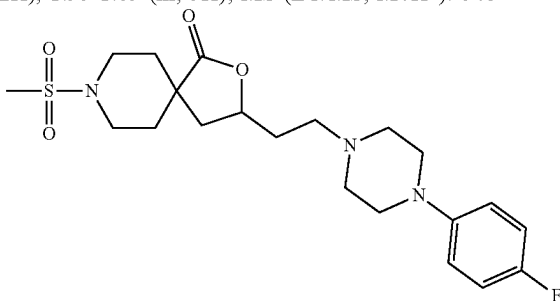
[0522] Example 60: Preparation of 8-(methylsulfonyl)-3-(2-(4-(4-nitrophenyl)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared and purified according to the procedure for 3-(2-(4-(4-methoxyphenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one except that it was microwaved for 1.5 hours at 120° C. and 1-(4-nitrophenyl)-piperazine was substituted for 1-(4-methoxyphenyl)-piperazine. ¹H NMR (400 MHz, CDCl₃) δ 8.13 (d, $J=9.3$ Hz, 2H), 6.82 (d, $J=9.3$ Hz, 2H), 4.62 (m, 1H), 3.66 (m, 1H), 3.46-3.36 (m, 4H), 3.36-3.28 (m, 3H), 2.8 (s, 3H), 2.67-2.5 (b, 6H), 2.28 (dd, $J=6.8, 5.9$ Hz, 1H), 2.08-1.98 (m, 2H), 1.95-1.72 (m, 5H); MS (LC/MS, M+H⁺): 467



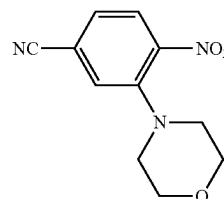
[0523] Example 61: Preparation of 3-(2-(4-(4-chlorophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared and purified according to the procedure for 3-(2-(4-(4-methoxyphenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one except that it was microwaved for 2 hours at 120° C. and 1-(4-chlorophenyl)-piperazine was substituted for 1-(4-methoxyphenyl)-piperazine. ¹H NMR (400 MHz, CDCl₃) δ 7.19 (d, J=8.9 Hz, 2H), 6.82 (d, J=8.9 Hz, 2H), 4.6 (m, 1H), 3.65 (m, 1H), 3.42-3.26 (m, 3H), 3.15 (t, J=4.9 Hz, 4H), 2.8 (s, 3H), 2.66-2.52 (m, 6H), 2.27 (dd, J=6.8, 5.9 Hz, 1H), 2.07-1.98 (m, 2H), 1.97-1.7 (m, 5H); MS (LC/MS, M+H⁺): 456



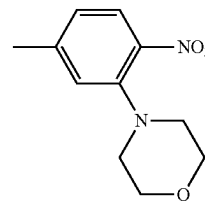
[0524] Example 62: Preparation of 3-(2-(4-(4-iodophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared and purified according to the procedure for 3-(2-(4-(4-chlorophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one except that and 1-(4-iodophenyl)-piperazine was substituted for 1-(4-methoxyphenyl)-piperazine. ¹H NMR (400 MHz, CDCl₃) δ 7.49 (d, J=8.8 Hz, 2H), 6.65 (d, J=8.8 Hz, 2H), 4.58 (m, 1H), 3.64 (m, 1H), 3.4-3.24 (m, 3H), 3.15 (t, J=4.8 Hz, 4H), 2.79 (s, 3H), 2.62-2.5 (m, 6H), 2.26 (dd, J=6.8, 6 Hz, 1H), 2.0-1.96 (m, 2H), 1.95-1.69 (m, 5H); MS (LC/MS, M+H⁺): 548



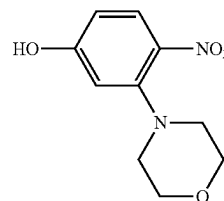
[0525] Example 63: Preparation of 3-(2-(4-(4-fluorophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared and purified according to the procedure for 3-(2-(4-(4-chlorophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one except that and 1-(4-fluorophenyl)-piperazine was substituted for 1-(4-methoxyphenyl)-piperazine. ¹H NMR (400 MHz, CDCl₃) δ 6.94 (m, 2H), 6.85 (m, 2H), 4.59 (m, 1H), 3.65 (m, 1H), 3.4-3.24 (m, 3H), 3.1 (t, J=4.8 Hz, 4H), 2.8 (s, 3H), 2.65-2.5 (m, 6H), 2.27 (dd, J=6.7, 5.9 Hz, 1H), 2.0 (m, 2H), 1.97-1.69 (m, 5H); MS (LC/MS, M+H⁺): 440.



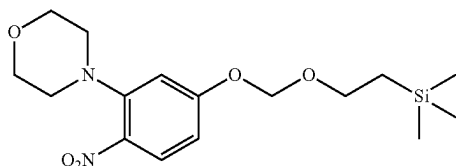
[0526] Preparation of 3-morpholino-4-nitrobenzonitrile: A solution of 3-fluoro-4-nitrobenzonitrile (0.6 g, 3.61 mmol, 1 eq.) and morpholine (0.629 g, 7.22 mmol, 2 eq.) in dimethyl sulfoxide (6.57 mL) was heated at 60° C. for 4 hours. The reaction solution was diluted with water 20 mL and extracted with ethyl acetate (3x20 mL). The combined organic phase was dried over Na₂SO₄ and concentrated in vacuo to give a crude product which was used in the next step without further purification. ¹H NMR (400 MHz, CDCl₃) δ 7.82 (d, J=8.3 Hz, 1H), 7.40 (d, J=1.5 Hz, 1H), 7.33 (dd, J=1.6, 8.3 Hz, 1H), 3.86 (m, 4H), 3.11 (m, 4H); MS (LC/MS, M+H⁺): 234



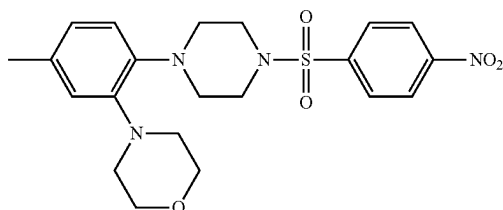
[0527] Preparation of 4-(5-methyl-2-nitrophenyl)morpholine: The title compound was prepared according to the procedure for 3-morpholino-4-nitrobenzonitrile, except 2-fluoro-4-methyl-1-nitrobenzene was substituted for 3-fluoro-4-nitrobenzonitrile. ¹H NMR (400 MHz, CDCl₃) δ 7.76 (d, J=8.2 Hz, 1H), 6.93 (b, 1H), 6.88 (d, J=8.3 Hz, 1H), 3.86 (m, 4H), 3.06 (m, 4H), 2.40 (s, 3H); MS (LC/MS, M+H⁺): 223



[0528] Preparation of 3-morpholino-4-nitrophenol: The title compound was prepared according to the procedure for 3-morpholino-4-nitrobenzonitrile, except 3-fluoro-4-nitrophenol was substituted for 3-fluoro-4-nitrobenzonitrile. ¹H NMR (400 MHz, MeOD) δ 7.90 (d, J=9.0 Hz, 1H), 6.56 (d, J=2.4 Hz, 1H), 6.49 (dd, J=2.5, 9.0 Hz, 1H), 3.83 (m, 4H), 3.02 (m, 4H); MS (LC/MS, M+H⁺): 225



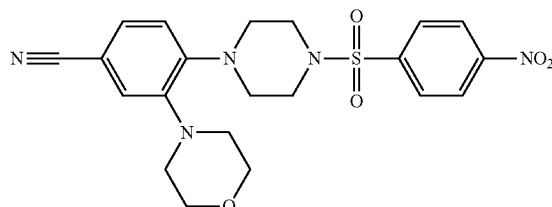
[0529] Preparation of 4-(2-nitro-5-((2-(trimethylsilyl)ethoxy)methoxy)phenyl)morpholine: A solution of 3-morpholino-4-nitrophenol (1.34 g, 5.98 mmol, 1 eq.), 2-(trimethylsilyl)ethoxymethyl chloride (1.05 g, 6.28 mmol, 1.05 eq.) and N,N-diisopropylethylamine (2.31 g, 17.9 mmol, 3 eq.) in dichloromethane (30.0 mL) was stirred at 25° C. for 16 hours. The reaction solution was diluted with 40 mL of water and extracted with dichloromethane (3×40 mL). The combined organic phase was dried over Na₂SO₄ and concentrated in vacuo to give a crude product which was used in the next step without further purification. ¹H NMR (400 MHz, CDCl₃) δ 7.93 (d, J=9.7 Hz, 1H), 6.68 (m, 2H), 5.24 (s, 2H), 3.85 (m, 4H), 3.74 (m, 2H), 3.04 (m, 4H), 0.94 (m, 2H), 0.00 (s, 9H); MS (LC/MS, M+H⁺): 355



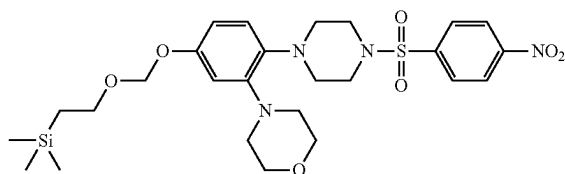
[0530] Preparation of 4-(5-methyl-2-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)phenyl)morpholine: A mixture of 4-(5-methyl-2-nitrophenyl)morpholine (1.58 g, 7.11 mmol, 1 eq.), Pd on carbon (316 mg, 20% wt) and methanol (72 mL) was stirred at 25° C. under 1 atm of H₂ (filled balloon) for 48 hours. The mixture was filtered through a plug of Celite, washed with methanol (50 mL) and concentrated in vacuo to give the crude intermediate, 4-methyl-2-morpholinoaniline.

[0531] (((4-nitrophenyl)sulfonyl)azanediyl)bis(ethane-2,1-diyl) bis(4-nitrobenzenesulfonate) (1.0 g, 1.5 mmol, 1 eq.), 4-methyl-2-morpholinoaniline (0.346 g, 1.8 mmol, 1.2 eq.), N,N-diisopropylethylamine (1.55 g, 12.0 mmol, 4 eq.) and acetonitrile (4.7 mL) were mixed in a microwave reaction vial (10 mL) fitted with a no-invasive vial cap. The reaction vials containing the mixture were reacted in the microwave for 1 h at 175° C. After 1 h, the solvent was removed under reduced pressure. The residue was dissolved in dichloromethane and washed with HCl (10%, 3×30 mL) and saturated NaHCO₃ (40 mL). The organic phase was dried over Na₂SO₄ and concentrated in vacuo to afford the crude product. The title compound was purified by column

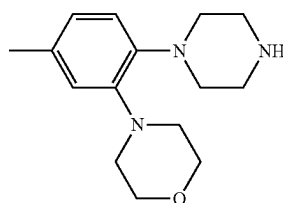
chromatography on a silica gel column. (ethyl acetate/dichloromethane, 0%~10%): ¹H NMR (400 MHz, CDCl₃) δ 8.44 (d, J=8.9 Hz, 2H), 8.02 (d, J=9.0 Hz, 2H), 6.82 (m, 2H), 6.74 (b, 1H), 3.71 (t, J=4.2 Hz, 4H), 3.24 (b, 8H), 3.05 (m, 4H), 2.29 (s, 3H); MS (LC/MS, M+H⁺): 447



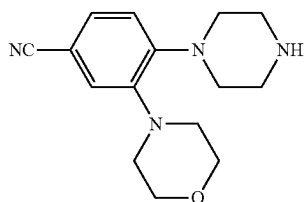
[0532] Preparation of 3-morpholino-4-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)benzonitrile: The title compound was prepared according to the procedure for 4-(5-methyl-2-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)phenyl)morpholine, except 3-morpholino-4-nitrobenzonitrile was substituted for 4-(5-methyl-2-nitrophenyl)morpholine and 4-amino-3-morpholinobenzonitrile for 4-methyl-2-morpholinoaniline. ¹H NMR (400 MHz, CDCl₃) δ 8.43 (d, J=8.8 Hz, 2H), 8.01 (d, J=8.8 Hz, 2H), 7.30 (dd, J=1.8, 8.2 Hz, 1H), 7.14 (d, J=1.8 Hz, 1H), 6.90 (d, J=8.3 Hz, 1H), 3.73 (t, J=4.5 Hz, 4H), 3.36 (m, 4H), 3.26 (m, 4H), 3.02 (t, J=4.3 Hz, 4H); MS (LC/MS, M+H⁺): 458



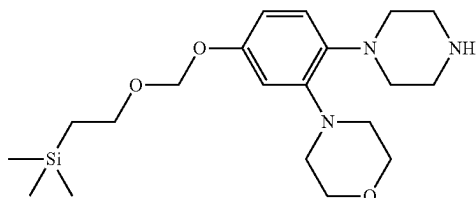
[0533] Preparation of 4-(2-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)-5-((2-(trimethylsilyl)ethoxy)methoxy)phenyl)morpholine: The title compound was prepared according to the procedure for 4-(5-methyl-2-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)phenyl)morpholine, except 4-(2-nitro-5-((2-(trimethylsilyl)ethoxy)methoxy)phenyl)morpholine was substituted for 4-(5-methyl-2-nitrophenyl)morpholine and 2-morpholino-4-((2-(trimethylsilyl)ethoxy)methoxy)aniline for 4-methyl-2-morpholinoaniline. ¹H NMR (400 MHz, CDCl₃) δ 8.41 (d, J=8.8 Hz, 2H), 8.01 (d, J=8.7 Hz, 2H), 6.80 (d, J=8.7 Hz, 1H), 6.67 (dd, J=2.7, 8.7 Hz, 1H), 6.60 (d, J=2.6 Hz, 1H), 5.14 (s, 2H), 3.74 (m, 2H), 3.68 (t, J=4.4 Hz, 4H), 3.23 (b, 4H), 3.16 (b, 4H), 3.03 (b, 4H), 0.95 (m, 2H), 0.00 (s, 9H); MS (LC/MS, M+H⁺): 579



[05334] Preparation of 4-(5-methyl-2-(piperazin-1-yl)phenyl)morpholine: Potassium carbonate (1.5 g, 10.8 mmol 12 eq.) was added to a mixture of acetonitrile and dimethyl-sulfoxide (CH₃CN/DMSO 49:1, 2.4 mL) and heated to 50° C. Thiophenol (0.988 g, 8.96 mmol, 10 eq.) was added dropwise via syringe to the mixture with stirring. After 30 minutes a solution of 4-(5-methyl-2-((4-nitrophenyl)sulfonyl)piperazin-1-yl)phenyl)morpholine (0.410 g, 0.896 mmol, 1 eq.) in acetonitrile and dimethyl sulfoxide (acetonitrile/dimethyl sulfoxide 49:1, 4.5 mL) was added dropwise. The reaction mixture was stirred for 3 hours, quenched with excess NaOH solution (40%) and concentrated under reduced pressure. The residue was extracted with dichloromethane (5×30 mL) and the organic phase was dried over MgSO₄, and concentrated in vacuo to give a crude oil. The oil was purified by reverse phase chromatography (acetonitrile in H₂O, gradient from 1%–100% with 0.1% formic acid) to afford the formic acid salt of the desired piperazine. The salt was dissolved in dichloromethane, washed with saturated NaHCO₃ solution, and the organic phase concentrated in vacuo to provide the product. ¹H NMR (400 MHz, CDCl₃) δ 6.87–6.76 (m, 2H), 6.71 (s, 1H), 3.84 (t, J=4.5 Hz, 4H), 3.18 (b, 4H), 3.07 (b, 4H), 2.98 (b, 4H), 2.29 (s, 3H); MS (LC/MS, M+H⁺): 262

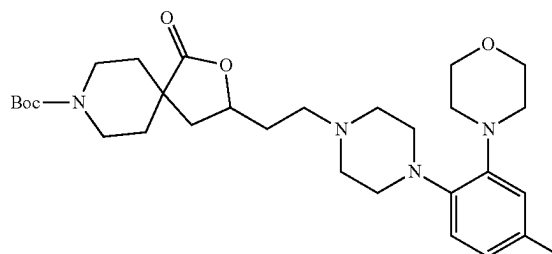


[0535] Preparation of 3-morpholino-4-(piperazin-1-yl)benzonitrile: The title compound was prepared according to the procedure for 4-(5-methyl-2-(piperazin-1-yl)phenyl)morpholine, except 3-morpholino-4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)benzonitrile was substituted for 4-(5-methyl-2-((4-nitrophenyl)sulfonyl)piperazin-1-yl)phenyl)morpholine. ¹H NMR (400 MHz, CDCl₃) δ 7.24 (dd, J=1.8, 8.2 Hz, 1H), 7.08 (d, J=1.8 Hz, 1H), 6.88 (d, J=8.4 Hz, 1H), 3.81 (t, J=4.6 Hz, 4H), 3.21 (b, 4H), 3.11 (b, 4H), 3.00 (b, 4H); MS (LC/MS, M+H⁺): 273

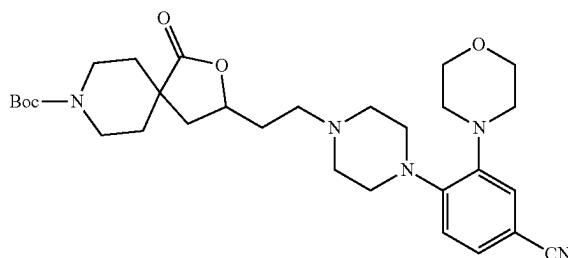


[0536] Preparation of 4-(2-(piperazin-1-yl)-5-((2-(trimethylsilyl)ethoxy)methoxy)phenyl)morpholine: The title compound was prepared according to the procedure for 4-(5-methyl-2-(piperazin-1-yl)phenyl)morpholine, except 4-(2-(4-((4-nitrophenyl)sulfonyl)piperazin-1-yl)-5-((2-(trimethylsilyl)ethoxy)methoxy)phenyl)morpholine was substituted for 4-(5-methyl-2-((4-nitrophenyl)sulfonyl)piperazin-1-yl)phenyl)morpholine. ¹H NMR (400 MHz,

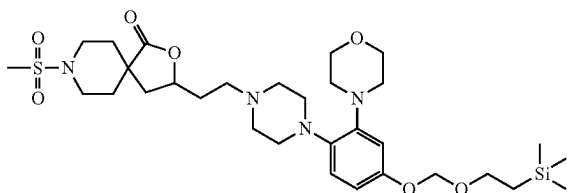
CDCl₃) δ 6.83 (d, J=8.6 Hz, 1H), 6.67 (dd, J=2.7, 8.5 Hz, 1H), 6.59 (d, J=2.8 Hz, 1H), 3.82 (t, J=4.7 Hz, 4H), 3.74 (m, 2H), 3.18 (b, 4H), 3.10–2.92 (b, 8H), 0.95 (m, 2H), 0.00 (s, 9H); MS (LC/MS, M+H⁺): 394



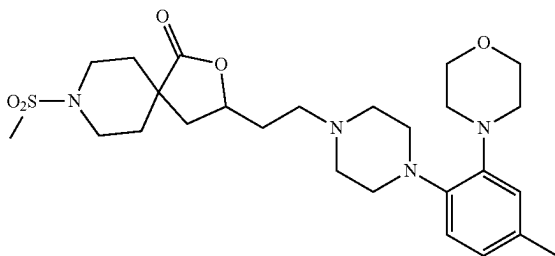
[0537] Preparation of tert-butyl 3-(2-(4-(4-methyl-2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: A solution of tert-butyl 3-(2-bromoethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate (0.05 g, 0.138 mmol, 1 eq.), 4-(5-methyl-2-(piperazin-1-yl)phenyl)morpholine (0.044 g, 0.166 mmol, 1.2 eq.), and triethylamine (0.070 g, 0.69 mmol, 5 eq.) in acetonitrile (2 mL) was microwaved at 120° C. for 1 hour. The reaction mixture was filtered and the filtrate was concentrated in vacuo to give a crude product. The title compound was purified by column chromatography on a silica gel column. (methanol/dichloromethane, 0%–10%) ¹H NMR (400 MHz, CDCl₃) δ 6.87–6.76 (m, 2H), 6.71 (s, 1H), 4.59 (m, 1H), 3.98 (m, 1H), 3.90–3.73 (m, 5H), 3.31–2.94 (m, 10H), 2.70–2.47 (m, 6H), 2.39 (dd, J=6.2, 12.8 Hz, 1H), 2.28 (s, 3H), 2.03–1.70 (m, 5H), 1.60 (m, 1H), 1.55–1.38 (m, 10H); MS (LC/MS, M+H⁺): 543



[0538] Preparation of tert-butyl 3-(2-(4-(4-cyano-2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate: The title compound was prepared according to the procedure for tert-butyl 3-(2-(4-(4-methyl-2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decane-8-carboxylate, except 3-morpholino-4-(piperazin-1-yl)benzonitrile was substituted for 4-(5-methyl-2-(piperazin-1-yl)phenyl)morpholine. ¹H NMR (400 MHz, CDCl₃) δ 7.26 (dd, J=1.8, 8.3 Hz, 1H), 7.09 (d, J=1.6 Hz, 1H), 6.89 (d, J=8.3 Hz, 1H), 4.57 (m, 1H), 3.96 (m, 1H), 3.89–3.73 (m, 5H), 3.40–2.98 (m, 10H), 2.72–2.45 (m, 6H), 2.37 (dd, J=6.1, 12.8 Hz, 1H), 2.03–1.67 (m, 5H), 1.58 (m, 1H), 1.54–1.38 (m, 10H); MS (LC/MS, M+H⁺): 554



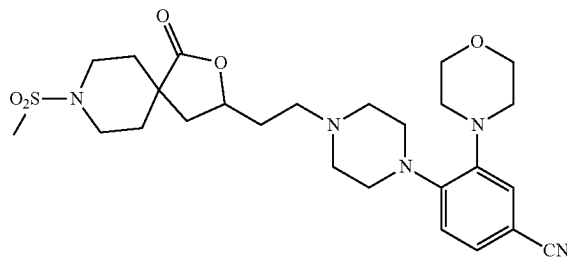
[0539] Preparation of 8-(methylsulfonyl)-3-(2-(4-(2-morpholino-4-((2-(trimethylsilyl)ethoxy)methoxy)phenoxy)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one: The title compound was prepared according to the procedure for tert-butyl 3-(2-(4-(4-methyl-2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate, except 4-(2-(piperazin-1-yl)-5-((2-(trimethylsilyl)ethoxy)methoxy)phenoxy)morpholine was substituted for 4-(5-methyl-2-(piperazin-1-yl)phenyl)morpholine and 3-(2-bromoethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one for tert-butyl 3-(2-bromoethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate. ^1H NMR (400 MHz, CDCl_3) δ 6.83 (d, $J=8.6$ Hz, 1H), 6.66 (dd, $J=2.7$, 8.6 Hz, 1H), 6.59 (d, $J=2.7$ Hz, 1H), 5.14 (s, 2H), 4.61 (m, 1H), 3.81 (t, $J=4.4$ Hz, 4H), 3.74 (m, 2H), 3.66 (m, 1H), 3.35 (m, 2H), 3.26 (m, 1H), 3.21-2.94 (b, 8H), 2.80 (s, 3H), 2.70-2.41 (m, 6H), 2.30 (dd, $J=5.6$, 12.9 Hz, 1H), 2.08-1.67 (m, 7H), 0.95 (m, 2H), 0.00 (s, 9H); MS (LC/MS, $\text{M}+\text{H}^+$): 653



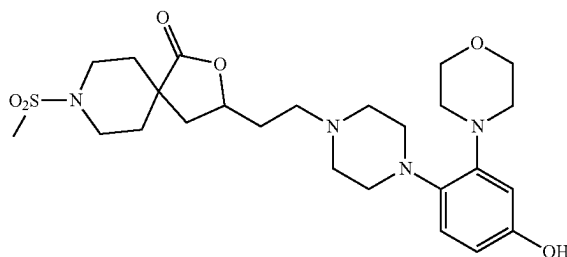
[0540] Preparation of 3-(2-(4-(4-methyl-2-morpholinophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: A solution of tert-butyl 3-(2-(4-(4-methyl-2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate (0.077 g, 0.142 mmol, 1 eq.) in trifluoroacetic acid:dichloromethane (1:3, 2 mL) was allowed to stir at 25° C. for 30 minutes. The reaction solution was diluted with methanol (2 mL) and concentrated in vacuo to give a crude intermediate as a trifluoroacetic acid salt. The resulting material was dissolved in dichloromethane (10 mL) and washed with sat. NaHCO_3 (aq.) solution (10 mL). The aqueous layer was backwashed with dichloromethane (2x10 mL) and the combined organic layers were dried over Na_2SO_4 and concentrated in vacuo to give a crude intermediate as a free base.

[0541] The resulting free base (0.052 g, 0.114 mmol, 1 eq.) was dissolved in dichloromethane (2 mL) and cooled to 0° C., and trimethylamine (0.058 g, 0.57 mmol, 5 eq.) and methanesulfonyl chloride (0.026, 0.228 mmol, 2 eq.) were added. The reaction solution was allowed to stir at 25° C. for 30 minutes and then concentrated in vacuo to give a crude solid. The title compound was purified by column chromatography on a silica gel column. (methanol/dichloromethane,

0%~10%) ^1H NMR (400 MHz, CDCl_3) δ 6.79-6.68 (m, 2H), 6.64 (s, 1H), 4.54 (m, 1H), 3.75 (t, $J=4.4$ Hz, 4H), 3.59 (m, 1H), 3.37-2.88 (b, 11H), 2.74 (s, 3H), 2.69-2.33 (b, 6H), 2.27-2.16 (m, 4H), 2.02-1.92 (m, 2H), 1.92-1.62 (m, 5H); MS (LC/MS, $\text{M}+\text{H}^+$): 521



[0542] Preparation of 4-(4-(2-(8-(methylsulfonyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-3-yl)ethyl)piperazin-1-yl)-3-morpholinobenzonitrile: The title compound was prepared according to the procedure for 3-(2-(4-(4-methyl-2-morpholinophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one, except tert-butyl 3-(2-(4-(4-cyano-2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate was substituted for 3-(2-(4-(4-methyl-2-morpholinophenyl)piperazin-1-yl)ethyl)-1-oxo-2-oxa-8-azaspiro[4.5]decan-8-carboxylate. ^1H NMR (400 MHz, CDCl_3) δ 7.29 (dd, $J=1.8$, 8.3 Hz, 1H), 7.12 (d, $J=1.7$ Hz, 1H), 6.91 (d, $J=8.3$ Hz, 1H), 4.63 (m, 1H), 3.85 (t, $J=4.3$ Hz, 4H), 3.67 (m, 1H), 3.55-2.96 (b, 11H), 2.83 (s, 3H), 2.75-2.40 (b, 6H), 2.30 (dd, $J=6.0$, 12.9 Hz, 1H), 2.12-1.99 (m, 2H), 1.99-1.71 (m, 5H); MS (LC/MS, $\text{M}+\text{H}^+$): 532



[0543] Preparation of 3-(2-(4-(4-hydroxy-2-morpholinophenyl)piperazin-1-yl)ethyl)-8-(methylsulfonyl)-2-oxa-8-azaspiro[4.5]decan-1-one: To a small vial, 8-(methylsulfonyl)-3-(2-(4-(2-morpholino-4-((2-(trimethylsilyl)ethoxy)methoxy)phenoxy)piperazin-1-yl)ethyl)-2-oxa-8-azaspiro[4.5]decan-1-one (0.190 g, 0.291 mmol, 1 eq.) was added and dissolved in hexamethylphosphoramide (1.5 mL). Tetra-n-butylammonium fluoride trihydrate (0.230 g, 0.873 mmol, 3 eq.) was added, followed by 300 mg of 4 Å molecular sieves. The reaction mixture was stirred at 60° C. for 48 hours before being filtered and concentrated in vacuo to give a crude oil. The title compound was purified by column chromatography on a silica gel column. (methanol/dichloromethane, 0%~10%) ^1H NMR (400 MHz, CDCl_3) δ 6.70 (d, $J=9.2$ Hz, 1H), 6.40-6.32 (m, 2H), 4.53 (m, 1H), 3.75 (t, $J=4.3$ Hz, 4H), 3.58 (m, 1H), 3.38-2.84 (b, 11H), 2.74 (s,

3H), 2.70-2.33 (b, 6H), 2.21 (dd, J=5.9, 12.8 Hz, 1H), 2.01-1.80 (m, 4H), 1.79-1.60 (m, 3H); MS (LC/MS, M+H⁺): 523.

Formulations

[0544] The present invention also relates to compositions or formulations which comprise the 5-hydroxytryptamine receptor 7 activity modulators according to the present invention. In general, the compositions of the present invention comprise an effective amount of one or more compounds of the disclosure and salts thereof according to the present invention which are effective for providing modulation of 5-hydroxytryptamine receptor 7 activity; and one or more excipients.

[0545] For the purposes of the present invention the term “excipient” and “carrier” are used interchangeably throughout the description of the present invention and said terms are defined herein as, “ingredients which are used in the practice of formulating a safe and effective pharmaceutical composition.”

[0546] The formulator will understand that excipients are used primarily to serve in delivering a safe, stable, and functional pharmaceutical, serving not only as part of the overall vehicle for delivery but also as a means for achieving effective absorption by the recipient of the active ingredient. An excipient may fill a role as simple and direct as being an inert filler, or an excipient as used herein may be part of a pH stabilizing system or coating to insure delivery of the ingredients safely to the stomach. The formulator can also take advantage of the fact the compounds of the present invention have improved cellular potency, pharmacokinetic properties, as well as improved oral bioavailability.

[0547] The present teachings also provide pharmaceutical compositions that include at least one compound described herein and one or more pharmaceutically acceptable carriers, excipients, or diluents. Examples of such carriers are well known to those skilled in the art and can be prepared in accordance with acceptable pharmaceutical procedures, such as, for example, those described in *Remington's Pharmaceutical Sciences*, 17th edition, ed. Alfonso R. Gennaro, Mack Publishing Company, Easton, PA (1985), the entire disclosure of which is incorporated by reference herein for all purposes. As used herein, “pharmaceutically acceptable” refers to a substance that is acceptable for use in pharmaceutical applications from a toxicological perspective and does not adversely interact with the active ingredient. Accordingly, pharmaceutically acceptable carriers are those that are compatible with the other ingredients in the formulation and are biologically acceptable. Supplementary active ingredients can also be incorporated into the pharmaceutical compositions.

[0548] Compounds of the present teachings can be administered orally or parenterally, neat or in combination with conventional pharmaceutical carriers. Applicable solid carriers can include one or more substances which can also act as flavoring agents, lubricants, solubilizers, suspending agents, fillers, glidants, compression aids, binders or tablet-disintegrating agents, or encapsulating materials. The compounds can be formulated in conventional manner, for example, in a manner similar to that used for known 5-hydroxytryptamine receptor 7 activity modulators. Oral formulations containing a compound disclosed herein can comprise any conventionally used oral form, including tablets, capsules, buccal forms, troches, lozenges and oral

liquids, suspensions or solutions. In powders, the carrier can be a finely divided solid, which is an admixture with a finely divided compound. In tablets, a compound disclosed herein can be mixed with a carrier having the necessary compression properties in suitable proportions and compacted in the shape and size desired. The powders and tablets can contain up to 99% of the compound.

[0549] Capsules can contain mixtures of one or more compound(s) disclosed herein with inert filler(s) and/or diluent(s) such as pharmaceutically acceptable starches (e.g., corn, potato or tapioca starch), sugars, artificial sweetening agents, powdered celluloses (e.g., crystalline and microcrystalline celluloses), flours, gelatins, gums, and the like.

[0550] Useful tablet formulations can be made by conventional compression, wet granulation or dry granulation methods and utilize pharmaceutically acceptable diluents, binding agents, lubricants, disintegrants, surface modifying agents (including surfactants), suspending or stabilizing agents, including, but not limited to, magnesium stearate, stearic acid, sodium lauryl sulfate, talc, sugars, lactose, dextrin, starch, gelatin, cellulose, methyl cellulose, microcrystalline cellulose, sodium carboxymethyl cellulose, carboxymethylcellulose calcium, polyvinylpyrrolidone, alginic acid, acacia gum, xanthan gum, sodium citrate, complex silicates, calcium carbonate, glycine, sucrose, sorbitol, dicalcium phosphate, calcium sulfate, lactose, kaolin, mannitol, sodium chloride, low melting waxes, and ion exchange resins. Surface modifying agents include nonionic and anionic surface modifying agents. Representative examples of surface modifying agents include, but are not limited to, poloxamer 188, benzalkonium chloride, calcium stearate, cetostearyl alcohol, cetomacrogol emulsifying wax, sorbitan esters, colloidal silicon dioxide, phosphates, sodium dodecylsulfate, magnesium aluminum silicate, and triethanolamine. Oral formulations herein can utilize standard delay or time-release formulations to alter the absorption of the compound(s). The oral formulation can also consist of administering a compound disclosed herein in water or fruit juice, containing appropriate solubilizers or emulsifiers as needed.

[0551] Liquid carriers can be used in preparing solutions, suspensions, emulsions, syrups, elixirs, and for inhaled delivery. A compound of the present teachings can be dissolved or suspended in a pharmaceutically acceptable liquid carrier such as water, an organic solvent, or a mixture of both, or a pharmaceutically acceptable oils or fats. The liquid carrier can contain other suitable pharmaceutical additives such as solubilizers, emulsifiers, buffers, preservatives, sweeteners, flavoring agents, suspending agents, thickening agents, colors, viscosity regulators, stabilizers, and osmo-regulators. Examples of liquid carriers for oral and parenteral administration include, but are not limited to, water (particularly containing additives as described herein, e.g., cellulose derivatives such as a sodium carboxymethyl cellulose solution), alcohols (including monohydric alcohols and polyhydric alcohols, e.g., glycols) and their derivatives, and oils (e.g., fractionated coconut oil and *arachis* oil). For parenteral administration, the carrier can be an oily ester such as ethyl oleate and isopropyl myristate. Sterile liquid carriers are used in sterile liquid form compositions for parenteral administration. The liquid carrier for pressurized compositions can be halogenated hydrocarbon or other pharmaceutically acceptable propellants.

[0552] Liquid pharmaceutical compositions, which are sterile solutions or suspensions, can be utilized by, for example, intramuscular, intraperitoneal or subcutaneous injection. Sterile solutions can also be administered intravenously. Compositions for oral administration can be in either liquid or solid form.

[0553] Preferably the pharmaceutical composition is in unit dosage form, for example, as tablets, capsules, powders, solutions, suspensions, emulsions, granules, or suppositories. In such form, the pharmaceutical composition can be sub-divided in unit dose(s) containing appropriate quantities of the compound. The unit dosage forms can be packaged compositions, for example, packeted powders, vials, ampoules, prefilled syringes or sachets containing liquids. Alternatively, the unit dosage form can be a capsule or tablet itself, or it can be the appropriate number of any such compositions in package form. Such unit dosage form can contain from about 1 mg/kg of compound to about 500 mg/kg of compound, and can be given in a single dose or in two or more doses. Such doses can be administered in any manner useful in directing the compound(s) to the recipient's bloodstream, including orally, via implants, parenterally (including intravenous, intraperitoneal and subcutaneous injections), rectally, vaginally, and transdermally.

[0554] When administered for the treatment or inhibition of a particular disease state or disorder, it is understood that an effective dosage can vary depending upon the particular compound utilized, the mode of administration, and severity of the condition being treated, as well as the various physical factors related to the individual being treated. In therapeutic applications, a compound of the present teachings can be provided to a patient already suffering from a disease in an amount sufficient to cure or at least partially ameliorate the symptoms of the disease and its complications. The dosage to be used in the treatment of a specific individual typically must be subjectively determined by the attending physician. The variables involved include the specific condition and its state as well as the size, age and response pattern of the patient.

[0555] In some cases it may be desirable to administer a compound directly to the airways of the patient, using devices such as, but not limited to, metered dose inhalers, breath-operated inhalers, multidose dry-powder inhalers, pumps, squeeze-actuated nebulized spray dispensers, aerosol dispensers, and aerosol nebulizers. For administration by intranasal or intrabronchial inhalation, the compounds of the present teachings can be formulated into a liquid composition, a solid composition, or an aerosol composition. The liquid composition can include, by way of illustration, one or more compounds of the present teachings dissolved, partially dissolved, or suspended in one or more pharmaceutically acceptable solvents and can be administered by, for example, a pump or a squeeze-actuated nebulized spray dispenser. The solvents can be, for example, isotonic saline or bacteriostatic water. The solid composition can be, by way of illustration, a powder preparation including one or more compounds of the present teachings intermixed with lactose or other inert powders that are acceptable for intrabronchial use, and can be administered by, for example, an aerosol dispenser or a device that breaks or punctures a capsule encasing the solid composition and delivers the solid composition for inhalation. The aerosol composition can include, by way of illustration, one or more compounds of the present teachings, propellants, surfactants, and co-sol-

vents, and can be administered by, for example, a metered device. The propellants can be a chlorofluorocarbon (CFC), a hydrofluoroalkane (HFA), or other propellants that are physiologically and environmentally acceptable.]

[0556] Compounds described herein can be administered parenterally or intraperitoneally. Solutions or suspensions of these compounds or a pharmaceutically acceptable salts, hydrates, or esters thereof can be prepared in water suitably mixed with a surfactant such as hydroxyl-propylcellulose. Dispersions can also be prepared in glycerol, liquid polyethylene glycols, and mixtures thereof in oils. Under ordinary conditions of storage and use, these preparations typically contain a preservative to inhibit the growth of microorganisms.

[0557] The pharmaceutical forms suitable for injection can include sterile aqueous solutions or dispersions and sterile powders for the extemporaneous preparation of sterile injectable solutions or dispersions. In some embodiments, the form can be sterile and its viscosity permits it to flow through a syringe. The form preferably is stable under the conditions of manufacture and storage and can be preserved against the contaminating action of microorganisms such as bacteria and fungi. The carrier can be a solvent or dispersion medium containing, for example, water, ethanol, polyol (e.g., glycerol, propylene glycol and liquid polyethylene glycol), suitable mixtures thereof, and vegetable oils.

[0558] Compounds described herein can be administered transdermally, i.e., administered across the surface of the body and the inner linings of bodily passages including epithelial and mucosal tissues. Such administration can be carried out using the compounds of the present teachings including pharmaceutically acceptable salts, hydrates, or esters thereof, in lotions, creams, foams, patches, suspensions, solutions, and suppositories (rectal and vaginal).

[0559] Transdermal administration can be accomplished through the use of a transdermal patch containing a compound, such as a compound disclosed herein, and a carrier that can be inert to the compound, can be non-toxic to the skin, and can allow delivery of the compound for systemic absorption into the blood stream via the skin. The carrier can take any number of forms such as creams and ointments, pastes, gels, and occlusive devices. The creams and ointments can be viscous liquid or semisolid emulsions of either the oil-in-water or water-in-oil type. Pastes comprised of absorptive powders dispersed in petroleum or hydrophilic petroleum containing the compound can also be suitable. A variety of occlusive devices can be used to release the compound into the blood stream, such as a semi-permeable membrane covering a reservoir containing the compound with or without a carrier, or a matrix containing the compound. Other occlusive devices are known in the literature.

[0560] Compounds described herein can be administered rectally or vaginally in the form of a conventional suppository. Suppository formulations can be made from traditional materials, including cocoa butter, with or without the addition of waxes to alter the suppository's melting point, and glycerin. Water-soluble suppository bases, such as polyethylene glycols of various molecular weights, can also be used.

[0561] Lipid formulations or nanocapsules can be used to introduce compounds of the present teachings into host cells either in vitro or in vivo. Lipid formulations and nanocapsules can be prepared by methods known in the art.

[0562] To increase the effectiveness of compounds of the present teachings, it can be desirable to combine a compound with other agents effective in the treatment of the target disease. For example, other active compounds (i.e., other active ingredients or agents) effective in treating the target disease can be administered with compounds of the present teachings. The other agents can be administered at the same time or at different times than the compounds disclosed herein.

[0563] Compounds of the present teachings can be useful for the treatment or inhibition of a pathological condition or disorder in a mammal, for example, a human subject. The present teachings accordingly provide methods of treating or inhibiting a pathological condition or disorder by providing to a mammal a compound of the present teachings including its pharmaceutically acceptable salt) or a pharmaceutical composition that includes one or more compounds of the present teachings in combination or association with pharmaceutically acceptable carriers. Compounds of the present teachings can be administered alone or in combination with other therapeutically effective compounds or therapies for the treatment or inhibition of the pathological condition or disorder.

[0564] Non-limiting examples of compositions according to the present invention include from about 0.001 mg to about 1000 mg of one or more compounds of the disclosure according to the present invention and one or more excipients; from about 0.01 mg to about 100 mg of one or more compounds of the disclosure according to the present invention and one or more excipients; and from about 0.1 mg to about 10 mg of one or more compounds of the disclosure according to the present invention; and one or more excipients.

Procedures

[0565] The following procedures can be utilized in evaluating and selecting compounds as 5-hydroxytryptamine receptor 7 activity modulators.

[0566] Radiolabel Binding Studies for Serotonin 5HT₇ receptors, method 1:

[0567] A solution of the compound of the disclosure to be tested is prepared as a 1-mg/ml stock in Assay Buffer or DMSO according to its solubility. A similar stock of the reference compound chlorpromazine is also prepared as a positive control. Eleven dilutions (5×assay concentration) of the compound of the disclosure and chlorpromazine are prepared in the Assay Buffer by serial dilution to yield final corresponding assay concentrations ranging from 10 pM to 10 μM.

[0568] A stock concentration of 5 nM [³H]LSD (lysergic acid diethyl amide) is prepared in 50 mM Tris-HCl, 10 mM MgCl₂, 1 mM EDTA, pH 7.4 (Assay Buffer). Aliquots (50 μl) of radioligand are dispensed into the wells of a 96-well plate containing 100 μl of Assay Buffer. Duplicate 50-μl aliquots of the compound of the disclosure test and chlorpromazine positive control reference compound serial dilutions are added.

[0569] Membrane fractions of cells expressing recombinant 5HT₇ receptors (50 L) are dispensed into each well. The membranes are prepared from stably transfected cell lines expressing 5HT₇ receptors cultured on 10-cm plates by harvesting PBS-rinsed monolayers, resuspending and lysing in chilled, hypotonic 50 mM Tris-HCl, pH 7.4, centrifuging at 20,000×g, decanting the supernatant and storing at -80°

C.; the membrane preparations are resuspended in 3 ml of chilled Assay Buffer and homogenized by several passages through a 26 gauge needle before using in the assay.

[0570] The 250-μl reactions are incubated at room temperature for 1.5 hours, then harvested by rapid filtration onto 0.3% polyethyleneimine-treated, 96-well filter mats using a 96-well Filtermate harvester. Four rapid 500-μl washes are performed with chilled Assay Buffer to reduce non-specific binding. The filter mats are dried, then scintillant is added to the filters and the radioactivity retained on the filters is counted in a Microbeta scintillation counter.

[0571] Raw data (dpm) representing total radioligand binding (i.e., specific+non-specific binding) are plotted as a function of the logarithm of the molar concentration of the competitor (i.e., test or reference compound). Non-linear regression of the normalized (i.e., percent radioligand binding compared to that observed in the absence of test or reference compound) raw data is performed in Prism 4.0 (GraphPad Software) using the built-in three parameter logistic model describing ligand competition binding to radioligand-labeled sites:

$$y = \text{bottom} + [(\text{top} - \text{bottom}) / (1 + 10^{\times - \log \text{IC}_{50}})]$$

where bottom equals the residual radioligand binding measured in the presence of 10 M reference compound (i.e., non-specific binding) and top equals the total radioligand binding observed in the absence of competitor. The log IC₅₀ (i.e., the log of the ligand concentration that reduces radioligand binding by 50%) is thus estimated from the data and used to obtain the K_i by applying the Cheng-Prusoff approximation:

$$K_i = \text{IC}_{50} / (1 + [\text{ligand}] / \text{KD})$$

where [ligand] equals the assay radioligand concentration and KD equals the affinity constant of the radioligand for the target receptor.

[0572] Compounds of the disclosure are also screened at a single concentration of 10 μM using the same method described for the Radiolabel Binding Studies for Serotonin 5HT₇ receptors to determine the percent inhibition of [³H] LSD binding.

Radiolabel Binding Studies for Serotonin 5-HT7 Receptors, Method 2:

[0573] A solution of the compound of the disclosure to be tested is prepared as a 1-mg/ml stock in Assay Buffer or DMSO according to its solubility. A similar stock of the reference compound chlorpromazine is also prepared as a positive control. Eleven dilutions (5×assay concentration) of the compound of the disclosure and chlorpromazine are prepared in the Assay Buffer by serial dilution to yield final corresponding assay concentrations ranging from 10 pM to 10 μM.

[0574] A stock concentration of 5 nM [³H]-5-Hydroxytryptamine ([³H]-5HT) is prepared in 50 mM Tris-HCl, 10 mM MgCl₂, 1 mM EDTA, pH 7.4 (Assay Buffer). Aliquots (50 μl) of radioligand are dispensed into the wells of a 96-well plate containing 100 μl of Assay Buffer. Duplicate 50-μl aliquots of the compound of the disclosure test and chlorpromazine positive control reference compound serial dilutions are added.

[0575] Membrane fractions of cells expressing recombinant 5HT₇ receptors (50 L) are dispensed into each well. The membranes are prepared from stably transfected cell lines expressing 5HT₇ receptors cultured on 10-cm plates by harvesting PBS-rinsed monolayers, resuspending and lysing in chilled, hypotonic 50 mM Tris-HCl, pH 7.4, centrifuging at 20,000×g, decanting the supernatant and storing at -80° C.; the membrane preparations are resuspended in 3 ml of chilled Assay Buffer and homogenized by several passages through a 26 gauge needle before using in the assay.

[0576] The 250-μl reactions are incubated at room temperature for 1.5 hours, then harvested by rapid filtration onto 0.3% polyethyleneimine-treated, 96-well filter mats using a 96-well Filtermate harvester. Four rapid 500-μl washes are performed with chilled Assay Buffer to reduce non-specific binding. The filter mats are dried, then scintillant is added to the filters and the radioactivity retained on the filters is counted in a Microbeta scintillation counter.

[0577] Raw data (dpm) representing total radioligand binding (i.e., specific+non-specific binding) are plotted as a function of the logarithm of the molar concentration of the competitor (i.e., test or reference compound). Non-linear regression of the normalized (i.e., percent radioligand binding compared to that observed in the absence of test or reference compound) raw data is performed in Prism 4.0 (GraphPad Software) using the built-in three parameter logistic model describing ligand competition binding to radioligand-labeled sites:

$$y = \text{bottom} + \frac{(\text{top} - \text{bottom})}{(1 + 10^{x - \log IC_{50}})}$$

where bottom equals the residual radioligand binding measured in the presence of 10 μM reference compound (i.e., non-specific binding) and top equals the total radioligand binding observed in the absence of competitor. The log IC₅₀ (i.e., the log of the ligand concentration that reduces radioligand binding by 50%) is thus estimated from the data and used to obtain the Ki by applying the Cheng-Prusoff approximation:

$$Ki = IC_{50} / (1 + [\text{ligand}] / KD)$$

where [ligand] equals the assay radioligand concentration and KD equals the affinity constant of the radioligand for the target receptor.

[0578] Compounds of the disclosure are also screened at a single concentration of 10 μM using the same method described for the Radiolabel Binding Studies for Serotonin 5HT₇ receptors to determine the percent inhibition of [³H]-5HT binding.

[0579] Results for representative compounds according to the present invention are listed in Table 11.

TABLE 11

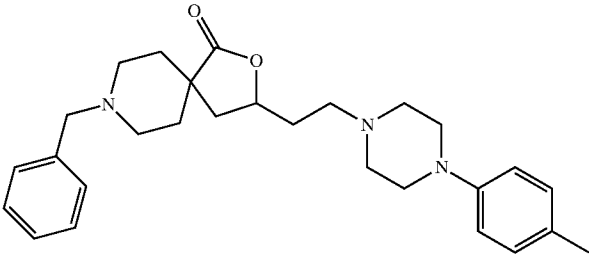
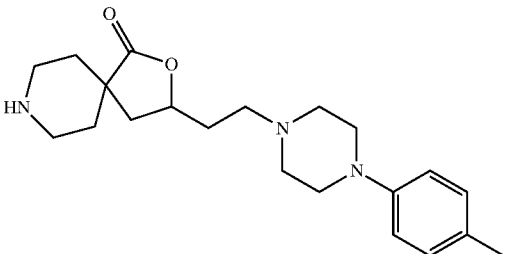
Radiolabel Binding Studies for Serotonin 5HT ₇ receptors results for exemplary compounds of the disclosure			
Entry	Structure	5-HT ₇ % inhib @ 10 uM	IC ₅₀ (nm)
1		96.5	18
2		88.8	149

TABLE 11-continued

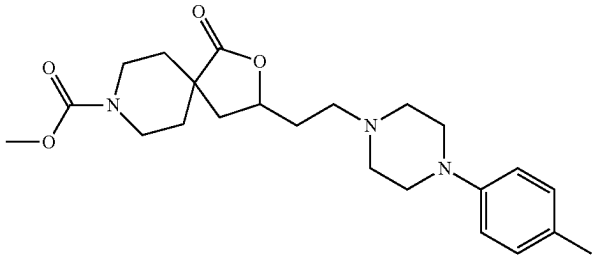
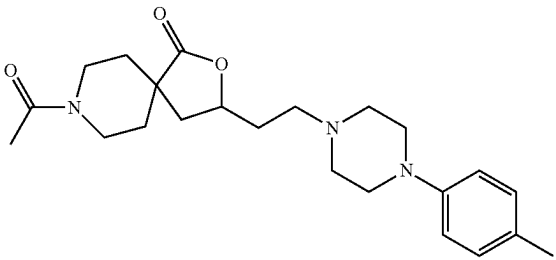
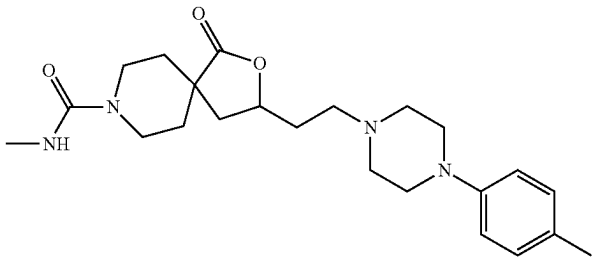
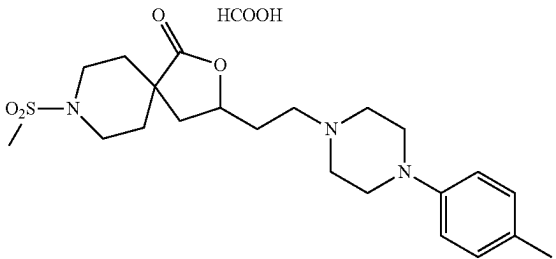
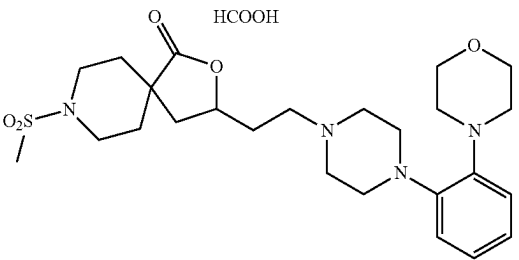
Radiolabel Binding Studies for Serotonin 5HT ₇ receptors results for exemplary compounds of the disclosure			
Entry	Structure	5-HT ₇ % inhib @10 uM	IC ₅₀ (nm)
3		90	81
4		88.5	122
5		91.6	102
6		90.5	93
7		N.D.	34

TABLE 11-continued

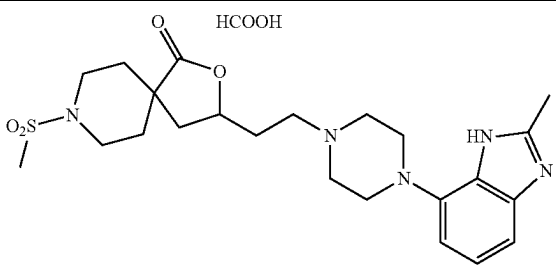
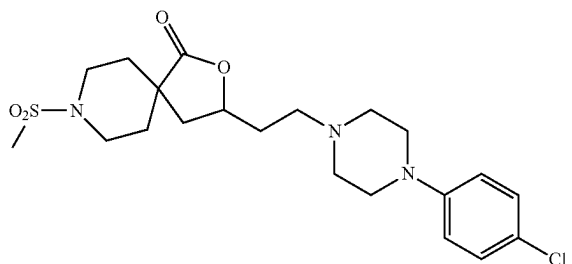
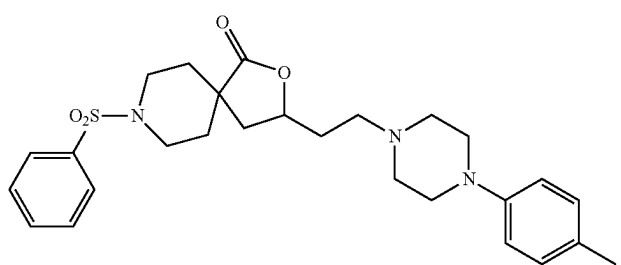
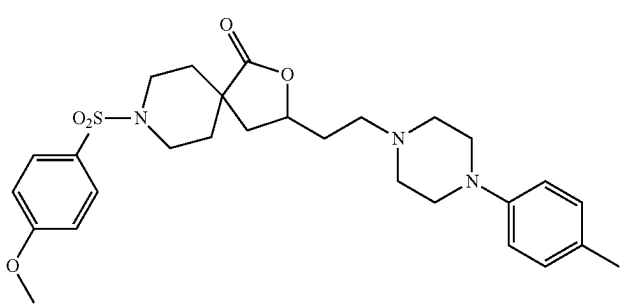
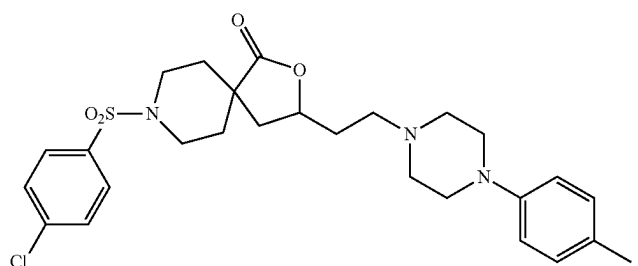
Radiolabel Binding Studies for Serotonin 5HT ₇ receptors results for exemplary compounds of the disclosure			
Entry	Structure	5-HT ₇ % inhib @10 uM	5-HT ₇ IC ₅₀ (nM)
8		100	46
9		N.D.	303
10		N.D.	64
11		N.D.	77
12		N.D.	108

TABLE 11-continued

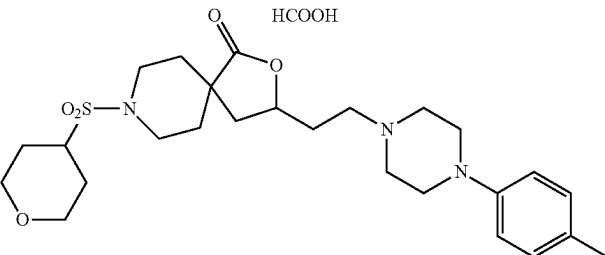
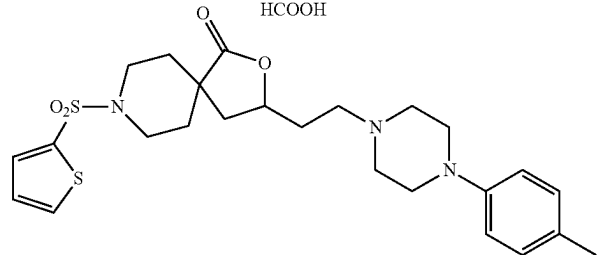
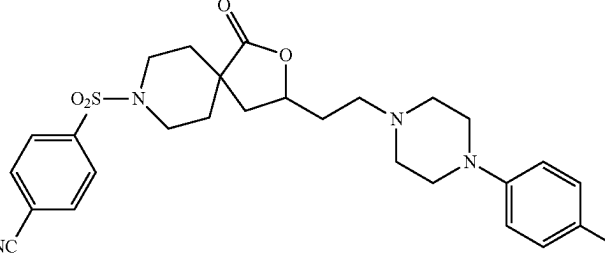
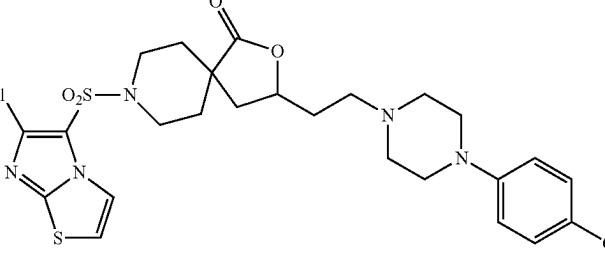
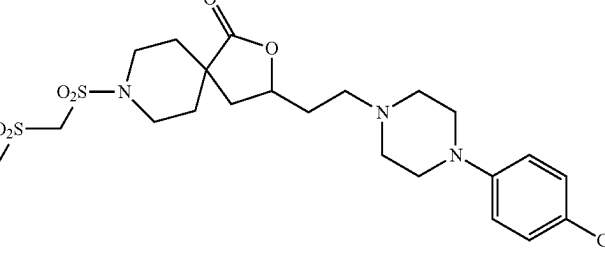
Radiolabel Binding Studies for Serotonin 5HT ₇ receptors results for exemplary compounds of the disclosure			
Entry	Structure	5-HT ₇ % inhib @10 uM	5-HT ₇ IC ₅₀ (nM)
13		N.D.	96
14		N.D.	47
15		N.D.	131
16		N.D.	56
17		N.D.	116

TABLE 11-continued

Radiolabel Binding Studies for Serotonin 5HT ₇ receptors results for exemplary compounds of the disclosure			
Entry	Structure	5-HT ₇ % inhib @10 uM	5-HT ₇ IC ₅₀ (nM)
18	 Chemical structure 18 is a spirocyclic compound. It features a spiro[3.3]heptanone core. One ring of the spiro system is substituted with a 4-chlorophenyl group via a piperazine linker. The other ring is substituted with a methanesulfonyl group and a nitrile group.	N.D.	86
19	 Chemical structure 19 is a spirocyclic compound. It features a spiro[3.3]heptanone core. One ring of the spiro system is substituted with a 4-cyanophenyl group via a piperazine linker. The other ring is substituted with a methanesulfonyl group and a methyl group. A morpholine ring is attached to the piperazine linker.	N.D.	461
20	 Chemical structure 20 is a spirocyclic compound. It features a spiro[3.3]heptanone core. One ring of the spiro system is substituted with a 4-methylphenyl group via a piperazine linker. The other ring is substituted with a methanesulfonyl group and a methyl group. A morpholine ring is attached to the piperazine linker.	N.D.	67
21	 Chemical structure 21 is a spirocyclic compound. It features a spiro[3.3]heptanone core. One ring of the spiro system is substituted with a 4-hydroxyphenyl group via a piperazine linker. The other ring is substituted with a methanesulfonyl group and a methyl group. A morpholine ring is attached to the piperazine linker.	N.D.	75
22	 Chemical structure 22 is a spirocyclic compound. It features a spiro[3.3]heptanone core. One ring of the spiro system is substituted with a 4-hydroxyphenyl group via a piperazine linker. The other ring is substituted with a methanesulfonyl group and a methyl group.	N.D.	202

N.D. = not determined

Functional Serotonin 5HT₇ assay, method 1:

[0580] Cell lines stably expressing human 5HT₇ receptors are seeded in 96-well, poly-L-lysine-coated plates 48 hours prior to the assay (40,000 cells per well) in Dulbecco's Modified Eagle Medium (DMEM) containing 5% dialyzed serum. Twenty hours prior to the assay, the medium is changed to serum-free DMEM. On the day of the assay, the DMEM is washed and replaced with 30 μ l of assay buffer (1 $\times$ Krebs-Ringer bicarbonate glucose buffer, 0.75 mM IBMX, pH 7.4). A 10-min pre-incubation is performed in a 37-degree centigrade, humidified incubator. Then, the cells are stimulated by addition of 30 μ l of 2 $\times$ dilutions of compounds of the disclosure or chlorpromazine (final concentrations ranging from 0.1 nM to 10 M, each concentration assayed in triplicate). A positive control (100 μ M forskolin) is also included. Accumulation of cAMP is allowed to continue for 15 min, after which the buffer is removed and the cells are lysed with Cell Lysis Buffer (CatchPoint cAMP Assay Kit, Molecular Devices). Next, the lysates are transferred to 96-well, glass-bottom plates coated with goat anti-rabbit IgG and adsorbed with rabbit anti-cAMP (Molecular Devices). Following a 5 minute incubation, horseradish peroxidase-cAMP conjugate is added (Molecular Devices) and a 2-hour incubation is performed at room temperature. Then, after three washes with Wash Buffer (Molecular Devices), Stoplight Red substrate (Molecular Devices), reconstituted in Substrate Buffer (Molecular Devices) containing freshly-added 1 mM H₂O₂, is added and, after a 15-min incubation at room temperature, fluorescence is measured (excitation 510-545 nm, emission 565-625 nm). For each assay, a cAMP calibration curve is generated and controls without lysate and without antibody are included.

[0581] For agonist tests, raw data (maximum fluorescence, fluorescence units) for each concentration of the compounds of the disclosure or chlorpromazine are normalized to the basal (vehicle-stimulated) fluorescence (reported as fold increase over basal) and plotted as a function of the logarithm of the molar concentration of the drug (i.e., test or reference compound). Non-linear regression of the normalized data is performed in Prism 4.0 (GraphPad Software) using the built-in three parameter logistic model (i.e., sigmoidal concentration-response) describing agonist-stimulated activation of one receptor population:

$$y = \text{bottom} + [(\text{top} - \text{bottom}) / (1 + 10^{\log EC_{50}})]$$

where bottom equals the best-fit basal fluorescence and top equals the best-fit maximal fluorescence stimulated by the compound of the disclosure or chlorpromazine. The log EC₅₀ (i.e., the log of the drug concentration that increases fluorescence by 50% of the maximum fluorescence observed for the compound of the disclosure or chlorpromazine) is thus estimated from the data, and the EC₅₀ (agonist potency) is obtained. To obtain an estimate of the relative efficacy of the test compound (Rel. Emax), its best-fit top is compared to and expressed as a ratio of that for the chlorpromazine (Rel. Emax of the reference agonist is 1.00).

[0582] To ascertain whether compounds of the disclosure are antagonists, a double-addition paradigm is employed. First, 30 μ l of a compound of the disclosure (20 M) is added (10 M final concentration) and a 15 minute incubation is performed. Then, 30 μ l of chlorpromazine (3 $\times$; EC₉₀) is added (final concentration of agonist is EC30) and cAMP accumulation is allowed to proceed for 15 minutes. The

samples are then processed for cAMP measurements as detailed above. Measurements of chlorpromazine-induced cAMP accumulation are compared to the signals elicited by the chlorpromazine following addition of vehicle instead of test compound and expressed as a ratio. 'Hits' (compounds that antagonize chlorpromazine-stimulated increases in baseline-normalized fluorescence by at least 50%) are then characterized by a modified Schild analysis.

[0583] For modified Schild analysis, a family of chlorpromazine concentration-response isotherms is generated in the absence and presence of graded concentrations of test compound (added 15 min prior to reference agonist). Theoretically, compounds that are competitive antagonists cause a dextral shift of agonist concentration-response isotherms without reducing the maximum response to agonist (i.e., surmountable antagonism). However, on occasion, factors such as non-competitive antagonism, hemiequilibria, and/or receptor reserve cause apparent insurmountable antagonism. To account for such deviations, we apply the modified Lew-Angus method to ascertain antagonist potency (Christopoulos et al., 1999). Briefly, equieffective concentrations of agonist (concentrations of agonist that elicit a response equal to the EC_{25%} of the agonist control curve) are plotted as a function of the compound of the disclosure concentration present in the wells in which they were measured. Non-linear regression of the baseline-normalized data is performed in Prism 4.0 using the following equation:

$$pEC_{25\%} = -\log([B] + 10^{-pK}) - \log c$$

where EC_{25%} equals the concentration of agonist that elicits a response equal to 25% of the maximum agonist control curve response and [B] equals the antagonist concentration; K, c, and s are fit parameters. The parameter s is equal to the Schild slope factor. If s is not significantly different from unity, pK equals pKB; otherwise, pA₂ is calculated (pA₂=pK/s). The parameter c equals the ratio EC_{25%}/[B].

Functional Efficacy Assay for 5-HT₇ Receptors Method 2:

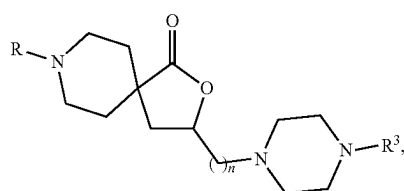
[0584] Functional efficacy of the compounds of the disclosure on 5-HT₇ serotonin receptors were measured in a cell based cAMP enzyme fragment complementation assay using the HitHunter cAMP assay (DiscoverRx). Cells stably expressing human 5HT₇ receptors were plated in 96-well plates at 4000 cells/well, 16-20 hours prior to assay in growth media (Ultraculture medium, 2 mM GlutaMax and G418 1 mg/mL. Serial dilutions of the agonist, 5-Carboxamidotryptamine (5-CT), were prepared in a final concentration range of 10 M to 10 nM. Compounds of the disclosure were prepared in 3-fold serial dilutions to obtain a final concentration range of 10 M to 0.1 nM. Compounds of the disclosure are tested for agonist activity in the absence of 5-CT and antagonist activity in the presence of 5-CT. For the cAMP assay, the protocol was followed according to the instructions provided by the supplier. Briefly, cells were incubated with a compound of the disclosure for 30 minutes at 37° C. prior to addition of EC₇₀ concentration of 5-CT. After an additional 30 minutes, cAMP antibody/cell lysis solution was added (20 L/well) and incubated for 60 minutes at room temperature. cAMP XS+EA reagent is added (20 L/well) and incubated for 2 hours at room temperature. Luminescence was read on the Envision Multilabel plate reader.

[0585] The disclosures of each and every patent, patent application, and publication cited herein are hereby incorporated herein by reference in their entirety.

[0586] While this invention has been disclosed with reference to specific embodiments, it is apparent that other embodiments and variations of this invention may be devised by others skilled in the art without departing from the true spirit and scope of the invention. The appended claims are intended to be construed to include all such embodiments and equivalent variations.

What is claimed is:

1. A compound having formula (I):



(XVII)

or a pharmaceutically acceptable salt thereof, wherein:

R is selected from the group consisting of COR^2 , CO_2R^{2a} , and SO_2R^{2d} ;

R^2 is selected from the group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;

R^{2a} is selected from the group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, and C_{3-7} cycloalkyl;

R^{2d} is selected from the group consisting of C_{1-6} linear alkyl, C_{3-7} branched alkyl, C_{3-7} cycloalkyl;

R^3 is phenyl or substituted phenyl, wherein the substituents are selected from the group consisting of hydroxyl, halo, cyano, C_{1-6} alkoxy, C_{1-6} linear alkyl, C_{3-7} branched alkyl, C_{1-6} haloalkyl, and heterocyclyl; and

n is 1, 2, 3, or 4.

2. The compound of claim 1, wherein R^2 , R^{2a} , or R^{2d} is C_{1-6} linear alkyl.

3. The compound of claim 1, wherein R is COR^2 .

4. The compound of claim 3, wherein R^2 is C_{1-6} linear alkyl.

5. The compound of claim 3, wherein R^2 is C_{3-7} branched alkyl.

6. The compound of claim 3, wherein R^2 is C_{3-7} cycloalkyl.

7. The compound of claim 1, wherein R is CO_2R^{2a} .

8. The compound of claim 7, wherein R^{2a} is C_{1-6} linear alkyl.

9. The compound of claim 7, wherein R^2 is C_{3-7} branched alkyl.

10. The compound of claim 7, wherein R^2 is C_{3-7} cycloalkyl.

11. The compound of claim 1, wherein R is SO_2R^{2d} .

12. The compound of claim 1, wherein R^{2d} is C_{1-6} linear alkyl.

13. The compound of claim 1, wherein R^{2d} is C_{3-7} branched alkyl.

14. The compound of claim 1, wherein R^{2d} is C_{3-7} cycloalkyl.

15. The compound of claim 1, wherein R^3 is selected from the group consisting of 4-hydroxyphenyl, 3-fluorophenyl, 4-fluorophenyl, 3-chlorophenyl, 4-chlorophenyl, 3,4-dichlorophenyl, 4-fluoro-3-chlorophenyl, 4-cyanophenyl, 2-methoxyphenyl, 2-methylphenyl, 3-methylphenyl, 4-methylphenyl, 2-isopropylphenyl, 4-trifluoromethylphenyl, 2-morpholinophenyl, and 4-methyl-2-morpholinophenyl.

16. The compound of claim 15, wherein R^3 is 4-fluorophenyl.

17. The compound of claim 1, wherein the compound is selected from a compound of Table 1, or a pharmaceutically acceptable salt thereof.

18. The compound of claim 1, wherein the compound is selected from a compound of Table 9, or a pharmaceutically acceptable salt thereof.

19. The compound of claim 1, wherein the compound is selected from a compound of Table 11, or a pharmaceutically acceptable salt thereof.

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